UNIORG: A CONFIGURABLE UNIVERSITY-WIDE STUDENT ORGANIZATION MANAGEMENT PLATFORM WITH OCR AND NLP FOR DOCUMENT PROCESSING AND AUTO-TAGGING

A Capstone Project

Presented to the Faculty

of College of Computer Studies

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LAGUNA STATE POLYTECHNIC UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Santa Cruz, Laguna

Philippines

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VISION

An ASEAN Polytechnic University by 2030

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3. Communicate effectively with a member or leader of the computing society with social, moral, and legal responsibilities to achieve a common goal.
4. Engage in life-long learning as a foundation for continuing professional advancement.

APPROVAL SHEET

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ABSTRACT

This study aimed to design, develop, test, and evaluate "UniOrg," a configurable university-wide student organization management platform for the Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC). Currently, the Office of Student Affairs and Services (OSAS) and the 39 recognized student organizations struggle with fragmented online tools and heavy manual paperwork for accreditation, event scheduling, and student feedback. To resolve these administrative bottlenecks, UniOrg features a multi-org architecture integrated with artificial intelligence. Specifically, it utilizes Optical Character Recognition (OCR) to automate text extraction from physical accreditation documents and Natural Language Processing (NLP) to auto-tag and summarize narrative reports. The system also features a master calendar, manual payment tracking, secure e-voting, and an interactive feedback hub. The study utilized a descriptive-developmental research design, applying the Agile methodology to systematically build and refine the web platform. The system's functionality was evaluated using black-box testing , while its user acceptance was measured using the Technology Acceptance Model (TAM). Through a mixed sampling design, key stakeholders including OSAS administrators, College Deans, organization advisers/officers, and regular students participated in the system evaluation. The implementation of UniOrg is expected to significantly streamline organizational workflows, promote campus-wide transparency, and modernize university governance.

Keywords: U*niOrg; Student Organization Management; Artificial Intelligence; OCR; NLP; TAM; Agile Methodology; Web Platform; Digital Governance; Campus Management*

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DEFINITION OF TERMS

The researchers describe the terminology used in the development of the system, which can be beneficial for a thorough understanding of the key terms used in the study

OPERATIONAL TERMS

The operational terms are not based on the general dictionary meaning but on how each term is applied in the actual operation of the system including accreditation, monitoring and evaluation of organizational activities.

UniOrg	The configurable, university-wide student organization management web platform developed specifically for Laguna State Polytechnic University - Santa Cruz Campus to streamline accreditation, activity monitoring, and document processing.
Accreditation	The administrative workflow within the system through which student organizations submit required compliance documents for verification and approval by the Office of Student Affairs and Services (OSAS).
OCR (Optical Character Recognition)	The automated text extraction feature integrated into the platform that reads physical and scanned accreditation documents to populate system fields without manual typing.

NLP (Natural Language Processing)	The intelligent text analysis module utilized in the platform to automatically categorize, tag, and summarize narrative reports and submitted organizational documents.
TAM (Technology Acceptance Model)	The evaluation framework applied in this study to measure the user acceptance and satisfaction of OSAS staff, advisers, officers, and students regarding the platform's usefulness and ease of use.
Multi-org Workspace	The isolated digital portal environment provided to each of the 39 recognized student organizations, allowing officers to manage their specific group's profiles, activities, and records independently.
Master Calendar	The centralized scheduling interface within the platform that consolidates all approved student organization activities to prevent date conflicts across the campus.

TECHNICAL TERMS

The technical terms are defined according to the frameworks, tools and processes utilized in UniOrg including OCR and NLP for document processing and auto-tagging.

UniOrg	A proper noun referring to the specific web-based software application name derived from the combination of 'University' and 'Organization.'
Accreditation	The formal process of granting official recognition or credit to an institution or organization that meets established standards.
OCR (Optical Character Recognition)	An electronic or mechanical technology that converts images of typed, handwritten, or printed text into machine-encoded text.
NLP (Natural Language Processing)	A subfield of computer science, information engineering, and artificial intelligence concerned with the interactions between computers and human languages.
TAM (Technology Acceptance Model)	An information systems theory that models how users come to accept and use a technology based on perceived usefulness and perceived ease of use.

Multi-org Workspace	A software multi-tenancy architecture where a single software instance serves multiple distinct user organizations with data isolation.
Master Calendar	A principal schedule or database that aggregates multiple individual calendar entries into a single, comprehensive timeline.

CHAPTER I

INTRODUCTION

1.1 Project Context

Efficient organizational governance and streamlined document processing are essential pillars of a well-functioning university student affairs system. At Laguna State Polytechnic University, administrators and student leaders have observed increasing challenges in managing organization activities, document accreditation, and student feedback. These issues are heavily influenced by the reliance on manual paper transactions and scattered online platforms, which lead to delayed approvals, disorganized records, and a lack of secure communication spaces for students. To understand the depth of this concern, the researchers conducted preliminary interviews with key university personnel and faculty members, namely OSAS Chairperson Christian P. San Luis, LPT, PhD, OSAS Secretary Nicole D. Bartolome, Organization Senior Adviser Rachiel R. Rivano, PhD, and Organization Junior Adviser Harlene G. Origines. The feedback gathered revealed a common observation: while the student organizations are highly active, the administrative burden of manually tracking accreditation documents, managing event records, and gathering honest student feedback requires a modernized, centralized digital intervention to sustain operational efficiency and improve student services.

In many Philippine universities, student affairs offices face the same shift toward digital record keeping, and several institutions already use centralized portals for organization transactions, which shows that manual paper based processing is no longer enough for a campus the size of LSPU.

The accreditation requirements show how heavy the manual work is. Each organization submits a letter of application for renewal and recognition, updated constitution and by-laws, renewal form, program of activities for one year, adviser commitment form, certification of officers, and list of officers and members with one by one picture. At the organization level, officers handle events, payments and elections through scattered online apps, students have no single place for information, and many hesitate to give feedback because there is no secure and private channel.

In response, the researchers proposed the development of UniOrg A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging. This system includes features such as a multi-org architecture for custom organization profiling, an accreditation document repository, an activity master calendar, and centralized payment and election management. To significantly reduce the administrative workload, UniOrg integrates document technologies like Optical Character Recognition (OCR) and Natural Language Processing (NLP) for automated text extraction, auto-tagging, and summarization of submitted requirements. By transforming tedious manual tasks into automated digital workflows, the proposed platform aimed to create a more organized, responsive, and data-driven governance environment.

The technology gap is the absence of a single university wide platform that combines accreditation tracking with OCR and NLP document processing, since existing tools in the campus handle only isolated tasks and cannot read, tag and summarize accreditation papers automatically. This need is supported by the gap analysis in Chapter II, where existing systems lack the combined accreditation and AI document features summarized in Table 2.1.

The platform is strictly accessible through a web-based application, ensuring a standardized and centralized experience for all users. User interactions are governed by strict role-based access OSAS administrators can monitor university-wide compliance and verify documents, organization officers can manage day-to-day operations and students can view announcements, access academic resources, and securely submit feedback through their respective web portals. The specific development tools and technologies used in this project are PHP (Laravel Framework), Supabase, Tesseract OCR, Cloud Vision API, Cloud Natural Language API, Groq AI models, Cloudflare R2 (S3-compatible storage), HTML, CSS, JavaScript, DomPDF, PHPSpreadsheet, and PDF Parser, and the system is built with scalability, usability, and security in mind.

Once fully implemented, UniOrg is expected to significantly enhance institutional transparency, streamline document processing, and improve communication, thereby contributing to a highly efficient and modern academic environment.

Beyond convenience, this digital shift matters for the quality of student services. By turning manual tasks into organized digital processes with strict checking and AI assisted tracking, the project supports SDG 4 on Quality Education and SDG 16 on Peace, Justice and Strong Institutions, so that organizations and administrators work with honesty, clear responsibility and readiness.

1.2 Research Objectives

1.2.1 General Objective

The general objective of this study is to design, develop, test and

evaluate UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging, a web based platform for the Office of Student Affairs and Services of LSPU Santa Cruz Campus.

1.2.2 Specific Objectives

This study evaluates UniOrg, a multi-org web platform featuring an integrated master calendar, OCR text extraction, and NLP document processing. Built for LSPU–OSAS, UniOrg automates accreditation reviews, centralizes data management, and streamlines student organization governance.

Specifically, this study aimed to:

1. To design and develop a configurable web platform (UniOrg) that centralizes student organization accreditation, document storage and activity monitoring for the Office of Student Affairs and Services. The platform provides organization profiling with role based access for OSAS, Organization advisers, Organization officers and students.
2. To test the functionalities of the developed UniOrg system based on the following ten (10) major features:

   2.1 Multi-Org Workspace and Organization Profiling

   2.2 Accreditation Tracking and Document Repository

   2.3 Activity Scheduling and Master Calendar Management

   2.4 Centralized Payment

2.5 Student Governance and Election System

2.6 Academic Resource Management

2.7 Communication and Feedback Hub

2.8 Role-Based Access Control (RBAC) and Security Management

2.9 Optical Character Recognition (OCR) for Automated Document Data Extraction

2.10 Natural Language Processing (NLP) for Text Summarization and Auto-Tagging

3. To evaluate the user acceptance of the developed UniOrg system among students, student organization officers and OSAS administrators of Laguna State Polytechnic University Santa Cruz Campus using the Technology Acceptance Model (TAM) in terms of:

3.1 Quality Factors

3.2 Perceived Ease of Use

3.3 Perceived Usefulness

3.4 Attitude Toward Using

3.5 Behavioral Intention to Use

3.6 Experience/Actual Use

1.3 Conceptual Framework

1.3.1 Components of the Conceptual Framework

The framework is composed of three linked parts. The Input gathers

user data from OSAS, officers and students with accreditation and financial records and current operational inefficiencies with system and technology requirements for OCR and NLP. The Process applies Agile Methodology in building OSAS level and organization level modules and use of OCR and NLP variables for extraction and auto tagging with TAM evaluation. The Output is the functional UniOrg System with a feedback loop that returns evaluation results to the inputs for refinement.

These inputs, processes and outputs were drawn from the synthesis of related literature and related systems in Chapter II, which showed that OCR and NLP features answer the manual processing gaps found in existing systems.

1.3.2 Conceptual Framework Diagram

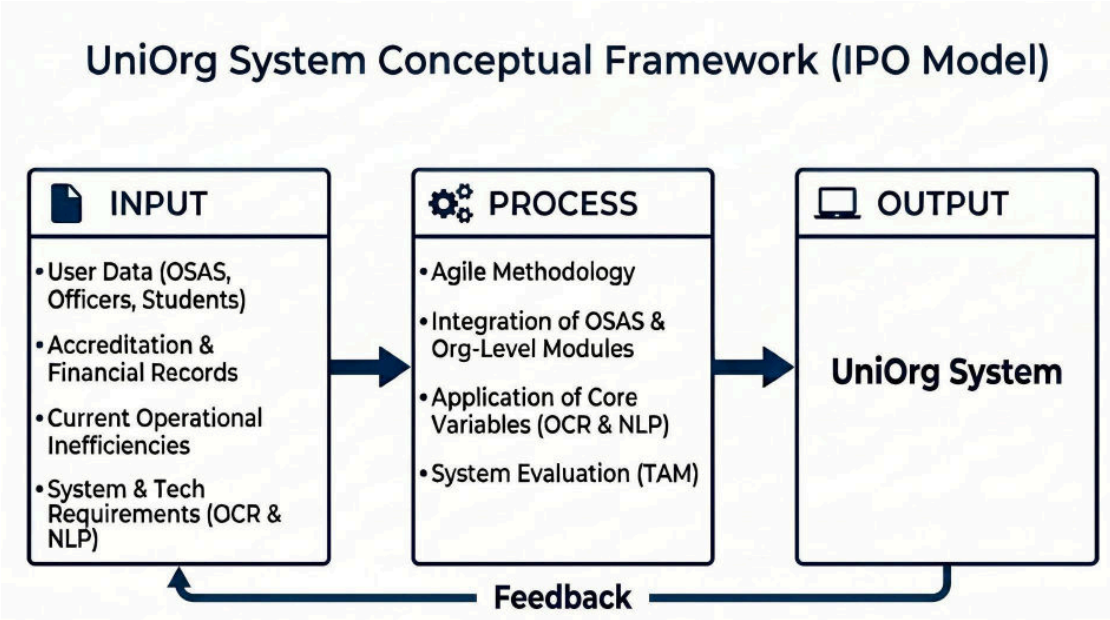


Figure 1.1. Conceptual Framework of UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging

As illustrated in Figure 1.1, the conceptual framework of this study utilizes the Input-Process-Output (IPO) model to explicitly demonstrate the research flow, which is anchored on the identified problem of manual student organization governance.

The Input phase identifies the current operational inefficiencies of the Office of Student Affairs and Services (OSAS). This phase also encompasses the raw organizational and financial records, user data, and the necessary system and technological requirements, specifically the integration of Optical Character Recognition (OCR) and Natural Language Processing (NLP).

The Process phase details the Agile Methodology utilized to develop UniOrg's core features, which are categorized into OSAS-Level Management and Configurable Organization-Level Modules. Crucially, building upon the synthesis of related literature, this phase incorporates the core research variables (OCR and NLP) to automate document data extraction and auto-tagging. Furthermore, this phase establishes the procedure for rigorous system evaluation using the Technology Acceptance Model (TAM).

Finally, the Output phase yields the fully functional UniOrg System. To ensure continuous improvement, a Feedback loop is integrated, connecting the Output back to the Input. This loop signifies that the evaluation results (based on TAM criteria) and any identified resolved or unresolved process bottlenecks are used to further refine the system's inputs and processes, ensuring the platform remains highly effective in addressing the defined research problems.

1.4 Project Purpose

The purpose of this study is to design, develop, and evaluate UniOrg, an innovative university-wide student organization management platform integrated with Optical Character Recognition (OCR) and Natural Language Processing (NLP). This study aimed to provide a centralized, configurable, and efficient digital hub that addresses the challenges of manual accreditation and document processing at Laguna State Polytechnic University. By automating document reading and summarizing narrative reports, the system intends to improve operational efficiency and transparency across all student organizations. Through the integration of AI-driven features, UniOrg aimed to foster a more organized and modern campus culture, ensuring that organizational governance is both accurate and accessible.

This study sought to improve university-wide engagement by providing a unified platform for tracking activities and maintaining a master calendar to prevent scheduling conflicts. The platform is designed to be highly configurable, allowing each student organization to have a dedicated workspace that aligns with their specific needs while remaining under the supervision of the Office of Student Affairs and Services (OSAS). By rewarding organizational transparency and streamlining the submission of requirements, UniOrg aimed to cultivate a more professional and collaborative environment among student leaders and administrators. Ultimately, the findings of this study contributed to the digital transformation of university governance and provide a replicable framework for other educational institutions.

1.4.1 Intended Beneficiaries

Office of Student Affairs and Services (OSAS)

The UniOrg system provides OSAS with a centralized platform to monitor all currently 39 student organizations effectively. It helps administrators verify accreditation documents faster through AI-driven OCR while utilizing NLP to search and summarize activity records. OSAS gains deeper insights into organizational performance, allowing them to provide better guidance and oversight.

Student Organizations (Org Officers and Org Advisers)

Organization leaders benefit from a professional workspace to manage their respective groups, streamline document submissions, and maintain digital records. The system helps officers motivate their members through a structured information hub while saving time on manual tracking of financial reports and activity proposals. It fosters accountability by giving them a clear view of their accreditation status and upcoming deadlines.

Students

Students benefit from UniOrg by having a more accessible and interactive way to engage with university organizations. The system motivates them to participate in events and stay informed about announcements through a unified platform. By providing a digital E-Library and collaborative spaces like EduShare, the platform supports their academic journey and strengthens

their connection with their chosen student organizations.

The School

For the university, the UniOrg system serves as a digital innovation that promotes a positive organizational culture and strengthens the link between administrators and the student body. It supports the institution's goal of enhancing educational quality through technology-driven governance. The system also provides data-driven insights that can be used for institutional performance analysis and accreditation purposes.

Researchers

This study provides the researchers with the opportunity to apply advanced technical skills in multi-org architecture and AI integration. By working on the system, the researchers enhance their expertise in solving real-world administrative challenges. The project allows them to gain valuable experience in software engineering and system evaluation using industry-standard models.

Future Researchers

This study acted as a guide and reference for future researchers who may be conducting investigations in the field of educational technology and administrative automation. It provides a foundational framework for similar initiatives, offering ideas and methodologies that can be adapted for other university-wide digital systems.

1.4.2 Expected Benefits

In general the expected benefits focus on faster accreditation checking with OCR and NLP support, organized records with a master calendar, clear payment and election tracking with officer user role based access, and a secure feedback channel for students. These improvements help OSAS supervision and daily work of organizations without claiming that every manual problem is removed.

1.5 Scope and Limitations of the Study

1.5.1 Scope of the Study

This study focuses on the design, development, testing, and evaluation of UniOrg, integrating Optical Character Recognition (OCR) and Natural Language Processing (NLP) to address the manual bottlenecks in student organization governance. The scope of the research is strictly confined to the Laguna State Polytechnic University (LSPU) - Santa Cruz Campus. Data collection and system evaluation involved a specific demographic of respondents: the Office of Student Affairs and Services (OSAS) administrators, the officers and organization advisers of the 39 recognized student organizations, the college Deans, and the enrolled students. Furthermore, the research utilizes the Technology Acceptance Model (TAM) to evaluate the respondents' user acceptance of the platform specifically in the context of accreditation, document processing, and activity tracking.

The system covers ten major modules from multi-org workspace and accreditation tracking to OCR and NLP processing, and it runs on a web based

environment using PHP Laravel, Supabase, Cloudflare R2, Tesseract OCR, Google Cloud Vision and Natural Language APIs, Groq AI models, DomPDF and PHPSpreadsheet. Development and evaluation were conducted at LSPU Santa Cruz Campus within Academic Year 2025 to 2026.

1.5.2 Limitations of the Study

However, the study presents certain limitations in its research boundaries. First, the investigation is strictly bounded to the administrative processes of student organization management; it explicitly excludes the observation and integration of other OSAS operations, such as student discipline, guidance and counseling, university scholarships, and lost and found services. Second, the study does not cover the evaluation of fully automated financial systems; the research scope for payment tracking is limited to manual routing via personal e-wallet accounts, as it does not include the integration of official payment gateway APIs or the university's central accounting office. Consequently, the study does not investigate the processing of university-wide tuition and miscellaneous fees. Lastly, the findings, user acceptance results, and conclusions formulated in this study are limited only to the LSPU Santa Cruz Campus context and may not be generalizable to other LSPU satellite campuses or different educational institutions.

The study is also limited by internet dependency since UniOrg is web based, by time constraints of the academic calendar, and by availability of complete accreditation records from the organizations.

Processes that are unique to certain organizations and not common to all student organizations are also excluded, since the system covers only the

general accreditation, document and activity workflows shared across the 39 organizations.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents related literature and studies gathered from credible academic journals and research articles that support the proposed UniOrg system. Reviewing these sources helps the researchers establish a strong theoretical foundation and identify gaps in the current manual management of student organizations.

The review covers local and foreign studies on key topics aligned with the system's core features, such as multi-org architecture, digital document repositories, Role-Based Access Control (RBAC), Optical Character Recognition (OCR), and Natural Language Processing (NLP). Ultimately, these insights guide the development of UniOrg in streamlining accreditation processes and fostering a modernized, data-driven campus environment.

2.1 Studies and Literature

Multi-Org Workspace and Organization Profiling

Managing data for multiple student groups within a single system requires strict boundaries and access rules. A recent study by Wang (2025) on campus service platforms highlights the importance of using a Software as a Service (SaaS) architecture equipped with multi-org data isolation and precise permission allocation. Implementing these technical features allows a system to handle large-scale data without overlapping or compromising user information. This structural approach directly mirrors the foundation of UniOrg’s Multi-org Workspace and Organization Profiling module. By adopting a multi-org design, the system ensures that the records,

activities, and documents of different student organizations remain completely separate and secure from one another, while customized profiling guarantees that only authorized officers can access and modify their specific group's workspace.

Creating secure, dedicated digital environments for different groups sharing the same infrastructure is a major technical challenge. Exploring this, Al-Jody et al. (2021) demonstrated that implementing a multi-org architecture with secure portal access provides a flexible and isolated workspace for different sets of users. This framework supports the core design of UniOrg's Multi-org Workspace and Organization Profiling module. By adopting a multi-org structure, the platform allows various student organizations to operate within the same system without their data overlapping. Each organization is provided with its own secure profile and dashboard, ensuring that their specific activities, members, and records remain private and independently managed.

Traditional management systems often struggle to adapt to the diverse requirements of different university groups. Addressing this limitation, Azouzi et al. (2025) found that transitioning to a cloud-based, multi-org architecture enables dynamic scalability and highly efficient resource sharing. This architectural approach is central to the design of UniOrg's Multi-org Workspace and Organization Profiling module. Rather than deploying separate applications for every student organization, the system utilizes a shared multi-org infrastructure. This setup ensures that the university can efficiently manage its technological resources while still giving each organization its own flexible and independent workspace to handle its specific activities and profiles.

Navigating through disconnected applications and multiple websites can easily frustrate college students trying to stay updated with campus activities. A recent study by Llicsi Angelo Miguel D et al. (2023) shows that building a centralized online campus platform effectively solves this by placing all necessary services in one location while strictly maintaining privacy and security. This concept is the main driving force behind UniOrg's Multi-org Workspace and Organization Profiling module. Instead of forcing students to use different apps for different campus clubs, the multi-org architecture brings all organizations into a single system. Because each organization has its own isolated and customized profile, students can easily interact with their specific groups in a safe digital environment, making their overall campus experience much more organized and engaging.

Managing vast amounts of student records requires a strong technical foundation that did not slow down during busy campus events. Research by Chen (2023) demonstrates that building an integrated student management platform using cloud computing effectively handles large volumes of data while keeping network performance fast and stable. This technical reliability is exactly why UniOrg relies on a shared cloud infrastructure for its Multi-org Workspace and Organization Profiling module. Because several student organizations operate within the same multi-org system, the platform must support simultaneous users without crashing. With this setup, student officers can smoothly log into their specific workspaces, manage their members' profiles, and update their group's information without experiencing frustrating delays, giving them more time to focus on actually leading their fellow students.

As university enrollments expand, the demand for fast and reliable

management systems becomes critical for smooth campus operations. Miao (2022) observes that integrating a distributed cloud computing system into educational platforms significantly improves processing speed and resource utilization compared to older, traditional setups. This technological advantage is highly relevant to the multi-org architecture of the UniOrg module. Since the platform hosts multiple student organizations within a single shared environment, utilizing efficient cloud-based processing ensures that the system remains stable even during peak usage. Because of this, student officers can quickly access their dedicated workspaces, update their organization profiles, and manage member records without dealing with slow load times, allowing them to dedicate more time to actual campus leadership rather than waiting on a loading screen.

Developing campus software that can quickly adapt to the unique needs of various student groups is a major technical challenge. Hadi et al. (2025) note that traditional applications often struggle to keep up with these changing demands, suggesting that adopting a highly modular architecture makes educational platforms much more flexible and independent. This structural approach perfectly supports the backend design of UniOrg's Multi-org Workspace and Organization Profiling module. Because the platform is built to be highly modular, it can simultaneously manage the isolated workspaces and profiles of multiple student organizations without data interference. Ultimately, this ensures that even if several campus clubs are heavily using and updating their individual portals at the same time, the entire system remains stable, providing a smooth and reliable online experience for all students.

As university populations grow and branch out into numerous campus groups,

keeping track of their respective activities and records becomes a logistical challenge. Niu (2021) explains that building a unified, cloud-based management system effectively organizes scattered information and strongly promotes autonomous management among different student groups. This principle directly supports the architecture of UniOrg's Multi-org Workspace and Organization Profiling module. Instead of dealing with fragmented tools, different student organizations operate within a single, unified platform. Because the system utilizes a multi-$_{org}$ design, each organization is equipped with its own distinct and isolated workspace. This setup empowers student officers to independently manage their own members and update their specific profiles, drastically reducing administrative confusion while ensuring that all campus groups stay organized.

Managing the activities and records of different university groups often leads to inconsistent and messy data if they do not share a standard platform. Research by Ma et al. (2021) highlights that building a unified digital campus system standardizes information while still allowing individual departments to manage their specific workflows flexibly. This framework directly applies to the objective of UniOrg's Multi-org Workspace and Organization Profiling module. By implementing a centralized multi-org structure, the system ensures that all student organizations operate under one secure infrastructure. At the same time, each group is provided with an independent workspace where student officers can customize their organization's profile and manage their own members, keeping their data organized without conflicting with other clubs.

Requiring students to navigate through several disconnected websites to

access different campus services often creates unnecessary confusion and administrative heavy lifting. Lin (2023) points out that the redundant construction of multiple systems with separate login entrances is a major flaw in traditional campus management, noting that integrating these into one unified platform greatly improves data handling. This directly highlights the value of UniOrg's Multi-org Workspace and Organization Profiling module. Rather than building separate applications for every student group, the system offers a single entry point for the entire campus. Because it operates on a multi-org architecture, different student organizations can share this main platform while still keeping their own private and customizable workspaces. This setup eliminates digital clutter, allowing student officers to manage their individual groups efficiently within one organized environment.

Accreditation Tracking and Document Repository

Handling physical paperwork for annual recognition processes can be extremely tiring and prone to document loss for both student officers and university administrators. Research by Kommey et al. (2022) demonstrates that shifting to a secure, web-based management system solves these manual challenges by establishing a central database for all required files. This digital shift is the exact purpose of UniOrg's Accreditation Tracking and Document Repository module. Instead of requiring student organizations to submit thick folders of physical documents to the administration, the system provides a secure online vault where they can safely upload their action plans, member rosters, and financial reports. Because the platform tracks the approval status of these submissions, student leaders no longer

have to manually follow up on the progress of their accreditation, making the entire compliance process much faster and far less stressful.

Preparing long lists of required documents often becomes chaotic when there is no clear way to track what has already been submitted and what is still missing. Gunawan et al. (2023) point out that the lack of proper monitoring systems and scattered data heavily burdens the accreditation process, noting that building a dedicated web-based dashboard effectively resolves these document preparation issues. This perfectly mirrors the goal of UniOrg's Accreditation Tracking and Document Repository module. When student officers prepare their yearly recognition requirements, keeping track of every single file can be confusing. By utilizing a centralized tracking dashboard, student organizations can easily check the exact status of their submission, whether pending, under review, or approved. Combined with the integrated repository, this clear visibility prevents last-minute scrambling for missing paperwork and allows the campus administration to evaluate the organizations efficiently.

Maintaining a high standard of performance year after year requires campus groups to keep meticulous records of their activities. Javed et al. (2023) emphasize that achieving sustainable accreditation relies heavily on having a comprehensive data management system capable of handling extensive records to track performance accurately. For student clubs, proving their active campus involvement means submitting volumes of documentary evidence. This aligns directly with the goal of UniOrg's Accreditation Tracking and Document Repository module. By providing a secure digital vault, the platform allows student officers to systematically upload and

organize their required documents. Because the system securely holds these files, important records are protected from getting lost even when a new batch of student leaders takes over the following year, making the entire evaluation process much easier for the campus administration.

Processing physical documents for student events and organizational approvals frequently causes severe administrative backlogs in universities. Aziz (2021) points out that managing the proposals and compliance reports of numerous student clubs traditionally relies on a heavy physical approach, noting that a web-based system effectively eliminates these delays by tracking and storing files online. This specific transition from physical paperwork to digital storage is the exact foundation of UniOrg's Accreditation Tracking and Document Repository module. Rather than forcing student officers to print thick folders for their activity reports and compliance requirements, the system allows them to upload all supporting documents into a secure digital vault. The platform then tracks the approval progress, allowing both the students and the campus administration to monitor submissions easily without the need for exhausting physical follow-ups.

Manual evaluations of campus compliance requirements often lead to long waiting times and misplaced paperwork. Mayulu et al. (2024) observe that implementing an automated accreditation system drastically reduces processing time by up to half while minimizing manual errors and improving transparency. For student organizations, waiting weeks just to find out if their renewal papers are approved significantly delays their ability to operate and hold events. UniOrg's Accreditation Tracking and Document Repository module directly addresses this

bottleneck. By replacing physical submissions with a secure digital vault, the platform allows the administration to review uploaded documents efficiently. Because the system provides clear, real-time status updates, student officers experience a transparent and much faster recognition process without the usual administrative delays.

Preparing for annual campus evaluations often exposes severe limitations in time and administrative support. Montano et al. (2025) reveal that despite shared efforts among campus stakeholders in the Philippines, poor documentation practices and a lack of proper digital infrastructure create significant operational barriers during the accreditation period. For student organizations, compiling a whole year's worth of paperwork with limited resources causes massive stress when deadlines approach. This is exactly where UniOrg's Accreditation Tracking and Document Repository module becomes essential. By acting as the required digital infrastructure, the system allows student officers to maintain consistent documentation throughout the semester. Instead of rushing to build thick physical folders right before the deadline, they can progressively upload their files into the repository, ensuring early preparation and minimizing administrative bottlenecks.

Providing a broader perspective on the bureaucratic hurdles of institutional compliance, Duarte and Vardasca (2023) investigated the accreditation processes in higher education institutions across multiple countries. Their comprehensive review highlighted that while accreditation frameworks exist to ensure operational quality, they often impose a significant bureaucratic burden on the involved entities. The study emphasized that the accreditation process becomes particularly complex and time-consuming when organizations sought approvals for the first time or introduce

new workflows. The authors concluded that to expedite accreditation, applying entities must strictly meet established criteria and provide all required documentation promptly and accurately, suggesting that streamlined communication and efficient document handling with accrediting bodies are absolutely essential.

Poor turnover of documents between changing batches of student leaders often leads to lost institutional memory and broken compliance records. Nailalmarom et al. (2025) highlight that implementing structured digital archives within student organizations significantly improves administrative efficiency and ensures that important knowledge regarding correspondence and compliance reporting is properly passed down. For campus groups, losing access to past accountability reports or budget formats makes yearly accreditation an exhausting process for incoming officers. UniOrg's Accreditation Tracking and Document Repository module directly addresses this by functioning as a centralized digital vault. Instead of misplacing files when new officers take over, student organizations can securely store their past and present compliance documents in the system, ensuring smooth continuity and an easier submission process to the campus administration.

Securing multiple departmental signatures for campus clearances traditionally forces student leaders to endure exhausting manual procedures. Eweoya et al. (2025) explain that replacing paper-based processes with a web-based clearance system allows for centralized document submission and real-time progress monitoring, which successfully eliminates misplaced files and administrative bottlenecks. For student organizations, trying to get clearance for their yearly accreditation often means walking from one office to another just to follow up on their pending papers. This

directly mirrors the function of UniOrg's Accreditation Tracking and Document Repository module. By digitizing the clearance workflow, student officers can simply upload their required documents into the repository and track exactly which stage of the approval process their papers are in, removing the need for stressful physical follow-ups.

Carrying physical forms to various campus offices for signatures often causes severe frustration and long lines during the compliance period. May et al. (2023) observed a similar situation in a Philippine academic institution, noting that transforming manual clearances into a web-based, paperless transaction completely eliminates the physical hassle of securing approvals. For student organizations, moving from building to building just to get their accreditation papers signed by the Student Affairs office is an exhausting routine that takes time away from actual event planning. This perfectly aligns with the core function of UniOrg's Accreditation Tracking and Document Repository module. By converting the manual signature hunt into an online tracking dashboard, the system allows the campus administration to approve or reject submitted documents directly on the platform, providing student officers with a clear and completely paperless mechanism to monitor their organizational clearance.

Activity Scheduling and Master Calendar Management

Coordinating campus activities without a unified platform often generates massive administrative workloads and miscommunication between student groups and the administration. Blancaflor et al. (2021) observed this within a Philippine

university, finding that developing a dedicated event management web application significantly reduces the burden of processing event applications and improves overall coordination. For student organizations, the lack of a shared system means they have to manually submit event proposals, making it difficult for the Student Affairs office to monitor which activities are happening and when. This highlights the necessity of UniOrg's Activity Scheduling and Master Calendar Management module. By providing a centralized digital space, the system allows student officers to submit their event applications directly on the platform. Once approved by the administration, these activities are automatically plotted onto a shared master calendar, ensuring that all campus events are organized and visible to everyone.

Relying on traditional pen and paper to schedule and announce campus activities often leads to miscommunication and poorly organized event timelines. Pugeda et al. (2024) note that while student groups exert effort in planning, the manual approach to event management creates unnecessary challenges and reveals a strong need for operational improvement. For student officers, passing around physical memos or writing on campus whiteboards to inform others about an upcoming activity is an outdated method that fails to reach the entire student body effectively. UniOrg's Activity Scheduling and Master Calendar Management module directly upgrades this process. By shifting away from manual announcements, the system provides a centralized digital calendar where all approved organizational events are instantly published. This guarantees that every student and campus club can view the exact schedule and details of an activity in real-time, replacing the slow and unreliable nature of paper-based event dissemination.

Navigating disorganized and fragmented approval workflows often discourages student leaders from proposing campus activities. Agpalo et al. (2025) observed this issue within the Philippine educational setting, proving that replacing analog coordination methods with a digital scheduling platform drastically reduces administrative delays and significantly increases the number of student-initiated extracurricular events. For university clubs, the tedious process of finding a vacant date without a clear guide drains their momentum before the actual planning even begins. UniOrg's Activity Scheduling and Master Calendar Management module acts as this definitive digital guide. By providing a transparent master calendar, student officers no longer have to guess which dates are available or undergo long administrative checks. They can simply look at the unified schedule, pick an open slot, and quickly submit their activity request, making the entire scheduling process fast and encouraging.

Missing out on important campus activities is a common frustration when event announcements are scattered across different bulletin boards and social media pages. Nimje et al. (2024) highlight that centralizing event information into a single platform prevents missed updates and significantly improves student participation. For regular students and club officers, keeping track of when and where an activity happens is difficult if there is no unified schedule to check. UniOrg's Activity Scheduling and Master Calendar Management module serves as this exact central information hub. By automatically displaying all approved activities on one shared calendar, the system ensures that the entire student body is aware of upcoming events, effectively replacing scattered online announcements with a single, reliable schedule.

Relying on scattered paper documents to track campus activities often creates massive administrative blind spots for both student leaders and university officials. Marzan et al. (2025) observed in a Philippine university that manual workflows and fragmented data storage lead to significant errors in monitoring student activities. For student officers, planning a new event becomes highly difficult when there is no centralized reference to check if a date or venue is actually available. UniOrg's Activity Scheduling and Master Calendar Management module resolves this fragmented setup. By centralizing all approved activity schedules into one digital master calendar, the system allows both the campus administration and the student clubs to accurately track upcoming events, completely replacing the outdated and error-prone manual tracking methods.

Manual event planning procedures frequently cause overlapping schedules and poor communication among campus groups. Sajor et al. (2025) observed this issue within a Philippine state university, highlighting that replacing manual scheduling with an integrated digital platform successfully reduces date conflicts and improves overall student attendance. For campus clubs, organizing an activity without knowing the plans of other organizations often leads to divided audiences and wasted resources. UniOrg's Activity Scheduling and Master Calendar Management module directly addresses this recurring problem. By utilizing a shared digital calendar, student officers can view existing event plots before submitting their proposals to the administration, ensuring their activities do not clash with others and protecting their events from low student turnout.

Student activities that are not integrated into a central system often suffer from

chaotic logistics, resulting in long queues and lost administrative data. Julian et al. (2021) examined an event management system for a student association and found that shifting from manual coordination to a digital platform significantly improves the governance of activities and overall service efficiency. For campus organizations, trying to manage event schedules without a unified tool leads to massive delays and a waste of manpower. This underscores the value of UniOrg's Activity Scheduling and Master Calendar Management module. By integrating every organizational activity into a single digital master calendar, the system removes the confusion of non-integrated scheduling. This ensures that every event is well-managed and visible to the administration, preventing the administrative bottlenecks and logistical errors common in manual student governance.

Manual scheduling in a large university often ignores the hierarchical importance of events, leading to disorganized calendars where minor meetings frequently clash with major institutional activities. Maina et al. (2024) highlight that utilizing mechanized scheduling systems significantly minimizes overlaps and improves the overall efficiency of managing campus events. For student affairs administrators, juggling requests from various offices without a clear, automated tool leads to constant logistical headaches and a waste of manpower. This underscores the necessity of UniOrg's Activity Scheduling and Master Calendar Management module. By providing a centralized digital space to plot all activities, the system ensures that organizational events are handled systematically. Once an event is submitted and processed, it is automatically added to a unified master calendar, preventing simultaneous bookings of the same time slots and providing a reliable scheduling experience for both student leaders and university staff.

Manually tracking school-wide activities often becomes chaotic for students and staff when there is no centralized visual tool to monitor upcoming dates. Khobragade (2024) explores this by developing an event management system that utilizes an integrated calendar to simplify how organizers schedule and track school activities. For student leaders, the absence of a visual calendar often makes it difficult to grasp the timeline of campus events, leading to poor participation and scheduling overlaps. UniOrg's Activity Scheduling and Master Calendar Management module applies this visual logic by displaying all approved organizational activities on a public dashboard. By featuring an integrated calendar, the system allows student officers to generate events that are instantly visible to the entire campus, ensuring that everyone stays informed about the exact dates and details of upcoming activities.

Limited awareness of campus activities often results in low student participation and a disconnected campus community. Rice et al. (2025) observe this within a university environment, noting that traditional communication channels fail to disseminate event information effectively, leading to poor student engagement in extracurricular life. For university clubs, having a well-planned activity is ineffective if the student body is unaware of its schedule or location. UniOrg's Activity Scheduling and Master Calendar Management module serves as the primary solution to this visibility problem. By providing an integrated calendar of events, the system acts as a centralized information hub where students can easily discover and track all organizational activities. This accessibility ensures that schedules are no longer hidden or scattered, directly increasing student involvement and creating a more active campus environment.

Centralized Payment

Manual financial management within student organizations often leads to issues with transparency and accountability due to unorganized record-keeping. Cahyanto (2024) observed that implementing digital tools for financial administration significantly improves the quality of governance and strengthens coordination among organization members. For student leaders, managing membership fees and event collections through paper-based logs or scattered files makes it difficult to track funds accurately, often causing mistrust within the organization. UniOrg's Centralized Payment module directly solves this by providing a dedicated digital hub for all financial transactions. By digitizing fee collections and fund management, the system ensures that every payment is recorded in real-time, promoting a more transparent and professional culture where financial reports can be generated and audited without the usual administrative confusion.

Inaccurate recording of educational payments frequently results from manual administrative procedures that are prone to human error. Chandra et al. (2025) note that implementing a web-based accounting information system improves the precision of payment administration and ensures more transparent financial reporting. For student organization officers, managing donations and membership dues without a structured digital tool often leads to lost records and difficulty in providing real-time updates to members. UniOrg's Centralized Payment module directly integrates these accounting principles by automating transaction verification and payment status

monitoring. By leveraging a centralized system, organizational treasurers can quickly access financial data and prepare automated reports, reducing administrative costs and ensuring that all stakeholders have clear visibility into the organization's funds.

Relying on traditional manual methods for collecting fees frequently creates a time-consuming and error-prone environment for financial administrators. Sipenji et al. (2024) observed that transitioning to an automated fee collection system significantly improves financial accountability by streamlining how transactions are tracked and managed. For student organization officers, manually receiving cash for membership dues or event tickets and recording them in physical logbooks often leads to miscalculations and a lack of transparency among members. UniOrg's Centralized Payment module directly addresses this operational flaw. By providing a structured digital platform for financial transactions, the system eliminates the need for manual cash handling, ensuring that all organizational funds are securely collected, accurately tracked, and easily auditable.

Recording contribution fees manually often subjects administrative departments to lengthy procedures and a high risk of erroneous data. Arifin et al. (2023) highlight that digitizing payment administration through a web-based system significantly decreases data inaccuracy and makes the recapitulation of funds much more efficient. For student organizations, treasurers face the exact same administrative burden when they rely on physical logbooks to track membership dues and event fees, often resulting in missing records and unbalanced ledgers. UniOrg's Centralized Payment module eliminates this manual bottleneck. By providing a dedicated digital ledger, the system allows student officers to record, verify, and

summarize payments online. This ensures that financial data recapitulation is handled efficiently, protecting the organization from the common recording errors found in traditional paper-based collection methods.

Verifying payments through physical receipts or scattered chat messages often results in duplicate records and highly delayed financial processing. Subhana (2025) highlights that transitioning from manual spreadsheets to a centralized web-based framework significantly resolves these administrative challenges by utilizing payment validation based on uploaded proofs of transfer. For student organizations, treasurers frequently struggle to confirm who has paid their membership dues because transaction screenshots are usually sent through disorganized messenger groups, making it difficult to match payments with student profiles. This highlights the exact core mechanism of UniOrg's Centralized Payment module. Instead of relying on expensive third-party payment gateways, the system provides a dedicated portal where students can securely upload their digital proof of payment. The organizational treasurer can then review and validate these submissions directly within the platform, ensuring faster administrative processing and reducing the risk of data inconsistencies in the organization's financial records.

Supervising the funds of student organizations frequently becomes a highly compromised administrative process when financial checking relies solely on disjointed physical evidence. Christyawan et al. (2021) highlight that the supervision of student funds often fails to run optimally because the validation phase is limited to manually checking the completeness of physical bills and receipts, creating major gaps in financial accountability. For student legislative bodies and organization

treasurers, auditing scattered physical receipts makes it incredibly difficult to verify the legitimacy of every transaction, often leading to unverified financial reports. UniOrg's Centralized Payment module directly resolves this supervisory gap. By mandating the digital upload of transaction receipts into a structured portal, the system allows student officers to systematically trace, cross-check, and validate every payment online. This ensures that all financial evidence is permanently attached to the digital ledger, replacing the flawed manual checking method with an organized and auditable digital trail.

Handling cash transactions poses significant administrative burdens and security risks, forcing many campus institutions to transition toward digital payment alternatives. Espeleta (2022) investigated the local educational landscape in the Philippines and found that while students heavily prefer using mobile wallets like GCash due to its accessibility, institutions still lack a systematic model to properly organize, control, and verify these online payments. For student organizations, treasurers face this exact dilemma daily: members naturally pay their fines or dues by sending money through e-wallets, but without a standard platform to collect the proof of these transfers, tracking who actually paid becomes completely chaotic. UniOrg's Centralized Payment module addresses this gap by serving as the dedicated systematic model for student clubs. Instead of relying on disjointed personal messages for payment confirmation, the system provides a structured digital portal where students can formally upload their e-wallet transaction receipts, allowing officers to safely and efficiently manage digital collections.

Relying on physical cash and face-to-face meetups to settle campus fees often

creates significant inconvenience for students navigating busy academic schedules. Shahril et al. (2025) demonstrated that the adoption of digital transaction methods, such as QR code payments, is highly driven by their perceived usefulness and ease of use, providing much-needed practicality and efficiency in campus environments. For student organization members, finding a common time to physically hand over cash to a treasurer is a recurring hassle that often leads to delayed payments. UniOrg's Centralized Payment module capitalizes on this student preference for digital efficiency. By allowing student officers to display their official e-wallet QR codes directly within the system, members can quickly pay their dues remotely and immediately upload the transaction screenshot, making the entire collection process highly accessible and user-friendly without requiring physical meetups.

Inconsistent fee collection tactics frequently disrupt the operational cash flow of educational institutions and obscure their financial transparency. Daniel et al. (2023) established that implementing strategic fee collection measures such as structured tracking and organized payment planning significantly improves the overall financial accountability of a school. For student organizations, treasurers often struggle with erratic cash flows because they rely on passive, unstructured methods of collecting membership dues or event fines, making it difficult to maintain transparent and updated financial records. UniOrg's Centralized Payment module directly acts as this needed structured fee collection strategy. By providing a dedicated platform where student officers can actively monitor pending dues, verify uploaded digital receipts, and maintain an organized digital ledger, the system ensures consistent collections and high financial transparency for the entire organization.

Handling multiple streams of incoming payments without a structured database often creates administrative bottlenecks that lead to highly inaccurate financial records. Ye (2025) highlights that designing a clearly structured financial management system, specifically incorporating distinct charging modules and automated report generation. Effectively resolves these complexities, ensuring high efficiency and accuracy in fee collections. For student organizations, treasurers act as the primary cashiers for their respective clubs, yet they are rarely equipped with structured tools to accurately record varying payments like membership dues, event tickets, and late fines, resulting in chaotic and unbalanced ledgers. UniOrg's Centralized Payment module directly applies this structured architectural approach. By providing a dedicated digital space with a centralized database for fee collection and report generation, the system equips student officers with a reliable tool to track all financial transactions accurately, moving the organization away from unstructured manual recording to a highly organized digital management solution.

Student Governance and Election System

Conducting student elections through traditional paper ballots frequently results in severe counting errors and prolonged administrative processing times. Susilo et al. (2026) emphasize that migrating to a web-based e-voting platform effectively eliminates these manual miscalculations, allowing incoming votes to be tallied immediately while significantly increasing overall student participation and transparency. For student governance bodies, manually counting hundreds of paper ballots often leads to delayed announcements and disputes regarding the accuracy of

the election results. UniOrg's Student Governance and Election System directly addresses this operational flaw. By providing a secure digital voting module, the system ensures that all casted votes are automatically tallied in real-time, completely removing the heavy administrative burden of manual counting and protecting the integrity of the student electoral process.

Depending on manual election procedures often exposes student government bodies to extreme inefficiencies, time-consuming ballot counting, and vulnerabilities regarding result manipulation. Pabilando et al. (2024) observed in a local university setting that developing a user-centric electronic voting application effectively mitigates these risks, providing a secure platform that ensures fair and transparent electoral processes. For student organizations, the fear of rigged elections or lost paper ballots creates profound distrust between the student body and the incumbent officers handling the transition of power. UniOrg's Student Governance and Election System directly eliminates this vulnerability. By enforcing strict access controls and secure login validations, the system ensures that only eligible members can cast their digital ballots, which are then tallied in real-time to prevent any manual tampering or delays in the announcement of newly elected officers.

Managing student elections using generic survey forms or physical ballots frequently compromises voter anonymity and fails to verify the true identity of the voters. Rashidov (2023) established that developing a dedicated electronic voting system for higher education institutions successfully resolves these vulnerabilities by enforcing strict voter identity verification while guaranteeing absolute anonymity during the voting process. For student organizations, using informal tools like Google

Forms or paper drop-boxes often leads to unauthorized voting, double entries, and a general lack of trust in the declared winners. UniOrg's Student Governance and Election System directly prevents these electoral breaches. By restricting ballot access exclusively to registered and verified club members within the platform, the system ensures that only authorized students can cast their votes. This protects the anonymity of the voter while producing transparent, tamper-proof results that the entire organization can trust.

Relying on manual procedures during campus elections frequently exposes the electoral process to human errors and potential vote manipulation. Azahra et al. (2025) demonstrated that implementing a dedicated electronic voting system using standard web databases significantly reduces these manual miscalculations while enhancing overall electoral transparency. For student organization officers, handling physical ballots creates severe administrative bottlenecks, making it difficult to count votes accurately and declare winners on time. UniOrg's Student Governance and Election System applies this exact technological approach. By utilizing a secure database architecture to record and tally votes automatically, the system entirely removes the risks of manual ballot tampering and provides student clubs with a highly transparent and accountable platform for their leadership transitions.

Conducting student elections through traditional paper-based methods frequently suffers from severe administrative inefficiencies, counting inaccuracies, and critical security concerns. Pugeda (2024) observed in a local Philippine university setting that transitioning to a web-based electronic voting platform directly resolves these bottlenecks, securing the electoral process while providing highly accurate,

real-time vote tallies. For student councils, relying on physical ballots creates a constant fear of miscounted votes and prolonged delays in announcing new officers, which heavily damages the student body's trust in the organization. UniOrg's Student Governance and Election System directly solves this operational problem. By deploying a centralized digital voting module, the platform allows student organizations to conduct highly transparent and accessible elections, ensuring that every casted vote is instantly and accurately recorded without the security risks associated with manual ballot handling.

Conducting college elections via traditional physical polling stations severely restricts voter accessibility and heavily drains administrative resources. Kanekar (2024) emphasized that replacing these manual setups with a web-based digital platform utilizing SQL databases and strict authentication protocols provides a highly accessible solution, allowing students to cast their votes securely from any location. For student organizations, requiring members to physically line up at designated booths often results in extremely low voter turnout, as students understandably prioritize their academic schedules over club elections. UniOrg's Student Governance and Election System directly solves this geographical limitation. By integrating secure user authentication into a responsive web-based module, the system empowers organization members to participate in the electoral process remotely at their own convenience, increasing overall voting engagement while completely removing the logistical burden of setting up physical ballot boxes.

The lack of standard data encryption and structured system architecture in campus elections often leaves voter data vulnerable to unexpected system failures and

security breaches. A 2023 study published in the International Research Journal of Modernization in Engineering Technology and Science established that utilizing a Model-View-Controller (MVC) architecture combined with integrated data encryption provides a highly secure, scalable, and fault-tolerant environment for digital voting. For student organizations, a major fear during online elections is the possibility of the system crashing due to heavy voter traffic or unauthorized users tampering with the database. UniOrg's Student Governance and Election System directly adapts these enterprise-level security measures. By separating the voting interface from the secure database logic, the platform ensures that the electoral process remains highly stable, capable of handling large volumes of voters simultaneously while strictly protecting ballot integrity through standard encryption protocols.

Utilizing simple physical sheets of paper for campus elections severely hinders administrative effectiveness and significantly increases the likelihood of electoral fraud. Nofrima et al. (2023) highlighted that implementing a dedicated web-based e-voting platform effectively eliminates these manual inefficiencies, minimizing information tampering while drastically simplifying the voting process for students. For student leaders managing organizational transitions, relying on manual paper ballots not only wastes critical time but also creates deep suspicions among candidates regarding the legitimacy of the vote-counting process. UniOrg's Student Governance and Election System directly resolves this procedural flaw. By fully digitizing the electoral process into a secure web portal, the platform prevents any physical tampering of ballots and ensures that student elections are conducted with maximum integrity, providing an undisputable transition of power for the organization.

Relying on outdated paper-based procedures often leaves student elections vulnerable to administrative chaos and opaque vote tracking, which ultimately discourages student participation. Deepika and Lal (2024) demonstrated that developing a modern, web-based election application using standard database frameworks effectively streamlines this process, providing a highly transparent and secure electoral environment through the clear separation of administrative management and student voting roles. For campus organizations, the logistical nightmare of setting up physical ballots and manually tallying results is a major reason why elections are frequently delayed or contested. UniOrg's Student Governance and Election System directly adopts this modern architectural approach. By providing distinct dashboards where student officers can securely manage candidates while regular members cast their votes remotely, the platform replaces manual inefficiencies with a fully digital system that actively boosts voter participation through real-time updates and an accessible interface.

Traditional voting methods frequently suffer from severe logistical inefficiencies and security vulnerabilities, particularly regarding unauthorized ballot access. Khade (2025) demonstrated that implementing a secure e-voting platform equipped with standard relational databases and strict authentication protocols effectively neutralizes these threats, guaranteeing that only legitimate voters can participate while maintaining encrypted data integrity. For student organizations, the risk of non-members infiltrating the voting process or students casting multiple votes is a constant threat that immediately delegitimatizes election results. UniOrg's Student Governance and Election System directly adopts this rigorous security framework. By

integrating strict user authentication measures before any student can access the digital ballot, the platform ensures that every recorded vote is accurately verified and securely stored, providing student clubs with an impenetrable and highly reliable electoral environment.

Academic Resource Management

According to Kato et al. (2021), the success of a university's digital library heavily depends on how easily students can access and use it. Their study pointed out that a simple interface and reliable access to information are the most critical factors for user satisfaction. This concept is exactly what the UniOrg system applies in its Academic Resource Management module. Instead of leaving students to rely on scattered files and links, the platform includes a centralized E-Library feature and EduShare. By providing a dedicated and hassle-free space for sharing academic resources, the system directly supports the students' learning needs while keeping them engaged with their respective organizations.

According to Zhou (2025), university libraries are currently undergoing an intelligent transformation to better support undergraduate education. Their study highlights that expanding digital platform services and building a "Virtual Library" are highly effective ways to boost student participation and create a healthy digital learning environment. This perspective directly applies to the Academic Resource Management feature of the UniOrg system. By integrating the EduShare, E-Library, UniOrg goes beyond basic organization management; it actively participates in the digital transformation of the campus by giving students a modern, shared space to access academic materials efficiently.

According to Punnoose et al. (2025), traditional methods of sharing academic resources, such as through email or scattered online links, often create barriers to accessibility and continuous learning. Their research demonstrated that implementing a centralized digital library platform significantly improves student engagement and makes finding reliable study materials much easier. This finding directly supports the implementation of UniOrg's Academic Resource Management module. By replacing fragmented file-sharing habits with the EduShare, E-Library, the system provides a single, organized hub where students can reliably access academic resources, ultimately enhancing their overall learning experience within the university.

As universities shift toward digital transformation, they often struggle with scattered educational materials. A recent study by Yang et al. (2025) suggests that building a single, unified platform to gather and organize these resources is the most effective solution. This concept perfectly aligns with the goal of UniOrg's Academic Resource Management module. Instead of letting student organizations rely on disconnected files, the platform provides a unified digital E-Library alongside the EduShare collaborative space. By centralizing these materials, the system makes resource sharing much easier and helps build a more organized digital environment for the campus.

Managing and distributing educational materials becomes much more efficient when institutions utilize cloud-based platforms. Research conducted by Wang (2023) highlights that designing a digital resource management and sharing system anchored on cloud storage not only organizes files effectively but also ensures fast and reliable

access for its users. This technical principle is a core foundation of UniOrg's Academic Resource Management module. By leveraging cloud storage technology, the system ensures that both the E-Library and the EduShare collaborative space can seamlessly handle the file uploads and downloads of various student organizations, providing a secure and scalable repository for all academic materials.

The effectiveness of a digital resource platform relies not only on file storage but also on how actively students engage with the materials. A recent evaluation by Zhang (2024) observed that traditional online platforms often suffer from low utilization rates because they lack interactive features. However, when materials are organized within a dedicated and dynamic digital platform, student engagement and learning behaviors significantly improve. This insight is highly relevant to the design of UniOrg’s Academic Resource Management module. To prevent the system from becoming just another static, ignored repository, UniOrg intentionally pairs its E-Library with the interactive EduShare space. This setup ensures that the academic materials shared within the student organizations are not only safely stored but also actively utilized by the students.

Uneven distribution and underutilization of study materials remain common challenges in modern education. Addressing these issues, Wang (2022) highlights that utilizing a cloud-based resource-sharing platform allows students to easily access, download, and share digital materials on demand. Such systems eliminate the need for excessive physical infrastructure while maximizing the reach of educational files. This strongly supports the technical foundation of UniOrg's Academic Resource Management module. Because the system utilizes cloud-compatible storage to power

both its E-Library and EduShare spaces, it guarantees that student organizations can efficiently upload and retrieve academic files anytime. Consequently, this setup eliminates the problem of inaccessible materials and promotes a highly collaborative learning environment across the campus.

Transitioning traditional library functions into a web-based format is a vital step in modernizing educational infrastructure. Manikumar et al. (2025) demonstrated that a digital library hub developed with standard web technologies, specifically PHP, JavaScript, and CSS. Significantly improves resource accessibility and administrative control. This mirrors the exact technological approach of the UniOrg platform, which utilizes the same programming languages to build its Academic Resource Management module. By establishing the E-Library and EduShare spaces through these web technologies, the system successfully centralizes educational materials while reducing paper dependency, promoting a more sustainable and organized academic environment for the student organizations.

One of the main hurdles in managing campus educational materials is ensuring that files are not just stored, but also quick to retrieve. Hui Xianghui et al. (2021) explored this issue and found that integrating digital teaching resources into a centralized cloud platform effectively solves problems related to file retrieval and sharing. This approach strongly aligns with how the UniOrg system is built. Through its cloud-backed module, the platform completely avoids the limitations of traditional, scattered file storage, creating instead a highly accessible digital cluster where student organizations can manage their shared resources without delay.

Navigating through disorganized academic files can cost students valuable time and effort. A recent study by Sabateen et al. (2025) highlighted that integrating digital library resources directly into a centralized student platform greatly improves resource discovery and overall efficiency. When educational materials are placed exactly where students are already interacting, it creates a much smoother learning experience. This concept strongly supports the architectural structure of UniOrg. By embedding a dedicated space for learning materials right within the main organization management system, the platform eliminates the need for students to jump between different applications, ensuring that academic files are highly accessible and immediately available to the users.

Communication and Feedback Hub

Establishing a systematic way to gather student insights is essential for continuous improvement within university organizations. A recent study by Bervell et al. (2024) examined an electronic feedback portal and found that students' willingness to continuously use the system depends heavily on its ease of use, reliable performance, and clear interface. If a platform is complicated or confusing, students simply did not participate in the evaluation process. This insight is highly applicable to the development of UniOrg's communication features. By designing a streamlined and user-friendly interface for its feedback channels, the platform ensures that members can easily voice their concerns and suggestions, promoting an open and continuous dialogue between the students and the organization officers.

Relying on delayed evaluations often limits an organization's ability to improve its current initiatives. Othman et al. (2025) observed that traditional evaluation methods are fundamentally flawed because they usually happen too late to benefit the present situation. To address this, their research highlighted the value of a real-time system where students can submit continuous feedback, which significantly strengthens communication between learners and leaders. This concept perfectly captures the purpose behind UniOrg's interactive features. By providing a dedicated channel for immediate and ongoing member feedback, the system empowers student officers to make timely adjustments to their events and programs, rather than waiting for outdated, end-of-semester reports.

Merely collecting evaluations is often insufficient if the respondents never see how their input is applied. Exploring this dynamic, Mair et al. (2023) found that students often feel disconnected from the evaluation process unless the feedback mechanism functions as an active, two-way dialogue rather than a one-sided survey. Their study emphasizes the need for continuous and transparent communication where members can actually see their suggestions being addressed. This principle strongly justifies the design of UniOrg’s communication features. Instead of functioning as a passive drop-box for complaints, the platform establishes an interactive hub where organization officers can directly acknowledge and act upon member insights, building a stronger sense of community and involvement within the campus.

Transitioning from physical forms to digital platforms often raises questions

about the actual quality of the responses gathered. Exploring this shift, a comprehensive review by Plante et al. (2022) revealed that while moving evaluations online might initially affect participation rates, it ultimately leads to longer and far more constructive written comments from the users. This evidence strongly supports the core objective of UniOrg to replace the traditional paper-based processes of student organizations. By digitizing the collection of suggestions and concerns through a dedicated communication portal, the platform is positioned to capture more detailed and actionable insights, equipping student leaders with better information to serve their members effectively.

The success of any digital feedback system relies heavily on the environment where it operates and how it affects user behavior. A comprehensive review by Taghizadeh Kerman et al. (2023) points out that effective online feedback mechanisms do more than just process information; they actively shape behavioral outcomes such as user engagement, teamwork, and overall communication. This principle strongly aligns with the collaborative nature of student organizations. Rather than simply setting up a one-way suggestion box, UniOrg's communication features are designed to cultivate a supportive digital environment. By facilitating structured and accessible interactions, the platform encourages active participation and strengthens the teamwork between student leaders and their peers.

Gathering feedback is only half the task; making sense of the collected data is just as important for decision-makers. Exploring this aspect, Patil et al. (2025) demonstrated that web-based feedback systems allow administrators to efficiently evaluate user responses by visualizing the data through charts and dashboards. When

feedback is organized visually, it becomes much easier to spot trends and areas needing improvement. This practical functionality is exactly what UniOrg's communication hub aimed to provide for student organizations. Rather than leaving officers to manually sort through piles of written suggestions, the platform offers a structured administrative view of member sentiments. This enables student leaders to quickly assess the success of their activities and make data-driven decisions for future initiatives.

A successful evaluation process does not end the moment a form is submitted. Examining this crucial next step, Novák (2023) highlighted that a truly efficient feedback system must go beyond simply collecting data; it should actively encourage follow-up conversations and practical improvements based on the results. If the collected insights are just stored and ignored, the whole process loses its value. This perspective perfectly justifies the interactive nature of UniOrg's communication features. Rather than functioning as a dead-end repository for complaints, the platform is structured to help student officers review and meaningfully respond to the concerns of their members. This setup ensures that every submitted suggestion can spark a productive discussion, ultimately translating member input into better organizational practices.

A robust evaluation system must do more than just record opinions; it requires a clear pathway from receiving feedback to taking actual steps for improvement. Exploring this necessity, Gautam et al. (2025) proposed a "Feedback-to-Action" approach, emphasizing that actively utilizing user input fosters a culture of accountability and continuous growth. Their research highlighted that students expect

clear reactions to their comments rather than having their insights ignored. This principle is a driving force behind the design of UniOrg's communication features. By providing a dedicated space where student leaders can systematically review and respond to member concerns, the platform effectively closes the feedback loop. This ensures that organizational decisions are consistently guided by the actual needs of the student body, promoting a higher standard of leadership accountability across the campus.

Effectively allocating time and resources requires a clear understanding of what students actually need. Exploring this assessment process, Kanwar et al. (2022) emphasized that structured satisfaction surveys are highly effective tools for pinpointing specific areas that hold high importance to users but currently suffer from low satisfaction. By systematically collecting this type of feedback, institutions can adapt faster and address the most pressing concerns. This strategic approach reflects the core function of UniOrg's feedback mechanisms. Rather than relying on guesswork to evaluate the success of organizational activities, the platform equips student leaders with targeted insights, allowing them to focus their future efforts and resources on the exact improvements that matter most to their member.

Role-Based Access Control (RBAC) and Security Management

Securing an educational platform requires strict boundaries on what different users are allowed to see and do. Exploring this security measure, Kabier et al. (2023) highlighted that utilizing a Role-Based Access Control (RBAC) approach effectively safeguards data by classifying users and authorizing specific roles. Because it

deliberately limits access based on a person's assigned level, this method is highly recommended for complex, multi-tier systems. This structural principle is exactly how UniOrg manages its user permissions. By integrating RBAC into the platform's architecture, it guarantees that Org Advisers, Officers, and regular Members are restricted only to the modules and records relevant to their specific duties, effectively preventing unauthorized actions within the organization's digital workspace.

Navigating administrative workflows within student groups often becomes chaotic without a clear digital hierarchy. Addressing this operational bottleneck, Destini et al. (2024) demonstrated that integrating a Hierarchical Role-Based Access Control (HRBAC) framework into a document management system drastically reduces processing delays. By mapping out specific privileges to match the exact chain of command, from student leaders up to faculty advisors, organizations can achieve secure and highly traceable document approvals. This hierarchical concept is central to the architecture of UniOrg. By assigning distinct digital permissions based on a user's actual position, the system guarantees that sensitive organizational records and approval requests are only accessible to the rightful officers or advisers, ensuring a secure and streamlined operational flow without the confusion of manual paperwork.

Maintaining transparency and data protection in academic portals requires a structured approach to user privileges. Examining this security requirement, Manik et al. (2025) developed a web-based academic system that utilizes Role-Based Access Control (RBAC) alongside activity tracking to create a highly secure and accountable digital environment. Their findings indicate that strictly defining user roles

significantly strengthens the system against unauthorized access while optimizing how information is managed. This multi-layered security concept directly aligns with the backend design of UniOrg. By implementing strict role classifications, the platform not only restricts users to their designated features but also ensures that administrative actions taken by officers or advisers are properly regulated, keeping sensitive organizational data safe from unwarranted modifications.

When organizations switch to a centralized digital platform for their daily tasks, it becomes highly important to manage who gets to access what. In a recent paper, Elakya (2025) pointed out that modern management systems rely heavily on role-based access control (RBAC). This feature allows different people, such as students, teachers, and admins, to log into the exact same website while keeping their dashboards separate and relevant only to their specific tasks. This setup perfectly describes the goal of UniOrg's security design. Because the platform was used simultaneously by regular members, student officers, and org advisers, RBAC acts as the system's primary filter. It ensures that while everyone is connected in one space, regular members cannot accidentally alter sensitive files, financial records, or event approvals that are strictly meant for the officers.

Keeping shared data secure is the biggest challenge of putting academic records online. In a recent study, Lavhe (2025) explained that integrating role-based access control (RBAC) into web platforms is a highly effective way to manage a central database safely. This feature makes sure that even if a system holds different types of records, users only saw what they are allowed to see. This concept perfectly matches how UniOrg is built. Instead of relying on messy physical records, the

platform uses RBAC to give officers, org advisers, and regular members their own specific permissions. Because of this, an ordinary member can easily download the files they need, but they are completely blocked from changing the administrative settings meant only for the organization's leaders.

Properly managing who gets to use a system is a major security requirement for any educational institution. A comparative study by Mollakuqe et al. (2024) highlighted that strong authorization practices depend on clearly defining what actions are allowed for specific user roles. Giving the exact right permissions prevents unauthorized individuals from accessing or altering important digital resources. This idea is a perfect match for how UniOrg handles its security. By setting up strict user levels, the platform makes sure that a regular student can only view basic announcements and files. Meanwhile, student officers and faculty org advisers are the only ones granted the authority to approve events, manage the organization's member roster, or edit official content.

Securing an online platform means making sure that sensitive records do not end up in the wrong hands. Exploring this necessity, Dada et al. (2025) emphasized that assigning specific user roles is crucial to prevent data breaches and maintain system integrity. By creating distinct interfaces for regular students and system administrators, their study proved that RBAC effectively stops unauthorized users from viewing or altering confidential records. This exact protective measure is built into the UniOrg platform. Because the system holds important files like event proposals, budget reports, and member directories, it cannot afford to give everyone equal access. Using RBAC, the platform provides separate, customized dashboards so

that regular members only see what they need to participate, while core officers and org advisers retain exclusive control over sensitive organizational documents.

Shifting away from messy paper records is a huge relief for any university, but it immediately creates a new challenge: keeping digital data safe. Falebita (2022) pointed out that while manual filing is slow and prone to errors, simply moving everything online is not enough if the web portal lacks a proper security architecture. Academic institutions must protect sensitive information, such as student profiles and staff details, from any unauthorized access. This exact concern is why UniOrg focuses heavily on secure user permissions. Since student organizations uploaded important files, like member directories and financial reports, instead of keeping them in physical folders, the system relies on strict access controls. This guarantees that only the right officers and org advisers can view or edit these documents, keeping the digital workspace safe from data leaks.

Handling various daily activities in an academic portal requires a clear division of tasks among its users. In a recent project, Mullapudi et al. (2025) developed a web-based management system and showed that role-based access control (RBAC) is the key to organizing students, faculty, and administrators. By assigning specific permissions to each group, the platform can process different operations, such as posting announcements or scheduling activities, smoothly and securely without confusing the users. This division of tasks is exactly how UniOrg intends to operate. Since the platform handles multiple organization activities, it relies on RBAC to keep everything organized. Because of this setup, a regular member can easily log in to view upcoming events, but only the designated officers and org

advisers have the system clearance to actually create those announcements or approve the activity schedules.

Building a single platform for an entire campus means different users needed their own specific workspaces. A recent study by Balage (2025) about a centralized campus web application highlighted that setting up customized dashboards for different roles is the best way to manage user actions securely. By restricting access based on a person's assigned role, the system keeps the workflow organized and prevents users from wandering into areas they should not see. This same logic is applied to the design of UniOrg. Since the platform serves regular members, student officers, and org advisers all at once, it relies on RBAC to give each group a personalized dashboard. Because of this setup, ordinary members only see the options they need to join events or submit feedback, while the officers are given the specific administrative tools required to run the organization.

Optical Character Recognition (OCR) for Automated Document Data Extraction

Speeding up repetitive administrative tasks is one of the biggest reasons why educational institutions adopt new technologies. In a recent study, Ghadekar et al. (2025) demonstrated this by automating the laborious process of grading exams. Their research showed that replacing manual work with automated system functions not only makes the process significantly faster but also heavily reduces human errors. This principle of automation is the exact foundation of UniOrg's core functionality. Rather than forcing student officers to physically chase after advisers for event approval signatures or manually sort through stacks of paper records, the platform

automates these daily workflows. By letting the system handle the routing and tracking of organizational documents, the officers can operate much faster and avoid the common mistakes tied to traditional paperwork.

Eliminating the hassle of manual data entry is a major step toward modernizing any campus system. Exploring this transition, Shaju et al. (2024) developed a software tool that digitizes physical school records into structured files, proving that digital solutions can significantly reduce manual labor and improve overall data management. This concept of digitization directly reflects the main functionality of the UniOrg platform. Instead of forcing student officers to manually type out member information from paper forms or physically compile documents for every event, the system takes care of the heavy lifting. By providing built-in features that automatically collect, organize, and store submitted records into a central database, the platform saves the officers countless hours of administrative work.

Processing student documents manually often leads to unnecessary delays and mistakes. Kharatmol et al. (2025) highlighted this issue by showing how colleges traditionally struggle with the time-consuming task of extracting data from educational credentials. Their study proved that shifting to an automated data-handling method creates a streamlined process that significantly reduces human error and speeds up administrative work. This drive for automation perfectly aligns with the core functionality of UniOrg. Instead of requiring student leaders to manually read and encode every member's registration details or printed application forms, the platform's built-in features do the heavy lifting. By automatically capturing and organizing submitted student information into the database, the system eliminates

tedious manual encoding, allowing officers to focus on actual organizational planning.

Safely keeping track of administrative paperwork over the years is a major challenge for any university. Synko (2024) addressed this by highlighting the importance of web-based tools in transforming physical university records into a structured digital archive. The study pointed out that moving documents into an electronic database equipped with specific search filters makes it significantly easier to store and retrieve important files. This archiving capability is a central functionality of the UniOrg platform. Since student organizations generate a lot of paperwork every semester, such as resolutions, meeting minutes, and activity reports, the system is designed to act as a secure digital cabinet. By providing built-in search and filtering tools, the platform allows OSAS staffs and future student officers to quickly pull up past records, ensuring that the organization's history is well-preserved and always easy to find.

Preserving historical academic records is often a struggle when they only exist in physical formats that easily deteriorate or get lost over time. In a recent study, Casas-Huamanta et al. (2024) addressed this by developing a system designed to extract and recover data from old academic cards. Their work demonstrated that using automated tools to digitize physical documents is a highly effective way to permanently save and upload historical records before the paper copies are completely ruined. This functionality of long-term digital preservation is highly relevant to UniOrg. Student organizations frequently lose their past documents, such as previous financial reports or old event proposals, every time a new batch of student leaders takes over. By allowing officers to digitally upload and permanently store these older

files into the database, the platform ensures that the organization's history is secured and smoothly passed down to the next academic year.

Manually typing out data from printed sheets is one of the most tedious tasks in campus administration. To solve this, Clementeena et al. (2025) developed a system that uses Optical Character Recognition (OCR) technology to extract information directly from physical templates. Their project proved that processing scanned documents to automatically convert printed text into digital data significantly cuts down manual data entry and reduces human typing errors. This specific data extraction functionality is a massive advantage for a platform like UniOrg. Instead of requiring student officers to manually retype long information from printed forms, physical memos, or official receipts, the system's OCR feature can automatically read and extract the necessary text from uploaded pdf. This automated extraction ensures that digital records are generated quickly and accurately, removing the heavy burden of manual encoding from the student leaders.

Typing out member details from identification cards or registration forms takes up a lot of valuable time for student leaders. To solve this inefficiency, Nashiruddin et al. (2025) implemented an Optical Character Recognition (OCR) system specifically designed to extract information directly from student records. Their research confirmed that using OCR to automatically read and separate text into specific data fields, such as a student's name and degree program, drastically improves both the speed and accuracy of campus administrative services. This specific automated extraction is exactly what makes the UniOrg platform efficient. Instead of having officers manually encode hundreds of member details into an Excel

sheet, the system allows them to simply upload PDF copies of student IDs or physical application forms. The OCR feature then scans the PDF document, extracts the relevant text, and automatically fills out the organization's database, eliminating the slow and error-prone process of manual typing.

Manually filling out administrative forms is a repetitive task that often leads to data entry errors. To address this, Simanjorang et al. (2025) developed an automated system for university document administration using Optical Character Recognition (OCR). Their study found that using OCR to automatically extract specific information from official records, such as names and dates from certificates, significantly speeds up the form-filling process and achieves high accuracy compared to manual typing. This concept of automated form-filling is exactly how the OCR feature functions within the UniOrg platform. Instead of typing out the details for every single event or member requirement, student officers simply upload the required PDF documents into the system. The platform then reads the PDF, extracts the necessary text, and automatically fills out the required database fields, completely removing the need to encode the information word for word.

Managing academic records usually requires a massive amount of manual labor that is highly prone to human error. In a recent study, Mushfiqul et al. (2025) addressed this by designing an intelligent framework that uses Optical Character Recognition (OCR) to automate the extraction of text from unstructured student records. Their findings revealed that using OCR to pull data from academic documents drastically decreases processing time and reduces the error rate to less than one percent. This high level of efficiency directly supports the data extraction feature

of the UniOrg platform. Because student officers handle various unstructured files throughout the semester, the system allows them to simply upload these documents as PDF files. The built-in OCR technology then scans the PDF, intelligently extracts the necessary text, and processes the information, sparing the officers from the tedious and error-prone task of typing out every detail by hand.

Natural Language Processing (NLP) for Text Summarization and Auto-Tagging

Categorizing academic materials correctly is essential for keeping a digital workspace organized. Khan et al. (2021) addressed this by developing a content classification system that utilizes Natural Language Processing (NLP) to accurately auto-tag educational materials. Their study showed that using NLP allows a system to understand the context of a document and organize it properly without relying solely on manual keyword searches. This auto-tagging capability is highly beneficial for organizing files within the UniOrg platform. Whenever student officers upload various PDF documents, such as event proposals, financial reports, or activity letters, the system applies NLP to scan the text and automatically tag these files into their correct categories. By implementing this intelligent tagging functionality, the platform ensures that the organization's digital archive remains neat and easily searchable, saving users from the hassle of manually sorting every single document.

Assigning the right labels to digital files is crucial for keeping an online database organized, but doing this by hand takes up too much time. Shahhoseini et al. (2025) addressed this by introducing an intelligent tagging platform that uses Natural Language Processing (NLP) to automatically annotate educational contents. Their research proved that NLP tools can accurately read text-based documents and

generate the necessary tags, effectively eliminating the slow and tedious process of manual labeling. This automatic tagging approach is directly applied to how UniOrg organizes its digital records. As student officers upload various PDF documents, such as meeting minutes, event guidelines, and activity reports, the platform utilizes NLP to analyze the text and automatically assign the relevant tags. By doing so, the system builds a highly organized and searchable archive without requiring the officers to manually type out categories for every single file.

Sorting through different types of organizational paperwork can quickly become overwhelming for student leaders. Kwak et al. (2024) explored how Natural Language Processing (NLP) tools can solve this through automated content tagging in educational environments. Their findings demonstrated that intelligent models can accurately read digital materials and assign the appropriate labels, proving that automated classifiers are highly effective for organizing content without manual intervention. This auto-tagging capability is exactly how the UniOrg platform manages a heavy influx of paperwork. Whenever officers upload their required PDF files, such as a Letter of Application, a Plan of Activities, or an Adviser Commitment Form, the system uses NLP to evaluate the text and automatically assign the correct tags. This ensures that every specific requirement is instantly sorted into its proper category, saving the officers the hassle of manually labeling each document.

Reading through lengthy organizational documents consumes a significant amount of time for student leaders and org advisers. Yan et al. (2023) highlighted this issue in their review of natural language processing tools in education, noting that analyzing text-based content is traditionally a laborious task. Their study confirmed that advanced NLP models can efficiently process massive amounts of text to

automate complex tasks, significantly reducing the manual effort required to understand long documents. This capability perfectly supports the text summarization feature of the UniOrg platform. Student organizations frequently upload lengthy PDF files, such as Narrative Reports, Accomplishment Reports, and Minutes of the Meeting. Instead of requiring OSAS staffs to read every single page to understand a student resolution, the platform uses NLP to automatically scan the PDF document and generate a concise summary. This allows decision-makers to quickly grasp the essential points of an activity, making the evaluation and approval process much faster.

Evaluating the quality of organizational paperwork by hand is often an overwhelming task that can be prone to human bias or error. Lukwaro et al. (2024) emphasized this in their review of educational data, noting that manual evaluation is highly burdensome. Their study confirmed that Natural Language Processing (NLP) is an ideal tool for automatically interpreting textual content, allowing systems to extract key information without forcing users to read everything from scratch. This automated comprehension directly supports the workflow within the UniOrg platform. Because the OSAS personnel receive a massive influx of submissions from various student groups throughout the academic year, it is not always feasible to examine every single page in detail. By utilizing NLP to scan and highlight the most critical points of these submissions, the system provides the staff with clear and objective summaries, making the overall review process much more efficient and less tedious.

Processing large volumes of textual data is a major administrative hurdle for universities. Aggarwal et al. (2024) explored how Natural Language Processing

(NLP) can solve this by automatically identifying key themes and classifying text in higher education settings. Their analysis demonstrated that NLP tools effectively categorize information and extract meaningful insights, significantly streamlining administrative workflows and improving decision-making. This capability aligns perfectly with how the UniOrg platform supports the Office of Student Affairs and Services (OSAS). Instead of manually sorting and reviewing every submitted file from various student groups, the system uses NLP to instantly classify the records and highlight their core topics. By automating the extraction of key themes and organizing the data intelligently, the platform empowers the OSAS staff to process organization requirements much faster, allowing them to focus on final approvals rather than tedious reading and sorting.

Integrating advanced technologies into academic workflows is rapidly changing how institutions handle heavy administrative tasks. Alqahtani et al. (2023) explored this transformation, emphasizing the emergent role of Natural Language Processing (NLP) in higher education. Their review highlighted that NLP applications, specifically in data interpretation, text analysis, and assessment, offer significant advantages in evaluating academic content and improving institutional outcomes. This potential for enhanced efficiency is the exact driving force behind the NLP features of the UniOrg platform. By utilizing intelligent text analysis to automatically summarize and categorize the continuous flow of organizational submissions, the system relieves the OSAS staff from manual reading and sorting. This automated support allows the administration to focus on providing constructive feedback and making quick, informed decisions, rather than getting bogged down by the sheer volume of paperwork.

Handling an unprecedented volume of organizational data requires advanced analytical tools to prevent administrative backlogs. Niu (2024) explored the innovative applications of Natural Language Processing (NLP) in analyzing massive amounts of educational content. The study demonstrated that leveraging NLP for text analysis and topic modeling significantly enhances the efficiency of data processing, uncovering important details that traditional manual methods often overlook. This finding directly supports the implementation of intelligent text analysis within the UniOrg platform. With the constant influx of requirements submitted by various student groups, the OSAS staff can rely on the system's NLP capabilities to instantly evaluate and categorize the contents of each file. By automating these time-consuming analytical tasks, the platform allows the administration to manage large volumes of paperwork effortlessly, paving the way for a faster and more data-driven approval process.

Locating specific files within a massive digital archive can be highly frustrating without an intelligent search and categorization system. Huang (2021) highlighted this by exploring how Natural Language Processing (NLP) technology can be applied to improve the retrieval of educational resources. The study detailed how NLP models process raw textual data and automatically extract key concepts, ultimately increasing the accuracy of search results and making document retrieval much easier for users. This intelligent retrieval capability is exactly what the auto-tagging feature of the UniOrg platform provides for the OSAS staff. As the administration accumulates a massive archive of organizational paperwork over multiple semesters, locating a specific past record can quickly become difficult. By

using NLP to analyze the uploaded PDF files and automatically generate relevant tags, the system ensures that the OSAS personnel can easily search for and retrieve the exact documents they need without having to dig through disorganized digital folders.

Managing a growing digital repository of organizational files demands an efficient categorization mechanism to keep records accessible. Joorabchi et al. (2025) addressed this challenge by developing an approach for the automatic classification of electronic educational materials. Their research demonstrated that utilizing intelligent algorithms eliminates the need to manually sort documents, providing a highly reliable way to organize institutional databases as they expand. This automated categorization aligns perfectly with the tagging functionality of the UniOrg platform. As the system continuously accumulates various PDF submissions from student groups over the years, relying on manual indexing is no longer practical. By applying intelligent text classification to automatically tag and sort these incoming files, the platform maintains a highly organized digital archive, ensuring that the OSAS staff can efficiently manage records without the heavy burden of manual file sorting.

Smoothly managing the continuous flow of university paperwork requires systems that can accurately sort files without human intervention. Tkachenko et al. (2022) emphasized this in their design of an electronic document management system tailored specifically for educational environments. Their research demonstrated that utilizing intelligent algorithms to automatically extract significant keywords from texts greatly improves the accuracy of document classification and significantly reduces the number of mistakenly sorted files. This level of precision is exactly what

the UniOrg platform brings to the Office of Student Affairs and Services (OSAS). Because human fatigue can easily lead to misplacing important organizational records into the wrong digital folders, the platform relies on Natural Language Processing to accurately evaluate and auto-tag incoming PDF submissions. By ensuring that every file is correctly classified based on its actual content, the system provides the OSAS staff with a highly reliable and error-free digital workspace.

2.2 Related Systems and Technology Gap Analysis

This section examines existing systems that are related to UniOrg. Ten related systems are compared in this review. The purpose is to determine what these systems can do and what they still fail to address. The result of the comparison is used to justify the features of the proposed system.

2.2.1 Identification of Related Systems

CampusGroups. This platform combines student groups, events, communication and co-curricular tracking in one engagement hub (Ready Education, 2026a). It was included because it represents the common baseline for event scheduling and community features. It has no OCR extraction and no NLP tagging, and it lacks the configurable multi-org hierarchy that OSAS requires.

Anthology Engage. This platform manages student engagement, events and organizations with co-curricular tracking and analytics (Anthology, 2023). It was included because it shows how a mature system handles

governance and approval processes. Document review in the platform is still manual and it has no AI processing for accreditation files.

Ready Education. This platform provides a mobile-first campus app for groups, events, announcements and student services (Ready Education, 2026b). It was included as the baseline for student facing mobile engagement. Its payment and feedback functions are only partial and it has no automation for accreditation documents.

Springly. This platform handles memberships, dues collection, accounting and events for associations (Springly, n.d.). It was included because its payment tracking is the closest baseline for the Centralized Payment module. It has no election module and no OCR or NLP support for documents.

Momentus Higher Education. This platform unifies venue booking and campus events in a single calendar system (Momentus, 2026). It was included as the closest baseline for the master calendar function. It has no organization hierarchy and no automated review of accreditation documents.

Eventbrite. This platform provides event pages, ticketing, checkout and organizer analytics (Eventbrite, 2026). It was included as the generic baseline for event scheduling and payment collection. It only partially covers calendar needs and it has no election, feedback or organization management features.

LSPU Nexus is included because it is the current online system used by the university and the OSAS (Laguna State Polytechnic University [LSPU],

2024). It is the most direct baseline since UniOrg operated in the same environment and serve the same offices. Its limitations show exactly what still needs improvement in student organization management.

The RSU Accreditation Web-based System was developed as a capstone project to accredit student organizations under the OSAS of Romblon State University (Tamboong, 2024). It was included because it addresses the same accreditation problem as UniOrg. Its document review process is still manual which makes it a good point of comparison.

The OSA Management System with OrgsUP is used by the Office of the Vice Chancellor for Student Affairs of UP Los Banos for organization registration, activity permits, and narrative reports (University of the Philippines Los Banos Office of the Vice Chancellor for Student Affairs. It was included because it is a mature system from a Philippine state university. It shows what a long running student affairs system can already do.

Experia for Universities is a Philippine made platform that provides member management, events, and dues collection for student organizations (Experia, n.d.). It was included to represent a modern commercial option built for local organizations. Its per organization setup shows the limits of tools that are not managed at the university level.

2.2.2 System Feature Comparison

The ten systems are compared using seven features taken from the objectives of UniOrg. These are AI OCR document processing, NLP auto tagging and summarization, multi-org workspace, master calendar sync,

payment tracking, election module, and secure feedback hub. Table 2.1 presents the comparison of all related systems against the proposed system.

Table 2.1 Technology Gap Analysis of Related Systems and UNIORG System

RELATED SYSTEM / PROPOSED SYSTEM	AI / OCR DOC.	NLP AUTO-TAG AND SUMMARIZATION	MULTI-ORG WORKSPACE	MASTER CALENDAR SYNC	PAYMENT TRACKING	ELECTION MODULE	SECURE FEEDBACK HUB
LSPU Nexus	X	X	X	X	X	X	–
CampusGroups	X	X	X	/	/	/	–
Anthology Engage	X	X	X	/	/	/	–
Ready Education	X	X	X	/	–	/	–
Springly	X	X	X	/	/	X	X
Momentus Higher Ed.	X	X	X	/	/	X	X
Eventbrite	X	X	X	–	/	X	X
RSU Accreditation	X	X	–	X	X	X	–
UPLB OSAM	X	X	–	–	X	X	–
Experia	X	X	–	/	/	X	–
UNIORG	/	/	/	/	/	/	/

Legend: / = Present, – = Partially Present, X = Absent

In Table 2.1, the reviewed systems provide solutions for student engagement, event management, and organization tracking but lack AI-powered document processing, customizable multi-organization structures, and features suited for Philippine state universities. UniOrg addresses these gaps through OCR and NLP integration, automated accreditation processing, auto tagging, and centralized modules for organization management, events, e-voting, payments, and feedback.

2.2.3 Analysis of Existing Systems

Most existing systems already handle event scheduling and community engagement well. CampusGroups, Anthology Engage, and the other commercial platforms all provide these common functions. Some systems only partially cover member structures and feedback channels. None of the reviewed systems provide AI powered document processing or NLP based auto tagging. The local systems such as Nexus, RSU accreditation, and OSAM focus on records and permits but their document review remains manual. Payment tracking and election functions appear in a few systems but they are not integrated with accreditation workflows.

2.2.4 Technology Gap

The technology gap is the absence of automated document processing inside student organization management. Existing systems require officers and staff to read, check, and sort accreditation documents by hand. There is also a lack of integration since events, payments, elections, and documents are handled by separate tools or offices. No reviewed system combines a configurable multi organization hierarchy with OCR extraction and NLP summarization in one platform.

2.2.5 Proposed Response to the Gap

UniOrg responds to these limitations through specific features. The manual document review is addressed by OCR for data extraction and NLP for auto tagging and summaries. The fragmented workflows are addressed by placing accreditation, events, payments, elections, and feedback inside one multi-org workspace. The needs of OSAS are addressed by approval tracking, a master calendar, and centralized reports. Each of these features directly answers a limitation found in the existing systems.

For the manual review problem the system provides OCR that reads scanned accreditation files and converts them into encoded data. This removes the need for officers to retype long details from printed forms. The NLP component then tags each file and generates short summaries so OSAS staff can understand a report without reading every page. For the fragmented setup the platform places accreditation tracking, event scheduling, payment monitoring, elections and feedback inside one workspace. Officers no longer switch between separate tools since records, approvals and schedules stay connected. The master calendar prevents date conflicts while the centralized payment ledger keeps collections transparent. The election module secures voting with verified member access and the feedback hub gives members a direct channel to officers. Each feature was selected to answer a specific limitation so the whole system directly responds to the gaps found in Table 2.1.

2.3 Synthesis

2.3.1 Integration of Findings

The reviewed studies agree that a cloud based multi-org architecture can isolate the data of different groups inside one system. RBAC is recognized as the standard way to secure user permissions. OCR reduces encoding errors while NLP makes long documents easier to classify and summarize. The studies also show that these technologies are usually applied separately and not inside one student organization platform.

2.3.2 Identified Gap

The identified gap is functional, technological, and process in nature. The functional gap is the missing automation for accreditation document review. The technological gap is the absence of OCR and NLP inside existing organization management systems. The process gap is the manual and scattered workflow from submission to approval. There is also a contextual gap since generic platforms are not configured for the OSAS processes of a Philippine state university.

2.3.3 Contribution of the Proposed Project

UniOrg contributes a centralized ecosystem that fills these gaps. It automates the extraction, tagging, and summarization of accreditation documents. It unifies organization management, events, payments, elections,

and feedback under OSAS supervision. The project provides a working model that other state universities with similar processes can adopt.

CHAPTER III

RESEARCH METHODOLOGY

This chapter discusses the methodology in conducting the study entitled "UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging" In this research, a descriptive-developmental research approach was applied to gather and analyze data. This chapter also includes the discussion of the Materials and Methods, Research Design, Population of the Study, Testers' Profile, Locale of the Study, Sampling Design, Data Collection Instruments, Project Design, Software Development Model, System Architecture, System Testing and Evaluation Procedure, System Testing, System User Evaluation, and Statistical Treatment.

3.1 Materials and Methods

The UniOrg platform was developed using a systematic approach that involved conducting fact-finding interviews with key university stakeholders to identify critical administrative needs and carrying out rigorous development activities supported by extensive resource gathering, by utilizing the Agile methodology the researchers successfully aligned the platform functionalities with the specific accreditation requirements of the Office of Student Affairs and Services while implementing modern technological tools to ensure that document processing and organizational management are streamlined effectively for the university

3.2 Research Design

This study utilizes a descriptive-developmental research design to effectively construct and evaluate the UniOrg platform. This method is specifically structured for projects that require the systematic design, development, and comprehensive evaluation of software systems or technological solutions. The research is fundamentally divided into two complementary phases: the developmental phase and the descriptive phase.

3.2.1 Rationale

Descriptive-developmental design was chosen because it covers both the development of the software and the evaluation of its acceptability in a real-world setting.

The developmental phase focuses on the actual technical creation of the centralized management platform. This involves the design and integration of specific features tailored for the Office of Student Affairs and Services (OSAS) and the thirty-nine (39) student organizations of Laguna State Polytechnic University, particularly the implementation of Optical Character Recognition (OCR) and Natural Language Processing (NLP) for document processing and auto-tagging.

Consequently, the descriptive phase assesses the system's acceptability and functionality among its actual end-users. Through quantitative data gathering, the researchers describe the perceptions of OSAS administrators, college Deans, org advisers, organization officers, and regular students. This evaluation is anchored on the Technology Acceptance Model (TAM), utilizing

survey questionnaires to measure five specific constructs: Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. This dual-phase approach ensures the platform is both technologically sound and highly acceptable to the campus community.

3.3 Population of the Study

The population of the study consisted of selected stakeholders from Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC), who are the intended end-users and primary evaluators of the proposed UniOrg platform. Since the project is designed specifically to streamline campus-wide organization management, document accreditation, and student engagement, the respondents were categorized into five distinct groups: the OSAS administrators, college Deans/Associate Deans, Org Advisers, student organization officers representing the nine (9) College Departments, and the regular students.

Locale of the Study: This research was conducted at the Laguna State Polytechnic University (LSPU) – Santa Cruz Campus, where the UniOrg platform is intended for implementation.

The regular students and organization officers represent the primary users of the UniOrg platform as shown in Table 3.2, as they were the ones who actively interacted with the multi-org dashboards, utilize the e-voting modules, and manage organization workflows. Respondents from various recognized student organizations at LSPU-SCC were selected to ensure that feedback accurately reflects the diverse demographic of the campus student body. For the purposes of this evaluation, the

researchers employed a mixed sampling design by explicitly selecting 428 specific individuals who are directly relevant to the system's operational implementation. This strategic distribution ensures that the practical perspectives of the main user groups (students and officers) are prioritized, while also incorporating highly valuable feedback from the Org Advisers, College Deans, and OSAS Administrators, who play critical supervisory and administrative roles in the project's campus-wide adoption and long-term sustainability.

3.4 System Testers

A team of IT professionals and technical staff evaluated the UniOrg system. Their positions and credentials are detailed in Table 3.1. They assessed the system based on their background in IT support, software development, and systems analysis.

3.4.1 Profile of System Testers

Table 3.1 outlines the demographic and professional profile of system testers who participated in functional and Black-Box testing of the UniOrg system.

Table 3.1 Profile of System Testers

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Jonas T. Santos	LSPU SCC BSIT SMP Graduate	MISO IT Staff	3 years
Janssen R. Espino	LSPU SCC BSIT Graduate	MISO IT Staff	1 year
John Marvin A. Tabo	LSPU SCC BSIT Graduate	MISO IT Staff	3 years
Kevin D. Adlaon	Data Scientist Malaysia	MISO IT Staff	2 years
Wilmar T. Morota	Freelance Encoder	MISO IT Staff	4 years
Justine Jhay Lopez	LSPU SCC BSIT Graduate	MISO Programmer	1 year
Robin Paolo L. Bona	Union College BSIT Graduate	MISO IT Staff	6 years
Mharjun V. Tobias	LSPU SCC Graduate	MISO IT Staff	1 year
Clydemarion P. Joya	LSPU SCC BSIT SMP Graduate	Video Editor	1 year
Christian Jay B. Tantan	LU BSIT Graduate	PTA(Quality Assurance)	2 years

Table 3.1 shows the profile of the ten system testers who conducted the black box testing. Most of them are MISO IT staff with BSIT background and experience ranges from one to six years. Their technical background qualified them to validate the functionality of each system module.

3.5 Respondents for User Evaluation

The evaluation involved OSAS administrators, College Deans, organization advisers, student organization officers, and regular students. They were chosen because they are directly involved in organizational management.

3.5.1 Distribution of Respondents

Table 3.2 presents the distribution of respondents by group. It shows the total population, sample size, and sampling design used for each group.

Table 3.2 Distribution of Respondents for User Evaluation

Respondent Group	Total Population	Sample Size	Sampling Design
OSAS Admins	2	2	Total enumeration
College Deans	9	9	Total enumeration
Org Advisers	78	56	Stratified random sampling
Organization Officers	768	161	Stratified random sampling
Regular Students	9,613	200	Stratified random sampling
Total	10,470	428	

The distribution of the survey questionnaire took place after the successful completion of the design, development and black-box testing phases of the UniOrg platform. Following the final Agile development sprint and system integration testing, the surveys were administered during the formal platform demonstration at the Laguna State Polytechnic University - Santa Cruz Campus. During this activity, the researchers distributed the questionnaires to the selected participants derived through stratified random sampling which encompassed the OSAS administrators college Deans, faculty advisers, organization officers, and regular students who actively participated in the testing phase. This direct approach ensured that the respondents comprehensively understood the purpose of the evaluation based on the Technology Acceptance Model (TAM) and the critical importance of their feedback. The researchers also explained how the gathered empirical data would be utilized to assess the platform's perceived usefulness ease of use, and overall effectiveness in streamlining student organization management, accreditation workflows and AI-assisted document processing.

Respondents were provided with adequate time to hands-on interact with the system's core modules, such as uploading narrative reports for OCR text extraction, testing the NLP auto-tagging, and exploring their respective multi-org dashboards, before completing the questionnaire. The researchers were physically present throughout the demonstration to address any inquiries and provide technical clarifications as needed. Upon the conclusion of the testing activity, the accomplished questionnaires were systematically collected immediately to ensure a high response rate and the completeness of the data required for statistical analysis.

3.6 Sampling Design

A mixed sampling design was used to select the participants for this study. Total enumeration was applied to the College Deans since there are only 9 of them, so all Deans were included as respondents. They oversee student organization activities under their colleges, making their evaluation of the system's scalability, functionality, and impact particularly critical for the university-wide deployment of UniOrg. For the Office of Student Affairs and Services administrators, total enumeration was employed. Since there are only two staff members in the office, selecting both of them as respondents ensured a complete and accurate understanding of the department requirements.

The study used Slovin's formula n = N / (1 + N(e)^2), where n is the sample size, N is the population, and e is the margin of error. Slovin was applied per group with a margin of error of 0.07. For organization officers with a population of 768, the sample was 161. For organization advisers with a population of 78, the sample was 56. For regular students with a population of 9,613, the sample was 200. The 2 OSAS administrators and 9 Deans were added through total enumeration. The final sample size was 428 respondents.

Stratified random sampling was utilized for the organization advisers, organization officers, and regular students. Slovin's formula with a margin of error of 0.07 set each group sample based on its own population. From the 78 organization advisers, a sample of 56 was taken. For the organization officers, 161 were selected from a total of 768. Lastly, 200 regular students were chosen from a population of 9,613. Together with total enumeration of OSAS administrators and Deans, the final sample reached 428 respondents, which allowed the researchers to capture diverse

perspectives from different levels of the university.

The population lists came from the registrar for students and from OSAS records for officers and advisers, all for AY 2025 to 2026 active organizations and enrolled students only. The 0.07 margin was set with the adviser to keep the total sample at a manageable 428 while keeping per group computation through Slovin. Inclusion covered active officers, advisers and enrolled students in that school year, while inactive organizations, graduated students and non members were excluded. Random selection was done per stratum by drawing from the group lists until each quota was met.

3.7 Data Collection Instruments

The process of data collection played a vital role in the development of this system. It ensured the gathering of reliable evidence, which was later transformed into comprehensive data analysis and helped in forming a convincing and credible response to the research questions. To achieve this, key stakeholders were directly consulted regarding the problem, process, and context in order to better understand and address their concerns through the system. The following outlines the techniques that the researchers employed in the data collection process.

The selection of appropriate data collection instruments is crucial for ensuring the validity and reliability of research findings. In this study, the researchers employed a mixed-methods approach, combining both qualitative and quantitative data collection techniques to gain a comprehensive understanding of the student organization management and accreditation challenges faced by the Office of Student

Affairs and Services (OSAS), as well as to thoroughly evaluate the UniOrg platform's effectiveness in addressing these challenges. The use of multiple data collection methods allows for triangulation, which strengthens the validity of the findings by corroborating evidence from different sources and perspectives.

3.7.1 Instrument Development

This study used researcher-made interview guides, brainstorming protocols and a TAM-based questionnaire for the UniOrg evaluation. The interview and brainstorming guides are described below.

Key Informant Interviews

The researchers conducted key informant interviews with OSAS Chairperson Christian P. San Luis, LPT, PhD, OSAS Secretary Nicole D. Bartolome, Organization Senior Adviser Rachiel R. Rivano, PhD, Organization Junior Adviser Harlene G. Origines, and various student organization officers to understand current operational challenges, pain points, and requirements for the UniOrg system. These interviews focused on identifying existing accreditation procedures, document handling protocols, and manual tracking challenges. Semi-structured interviews allowed flexibility for participants to share their experiences and insights comprehensively.

Key informant interviews are a qualitative research method that involves in-depth conversations with individuals who have specialized knowledge or experience relevant to the research topic. In this study, key

informants included the OSAS administrators, senior org advisers with extensive experience in campus governance, student officers who have managed organizational activities, and regular students who actively participate in campus events. These individuals were selected because they possess deep insights into the current manual system's strengths and weaknesses, as well as clear ideas about what digital improvements would be most beneficial.

The interview protocol was designed to explore several key themes: (1) the current process for submitting, reviewing, and approving accreditation documents; (2) challenges and bottlenecks in the existing manual filing system; (3) communication and coordination issues between OSAS and student officers; (4) factors affecting event scheduling and master calendar management; (5) student satisfaction with current organizational activities and payment collections; (6) desired features for a new system, particularly the integration of Optical Character Recognition (OCR) and Natural Language Processing (NLP); and (7) concerns about technology adoption and implementation. The semi-structured format allowed the researchers to ensure that all key topics were covered while also providing flexibility to explore unexpected themes that emerged during the conversations.

Each interview lasted approximately 45-60 minutes and was conducted in a private setting to ensure confidentiality and encourage candid responses. With the participants' permission, interviews were audio-recorded and later transcribed for analysis. The researchers also took detailed field notes during the interviews to capture non-verbal cues and contextual

information. Following each interview, the researchers conducted member checking by sharing a summary of the key points with the participant and asking them to verify the accuracy of the researchers' interpretation.

The data from key informant interviews was analyzed using thematic analysis, a qualitative method for identifying, analyzing, and reporting patterns (themes) within data. The researchers read through the interview transcripts multiple times to become familiar with the data, then systematically coded the data by identifying segments of text that related to specific topics or concepts. These codes were then grouped into broader themes that captured important patterns in the data. The themes that emerged from this analysis informed the design of the UniOrg platform, ensuring that it addressed the real needs and concerns of its intended users.

Brainstorming Sessions

Collaborative brainstorming sessions were conducted among the researchers and with OSAS personnel and student leaders to generate ideas and identify system features. These sessions created an open environment for sharing insights about administrative bottlenecks, multi-organization coordination strategies, student communication needs, and document preservation requirements. Feedback on Role-Based Access Control (RBAC), automated text extraction, and digital workspaces was incorporated into the system design.

Brainstorming is a group creativity technique designed to generate a

large number of ideas for the solution of a problem. In this study, brainstorming sessions served multiple purposes: (1) generating innovative ideas for system features and functionality; (2) identifying potential challenges and obstacles to system implementation across 39 distinct organizations; (3) building consensus among stakeholders about system priorities and design decisions; and (4) fostering a sense of ownership and investment in the project among OSAS staff and officers who would ultimately use the system.

The researchers conducted a series of brainstorming sessions at different stages of the project. Initial sessions focused on identifying problems with the current paper-based system and envisioning an ideal digital solution. Later sessions focused on specific aspects of the system design, such as the user interface, NLP auto-tagging scheme, multi-org database architecture, and reporting features. Each session typically involved 5-10 participants and lasted 1-2 hours. The researchers used various brainstorming techniques, including mind mapping, the nominal group technique, and SWOT analysis (Strengths, Weaknesses, Opportunities, Threats), to stimulate creative thinking and ensure that all participants had an opportunity to contribute.

During brainstorming sessions, the researchers emphasized the importance of generating a large quantity of ideas without immediate evaluation or criticism, following the principle that quantity breeds quality. All ideas were recorded on flip charts or whiteboards so that participants could see and build upon each other's suggestions. After the idea generation phase, the group worked together to evaluate and prioritize the ideas,

considering factors such as feasibility, cost, potential impact, and alignment with project goals. The most promising ideas were then incorporated into the system design specifications.

The brainstorming sessions also served an important function in change management and user adoption. By involving OSAS administrators and student leaders in the design process from the beginning, the researchers helped to build support for the new platform and reduce resistance to change. Participants in the brainstorming sessions became champions for the UniOrg system within the campus, helping to communicate its benefits to their peers and encouraging adoption.

Online Research and Literature Review

The researchers conducted extensive online research to gather information on multi-org student management platforms, Optical Character Recognition (OCR) for document processing, Natural Language Processing (NLP) technologies for text summarization, and campus e-governance solutions. Academic sources such as Google Scholar, research journals, and industry publications were consulted to establish the theoretical foundation and identify best practices in educational management systems.

The literature review served several important functions in this study. First, it provided a theoretical and conceptual framework for understanding student organization management and accreditation tracking in the context of higher education. Second, it identified best practices and lessons learned from

similar systems implemented in other universities or institutions. Third, it informed the selection of appropriate technologies and methodologies for the UniOrg system. Fourth, it helped to position this study within the broader context of research on digital campus transformation, student engagement, and administrative automation.

The researchers searched multiple academic databases including Google Scholar, IEEE Xplore, ACM Digital Library, and ScienceDirect, using keywords such as "student organization management system," "university accreditation tracking," "e-governance in higher education," "Role-Based Access Control (RBAC)," "multi-org architecture," "OCR data extraction," and "NLP text summarization." The search was limited to publications from the past five to ten years to ensure that the information was current and relevant to contemporary technological capabilities and academic practices.

In addition to academic sources, the researchers also reviewed gray literature including CHED memorandums, white papers from technology companies, case studies of similar university systems, and documentation from open-source projects. This gray literature provided practical insights into implementation challenges, scalability considerations, and user adoption strategies that complemented the more theoretical perspectives found in academic publications.

The online research also included an investigation of relevant technologies and platforms that could be used to build the UniOrg system. The researchers evaluated different web development frameworks, backend technologies, database systems, OCR libraries, natural language processing

models, and cloud storage options. This technical research informed the selection of the technology stack, specifically PHP (Laravel), Supabase, Tesseract OCR, and Cloudflare R2, ensuring that the system was built using appropriate, reliable, and highly scalable technologies.

3.7.2 Instrument Validation.

The thesis adviser and a panel of IT professionals and domain experts checked the instruments through face and content validation. The full procedure is detailed in Section 3.13.2.

3.7.3 Administration of the Instrument.

The instruments were administered in person. Interviews and brainstorming sessions were held with stakeholders, and the TAM questionnaire was answered during the system demonstration at LSPU-SCC. The full procedure is detailed in Section 3.13.3.

Table 3.3 displays the five-point Likert scale used to evaluate the effectiveness and user acceptability of UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging. This evaluation focuses on the perceptions of the respondents regarding the system's perceived usefulness, perceived ease of use, attitude toward using, behavioral intention to use, and experience based on the Technology Acceptance Model (TAM).

The research utilizes descriptive statistics to analyze the gathered data. Specifically, the summation of data counts and the computation of the weighted mean are applied to determine the overall perception of the

respondents. These measures also help identify any significant differences between the responses of the various stakeholder groups involved in the survey.

Table 3.3. Likert Scale for User Acceptance

Scale	Range	Verbal Interpretation
5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly Disagree

Table 3.3 presents the five point Likert scale used in evaluating user acceptance of the system. Scores were interpreted from 5 Strongly Agree down to 1 Strongly Disagree with their corresponding ranges. This scale served as basis in determining the perception of respondents toward UNIORG.

3.8 Project Design

This section describes how the UniOrg platform was planned and structured before and during development. The design was based on the requirements gathered from interviews with OSAS staff and student organization officers, who described the slow manual accreditation process, scattered files, and delayed approvals. From these findings, the researchers identified the major components, processes, resources, interfaces, and technical considerations needed for the solution. Each part of the

design answers a specific manual bottleneck found during the preliminary investigation.

3.8.1 System Requirements

The functional requirements are OCR-based text extraction, NLP auto-tagging of documents and AI summarization. They also include multi-org workspaces with profiles and branding, online accreditation application and renewal with status tracking, event proposals with Dean and OSAS approval plus master calendar sync, centralized payment tracking with GCash receipt upload and verification, secure e-voting with one-time ballots and tally publishing, EduShare folders and e-library search, and a feedback hub with complaints, chatbot, and live chat. Role-based dashboards control what each user type sees. The non-functional requirements are system security, good performance, and responsive UI. Security means strict login controls and role-based access in line with the Data Privacy Act of 2012. Performance means the system stays fast even when hundreds of students use it at the same time. Responsive UI means the pages work on both desktop and mobile browsers.

3.8.2 System Design

The platform uses a multi-tenant use case model where each organization has an isolated workspace under one shared system. The main users are OSAS administrators, Deans, org advisers, organization officers, and regular students, and each has defined interactions with the modules. Accreditation flows from officer submission to OSAS review and approval, while event proposals pass through Dean approval before OSAS approval.

Data design uses relational models that connect user profiles, organization records, documents, events, elections, and payment logs. Process flows, the context diagram, data flow diagrams, and the entity-relationship diagram documented in the appendices show how data moves between users and modules. Interface wireframes were prepared first so the layout, navigation, and dashboards matched what each role needed before coding started.

3.8.3 System Architecture:

The system uses a client-server architecture with four layers. The presentation layer is the responsive web interface that all user roles open in their browsers. The application layer holds the Laravel backend logic, including role-based access control, accreditation routing, and document generation. The data layer keeps structured records in Supabase and heavy files in Cloudflare R2. The external services layer connects Tesseract OCR, Google Cloud Vision, Google Cloud Natural Language, and Groq AI models for text extraction, auto-tagging, and summarization. Requests flow from the interface through access checks to the backend and storage, then back with the needed data. Refer to the system architecture diagram for the visual representation of these components and their interactions.

3.9 Software Development Model



Figure 3.1. This figure illustrates the Agile Development methodology
Image Source:
https://aguayo-co.s3.amazonaws.com/public/images/Agile-methodology-user-experience.width-490.png

As depicted in figure 3.1, the development of UniOrg followed the Agile methodology, an iterative approach to software development that organizes work into short, time-boxed cycles called sprints, typically lasting one to four weeks, where each sprint produces a working and testable increment of the system (Binboga & Altin Gumussoy, 2024). Unlike traditional waterfall development, Agile prioritizes continuous stakeholder collaboration, rapid feedback integration, and flexible response to evolving requirements, making it particularly suitable for systems whose full functional scope cannot be entirely defined at the (Tetteh, 2024). This methodology was selected for UniOrg because the system's requirements, spanning a configurable multi-org architecture, strict Role-Based Access Control (RBAC), Optical Character Recognition (OCR) for document extraction, and NLP-driven

auto-tagging and summarization, required iterative refinement through actual user feedback rather than rigid pre-planned specification.

Specifically, Agile was applied in this study through the following sprint structure based on UniOrg's project Gantt chart. Sprint 1 (Planning, March–April 2026) covered requirements elicitation through interviews with OSAS administrators, org advisers, and student organization leaders, definition of all functional and non-functional requirements, and production of the System Technical Specification (STS). Sprint 2 (Design, May–July 2026) encompassed the system architecture design, entity-relationship diagram, use case specifications, and UI/UX wireframes for the centralized web platform built with PHP (Laravel) and Supabase. Sprints 3 and 4 (Development, July–October 2026) involved the incremental coding and integration of UniOrg's configurable organization modules and advanced AI features using Tesseract OCR, Groq AI models, Google Cloud Vision API, and Google Cloud Natural Language API, with each sprint delivering testable features reviewed against the STS. Sprint 5 (Testing, October–November 2026) covered functional testing, integration testing, technical evaluation, and user acceptance testing utilizing the Technology Acceptance Model (TAM). Sprint 6 (Deployment, November 2026) finalized system deployment and documentation. At each sprint boundary, outputs were reviewed by the project adviser and technical experts, and feedback was incorporated into the next sprint, ensuring that the final UniOrg platform accurately reflects the real operational needs of the Office of Student Affairs and Services (OSAS) and the thirty-nine (39) student organizations of Laguna State Polytechnic University - Santa Cruz Campus.

Development Phases:

Phase 1 - Planning The planning phase involved conducting interviews with key stakeholders, identifying system requirements, and defining the project scope and deliverables. During this phase, the researchers worked closely with the Office of Student Affairs and Services (OSAS) administrators and org advisers to understand the current manual accreditation and student organization management processes, identify administrative bottlenecks, and envision how a centralized digital platform could improve campus operations. The researchers also conducted a feasibility study to assess the technical, operational, and economic viability of the proposed system.

The planning phase began with a comprehensive needs assessment. The researchers asked OSAS personnel about their daily workflows, the volume of accreditation documents they receive, and the challenges in tracking multi-organization compliance. Organization officers were asked about their experiences handling event proposals, managing digital files, and what features would make their leadership roles more efficient. Regular students were interviewed regarding their experiences with campus announcements, organizational feedback, and digital voting processes.

Based on these interviews, the researchers developed a comprehensive list of functional and non-functional requirements for the UniOrg platform. Functional requirements specified critical system capabilities, such as "The system shall automatically extract text from uploaded accreditation documents using Optical Character Recognition (OCR)" and "The system shall summarize narrative reports using Natural Language Processing (NLP)." Non-functional requirements specified performance metrics, ensuring the system responds efficiently under normal load and

maintains strict data security across different user roles. The researchers then created a prioritized product backlog to guide development.

Phase 2 - Designing The designing phase involved creating the system architecture, wireframes, and high-fidelity mockups for the web interfaces. The researchers worked with OSAS and student leaders to design user interfaces that would be intuitive and efficient for their specific organizational workflows.

The design phase began with the creation of high-level system architecture diagrams. The researchers established a scalable, multi-org web architecture powered by a PHP (Laravel) backend, a Supabase database, and Cloudflare R2 for file storage. The architecture was specifically mapped to handle strict Role-Based Access Control (RBAC), ensuring that OSAS, Deans, Org Advisers, Officers, and Students only access modules pertinent to their permissions. Furthermore, the integration points for Tesseract OCR, Groq AI models, Google Cloud Vision API, and Google Cloud Natural Language API were mapped out within the system's data pipeline.

Next, the researchers created wireframes focusing on layout, navigation, and functionality. After several rounds of review with the stakeholders, these were upgraded into high-fidelity mockups reflecting modern web design principles. The database schema was also finalized, defining relationships between users, organizations, events, documents, elections, and manual payment tracking logs.

Phase 3 - Development The development phase involved building the core modules of UniOrg, including the OSAS Admin Dashboard, Organization Workspaces, Document Repository, E-Voting System, and the AI-powered processing features. Following Agile principles, the researchers developed the system in short, two-week sprints, delivering working software increments continuously.

The first sprint focused on setting up the Laravel development environment, connecting the Supabase backend, and implementing the fundamental user authentication and RBAC infrastructure. Subsequent sprints incrementally added functionalities. Early sprints covered basic user profile management and document uploads, while later sprints integrated the complex features, such as the Google Cloud Vision API for accurate text extraction and the Google Cloud NLP API for intelligent document auto-tagging.

Throughout the development process, the team followed strict coding best practices, including version control using Git, continuous integration, and peer code reviews. This approach ensured that the integration of advanced AI APIs did not compromise the speed or security of the core web platform.

Phase 4 - Testing The testing phase involved conducting functional, performance, and security testing. At the end of each sprint, the development team demonstrated the new features to stakeholders and conducted user acceptance testing to verify that the modules worked as expected.

Testing was conducted at multiple levels. Unit testing verified isolated functions (e.g., checking if the NLP API correctly summarizes a given text), while system testing evaluated the entire multi-org platform. The researchers employed a black-box testing methodology, focusing on the platform from the end-user's perspective without analyzing the internal code. Testers provided various inputs, such as uploading blurred accreditation forms or simulating high-traffic digital elections, to verify system accuracy and stability. Performance and load testing were also conducted to ensure that the system could handle simultaneous access by hundreds of students across 39 organizations without crashing.

Phase 5 - User Evaluation The user evaluation phase involved deploying the system to the target population and gathering quantitative and qualitative feedback. Utilizing the Technology Acceptance Model (TAM), the researchers conducted formal evaluation sessions where representatives from all stakeholder groups (OSAS administrators, Deans, Officers, Org Advisers, and Students) interacted with their respective dashboards.

Each group was asked to complete specific tasks. For instance, OSAS administrators tested the approval of extracted OCR documents, while students navigated the e-Library and event registration. After interacting with the system, participants completed the structured TAM questionnaire to measure perceived usefulness and ease of use. The evaluation sessions revealed valuable insights, such as minor adjustments needed in the NLP auto-tagging labels and UI enhancements for the mobile-responsive view of the web application.

Phase 6 - Refinement The refinement phase involved making final adjustments based on the TAM evaluation feedback. The researchers prioritized the issues and suggestions that emerged, immediately addressing any critical usability problems before the final deployment.

Modifications were made to optimize the OCR scanning accuracy for specific university document formats, and navigation menus were refined for better user experience. The refinement phase also included comprehensive regression testing to ensure that recent changes did not break existing functionalities within other organization modules. Through the Agile methodology's iterative nature, mistakes were corrected quickly, ensuring that the final UniOrg system successfully addresses the complex organizational and accreditation challenges faced by Laguna State

Polytechnic University.

Figure 3.2. Software Development Life Cycle (SDLC)

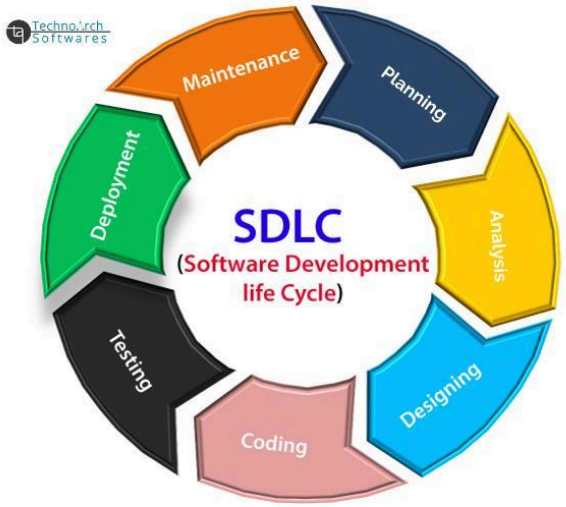


Image Source: https://tas-dp-prod-media.s3.amazonaws.com/blog/sdlc-image.jpg

As depicted in figure 3.2, the Software Development Life Cycle (SDLC) functions as a structured framework that guides the creation of high-quality software through distinct phases: planning, design, development, testing, deployment, and maintenance (Kasım, 2024). For this research, the SDLC was operationalized utilizing the Agile methodology, wherein these specific phases were executed iteratively across short development sprints. This approach enabled the researchers to continuously refine the UniOrg platform based on active, ongoing feedback from its target university stakeholders.

The SDLC was applied to the study as follows. During the planning phase (March–April 2026), the developers conducted key informant interviews with the Office of Student Affairs and Services (OSAS) personnel and student leaders to establish operational requirements, define the platform's core modules, and formulate the System Technical Specification (STS). This ensured compliance with university

accreditation standards and the Data Privacy Act of 2012 (RA 10173) for handling student information. In the design phase (May–July 2026), the overarching system architecture, Supabase database schema, entity-relationship diagrams (ERD), use case specifications, and UI/UX wireframes for the centralized PHP (Laravel) web application were drafted based on the elicited requirements.

Within the development phase (July–October 2026), the platform's core functionalities were built incrementally. This included the integration of Tesseract OCR and Google Cloud Vision API for document text extraction, Google Cloud NLP for the auto-tagging of narrative reports, the multi-org organization workspaces, and the centralized OSAS admin dashboard. Each development sprint yielded testable features that were thoroughly reviewed before advancing. In the testing phase (October–November 2026), initial functional testing was executed based on test case scenarios, followed by full system integration testing and user acceptance evaluations utilizing the Technology Acceptance Model (TAM). This involved respondents from the four primary user groups: OSAS administrators, Deans, organization officers/advisers, and regular students. The deployment phase (November 2026) covers the turnover of the production build and documentation of the TAM evaluation results, while the maintenance phase encompasses post-evaluation refinements and system updates derived from the final findings.

This iterative Agile framework was crucial for UniOrg because it guaranteed that the appropriate stakeholders were involved at the correct development milestones. It allowed technical issues to be identified and mitigated early in the process, ultimately producing a comprehensive system whose features directly address the real-world operational and accreditation requirements of the OSAS and

the thirty-nine (39) student organizations of Laguna State Polytechnic University - Santa Cruz Campus.

3.10 System Architecture

UniOrg's system architecture defines the overall structure and organization of the platform, including its components, modules, and their interactions. The architecture was designed to ensure that the system meets its functional requirements (automated document text extraction, intelligent auto-tagging, centralized dashboard monitoring, and multi-org organization management) as well as its non-functional requirements (security, strict role-based access control, scalability, and performance).

The system follows a modern client-server architecture centered around a responsive web-based platform that caters to all user roles: OSAS Administrators, College Deans, Organization Officers/Org Advisers, and Regular Students. It utilizes a centralized backend server that handles complex data processing, secure multi-org routing, and AI-powered document analysis. The architecture consists of the following key components:

Presentation Layer: A responsive Web Interface built using HTML, CSS, JavaScript, and Laravel's Blade templating engine, designed to adapt seamlessly across desktop and mobile browsers for all stakeholder dashboards.

Application Layer: The core backend logic built with the PHP (Laravel) framework. This layer enforces strict Role-Based Access Control (RBAC), handles the routing of accreditation documents, and utilizes libraries such as DomPDF, PDF Parser, and PHPSpreadsheet for internal document generation and parsing.

Data Layer: Supabase serves as the primary database for structured data storage, handling user credentials, organization records, and election tallies. Cloudflare R2 is utilized as a highly scalable cloud storage solution for safekeeping heavy multimedia files, event evidence, and uploaded accreditation documents.

External Services: Integration of the Google Cloud Vision API and Tesseract OCR for advanced text extraction from images and scanned documents; alongside the Google Cloud Natural Language API for intelligent auto-tagging, and the automated summarization of organizational narrative reports.

3.10.1 Hardware and Software Requirements

The development and deployment setup needs a web server that can run PHP applications with a Laravel backend, a Supabase project for the database, and Cloudflare R2 storage for uploaded files and documents. On the client side, users only need a device with an updated web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge, since UniOrg runs fully in the browser. The AI features need active API access to Tesseract OCR, Google Cloud Vision and Natural Language APIs, and Groq AI models. DomPDF and PHPSpreadsheet handle document and spreadsheet generation on the server.

3.10.2 System Components and Modules

The system is organized into modules that match the main features.

User authentication and role-based access control handles login and view switching between roles. Account management covers user accounts, the organization directory, appointment of deans and officers, and organizational lifecycle records. Organization workspaces give each student organization its own profile, branding, positions, and permission settings. Accreditation covers online application, renewal, document review, and audit logs for OSAS. Document intelligence handles OCR text extraction, NLP auto-tagging and summarization, the document library, and smart search. Events cover proposals, Dean and OSAS approvals, the master calendar, registrations, and financial reports. Elections cover candidate setup, secure voting, and results tabulation. Payments cover membership and events fee, GCash receipt uploads, verification, digital receipts, and collection reports. EduShare and e-library handle folders, file sharing, previews, and resource search. The feedback hub covers complaints, chat FAQs, announcements, and realtime chat. Each role opens its own dashboard: OSAS, Dean, organization admin, and student.

For clarity, the authoritative stack used in the build is PHP Laravel with Supabase and Cloudflare R2 for data and files. Text extraction uses Tesseract OCR with Google Cloud Vision API while tagging and summarization use Groq gpt-oss-20b cloud API with Google Cloud Natural Language API for entity signals. OCR checking relied on softcopy documents from one organization (SOCCS) plus COR samples for registration extraction. The check was functional through black box cases and no formal character level accuracy was measured which is stated here as a limit. Cloud processing also needs stable internet since Groq and Google APIs run online.

3.10.3 Database Structure

The database of UniOrg is relational and hosted on Supabase Postgres. User accounts, organization records, documents, events, elections, payments, and feedback are stored in separate tables. The tables are connected through foreign keys. The full database has 53 tables. For presentation, the schema is grouped into seven figures by module. The users and organizations tables appear again in each figure because almost all records belong to a user and an organization. System tables such as migrations, sessions, cache, and jobs are not shown because they only support the framework.

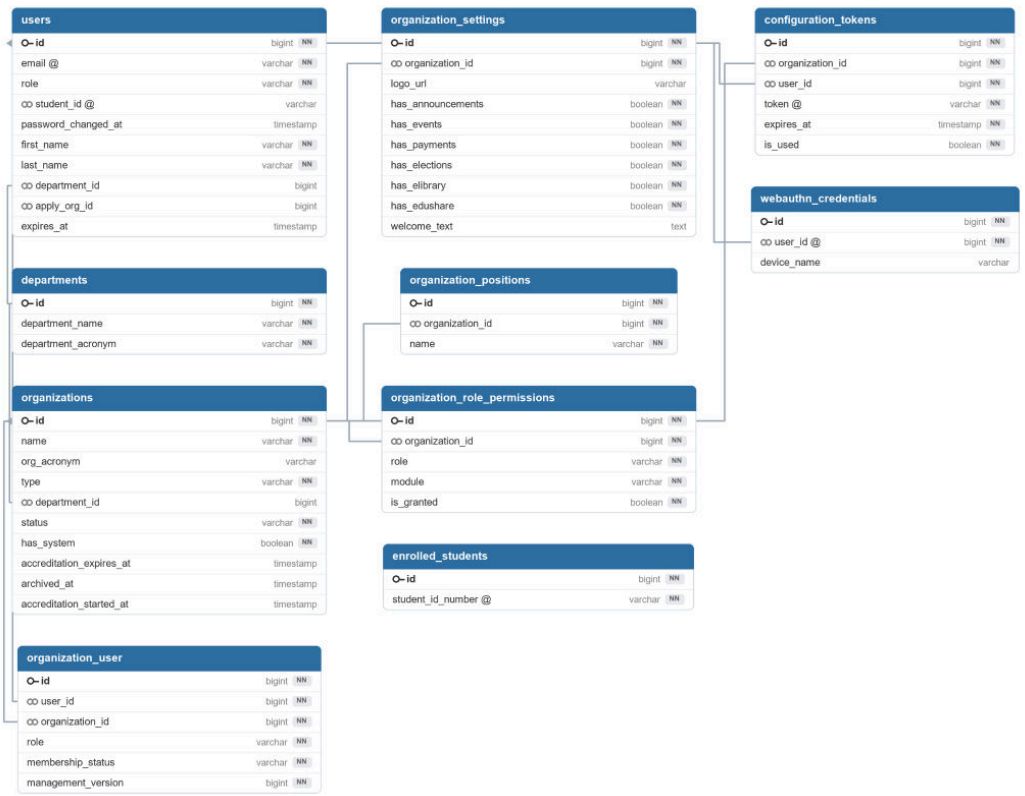


Figure 3.3. Users and Organizations database schema.

This group holds accounts and org structure. The users table connects to departments and to pending applications. The organization_user table links users to organizations with role and membership status. Settings, positions, and role permissions per organization are stored in their own tables

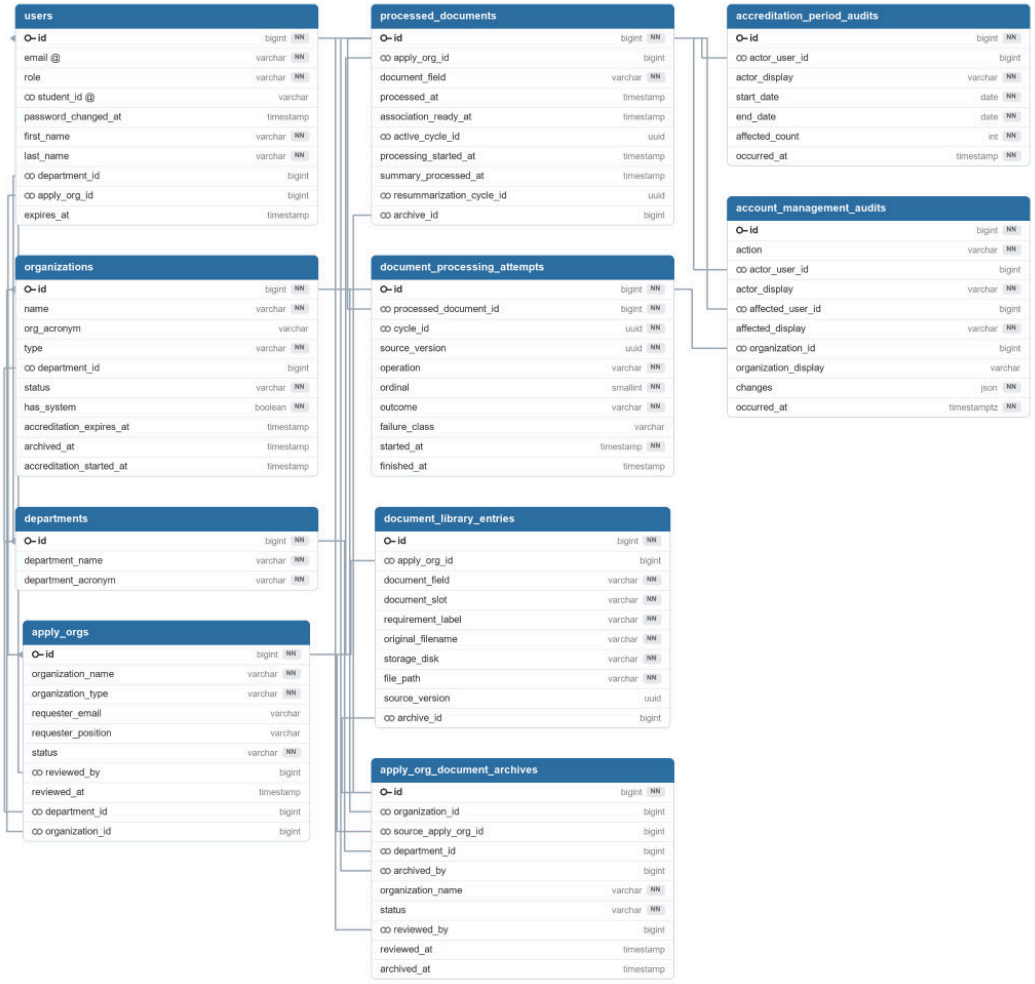


Figure 3.4. Accreditation pipeline database schema.

This group handles org application and renewal. An apply_orgs record holds the request and its status. Each uploaded requirement becomes a processed_documents row with extracted text, summary, and tags. Each AI

attempt is logged in document_processing_attempts. Final copies sit in document_library_entries. Past cycles are kept in apply_org_document_archives. Audit tables record who acted and when.

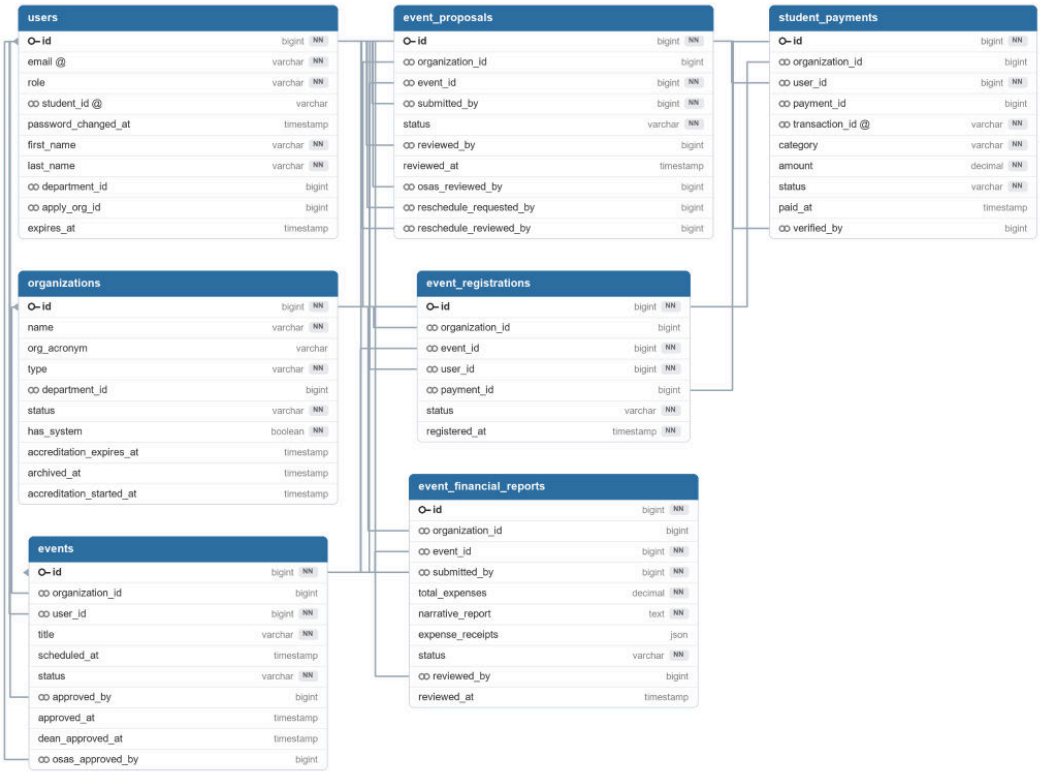


Figure 3.5. Events database schema.

This group handles activities. The events table holds the activity and its approval status from Dean and OSAS. Proposals, registrations, and financial reports point back to the event. Registrations connect to student payments when a fee is required.

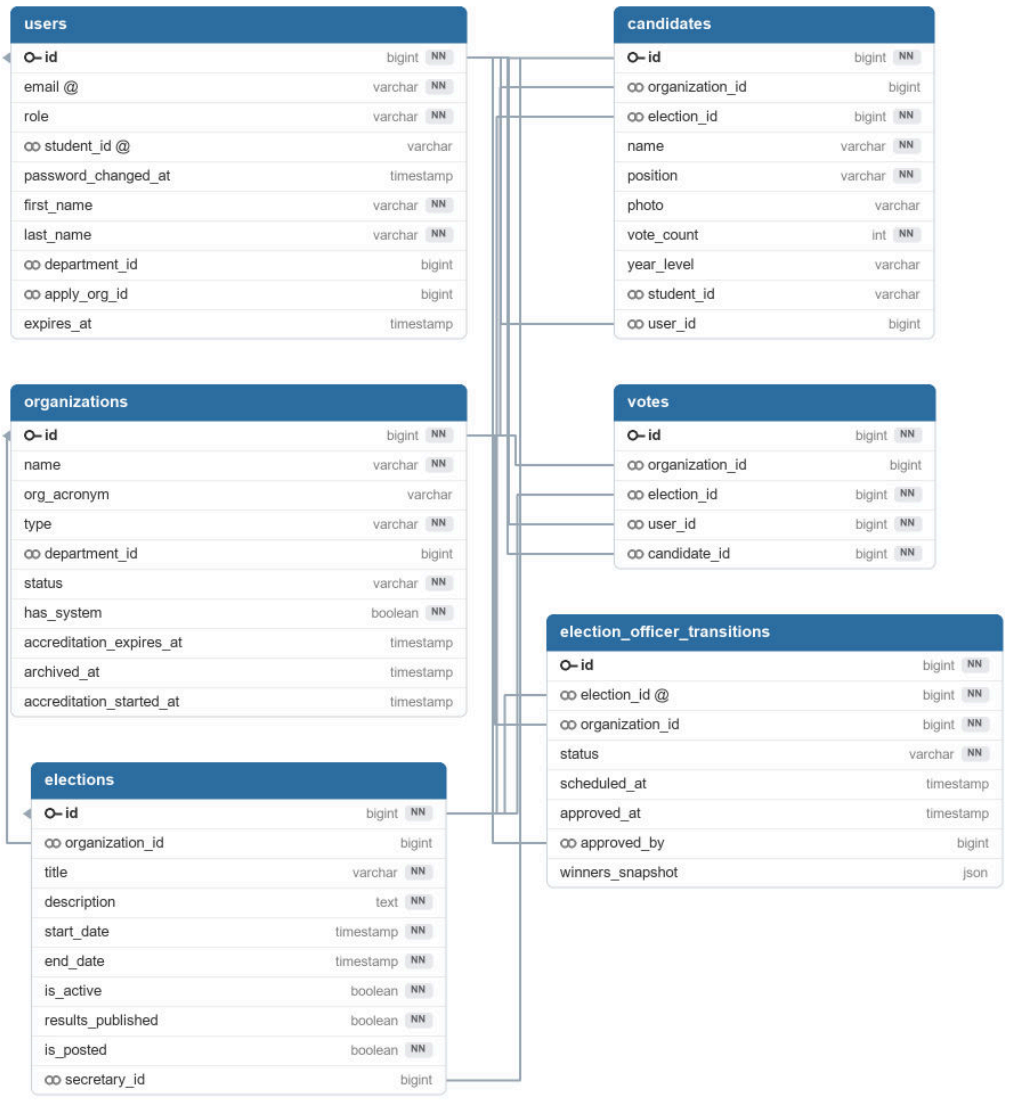


Figure 3.6. Elections database schema.

This group handles e-voting. One election has many candidates and many votes. Each vote links a user to a candidate. Officer turnover after results is recorded in election_officer_transitions.

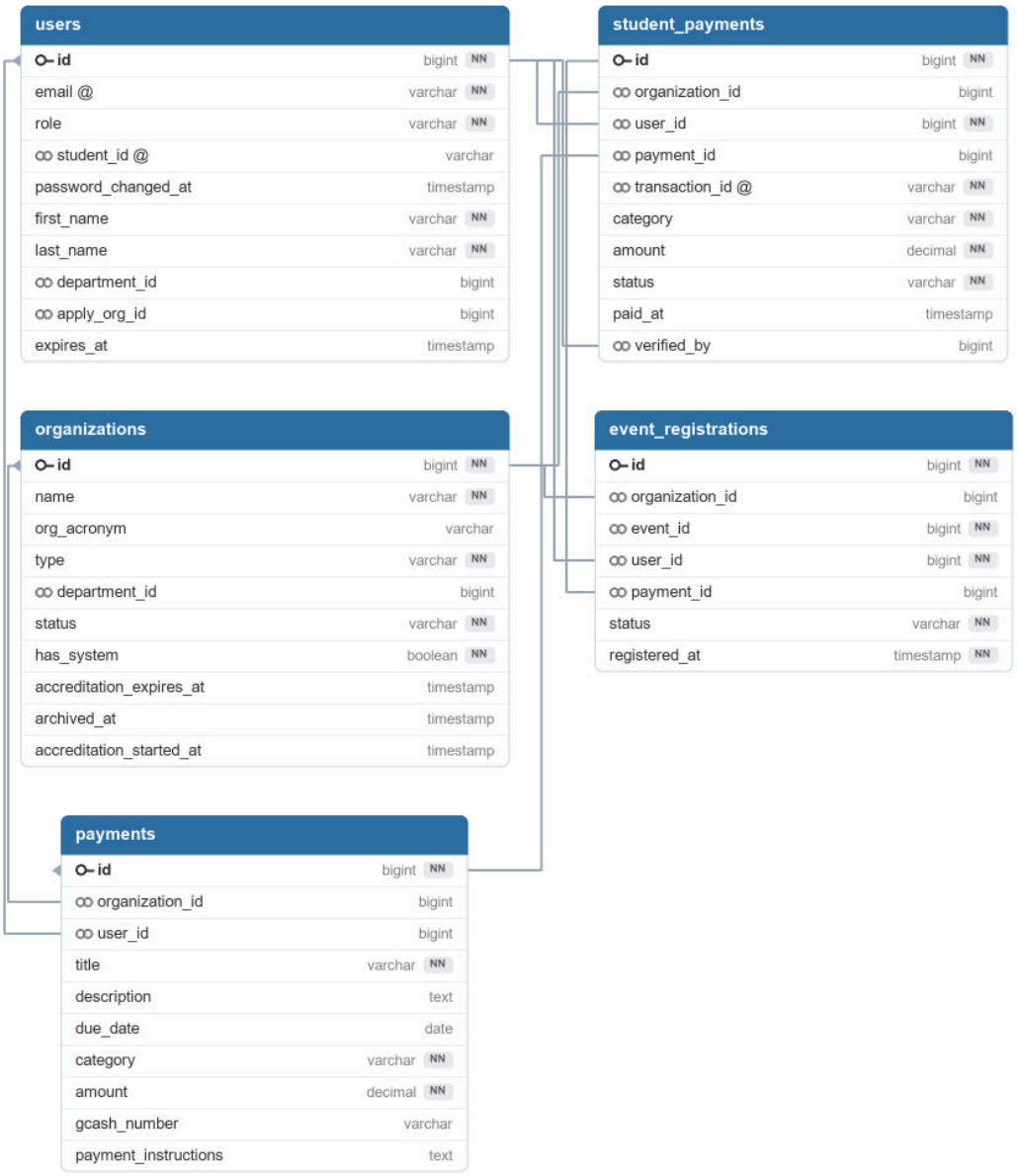


Figure 3.7. Payments database schema.

This group handles collections. The payments table defines a fee of an organization. Each student_payments row is one student payment with proof, status, and verifier. Event registrations can point to a student payment.

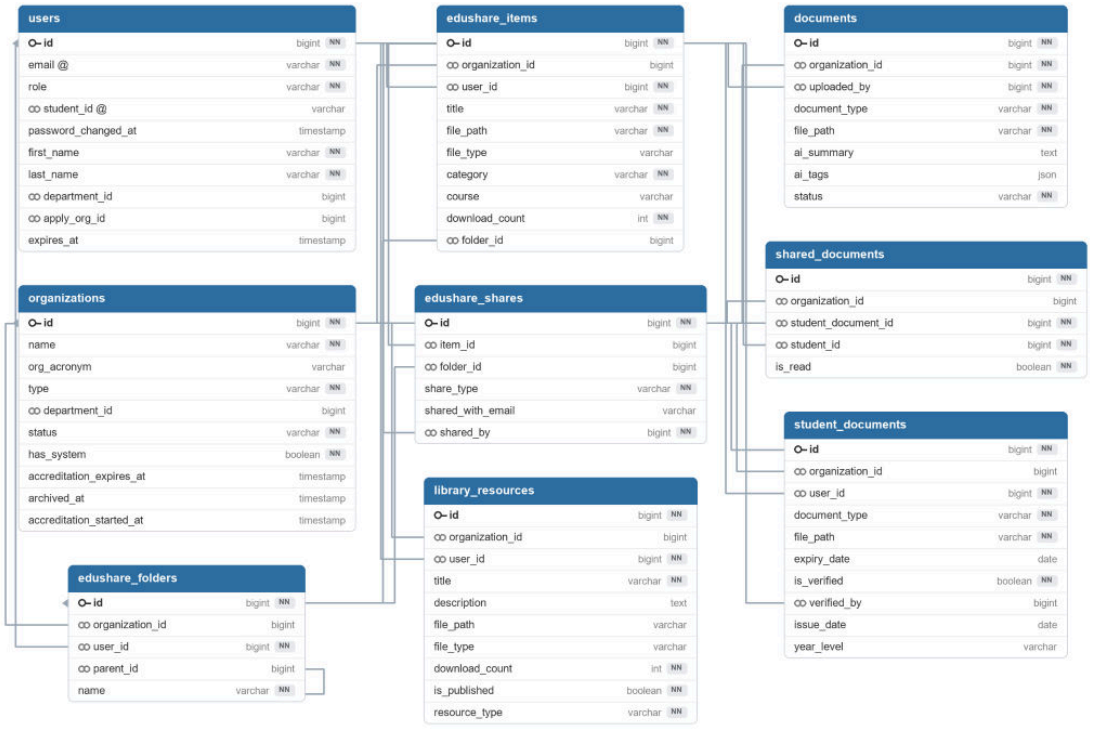


Figure 3.8. EduShare, e-library, and documents database schema.

This group handles files and resources. EduShare folders hold items and sharing rules. Library resources are published materials. The documents, shared_documents, and student_documents tables track uploads and student requirements.

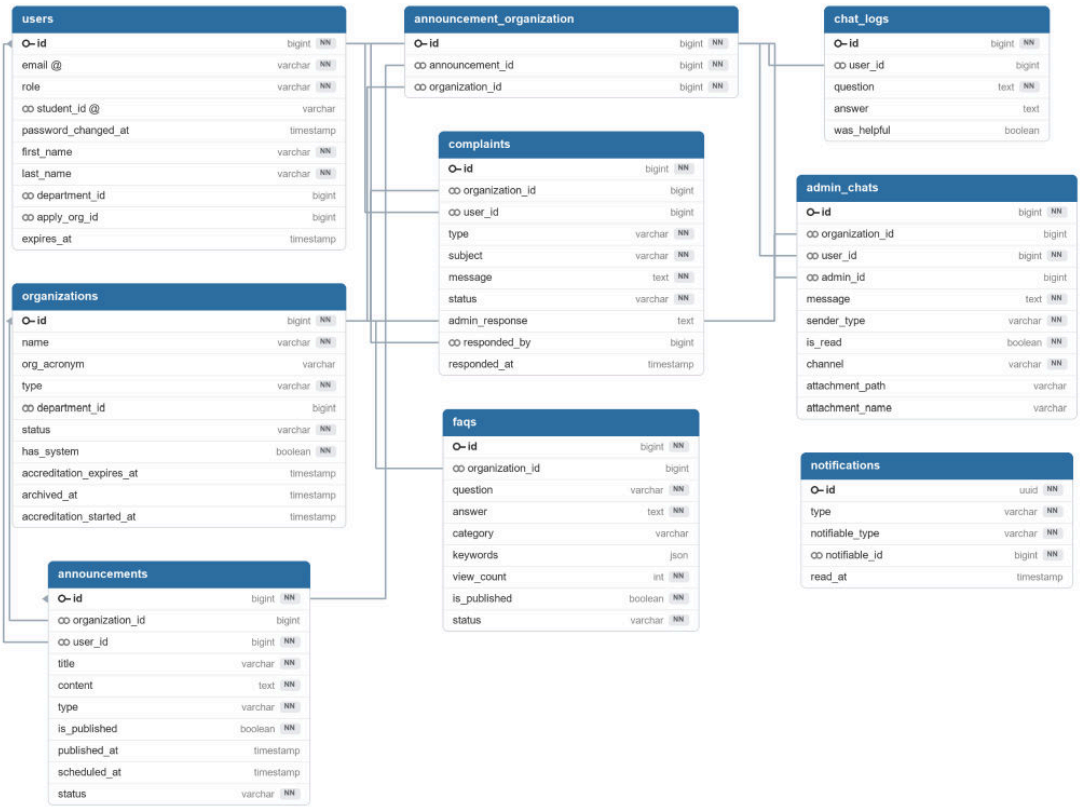


Figure 3.9. Communications and feedback database schema.

This group handles announcements, complaints, chat, and notifications. Announcements reach many organizations through announcement_organization. Complaints and admin_chats link users and organizations. FAQs, chat logs, and notifications support help and updates.

3.10.4 User Interface Design

The user interface is built with Laravel Blade templates using HTML,

CSS, and JavaScript. It is responsive, so pages adjust to desktop and mobile screens. Each role sees its own dashboard with only the menus and data allowed by its permissions. OSAS administrators see university-wide monitoring, Deans see college-level approvals, organization officers see their workspace tools, and students see announcements, events, resources, and feedback forms. Navigation stays consistent across pages so users always know where they are.

3.10.5 System Communication and Interaction

Secure RESTful APIs connect the web interface, the Laravel backend, and external services. Each request passes through role-based access checks so users only reach data and actions allowed for their role. File uploads go to Cloudflare R2 while records stay in Supabase. Realtime features such as chat, notifications, and e-library updates use Supabase Realtime and Pusher channels. Document processing calls Tesseract OCR, Google Cloud Vision, Google Cloud Natural Language, and Groq AI models, then returns extracted text, tags, and summaries to the platform.

3.11 System Testing and Evaluation Procedure

The developed UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging was initially reviewed by IT specialists who evaluated its technical features, database architecture, and AI integrations to identify areas for improvement. Subsequently, feedback was gathered from OSAS administrators, college Deans, org

advisers, student organization officers, and regular students at Laguna State Polytechnic University - Santa Cruz Campus to determine the level of user acceptance and satisfaction with the platform. Comments and suggestions were systematically collected using the Technology Acceptance Model (TAM) based on actual user experience to ensure that the system effectively streamlines student organization management and accreditation tracking. This section presents the process followed by the researchers, from technical testing to end-user evaluation, to validate the system's overall functionality, usability, and effectiveness.

System Testing

System testing was conducted to ensure that UniOrg meets all functional and non-functional requirements specified in the project objectives. The rigorous testing process involved various software evaluation methods, including functional testing, performance testing, security testing, usability testing, and compatibility testing across different viewing environments.

Test cases and test scenarios were meticulously created based on the initial system requirements, covering all major functionalities such as document upload and repository management, multi-org workspace isolation, Optical Character Recognition (OCR) text extraction, Google Cloud NLP-powered auto-tagging and summarization, e-voting, and real-time dashboard monitoring. Each test case was executed systematically to identify, document, and resolve any defects or bottlenecks in the platform.

Testing Types Performed:

3.11.1 Testing Plan

Testing followed a structured plan across the Agile sprints. Each sprint output was checked before moving to the next phase. Functional testing verified every feature against the system requirements. Performance testing checked response time and load capacity during simultaneous access. Security testing checked login controls, data protection, and role-based access limits. Usability testing checked navigation and ease of use across the different dashboards. Compatibility testing checked the platform on desktop browsers like Chrome, Firefox, and Edge, as well as mobile browsers. Formal black-box testing was done by 10 IT testers, and user acceptance evaluation was done by 428 respondents using the TAM questionnaire.

3.11.2 Test Cases

Test cases were designed to check specific inputs and expected outputs across all system features. They covered ten areas matching the black-box testing sheet: user authentication and role control (TC-01), multi-org workspaces (TC-02), accreditation and renewal (TC-03), OCR extraction (TC-04), NLP summarization and auto-tagging (TC-05), event proposals and calendar sync (TC-06), payment tracking and receipt verification (TC-07), e-voting (TC-08), EduShare and e-library (TC-09), and the feedback hub with chatbot and live chat (TC-10). Each case listed the steps, the expected result, and space for the actual result and status.

3.11.3 Testing Data

The testing data were mock student profiles, sample accreditation documents such as constitutions, plans of activities, and officer certifications, scanned COR copies for OCR trials including blurred and corrupted images for negative tests, simulated election candidates and voter accounts, dummy GCash receipts with reference numbers, and sample event proposals. These data resembled real records so the tests reflected actual operating conditions.

3.11.4 Testing Procedure

Tests ran from unit checks of single functions up to full system runs of the whole platform. The 10 IT testers executed the black-box cases from an end-user view and logged inputs, outputs, and remarks. After black-box testing, the 428 respondents tried the system hands-on during the demonstration and answered the TAM questionnaire. Issues found during testing were fixed and retested before final deployment.

3.12 System Testing.

System testing checked if UniOrg meets its functional and non-functional requirements under normal operating conditions. Functional testing covered all ten feature areas while non-functional testing covered performance, security, usability, and compatibility. Black-box method was used so the testers only checked inputs and outputs without looking at the internal code. They tried valid inputs like standard PDF reports and invalid ones like bad image formats or blocked pages, then wrote down what the system did for each step. The testing team consisted of 10 IT professionals and technical staff. All executed test cases passed.

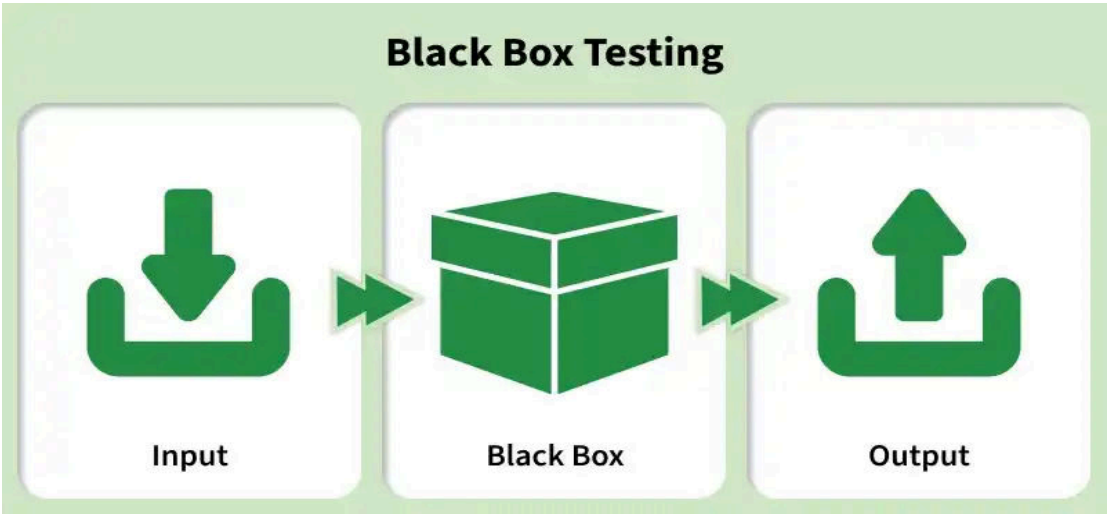


Figure 3.10. GeeksforGeeks (2024). Black Box Testing in Software Testing.: The tool that the researcher had used.

Black box testing is a software testing method that examines functionality without knowledge of internal code structures or implementation details. It focuses solely on inputs and expected outputs to validate system behavior from the user perspective (Ayuningtyas et al., 2023). The researchers utilized this methodology to assess the UniOrg system functionality by providing test inputs and observing outputs without examining the underlying code.

3.12.1 Test Cases

Test cases were based on the system requirements and covered user authentication and role control, multi-org workspaces, accreditation, OCR and NLP document processing, event scheduling, payment tracking, e-voting, academic resources, and the feedback hub. The table below summarizes the test cases and their results.

Table 3.4. System Test Cases

Feature / Module	Test Condition	Expected Result	Actual Result	Status
OCR text extraction	Scanned accreditation form	Printed text is recognized with confidence scores and entities are parsed into form fields	Text, confidence scores, and entities were extracted and parsed as expected	Passed
NLP auto-tagging	Narrative report upload	Entities are extracted, a 2-3 sentence summary is generated, and 3-10 tags are assigned	Entities, summary, and tags were generated and searchable as expected	Passed
E-voting	Verified member casts a vote	Ballot is encrypted and cast once, double voting is blocked, and tallies are published	Vote was cast once, double voting was blocked, and results published as expected	Passed
Payment tracking	E-wallet receipt upload	GCash receipt is submitted, verified or rejected with remarks, and records are exportable	Receipt submission, verification, and export worked as expected	Passed
User authentication and RBAC	Login with valid credentials and role switching	Users log in, reach role dashboards, and blocked URLs redirect to login	Login, role routing, and unauthorized-access redirect behaved as expected	Passed

Table 3.4—continued.

Feature / Module	Test Condition	Expected Result	Actual Result	Status
Multi-org workspace	Officer opens workspace and switches organizations	Portal, branding, and permissions display and switch correctly	Workspace display and organization switching worked as expected	Passed
Accreditation tracking	Submit accreditation package and OSAS review	Tracking ID is issued and status moves from Pending to Approved or Rejected with remarks	Submission, tracking, and approval workflow behaved as expected	Passed
Event scheduling and master calendar	Propose event through Dean and OSAS approval	Proposal flows to approval and the approved event appears on the master calendar	Proposal flow and calendar sync behaved as expected	Passed
Academic resources (EduShare and e-Library)	Upload, share, preview, and search resources	Files, sharing, previews, and live search work correctly	Upload, sharing, previews, and search worked as expected	Passed
Feedback hub and live chat	Submit complaint and use chatbot and live chat	Tickets, responses, chatbot answers, and realtime messages work correctly	Complaints, chatbot, and realtime messaging worked as expected	Passed

Table 3.4 summarizes the system test cases conducted across the ten core modules of UNIORG. Each feature was checked against its expected result and all were marked as Passed. The results indicate that the system functions operated according to their requirements.

3.12.2 Testing Instrument

A black-box testing checklist was used to record the inputs, observed outputs, and remarks for each test case. The checklist followed the ten test cases TC-01 to TC-10, from authentication and workspaces up to resources and the feedback hub. Testers entered valid and invalid inputs, including blurred documents and unauthorized access attempts, and wrote down what the system actually did for each step. Black box testing is a software testing method that examines functionality without knowledge of internal code structures or implementation details. It focuses solely on inputs and expected outputs to validate system behavior from the user perspective.

The researchers utilized this methodology to assess the UniOrg system functionality by providing test inputs and observing outputs without examining the underlying code.

3.12.3 Validation of Testing Instrument

The checklist was reviewed by the thesis adviser and IT experts for clarity and coverage before use.

3.12.4 Interpretation of Test Results

A test case is marked Passed if the actual result matches the expected result. Otherwise it is marked Failed and returned for fixing and retesting. In this study, all executed test cases passed, so no major rework was needed. Minor issues found during the runs were corrected and verified through

regression checks.

Functional Testing: Verified that all platform features work as intended, particularly the uploading of accreditation documents, the accuracy of OCR and Google Cloud Vision in extracting text, the reliability of the NLP summarization, and the proper execution of approval workflows between OSAS and student organizations.

Performance Testing: Assessed the system's response time and load handling capacity, specifically ensuring that the web server (Laravel) and database (Supabase) remain stable even when hundreds of students simultaneously access the system during campus-wide e-voting or major event registrations.

Security Testing: Evaluated user authentication protocols, data encryption, and the strict enforcement of Role-Based Access Control (RBAC) to guarantee that organization officers cannot access files belonging to other organizations and that sensitive student data is securely protected.

Usability Testing: Tested the intuitiveness of the user interface, the ease of navigation within the multi-level dashboards (OSAS, Deans, Org Advisers, Officers, Students), and the overall user experience based on the Technology Acceptance Model (TAM) constructs.

Compatibility Testing: Verified the responsive web functionality of the platform across different desktop web browsers (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge) and mobile web browsers to ensure accessibility on various devices.

3.13 System User Evaluation

System User Evaluation is a critical assessment phase that measures the usability, functionality, and overall user satisfaction of a newly deployed software platform. In the context of this study, System User Evaluation refers to the process of rigorously evaluating the UniOrg platform from the diverse perspectives of its actual end-users, specifically the OSAS administrators, college Deans, organization officers, org advisers, and regular enrolled students. The primary purpose of this evaluation is to gather direct feedback regarding the system's effectiveness in centralizing student organization management and streamlining accreditation processes, as well as to identify any technical or user interface areas requiring further improvement.

This comprehensive evaluation was conducted through a mixed-methods approach, utilizing structured user surveys, key informant interviews, and direct usability testing. The structured surveys, firmly anchored on the Technology Acceptance Model (TAM), were deployed to collect empirical, quantitative data regarding the users' perceived usefulness, perceived ease of use, and overall behavioral intention to utilize the platform. Concurrently, interviews and usability testing allowed the researchers to observe exactly how stakeholders interact with the platform's core modules, including the AI-powered document extraction and NLP auto-tagging features, thereby capturing detailed qualitative insights and identifying any specific usability bottlenecks encountered during actual operation.

The succeeding subsections detail the specific survey questionnaire and its source content, the validation of the research instrument, and the step-by-step procedure implemented during the system user evaluation.

3.13.1 User Evaluation Instrument

This aspect focuses on evaluating how effectively UniOrg addresses the specific operational bottlenecks in student organization governance.

The questionnaire has 15 main items, three for each of the five TAM constructs: Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. It also has 3 Quality Factors items about layout, tools, and visual design, plus respondent background questions and a comment box for suggestions. Answers use a five-point Likert scale.

Figure 3.11. Technology Acceptance Model (TAM) Framework

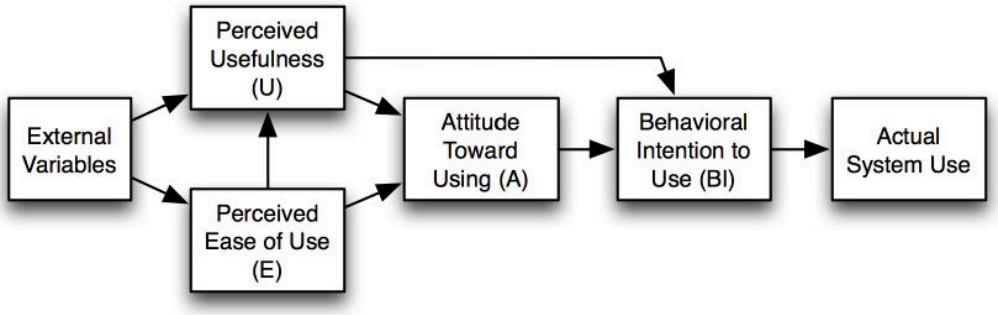


Image Source: https://en.wikipedia.org/wiki/Technology_acceptance_model

As depicted in figure 3.11, the UniOrg platform adopts the Technology Acceptance Model (TAM) as its primary evaluation framework. Originally developed to understand user adoption of information systems, TAM has been continuously refined to suit contemporary software implementations,

including multi-org educational platforms and AI-integrated web applications. Porkodi and Tabash (2024), in their comprehensive analysis of technological acceptance in higher education, confirmed that TAM remains one of the most robust and widely adopted frameworks for evaluating e-learning and campus management systems. Furthermore, recent interdisciplinary studies have demonstrated the operationalization of TAM across diverse university stakeholder groups, such as administrators, faculty, and students, confirming that the model's core constructs can be effectively translated into measurable questionnaire instruments for assessing the real-world acceptance of platforms like UniOrg.

Following this framework, the evaluation questionnaire for UniOrg was meticulously designed to assess user perception across the core constructs of TAM applicable to the system: Perceived Usefulness (PU), which assesses whether stakeholders believe the system enhances their productivity (e.g., whether the OCR text extraction and NLP auto-tagging significantly reduce the manual workload of OSAS in checking accreditation documents); Perceived Ease of Use (PEOU), which evaluates the intuitiveness of the user interface, ease of navigation across the multi-org dashboards, and the simplicity of the document upload process; Attitude Toward Using (ATU), which measures the overall positive or negative disposition of the users (Deans, officers, and students) toward shifting from manual paperwork to a centralized digital platform; and Behavioral Intention to Use (BI), which determines the likelihood that the university stakeholders continuously utilized and mandated the platform for future organization management, e-voting, and manual payment logging.

3.13.2 Instrument Source and Validation

The researchers developed a structured survey questionnaire for user assessment anchored on the Technology Acceptance Model (TAM) as the primary evaluation framework. To guarantee the research instrument's effectiveness, it underwent rigorous validation to confirm that each item is appropriate and strictly aligned with the study's customized evaluation criteria (Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience). The researchers actively consulted with their thesis adviser to implement necessary preliminary revisions. Face and content validity were employed to assess the reliability of the questionnaire. A panel of experts meticulously reviewed the instrument to ensure that it accurately captures the concepts being measured prior to its actual deployment.

The formal validation process involved presenting the drafted questionnaire to a panel consisting of IT professionals, research advisers, and domain experts. These evaluators reviewed each specific item for clarity, relevance, and strict alignment with the TAM constructs, ensuring that the questions are highly comprehensible to the target respondents, namely the OSAS administrators, college Deans, organization officers, and regular students. All gathered feedback and expert recommendations were systematically incorporated to refine the questionnaire. This comprehensive validation guaranteed that the final instrument effectively measures the users' acceptance, practical experience, and overall satisfaction regarding UniOrg's

AI-integrated document processing and centralized organizational workflows.

The questionnaire is researcher made and it was checked through face and content validation by the thesis adviser, Rachiel R. Rivano. Comments on clarity and alignment to TAM were addressed in the revision of items before deployment. No Cronbach alpha or other reliability testing was performed, so reliability is not claimed here and this is noted as a limitation of the study.

3.13.3 User Evaluation Procedure

The system was presented through a live demonstration at LSPU-SCC showing the core modules such as OCR upload, NLP auto-tagging, and the dashboards. Respondents tried the system hands-on before answering, and the researchers stayed present for questions. The questionnaires were collected right after the session and checked for completeness. The full distribution detail is in Section 3.5.1.

3.13.4 Evaluation Criteria:

The system was evaluated using five criteria: Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. Responses were measured on a five-point Likert scale, where higher mean scores indicate stronger acceptance. Completed instruments were retrieved, verified for completeness, and compiled for analysis.

Table 3.5. ***Likert Scale for Weighted Mean***

Scale	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Mean scores were interpreted using a 5-point Likert scale continuum 4.21 to 5.00 (Strongly Agree / Excellent) 3.41 to 4.20 (Agree / Very Good) 2.61 to 3.40 (Neutral / Fair) 1.81 to 2.60 (Disagree / Poor) and 1.00 to 1.80 (Strongly Disagree / Very Poor) The weighted mean was calculated for each questionnaire item to determine the overall level of agreement across the respondent groups while standard deviation was used in measuring the consistency of responses. These measures also helped in identifying significant differences between respondent categories

3.14 Statistical Treatment

This section details the specific statistical techniques utilized in the processing, analysis, and presentation of the gathered survey data. Statistical treatment encompasses the comprehensive range of methods applied in quantitative research to examine empirical data, describe it using quantifiable measures, and compare the results against established theoretical frameworks, which, in this study, is the Technology Acceptance Model (TAM). Statistical analysis is vital for adequately sampling the target population, determining relationships or correlations

between different system attributes, and measuring the concrete differences between sets of empirical data based on the study's stated objectives.

The careful selection of appropriate statistical methods is paramount to ensuring that the research findings are valid, reliable, and practically meaningful. For this study, the researchers mainly used descriptive statistical techniques to analyze the quantitative data from the evaluation questionnaires. Cross-tabulation and correlation analysis were added to explore relationships between variables, but no hypothesis testing was done. These techniques were deliberately chosen based on the ordinal nature of the data derived from the Likert scale responses. The analysis directly addresses the study's primary objectives, providing clear and interpretable results regarding the end-users' Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and overall Experience with the UniOrg platform.

The formal statistical analysis was conducted utilizing the Statistical Package for the Social Sciences (SPSS), a robust and widely used software package for statistical analysis in technical and social science research. SPSS provides a comprehensive suite of statistical tools capable of efficiently processing the TAM evaluation data, making it exceptionally well-suited for this study. Additionally, Microsoft Excel was utilized for the initial data entry, preliminary data screening, and the generation of clear visual representations such as charts and graphs to effectively illustrate the evaluation results of the platform's AI integrations and centralized organizational workflows.

Frequency and Percentage Distribution

The Frequency and Percentage Distribution method was utilized to appropriately organize, analyze, and interpret the survey data gathered from the respondents evaluating the UniOrg platform. This fundamental statistical tool was applied to determine the proportion of responses falling under each specific category, particularly the demographic profiles of the users and their corresponding answers across the Likert scale options. By calculating these distributions, the researchers were able to generate a clear, quantitative overview of the participants' feedback and summarize the evaluation results based on the Technology Acceptance Model (TAM).

Equation 2. Frequency and Percentage Distribution[10]

Formula: % = (f / N) Ã— 100

Where:

- % = Percentage of responses
- f = Frequency of responses
- N = Total Number of Respondents

Frequency distribution is one of the most basic yet fundamental statistical techniques, providing highly valuable insights into empirical data. By accurately counting how many respondents selected each Likert scale option for every questionnaire item, the researchers could quickly identify emerging patterns regarding user acceptance. For example, if a significant majority of the respondents strongly agreed that the UniOrg platform was easy to navigate (Perceived Ease of Use), this would indicate a high level of user satisfaction with the system's interface design.

Conversely, if responses were evenly distributed across all five response options regarding a specific functionality, such as the NLP-powered auto-tagging, this would suggest that the stakeholders had mixed opinions and varied experiences regarding that particular feature.

Percentage distribution is particularly useful for comparing responses across different stakeholder groups or various TAM questionnaire items. For instance, the researchers could compare the percentage of OSAS administrators who found the OCR text extraction highly beneficial (Perceived Usefulness) to the percentage of student organization officers who shared the same sentiment. This comparative analysis identifies whether technological acceptance levels significantly differed between distinct user roles. Similarly, the researchers could compare the percentage of respondents who were satisfied with different platform modules, such as the e-voting system versus the manual payment logging, identifying which features were highly effective and which areas required further technical refinement.

The researchers systematically presented the frequency and percentage distributions through comprehensive tables that displayed, for each questionnaire item, the exact number of respondents who selected a specific response option and the corresponding percentage of the total sample it represented. To enhance data interpretability, these tabular presentations were accompanied by appropriate visual representations, such as bar charts and pie charts, making it significantly easier to quickly grasp the overall pattern of the university stakeholders' responses.

Weighted Mean

The Weighted Mean formula was employed to accurately interpret the degree of user acceptance of the UniOrg platform based on the empirical responses gathered

from the OSAS administrators, college Deans, faculty advisers, organization officers, and regular students. Since the TAM-based evaluation utilized a structured five-point Likert scale to measure the core constructs (Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience), each response category was assigned a specific numerical weight to reflect its relative importance and to compute the central tendency of the stakeholders' feedback.

Formula:

$$\bar{X}w = \sum(f \times x) / N$$

Where:

$\bar{X}w$ = Weighted Mean (average Likert score)

f = Frequency of responses for each Likert scale option

x = Numerical weight assigned to each option (5 = Strongly Agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly Disagree)

N = Total number of respondents

The weighted mean is a measure of central tendency that takes into account both the frequency of each response and the numerical value assigned to that response. It provides a single number that summarizes the overall level of agreement or satisfaction for a particular questionnaire item. The weighted mean is particularly useful for Likert scale data because it treats the response options as if they were on an interval scale, allowing for mathematical operations that would not be appropriate for purely categorical data.

Standard Deviation Formula

The Standard Deviation formula was used to measure the variability or dispersion of the survey responses gathered from the university stakeholders in evaluating the UniOrg platform. It mathematically indicates how much the individual Likert scale responses differ from the computed weighted mean, providing a clearer picture of the consistency and consensus of the participants' perceptions regarding the system's perceived usefulness, perceived ease of use, and overall experience. A low standard deviation indicates that the users' responses are highly clustered around the mean (showing strong agreement among the stakeholders), while a high standard deviation indicates a wider divergence in their feedback.

$$SD = s = n - 1\sum(xi - X)2$$

Where:

s = Sample Standard Deviation

x_i = Each individual response value

$\bar{X}$ = Mean of all responses

n = Total number of respondents

Standard deviation serves as a critical measure of variability that indicates how spread out the survey responses are around the computed weighted mean. A lower standard deviation signifies that responses are closely clustered around the mean, suggesting a high level of consistency and consensus among the respondents. This implies that the university stakeholders share similar perceptions regarding that specific aspect of the platform, whether positive or negative. Conversely, a higher

standard deviation indicates greater variability in the responses, revealing diverse opinions about the system's features and overall performance. This might suggest that the system workflows seamlessly for certain user groups but not for others, or that different stakeholders have varying administrative expectations.

The standard deviation is particularly insightful when interpreted in conjunction with the weighted mean. For example, if a specific questionnaire item yields a high mean, indicating overall satisfaction with the NLP auto-tagging feature, but also presents a high standard deviation, this reveals that while the majority of users are highly satisfied, a significant minority holds dissenting views. Such instances warrant further investigation to understand why specific users encountered different experiences. On the other hand, a high mean combined with a low standard deviation indicates a robust consensus that the system performs excellently on that given criterion.

The researchers utilized the standard deviation to identify specific questionnaire items where there was significant disagreement among the respondents. These flagged items were subjected to further analysis, often involving the examination of qualitative feedback or open-ended comments to understand the root causes of the divergent opinions. Analyzing these variances helps determine if different user roles (e.g., OSAS administrators versus regular students) hold systematically different views regarding certain platform functionalities, providing valuable insights on how the system can be improved to better serve the entire campus ecosystem.

Furthermore, the standard deviation was employed to assess the internal reliability of the evaluation instrument itself. Items yielding abnormally high standard

deviations might indicate that the question was phrased ambiguously or was confusing to non-technical users, leading respondents to interpret it in varied ways. In such cases, the researchers would note these discrepancies for the refinement of future evaluations.

In addition to these primary statistical techniques, the researchers employed supplementary analyses to extract deeper insights from the quantitative data. Cross-tabulation was utilized to examine relationships between distinct variables, such as whether technological acceptance differed based on respondent characteristics like their designated user role, college department, or prior technological experience. Correlation analysis was also conducted to identify significant relationships between the different Technology Acceptance Model (TAM) constructs, such as determining whether stakeholders who rated the platform highly on *Perceived Ease of Use* also exhibited a strong *Behavioral Intention* to continue utilizing the system. These supplementary statistical treatments enabled the researchers to develop a highly nuanced understanding of user perceptions and accurately identify the key factors influencing overall satisfaction with the UniOrg platform.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results of the design, development, testing and evaluation of UNIORG: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging. It covers how the system works for accreditation and document management of the Office of Student Affairs and Services (OSAS) in Laguna State Polytechnic University Santa Cruz Campus, which handles 39 recognized student organizations. The architecture, the interface outputs and the evaluation results from the OSAS administrators, Deans, organization advisers and officers, and students are presented and interpreted to check if the objectives were met.

4.1 General Presentation of Results

This section outlines how the findings are arranged from the design, development, testing and evaluation of UNIORG: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging. The results follow the order of the specific objectives in Chapter I, so each goal has its own evidence from the study.

To keep the flow clear, each objective is discussed with the following parts:

- Objective: The specific goal being answered, from module development to testing and user evaluation.

- Activity or Procedure Performed: The steps done to meet the objective, such as Agile development sprints, Black-Box testing, or the Technology Acceptance Model (TAM) survey.

- Data or Evidence Obtained: The actual outputs from the steps, including the deployed features, test case results, and survey answers from the OSAS administrators, Deans, organization advisers and officers, and students.

- Visual and Statistical Presentation: The architecture diagrams, interface screenshots, testing summaries, and statistical tables that support the findings.

- Interpretation of the Result: What the data and the built modules mean based on the system requirements.

- Implication of the Finding: How the result answers the objective, specifically how it fixes the manual accreditation and document handling of OSAS with its 39 organizations.

4.2 System Architecture and Development Results

This section presents how UNIORG was designed, developed and built to meet Objective 1. It shows the actual outputs from turning the manual, paper-based accreditation of the Office of Student Affairs and Services (OSAS) in Laguna State Polytechnic University Santa Cruz Campus into a working web platform. Based on the interviews with the OSAS Chairperson and the organization advisers, development followed Agile with a Laravel web app for all user roles, backed by Supabase for records and Cloudflare R2 for files, with Tesseract OCR and Cloud Vision for text reading and Cloud Natural Language with Gemini for tagging and

summaries. To show the full build, the next subsections cover the system overview and architecture, the hardware and software requirements, the database structure, and the interfaces made for each user group.

4.2.1 System Overview

UNIORG is built as a configurable university-wide platform made to modernize how the Office of Student Affairs and Services (OSAS) and the student organizations of Laguna State Polytechnic University Santa Cruz Campus handle accreditation, documents and activities. At its core, the system replaces the manual submission of hard copies and logbook tracking with one web platform. Its main purpose is to connect OSAS and the 39 organizations in a single workspace, so papers are submitted, tagged and approved with less delay.

The platform serves five user groups with pages per role: Students, who register with their COR, join organizations and view events and payments; Organization Officers, who submit accreditation papers, propose events and manage members; Org Advisers, who guide and check their organization records; Deans, who review event proposals; and the OSAS administrators, who monitor compliance and approve accreditation with Role-Based Access Control (RBAC). Major functions cover accreditation tracking, OCR text extraction with NLP auto-tagging and summaries, master calendar, centralized payments with receipt upload, e-voting with live tally, and EduShare with e-library. The flow runs as a loop: an officer uploads a requirement, the system reads and tags it, and OSAS reviews the result for approval or return for compliance.

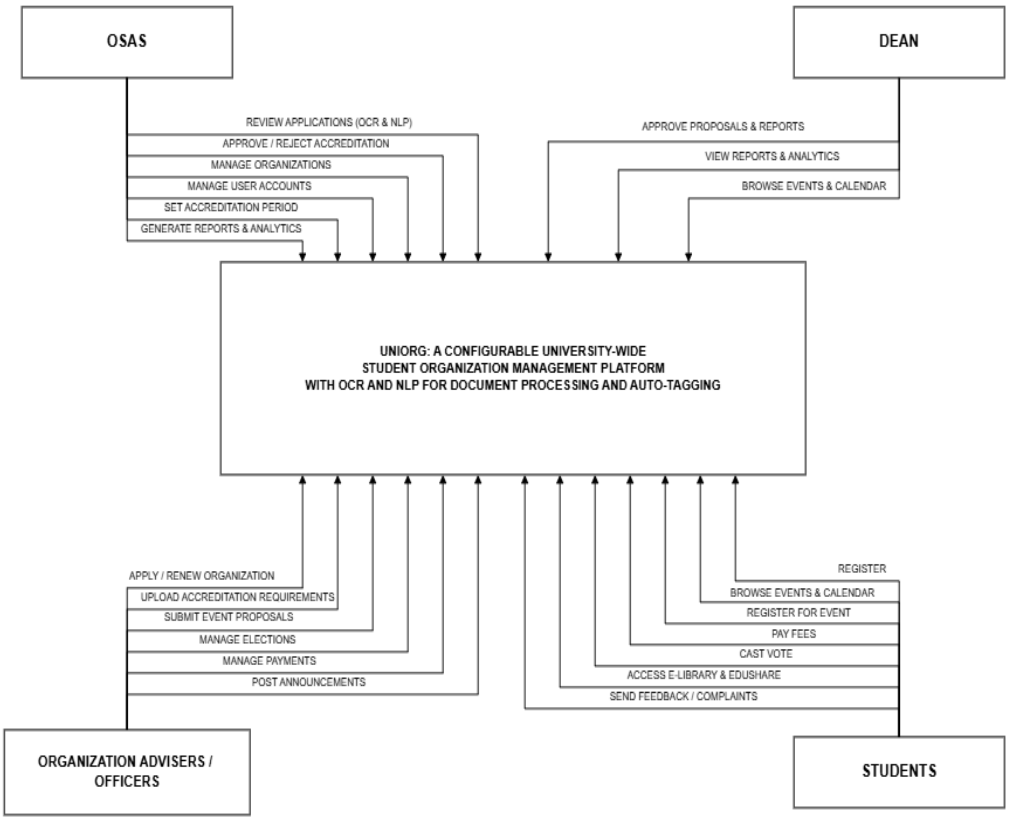


Figure 4.1. System Context Diagram

4.2.2 System Architecture

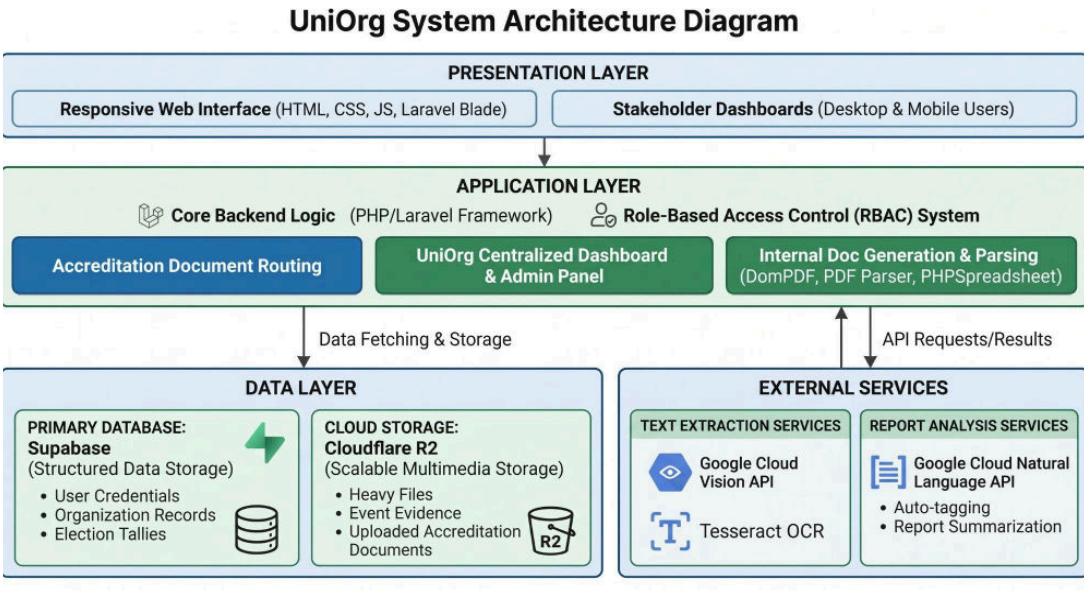


Figure 4.2. System Architecture

The build of UNIORG uses a client-server setup that keeps all users in one web app with pages per role. As shown in Figure 4.1, the parts are split into the Presentation Layer, the Application Layer, the Data Layer and the External Services, so browsers, backend, storage and AI tools work as one flow.

The Presentation Layer is the responsive web interface that all roles open in browsers. It uses HTML, CSS, javaScript and laravel blade templates. Students, officers, advisers, deans and OSAS see their own dashboards for accreditation, events, payments, elections and files. The pages adjust to desktop and mobile browsers.

The Application Layer holds the core logic in PHP (Laravel). It checks the role of each user before opening a module, routes accreditation papers from submission to OSAS review, and handles event proposals through dean approval. Libraries such as DomPDF, PDF Parser and PHPSpreadsheet support report generation and file parsing.

The Data Layer keeps records and files apart. Supabase stores accounts, organizations, events, votes and payment logs. Cloudflare R2 keeps the uploaded accreditation papers and other heavy files. Every paper, payment, vote and registration points to one organization and one user role.

The External Services handle the document reading. Tesseract OCR and Google Cloud Vision API extract text from scanned papers. Google Cloud Natural Language API and Gemini models tag the files and summarize narrative reports. Requests pass from the interface through access checks to the backend and storage, then return with the needed data.

4.2.3 Hardware and Software Requirements

To deploy UNIORG without adding heavy cost on the school, the system runs on standard computers and browsers. As listed in Table 4.1, the hardware covers the minimum setup and the suggested setup for daily use. For OSAS, deans, advisers and officers, a standard laptop or desktop with at least a dual-core processor and 4 GB of RAM is enough to open the dashboards and review papers. For students, any device that can run a modern browser works, including shared computer units in school. All specs listed are already available in the campus.

For software, the platform needs only a modern web browser such as Google Chrome, Microsoft Edge or Mozilla Firefox. There is no heavy install since UNIORG is web-based for all roles. A stable internet connection is required in all endpoints so records sync through Supabase, files load from Cloudflare R2, and the OCR and NLP services respond.

Table 4.1. Hardware Requirements

HARDWARE REQUIREMENTS	EXISTING
Laptop or Desktop Computer	YES
Smartphone (Android 11.0+)	YES
Stable Internet Connection	YES
Dual-Core Processor or Higher	YES
At least 4 GB RAM or Higher	YES

Table 4.2. Software Requirements

SOFTWARE REQUIREMENTS	EXISTING
Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)	YES
Internet Connection	YES
Android 11.0+ (for mobile users)	YES

4.2.4 Database Structure

The database of UniOrg is relational and hosted on Supabase Postgres. User accounts, organization records, documents, events, elections, payments, and feedback are stored in separate tables. The tables are connected through foreign keys. The full database has 53 tables. For presentation, the schema is grouped into seven figures by module. The users and organizations tables appear again in each figure because almost all records belong to a user and an organization. System tables such as migrations, sessions, cache, and jobs are not shown because they only support the framework.

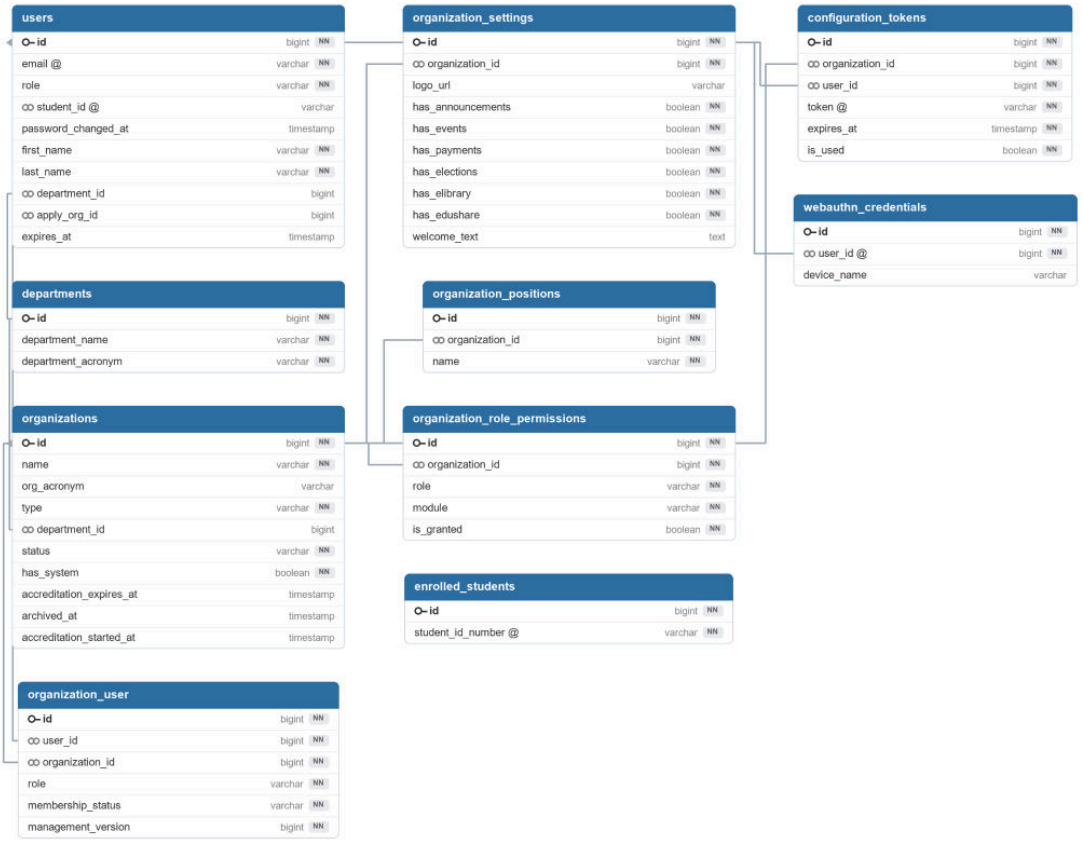


Figure 4.3. Users and Organizations database schema.

This group holds accounts and org structure. The users table connects to departments and to pending applications. The organization_user table links users to organizations with role and membership status. Settings, positions, and role permissions per organization are stored in their own tables

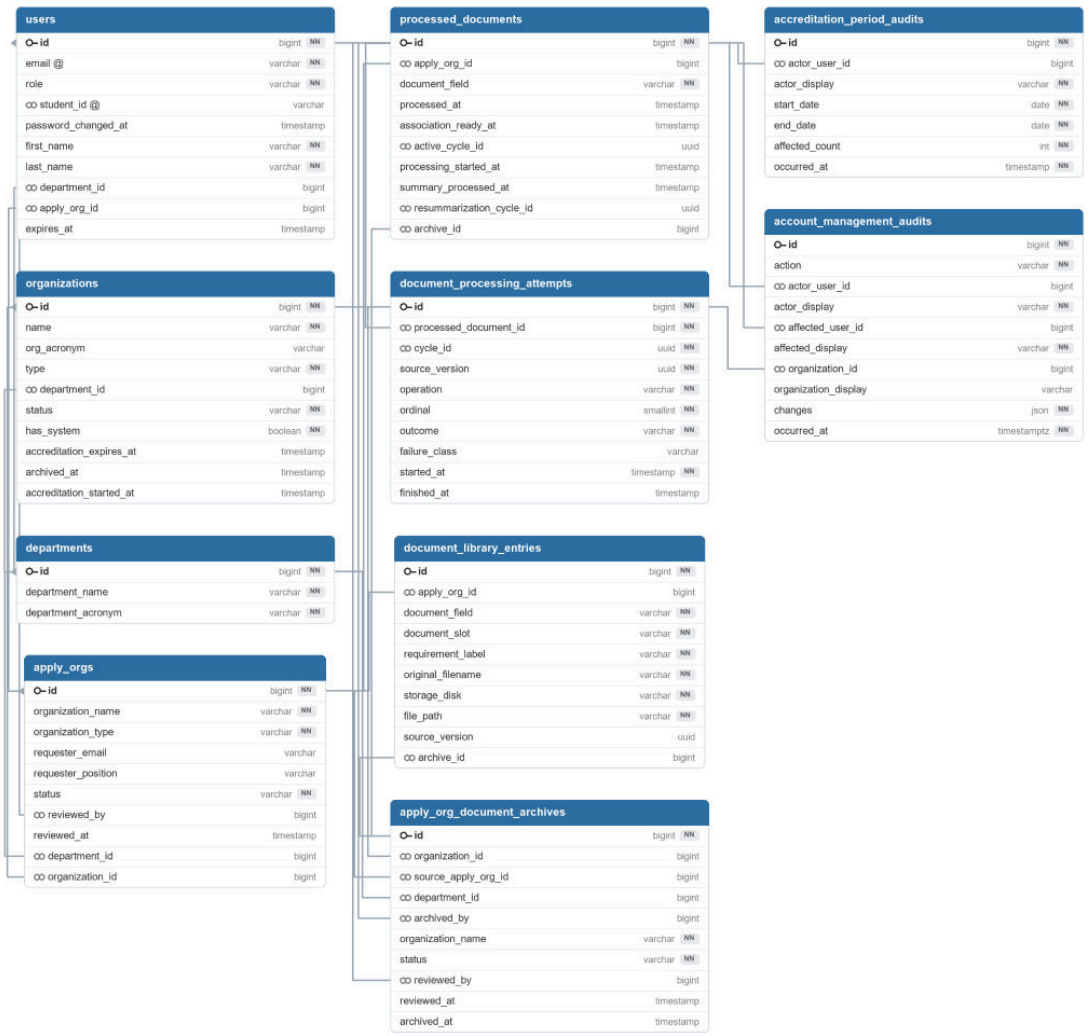


Figure 4.4. Accreditation pipeline database schema.

This group handles org application and renewal. An apply_orgs record holds the request and its status. Each uploaded requirement becomes a processed_documents row with extracted text, summary, and tags. Each AI attempt is logged in document_processing_attempts. Final copies sit in document_library_entries. Past cycles are kept in apply_org_document_archives. Audit tables record who acted and when.

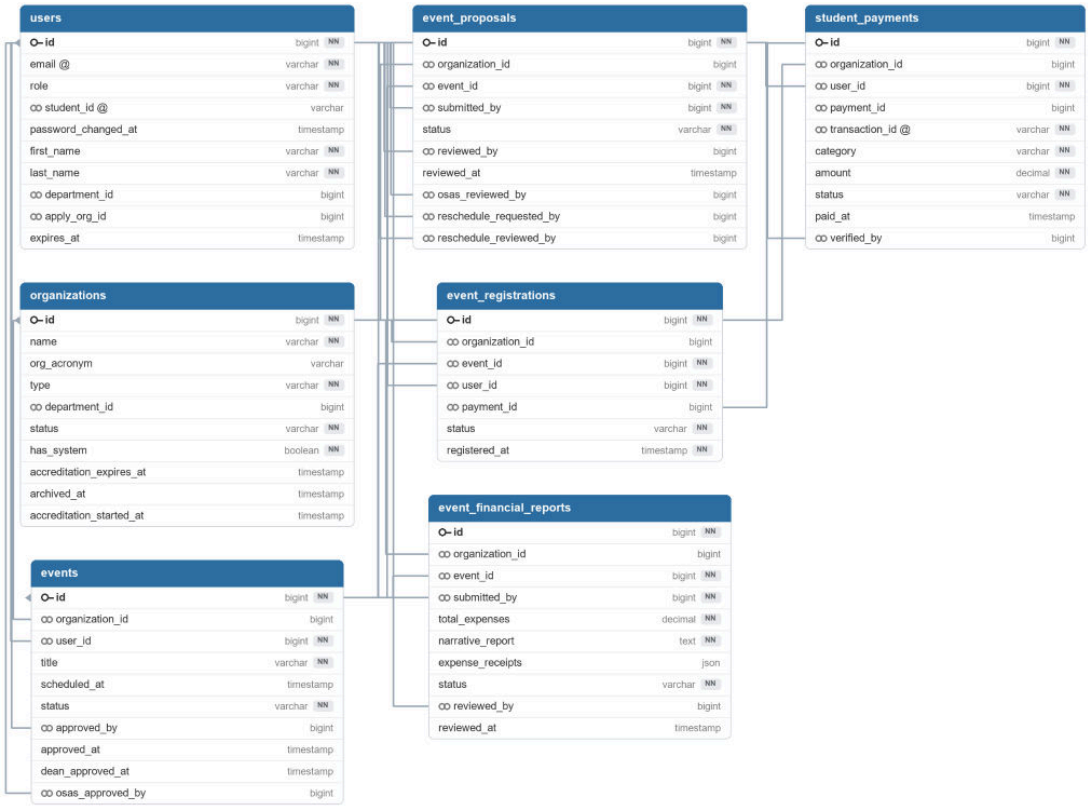


Figure 4.5. Events database schema.

This group handles activities. The events table holds the activity and its approval status from Dean and OSAS. Proposals, registrations, and financial reports point back to the event. Registrations connect to student payments when a fee is required.

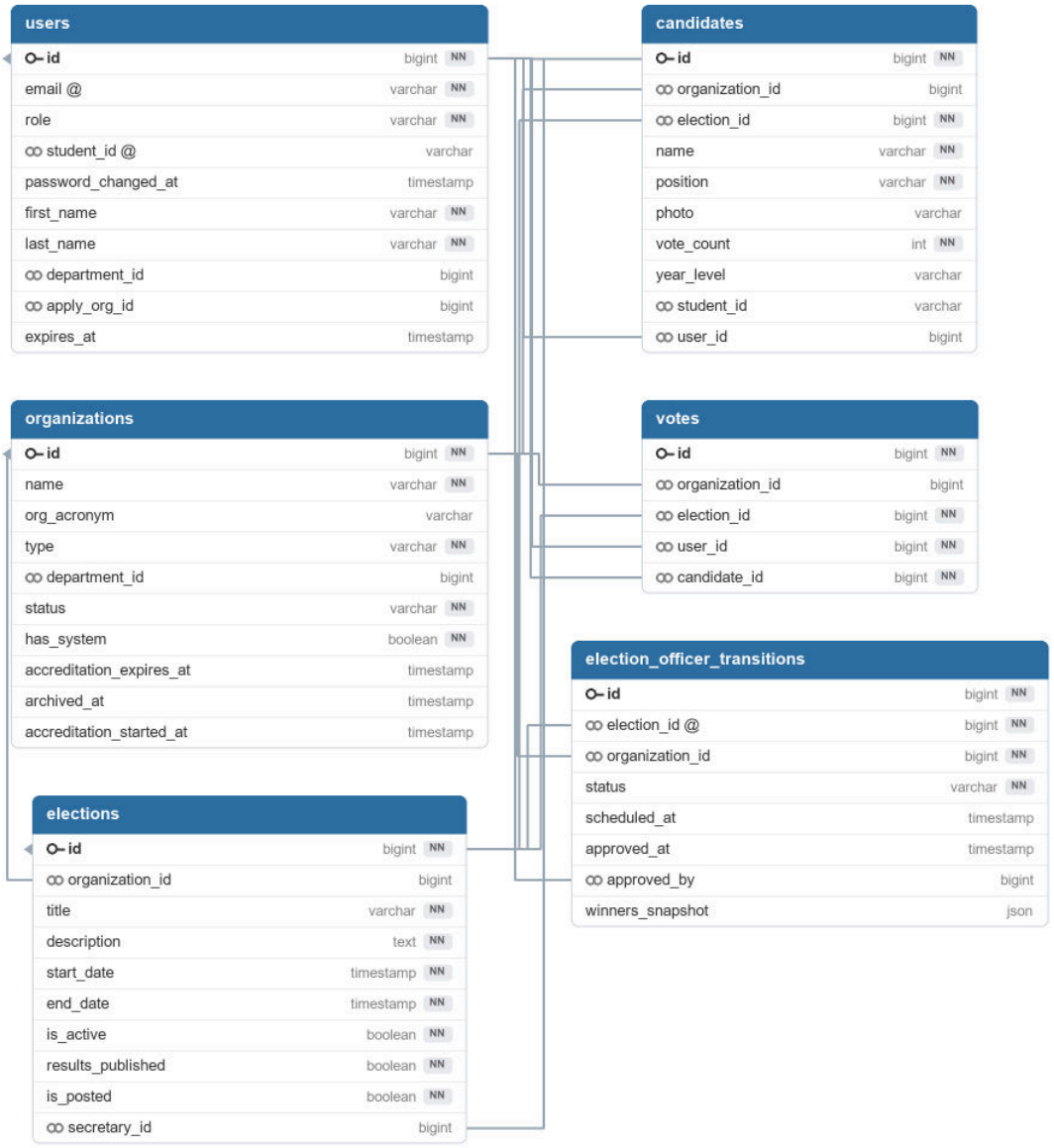


Figure 4.6. Elections database schema.

This group handles e-voting. One election has many candidates and many votes. Each vote links a user to a candidate. Officer turnover after results is recorded in election_officer_transitions.

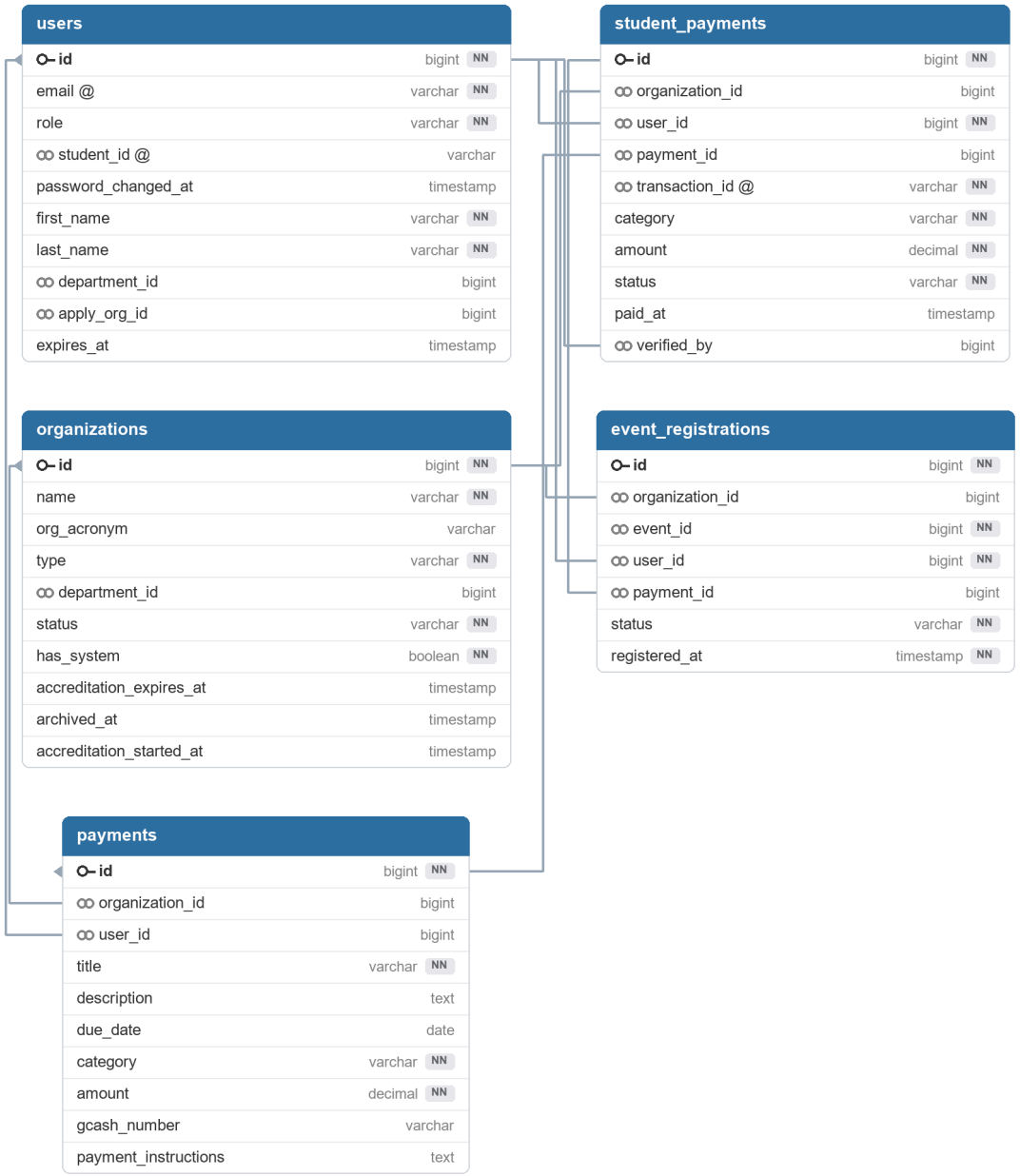


Figure 4.7. Payments database schema.

This group handles collections. The payments table defines a fee of an organization. Each student_payments row is one student payment with proof, status, and verifier. Event registrations can point to a student payment.

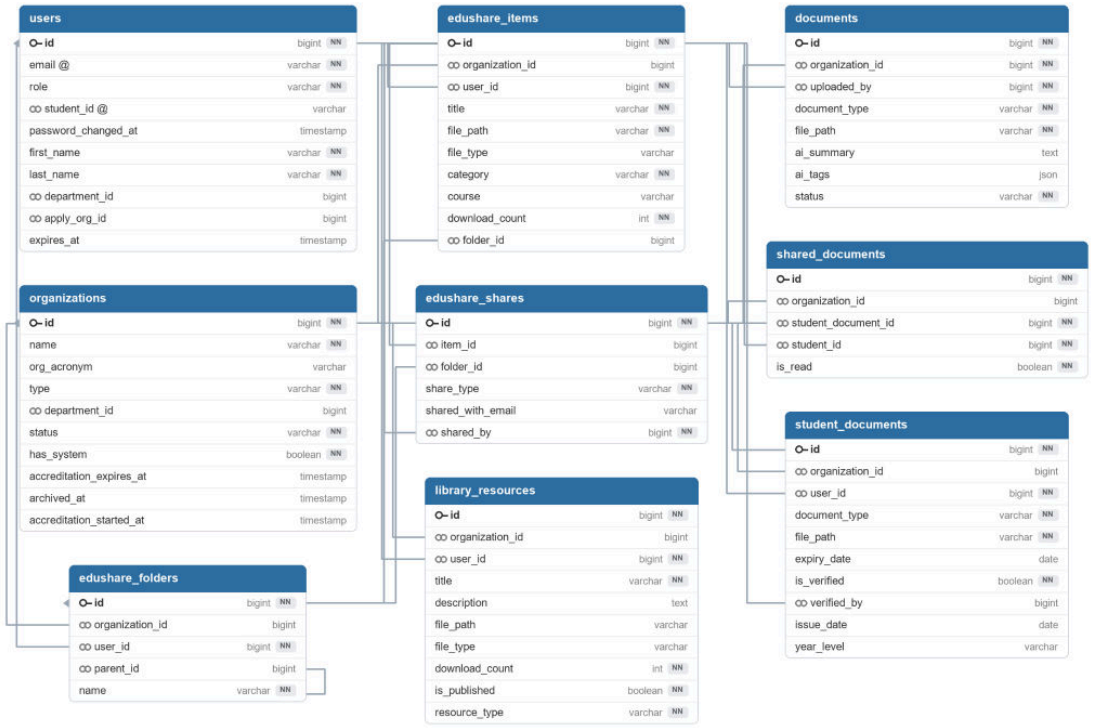


Figure 4.8. EduShare, e-library, and documents database schema.

This group handles files and resources. EduShare folders hold items and sharing rules. Library resources are published materials. The documents, shared_documents, and student_documents tables track uploads and student requirements.

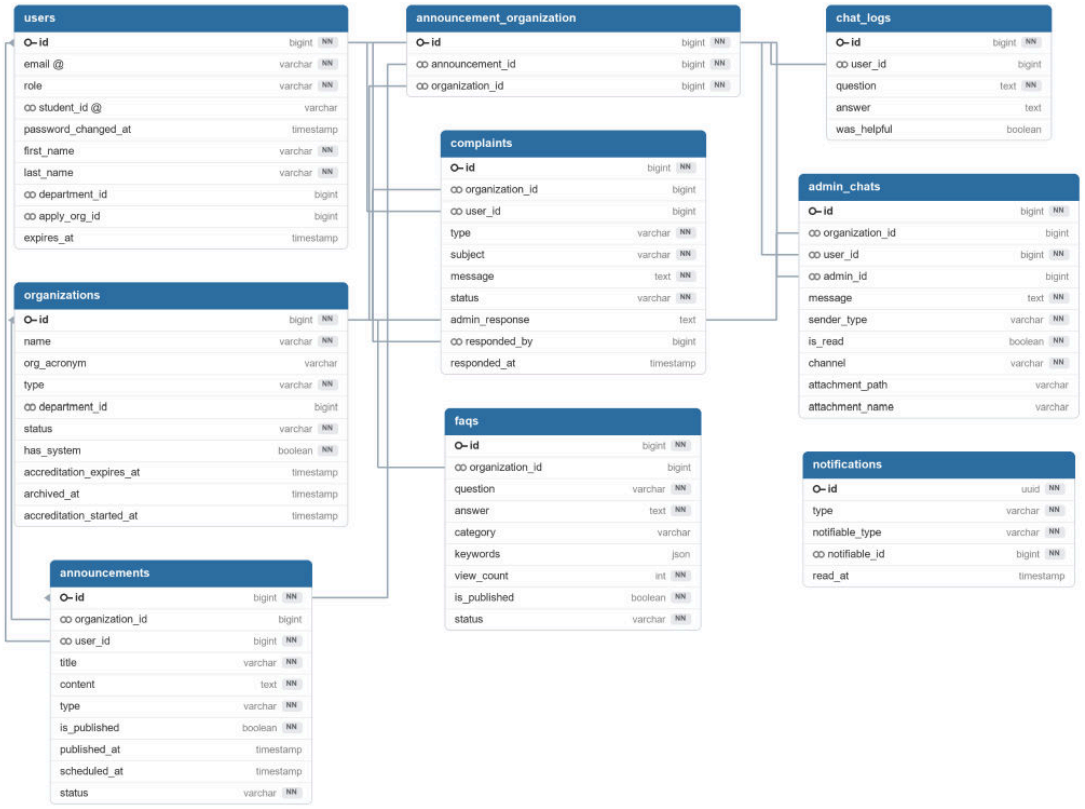


Figure 4.9. Communications and feedback database schema.

This group handles announcements, complaints, chat, and notifications. Announcements reach many organizations through announcement_organization. Complaints and admin_chats link users and organizations. FAQs, chat logs, and notifications support help and updates.

These tables are interconnected through robust primary and foreign key relationships, ensuring that every transaction such as linking a student payment to a specific organization event or associating an accreditation audit with an OSAS administrator is fully traceable. Real-time data synchronization is achieved through Supabase built-in realtime triggers, allowing database updates to reflect instantly on the OSAS dashboard and organization workspaces accessed through web and mobile

browsers without manual refreshing. The comprehensive attribute definitions, data types, and constraints governing these tables are detailed in the data dictionary.

Table 4.3. The table below displays the data dictionary of User Information.

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each user in the system
email	varchar	Not Null, Unique	Official email address used for login and account identification
password	varchar	Not Null	Encrypted password for authentication
role	varchar	Not Null	System role assigned to user (osas, dean, adviser, officer, student)
student_id	varchar	Null, Unique	Official student number for student accounts
course	varchar	Null	Degree program of the student
year_level	varchar	Null	Year level of the student
section	varchar	Null	Class section of the student
Photo	varchar	Null	File path of profile photo
is_active	bool	Not Null	Flag indicating if account is currently active
created_at	timestamptz	Not Null	Date and time when the user account was registered
updated_at	timestamptz	Null	Date and time when user profile was last updated
first_name	varchar	Not Null	First name of the user
password_changed_at	timestamp	Null	Date and time when password was last changed

Table 4.3—continued.

Field	Data Type	Constraints	Description
suffix	varchar	Null	Name suffix such as Jr or III
last_name	varchar	Not Null	Last name of the user
department_id	int8	Null, Foreign Key	ID of department the user belongs to
middle_name	varchar	Null	Middle name of the user
middle_initial	varchar	Null	Middle initial of the user
apply_org_id	int8	Null, Foreign Key	ID of pending accreditation application linked to user
active_view	varchar	Null	Current organization workspace view selected by user
management_version	int8	Not Null	Version of organization management assigned to user
temp_position	varchar	Null	Temporary officer position during transition
expires_at	timestamp	Null	Date and time when temporary access expires

Table 4.4. The table below displays the data dictionary of Organizations.

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each student organization
name	varchar	Not Null	Official name of the organization

Table 4.4 - continued.

Field	Data Type	Constraints	Description
org_acronym	varchar	Null	Acronym or short name of the organization
type	varchar	Not Null	Type of organization (academic, non-academic)
department_id	int8	Null, Foreign Key	ID of department handling the organization
status	varchar	Not Null	Accreditation status (pending, accredited, expired)
has_system	bool	Not Null	Flag indicating if organization workspace is enabled
created_at	timestamp	Null	Date and time when organization record was created
updated_at	timestamp	Null	Date and time when organization record was last updated
accreditation_expires_at	timestamp	Null	Date and time when accreditation expires
archived_at	timestamp	Null	Date and time when organization was archived
management_version	int8	Not Null	Current officer management version
accreditation_started_at	timestamp	Null	Date and time when current accreditation started

Table 4.5. The table below displays the data dictionary of Organization Members

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each membership record
user_id	int8	Not Null, Foreign Key	ID of user who is a member
organization_id	int8	Not Null, Foreign Key	ID of organization joined
role	varchar	Not Null	Member role within the organization
membership_status	varchar	Not Null	Membership status (active, pending, removed)
created_at	timestamp	Null	Date and time when membership was created
updated_at	timestamp	Null	Date and time when membership was last updated
management_version	int8	Not Null	Management version during membership

Table 4.6. The table below displays the data dictionary of Organization Settings

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each settings record
organization_id	int8	Not Null, Foreign Key	ID of organization owning the settings
logo_url	varchar	Null	URL of organization logo
has_announcements	bool	Not Null	Flag if announcements module is enabled
has_events	bool	Not Null	Flag if events module is enabled

Table 4.6 - continued.

Field	Data Type	Constraints	Description
has_payments	bool	Not Null	Flag if payments module is enabled
has_elections	bool	Not Null	Flag if elections module is enabled
has_elections	bool	Not Null	Flag if elections module is enabled
has_elibrary	bool	Not Null	Flag if e-library module is enabled
has_edushare	bool	Not Null	Flag if EduShare module is enabled
welcome_text	text	Null	Welcome message shown on organization workspace
created_at	timestamp	Null	Date and time when settings were created
updated_at	timestamp	Null	Date and time when settings were last updated
primary_color	varchar	Null	Primary theme color of workspace
complementary_color	varchar	Null	Secondary theme color of workspace
has_chatbot	bool	Not Null	Flag if chatbot help is enabled
has_complaints	bool	Not Null	Flag if feedback hub is enabled
has_documents	bool	Not Null	Flag if documents module is enabled

Table 4.7. The table below displays the data dictionary of Organization Positions

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each officer position
organization_id	int8	Not Null, Foreign Key	ID of organization owning the position

Table 4.7 - continued.

Field	Data Type	Constraints	Description
created_at	timestamp	Null	Date and time when position was created
updated_at	timestamp	Null	Date and time when position was last updated

Table 4.8. The table below displays the data dictionary of Organization Role Permissions

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each permission record
organization_id	int8	Not Null, Foreign Key	ID of organization owning the permission
role	varchar	Not Null	Role name granted with access
module	varchar	Not Null	Module name controlled by permission
is_granted	bool	Not Null	Flag indicating if access is granted
created_at	timestamp	Null	Date and time when permission was created
updated_at	timestamp	Null	Date and time when permission was last updated

Table 4.9. The table below displays the data dictionary of Departments

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each department
department_name	varchar	Not Null	Official name of the department
department_acronym	varchar	Not Null	Short code of the department
created_at	timestamp	Null	Date and time when department was created
updated_at	timestamp	Null	Date and time when department was last updated

Table 4.10. The table below displays the data dictionary of Enrolled Students

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each enrolled student record
student_id_number	varchar	Not Null, Unique	Official student number from registrar
created_at	timestamp	Null	Date and time when record was created
updated_at	timestamp	Null	Date and time when record was last updated

Table 4.11. The table below displays the data dictionary of Accreditation Applications

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each accreditation application
organization_name	varchar	Not Null	Name of organization applying

Table 4.11 - continued.

Field	Data Type	Constraints	Description
organization_type	varchar	Not Null	Type of organization applying
requester_email	varchar	Null	Email of student who filed the request
requester_position	varchar	Null	Position of requester in the organization
expected_members	int4	Null	Expected number of members
accreditation_form	varchar	Null	File path of accomplished accreditation form
status	varchar	Not Null	Application status (pending, approved, rejected)
admin_notes	text	Null	Remarks entered by OSAS reviewer
reviewed_by	int8	Null, Foreign Key	ID of OSAS user who reviewed
reviewed_at	timestamp	Null	Date and time when application was reviewed
created_at	timestamp	Null	Date and time when application was filed
updated_at	timestamp	Null	Date and time when application was last updated
requester_first_name	varchar	Null	First name of requester
requester_last_name	varchar	Null	Last name of requester
request_type	varchar	Not Null	Request type (new, renewal)
letter_of_application	varchar	Null	File path of letter of application

Table 4.11 - continued.

Field	Data Type	Constraints	Description
constitution_bylaws	varchar	Null	File path of constitution and bylaws
renewal_form	varchar	Null	File path of renewal form for renewing orgs
certification_of_officers	varchar	Null	File path of certified officer list
current_step	int4	Not Null	Current step in accreditation workflow
organization_acronym	varchar	Null	Acronym of applying organization
department_id	int8	Null, Foreign Key	ID of department handling the application
application_type	varchar	Null	Category of application submitted
adviser_commitment	varchar	Null	File path of adviser commitment form
application_form	varchar	Null	File path of standard application form
plan_of_activities	varchar	Null	File path of plan of activities
certification_dean	varchar	Null	File path of dean certification
financial_report	varchar	Null	File path of financial report
list_of_members	text	Null	List of members submitted
accomplishment_reports	varchar	Null	File path of accomplishment reports
organization_id	int8	Null, Foreign Key	ID of organization linked after approval

Table 4.11 - continued.

Field	Data Type	Constraints	Description
auto_generated	bool	Not Null	Flag if record was auto generated by system
accreditation_year	varchar	Null	Academic year covered by accreditation
other_files	text	Null	Additional supporting files submitted

Table 4.12. The table below displays the data dictionary of Processed Documents

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each processed document
apply_org_id	int8	Null, Foreign Key	ID of accreditation application source
document_field	varchar	Not Null	Form field the document was uploaded to
original_filename	varchar	Not Null	Original filename upon upload
file_path	varchar	Not Null	Storage path of uploaded file
document_type	varchar	Null	System code of detected document type
document_type_name	varchar	Null	Readable name of detected document type
summary	text	Null	AI generated summary of document content
tags	json	Null	JSON array of NLP auto generated tags

Table 4.12–continued.

Field	Data Type	Constraints	Description
key_information	json	Null	JSON object of extracted key details
extracted_text	text	Null	Full OCR extracted text from document
confidence_score	numeric	Null	Confidence score of AI classification
processing_status	varchar	Not Null	Processing status (pending, completed, failed)
error_message	text	Null	Error details if processing failed
processed_at	timestamp	Null	Date and time when processing finished
created_at	timestamp	Null	Date and time when document was uploaded
updated_at	timestamp	Null	Date and time when record was last updated
document_slot	varchar	Not Null	Slot assigned for accreditation requirement
storage_disk	varchar	Null	Storage disk where file is kept
source_version	uuid	Null	Version ID of source file
source_fingerprint	bpchar	Null	Hash fingerprint of source file
association_ready_at	timestamp	Null	Date and time when ready for association
active_cycle_id	uuid	Null	ID of active processing cycle
cycle_kind	varchar	Null	Kind of processing cycle executed

Table 4.12–continued.

Field	Data Type	Constraints	Description
cycle_attempt_count	int2	Not Null	Number of processing attempts in cycle
processing_started_at	timestamp	Null	Date and time when processing started
processing_duration_ms	int8	Null	Processing time in milliseconds
output_source_version	uuid	Null	Version ID of processed output
failure_code	varchar	Null	System code of processing failure
summary_version	int4	Not Null	Version number of generated summary
summary_processed_at	timestamp	Null	Date and time when summary was generated
resummarization_status	varchar	Not Null	Status of summary regeneration
resummarization_cycle_id	uuid	Null	ID of resummarization cycle
resummarization_error_message	text	Null	Error details if resummarization failed
last_resummarization_changed	bool	Null	Flag if last resummarization changed output
archive_id	int8	Null, Foreign Key	ID of archive where document was moved

Table 4.13. The table below displays the data dictionary of Document Library Entries.

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each library entry
apply_org_id	int8	Null, Foreign Key	ID of accreditation application source
document_field	varchar	Not Null	Form field the document belongs to
document_slot	varchar	Not Null	Requirement slot fulfilled by document
requirement_label	varchar	Not Null	Label of accreditation requirement
original_filename	varchar	Not Null	Original filename upon upload
storage_disk	varchar	Not Null	Storage disk where file is kept
file_path	varchar	Not Null	Storage path of document file
source_version	uuid	Null	Version ID of source file
source_fingerprint	bpchar	Null	Hash fingerprint of source file
created_at	timestamp	Null	Date and time when entry was created
updated_at	timestamp	Null	Date and time when entry was last updated
archive_id	int8	Null, Foreign Key	ID of archive where entry was moved

Table 4.14. The table below displays the data dictionary of Organization Documents

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each organization document
organization_id	int8	Not Null, Foreign Key	ID of organization owning the document
uploaded_by	int8	Not Null, Foreign Key	ID of user who uploaded the document
document_type	varchar	Not Null	Type of document uploaded
file_path	varchar	Not Null	Storage path of document file
ai_summary	text	Null	AI generated summary of document
ai_tags	json	Null	JSON array of AI generated tags
status	varchar	Not Null	Document status (pending, approved, archived)
created_at	timestamp	Null	Date and time when document was uploaded
updated_at	timestamp	Null	Date and time when document was last updated

Table 4.15. The table below displays the data dictionary of Student Documents.

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each student document
organization_id	int8	Null, Foreign Key	ID of organization receiving the document

Table 4.15—continued.

Field	Data Type	Constraints	Description
user_id	int8	Not Null, Foreign Key	ID of student who submitted
document_type	varchar	Not Null	Type of document submitted
file_path	varchar	Not Null	Storage path of document file
expiry_date	date	Null	Expiry date of submitted document
is_verified	bool	Not Null	Flag indicating if document was verified
verified_by	int8	Null, Foreign Key	ID of officer who verified
created_at	timestamp	Null	Date and time when document was submitted
updated_at	timestamp	Null	Date and time when document was last updated
issue_date	date	Null	Date when document was issued
year_level	varchar	Null	Year level of student upon submission
semester	varchar	Null	Semester covered by document

Table 4.16. The table below displays the data dictionary of Shared Documents

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each shared document record
organization_id	int8	Null, Foreign Key	ID of organization sharing the document
student_document_id	int8	Not Null, Foreign Key	ID of student document shared

Table 4.16 —continued.

Field	Data Type	Constraints	Description
student_id	int8	Not Null, Foreign Key	ID of student owner of document
is_read	bool	Not Null	Flag indicating if recipient opened the file
created_at	timestamp	Null	Date and time when document was shared
updated_at	timestamp	Null	Date and time when share record was updated

Table 4.17. The table below displays the data dictionary of Document Templates

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each template
title	varchar	Not Null	Title of document template
description	text	Null	Description of template use
file_path	varchar	Not Null	Storage path of template file
category	varchar	Null	Category of template
download_count	int4	Not Null	Number of times template was downloaded
created_at	timestamp	Null	Date and time when template was uploaded
updated_at	timestamp	Null	Date and time when template was last updated

Table 4.18. The table below displays the data dictionary of Announcements

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each announcement
organization_id	int8	Null, Foreign Key	ID of organization posting the announcement
user_id	int8	Not Null, Foreign Key	ID of user who posted
title	varchar	Not Null	Headline title of announcement
content	text	Not Null	Full body text of announcement
type	varchar	Not Null	Announcement type
is_published	bool	Not Null	Flag indicating if announcement is published
published_at	timestamp	Null	Date and time when announcement was published
created_at	timestamp	Null	Date and time when announcement was created
updated_at	timestamp	Null	Date and time when announcement was last updated
images	json	Null	JSON array of attached image paths
scheduled_at	timestamp	Null	Scheduled date and time for posting
status	varchar	Not Null	Posting status
visibility	varchar	Not Null	Audience scope of announcement

Table 4.19. The table below displays the data dictionary of Announcement Distribution

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each distribution record
announcement_id	int8	Not Null, Foreign Key	ID of announcement distributed
organization_id	int8	Not Null, Foreign Key	ID of recipient organization
created_at	timestamp	Null	Date and time when distribution was created
updated_at	timestamp	Null	Date and time when distribution was updated

Table 4.20. The table below displays the data dictionary of Events

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each event
organization_id	int8	Null, Foreign Key	ID of organization hosting the event
user_id	int8	Not Null, Foreign Key	ID of user who created the event
title	varchar	Not Null	Title of the event
description	text	Not Null	Full description of the event
start_date	timestamp	Not Null	Starting date and time of event
end_date	timestamp	Not Null	Ending date and time of event
venue	varchar	Not Null	Venue of the event

Table 4.20 —continued.

Field	Data Type	Constraints	Description
is_published	bool	Not Null	Flag indicating if event is shown in calendar
created_at	timestamp	Null	Date and time when event was created
updated_at	timestamp	Null	Date and time when event was last updated
images	json	Null	JSON array of event image paths
scheduled_at	timestamp	Null	Scheduled date and time for publishing
status	varchar	Not Null	Event status
planned_budget	numeric	Null	Planned budget for the event
approval_status	varchar	Not Null	Dean approval status
approved_by	int8	Null, Foreign Key	ID of dean who approved
approved_at	timestamp	Null	Date and time when dean approved
dean_approved_at	timestamp	Null	Date and time of dean approval stamp
osas_status	varchar	Null	OSAS approval status
osas_approved_by	int8	Null, Foreign Key	ID of OSAS user who approved
osas_approved_at	timestamp	Null	Date and time when OSAS approved
hardcopy_received_at	timestamp	Null	Date and time when hardcopy was received
category	varchar	Null	Category of event

Table 4.21. The table below displays the data dictionary of Event Proposals

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each event proposal
organization_id	int8	Null, Foreign Key	ID of organization submitting proposal
event_id	int8	Not Null, Foreign Key	ID of event proposed
submitted_by	int8	Not Null, Foreign Key	ID of officer who submitted
status	varchar	Not Null	Proposal status (pending, approved, rejected)
rejection_reason	text	Null	Reason given if proposal was rejected
reviewed_by	int8	Null, Foreign Key	ID of dean reviewer
reviewed_at	timestamp	Null	Date and time when proposal was reviewed
created_at	timestamp	Null	Date and time when proposal was submitted
updated_at	timestamp	Null	Date and time when proposal was last updated
request_letter	varchar	Null	File path of request letter
osas_rejection_reason	text	Null	Reason given if OSAS rejected proposal
osas_reviewed_by	int8	Null, Foreign Key	ID of OSAS reviewer
osas_reviewed_at	timestamp	Null	Date and time when OSAS reviewed
reschedule_status	varchar	Null	Status of reschedule request

Table 4.21—continued.

Field	Data Type	Constraints	Description
reschedule_reason	text	Null	Reason for reschedule request
reschedule_requested_by	int8	Null, Foreign Key	ID of user who requested reschedule
reschedule_requested_at	timestamp	Null	Date and time when reschedule was requested
reschedule_reviewed_by	int8	Null, Foreign Key	ID of reviewer of reschedule
reschedule_reviewed_at	timestamp	Null	Date and time when reschedule was reviewed
reschedule_rejection_reason	text	Null	Reason given if reschedule was denied

Table 4.22. The table below displays the data dictionary of Event Registrations

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each registration
organization_id	int8	Null, Foreign Key	ID of organization hosting the event
event_id	int8	Not Null, Foreign Key	ID of event registered to
user_id	int8	Not Null, Foreign Key	ID of student registered
payment_id	int8	Null, Foreign Key	ID of linked payment if required
status	varchar	Not Null	Registration status
registered_at	timestamp	Not Null	Date and time when student registered
created_at	timestamp	Null	Date and time when record was created
updated_at	timestamp	Null	Date and time when record was last updated

Table 4.23. The table below displays the data dictionary of Event Financial Reports

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each financial report
organization_id	int8	Null, Foreign Key	ID of organization submitting report
event_id	int8	Not Null, Foreign Key	ID of event covered by report
submitted_by	int8	Not Null, Foreign Key	ID of officer who submitted
total_expenses	numeric	Not Null	Total expenses incurred for event
narrative_report	text	Not Null	Narrative accomplishment and expense summary
expense_receipts	json	Null	JSON array of receipt file paths
approved_letter	varchar	Null	File path of approved request letter
status	varchar	Not Null	Report status (pending, approved, rejected)
remarks	text	Null	Reviewer remarks on report
reviewed_by	int8	Null, Foreign Key	ID of reviewer
reviewed_at	timestamp	Null	Date and time when report was reviewed
created_at	timestamp	Null	Date and time when report was submitted
updated_at	timestamp	Null	Date and time when report was last updated

Table 4.24. The table below displays the data dictionary of Payments

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each payment collection
organization_id	int8	Null, Foreign Key	ID of organization collecting payment
user_id	int8	Null, Foreign Key	ID of officer who created collection
title	varchar	Not Null	Title of payment collection
description	text	Null	Description of payment purpose
due_date	date	Null	Deadline for payment
category	varchar	Not Null	Category of collection
amount	numeric	Not Null	Amount to be collected
gcash_number	varchar	Null	GCash number for manual routing
payment_instructions	text	Null	Instructions for paying members
created_at	timestamp	Null	Date and time when collection was created
updated_at	timestamp	Null	Date and time when collection was last updated
qr_code	text	Null	QR code data for e-wallet payment
is_required_membership_fee	bool	Not Null	Flag if payment is required membership fee
membership_cycle_start	date	Null	Start date of membership cycle covered

Table 4.25. The table below displays the data dictionary of Student Payments

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each payment
organization_id	int8	Null, Foreign Key	ID of organization receiving payment
user_id	int8	Not Null, Foreign Key	ID of student who paid
payment_id	int8	Null, Foreign Key	ID of payment collection settled
transaction_id	varchar	Not Null, Unique	Unique transaction reference number
category	varchar	Not Null	Category of payment made
amount	numeric	Not Null	Amount paid by student
payment_method	varchar	Not Null	Method used (GCash, cash)
status	varchar	Not Null	Payment status (pending, verified, rejected)
reference_number	varchar	Null	E-wallet reference number
receipt_path	varchar	Null	File path of uploaded receipt
paid_at	timestamp	Null	Date and time when payment was made
verified_by	int8	Null, Foreign Key	ID of officer who verified payment
created_at	timestamp	Null	Date and time when record was created
updated_at	timestamp	Null	Date and time when record was last updated
gcash_receipt_path	varchar	Null	File path of GCash receipt screenshot
verification_status	varchar	Not Null	Verification stage of payment
rejection_reason	text	Null	Reason given if payment rejected

Table 4.26. The table below displays the data dictionary of Elections

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each election
organization_id	int8	Null, Foreign Key	ID of organization holding election
title	varchar	Not Null	Title of election
description	text	Not Null	Description and guidelines of election
start_date	timestamp	Not Null	Start date and time of voting
end_date	timestamp	Not Null	End date and time of voting
is_active	bool	Not Null	Flag indicating if election is open
results_published	bool	Not Null	Flag indicating if results are published
created_at	timestamp	Null	Date and time when election was created
updated_at	timestamp	Null	Date and time when election was last updated
is_posted	bool	Not Null	Flag indicating if election is posted publicly
secretary_id	int8	Null, Foreign Key	ID of secretary managing election
start_notified	bool	Not Null	Flag indicating if start notice was sent

Table 4.27. The table below displays the data dictionary of Candidates

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each candidate
organization_id	int8	Null, Foreign Key	ID of organization holding election
election_id	int8	Not Null, Foreign Key	ID of election joined
name	varchar	Not Null	Full name of candidate
position	varchar	Not Null	Position aspired for
photo	varchar	Null	File path of candidate photo
vote_count	int4	Not Null	Total votes received
created_at	timestamp	Null	Date and time when candidacy was filed
updated_at	timestamp	Null	Date and time when candidacy was last updated
year_level	varchar	Null	Year level of candidate
student_id	varchar	Null	Student number of candidate
	user_idint8	Null, Foreign Key	ID of user account of candidate

Table 4.28. The table below displays the data dictionary of Votes

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each vote cast
organization_id	int8	Null, Foreign Key	ID of organization holding election
election_id	int8	Not Null, Foreign Key	ID of election voted in
user_id	int8	Not Null, Foreign Key	ID of student voter

Table 4.28—continued.

Field	Data Type	Constraints	Description
candidate_id	int8	Not Null, Foreign Key	ID of candidate voted for
created_at	timestamp	Null	Date and time when vote was cast
updated_at	timestamp	Null	Date and time when vote record was updated

Table 4.29. The table below displays the data dictionary of Election Officer Transitions

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each transition record
election_id	int8	Not Null, Unique, Foreign Key	ID of election completed
organization_id	int8	Not Null, Foreign Key	ID of organization owning election
status	varchar	Not Null	Transition status (pending, approved, completed)
scheduled_at	timestamp	Null	Scheduled date and time of transition
approved_at	timestamp	Null	Date and time when transition was approved
approved_by	int8	Null, Foreign Key	ID of adviser or OSAS approver
winners_snapshot	json	Null	JSON snapshot of election winners
created_at	timestamp	Null	Date and time when transition was created
updated_at	timestamp	Null	Date and time when transition was last updated

Table 4.30. The table below displays the data dictionary of Library Resources

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each library resource
organization_id	int8	Null, Foreign Key	ID of organization sharing resource
user_id	int8	Not Null, Foreign Key	ID of user who uploaded
title	varchar	Not Null	Title of resource
description	text	Null	Description of resource content
file_path	varchar	Null	Storage path of resource file
file_type	varchar	Null	File type of resource
download_count	int4	Not Null	Number of times resource was downloaded
is_published	bool	Not Null	Flag indicating if resource is published
created_at	timestamp	Null	Date and time when resource was uploaded
updated_at	timestamp	Null	Date and time when resource was last updated
resource_type	varchar	Not Null	Type of library material
url	varchar	Null	External link if resource is online

Table 4.31. The table below displays the data dictionary of EduShare Items

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each shared learning item
organization_id	int8	Null, Foreign Key	ID of organization sharing the item
user_id	int8	Not Null, Foreign Key	ID of user who uploaded
title	varchar	Not Null	Title of learning material
file_path	varchar	Not Null	Storage path of material file
file_type	varchar	Null	File type of material
category	varchar	Not Null	Category of learning material
course	varchar	Null	Course related to material
download_count	int4	Not Null	Number of times material was downloaded
is_public	bool	Not Null	Flag indicating if item is public
created_at	timestamp	Null	Date and time when item was uploaded
updated_at	timestamp	Null	Date and time when item was last updated
folder_id	int8	Null, Foreign Key	ID of folder containing the item

Table 4.32. The table below displays the data dictionary of EduShare Folders

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each folder
organization_id	int8	Null, Foreign Key	ID of organization owning the folder

Table 4.32—continued.

Field	Data Type	Constraints	Description
user_id	int8	Not Null, Foreign Key	ID of user who created the folder
parent_id	int8	Null, Foreign Key	ID of parent folder if nested
name	varchar	Not Null	Name of folder
created_at	timestamp	Null	Date and time when folder was created
updated_at	timestamp	Null	Date and time when folder was last updated

Table 4.33. The table below displays the data dictionary of EduShare Shares

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each share record
item_id	int8	Null, Foreign Key	ID of item shared
folder_id	int8	Null, Foreign Key	ID of folder shared
share_type	varchar	Not Null	Type of sharing applied
shared_with_email	varchar	Null	Email of recipient user
shared_by	int8	Not Null, Foreign Key	ID of user who shared
created_at	timestamp	Null	Date and time when share was created
updated_at	timestamp	Null	Date and time when share was last updated

Table 4.34. The table below displays the data dictionary of Feedback and Complaints

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each feedback record
organization_id	int8	Null, Foreign Key	ID of organization concerned
user_id	int8	Null, Foreign Key	ID of student who filed feedback
type	varchar	Not Null	Type of feedback submitted
subject	varchar	Not Null	Subject line of feedback
message	text	Not Null	Full message content of feedback
status	varchar	Not Null	Status (pending, reviewed, resolved)
admin_response	text	Null	Response entered by officer or OSAS
responded_by	int8	Null, Foreign Key	ID of user who responded
is_anonymous	bool	Not Null	Flag indicating if sender is hidden
created_at	timestamp	Null	Date and time when feedback was filed
updated_at	timestamp	Null	Date and time when feedback was last updated
responded_at	timestamp	Null	Date and time when response was given

Table 4.35. The table below displays the data dictionary of FAQs

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each FAQ entry
organization_id	int8	Null, Foreign Key	ID of organization owning the FAQ

Table 4.35—continued.

Field	Data Type	Constraints	Description
question	varchar	Not Null	Frequently asked question
answer	text	Not Null	Official answer to question
category	varchar	Null	Category of FAQ
keywords	json	Null	JSON array of search keywords
view_count	int4	Not Null	Number of times FAQ was viewed
is_published	bool	Not Null	Flag indicating if FAQ is published
created_at	timestamp	Null	Date and time when FAQ was created
updated_at	timestamp	Null	Date and time when FAQ was last updated
status	varchar	Not Null	Publishing status of FAQ

Table 4.36. The table below displays the data dictionary of Admin Chats

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each chat message
organization_id	int8	Null, Foreign Key	ID of organization in conversation
user_id	int8	Not Null, Foreign Key	ID of student sender
admin_id	int8	Null, Foreign Key	ID of officer or admin participant
message	text	Not Null	Message text transmitted
sender_type	varchar	Not Null	Sender type (student, admin)

Table 4.36—continued.

Field	Data Type	Constraints	Description
is_read	bool	Not Null	Flag indicating if message was read
created_at	timestamp	Null	Date and time when message was sent
updated_at	timestamp	Null	Date and time when message was last updated
channel	varchar	Not Null	Conversation channel identifier
attachment_path	varchar	Null	File path of attached file
attachment_name	varchar	Null	Original name of attached file

Table 4.37. The table below displays the data dictionary of Chatbot Logs

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each chatbot interaction
user_id	int8	Null, Foreign Key	ID of user who asked
question	text	Not Null	Question entered by user
answer	text	Null	Answer generated by chatbot
was_helpful	bool	Null	Flag indicating if answer was helpful
created_at	timestamp	Null	Date and time when question was asked
updated_at	timestamp	Null	Date and time when log was updated

Table 4.38. The table below displays the data dictionary of Notifications

Field	Data Type	Constraints	Description
id	uuid	Not Null, Primary Key	Unique identifier for each notification
type	varchar	Not Null	Type of notification sent
notifiable_type	varchar	Not Null	Model type of recipient
notifiable_id	int8	Not Null	ID of recipient user
data	text	Not Null	JSON content of notification
read_at	timestamp	Null	Date and time when notification was read
created_at	timestamp	Null	Date and time when notification was created
updated_at	timestamp	Null	Date and time when notification was last updated

Table 4.39. The table below displays the data dictionary of Accreditation Archives

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each archived application
organization_id	int8	Null, Foreign Key	ID of accredited organization
source_apply_org_id	int8	Null, Foreign Key	ID of original application
department_id	int8	Null, Foreign Key	ID of department handling archive
archived_by	int8	Null, Foreign Key	ID of OSAS user who archived

Table 4.39—continued.

Field	Data Type	Constraints	Description
organization_name	varchar	Not Null	Name of archived organization
organization_acronym	varchar	Null	Acronym of archived organization
organization_type	varchar	Not Null	Type of archived organization
request_type	varchar	Not Null	Request type of archived application
application_type	varchar	Null	Category of archived application
auto_generated	bool	Not Null	Flag if archive was auto generated
requester_first_name	varchar	Null	First name of original requester
requester_last_name	varchar	Null	Last name of original requester
requester_email	varchar	Null	Email of original requester
requester_position	varchar	Null	Position of original requester
accreditation_form	varchar	Null	File path of archived accreditation form
letter_of_application	varchar	Null	File path of archived letter
constitution_bylaws	varchar	Null	File path of archived constitution
renewal_form	varchar	Null	File path of archived renewal form
adviser_commitment	varchar	Null	File path of archived adviser form
application_form	varchar	Null	File path of archived application form
certification_of_officers	varchar	Null	File path of archived officer list

Table 4.39—continued.

Field	Data Type	Constraints	Description
plan_of_activities	varchar	Null	File path of archived plan
certification_dean	varchar	Null	File path of archived dean certification
financial_report	varchar	Null	File path of archived financial report
list_of_members	text	Null	Archived list of members
accomplishment_reports	varchar	Null	File path of archived accomplishment reports
other_files	text	Null	Other archived supporting files
current_step	int4	Not Null	Final step reached before archiving
status	varchar	Not Null	Final status of archived application
accreditation_year	varchar	Null	Academic year covered
admin_notes	text	Null	Final reviewer notes
reviewed_by	int8	Null, Foreign Key	ID of reviewer
reviewed_at	timestamp	Null	Date and time when reviewed
archived_at	timestamp	Null	Date and time when archived
created_at	timestamp	Null	Date and time when original was filed
updated_at	timestamp	Null	Date and time when archive was last updated

Table 4.40. The table below displays the data dictionary of Archived Document Library.

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each archived library entry
archive_id	int8	Not Null, Foreign Key	ID of accreditation archive owner
original_apply_org_id	int8	Null, Foreign Key	ID of original application
document_field	varchar	Not Null	Form field of archived document
document_slot	varchar	Not Null	Requirement slot fulfilled
requirement_label	varchar	Not Null	Label of requirement fulfilled
original_filename	varchar	Not Null	Original filename of archived file
storage_disk	varchar	Not Null	Storage disk where file is kept
file_path	varchar	Not Null	Storage path of archived file
source_version	uuid	Null	Version ID of source file
source_fingerprint	bpchar	Null	Hash fingerprint of source file
created_at	timestamp	Null	Date and time when archived
updated_at	timestamp	Null	Date and time when archive was updated

Table 4.41. The table below displays the data dictionary of Archived Processed Documents

Field	Data Type	Constraints	Description
id	int8	Not Null, Primary Key	Unique identifier for each archived processed document
archive_id	int8	Not Null, Foreign Key	ID of accreditation archive owner
original_apply_org_id	int8	Null, Foreign Key	ID of original application
original_processed_document_id	int8	Null, Foreign Key	ID of original processed document
document_field	varchar	Not Null	Form field of archived document
document_slot	varchar	Not Null	Requirement slot fulfilled
original_filename	varchar	Not Null	Original filename of archived file
file_path	varchar	Not Null	Storage path of archived file
storage_disk	varchar	Null	Storage disk where file is kept
source_version	uuid	Null	Version ID of source file
source_fingerprint	bpchar	Null	Hash fingerprint of source file
document_type	varchar	Null	Detected document type code
document_type_name	varchar	Null	Readable document type name
summary	text	Null	Archived AI summary
tags	json	Null	Archived JSON tags
key_information	json	Null	Archived extracted key details
extracted_text	text	Null	Archived OCR text

Table 4.41—continued.

Field	Data Type	Constraints	Description
confidence_score	numeric	Null	Archived confidence score
processing_status	varchar	Not Null	Final processing status
error_message	text	Null	Archived error details if any
processed_at	timestamp	Null	Date and time when originally processed
association_ready_at	timestamp	Null	Date and time when ready for association
active_cycle_id	uuid	Null	ID of processing cycle used
cycle_kind	varchar	Null	Kind of cycle executed
cycle_attempt_count	int2	Not Null	Number of attempts in cycle
processing_started_at	timestamp	Null	Date and time when processing started
processing_duration_ms	int8	Null	Processing time in milliseconds
output_source_version	uuid	Null	Version ID of processed output
failure_code	varchar	Null	Failure code if processing failed
summary_version	int4	Not Null	Version of archived summary
summary_processed_at	timestamp	Null	Date and time when summary was generated
resummarization_status	varchar	Not Null	Status of resummarization
resummarization_cycle_id	uuid	Null	ID of resummarization cycle

Table 4.41—continued.

Field	Data Type	Constraints	Description
resummarization_error_message	text	Null	Error details if resummarization failed
last_resummarization_changed	bool	Null	Flag if last resummarization changed output
created_at	timestamp	Null	Date and time when archived
updated_at	timestamp	Null	Date and time when archive was updated

4.2.5 User-Interface Design and System Modules

The screens are arranged by research objective. Each objective has one screenshot per user role where the role has a screen, so the images can stay large and readable. The full set of screenshots is in the Appendices.

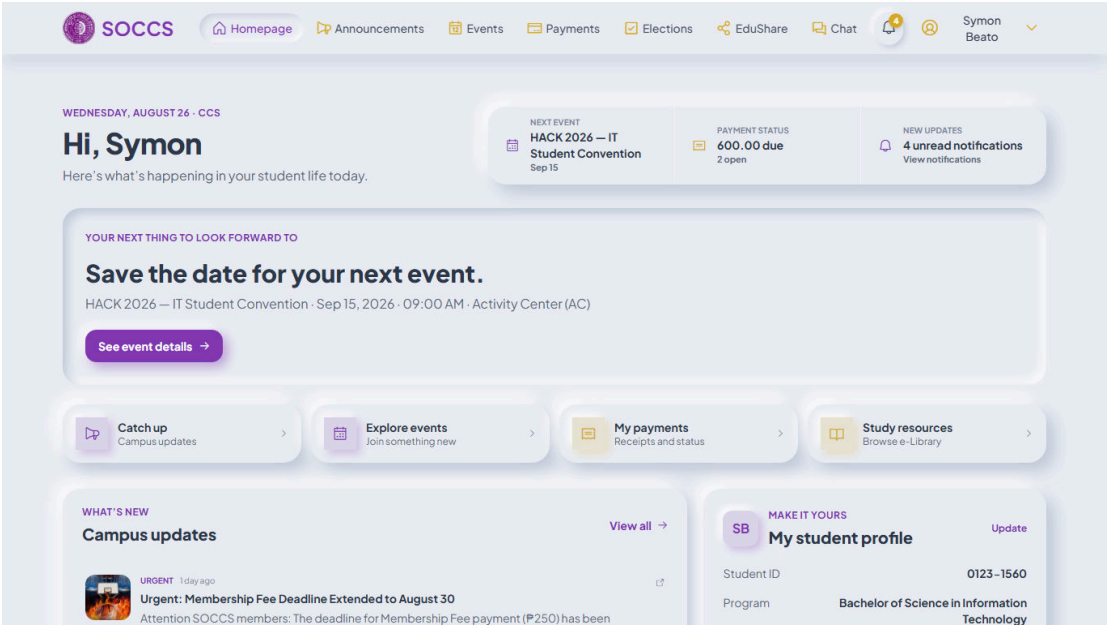


Figure 4.10. Student dashboard.

Figure 4.10 shows the student dashboard. It has cards for announcements, events, payments, and elections so the student sees every update in one page. This is the student entry point to the centralized platform in Objective 1.

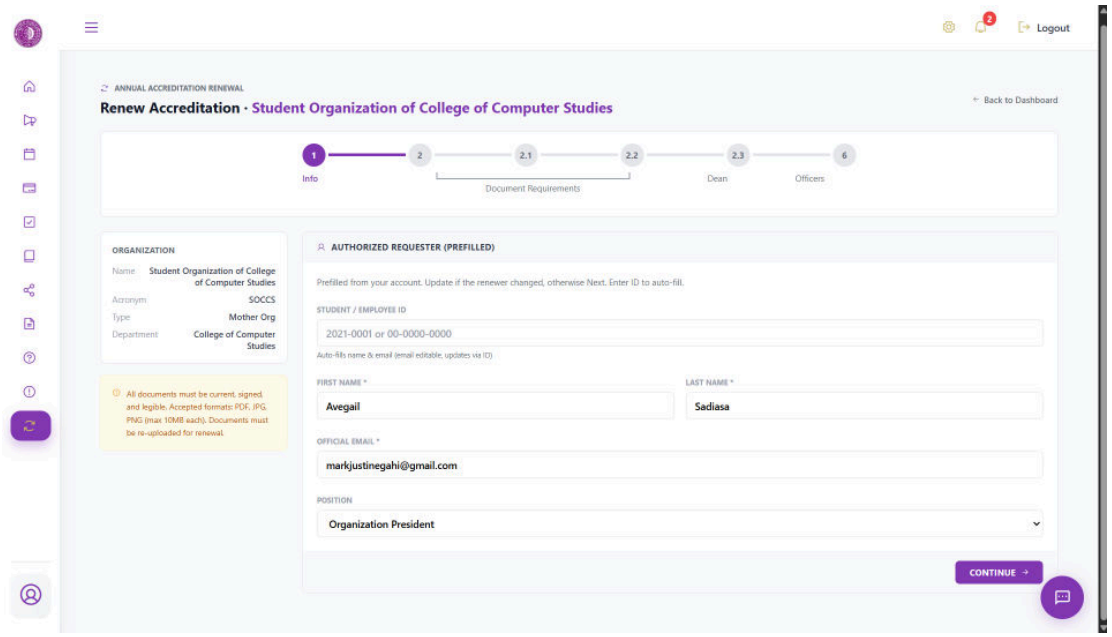


Figure 4.11. Accreditation renewal request form (organization admin).

Figure 4.11 is the accreditation renewal request form filled by the organization admin. The requester details and organization profile are encoded here before the supporting papers are attached. This starts the online accreditation flow in Objective 1 that replaces printed folders.

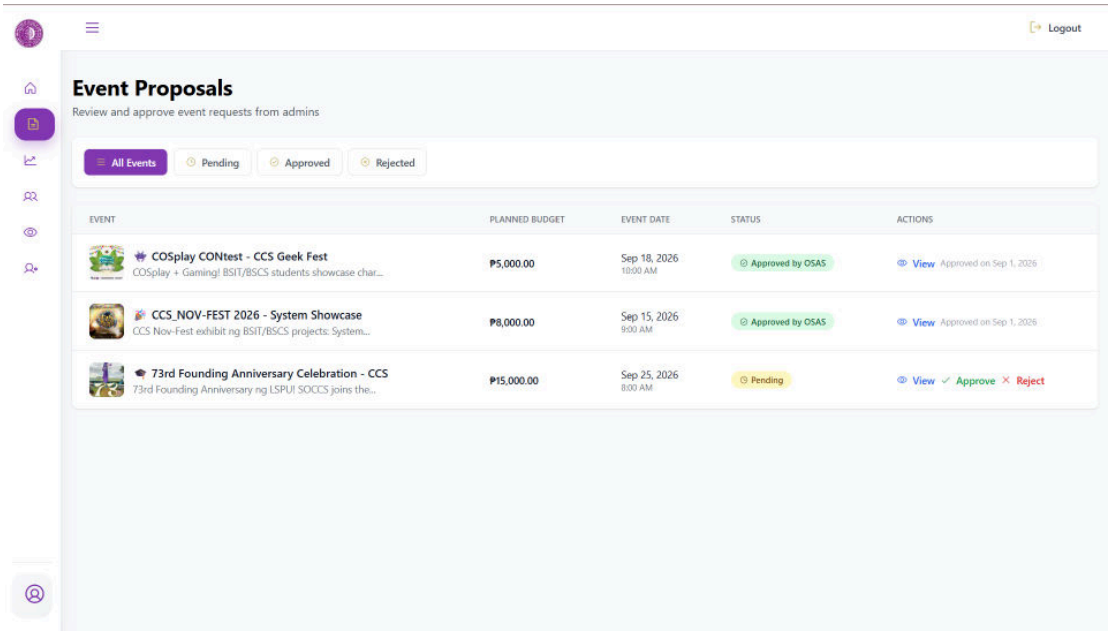


Figure 4.12. Event proposals list (dean).

Figure 4.12 shows the dean event proposals list. All proposals from organizations under the college appear here for checking and approval. This covers the activity monitoring side of Objective 1.

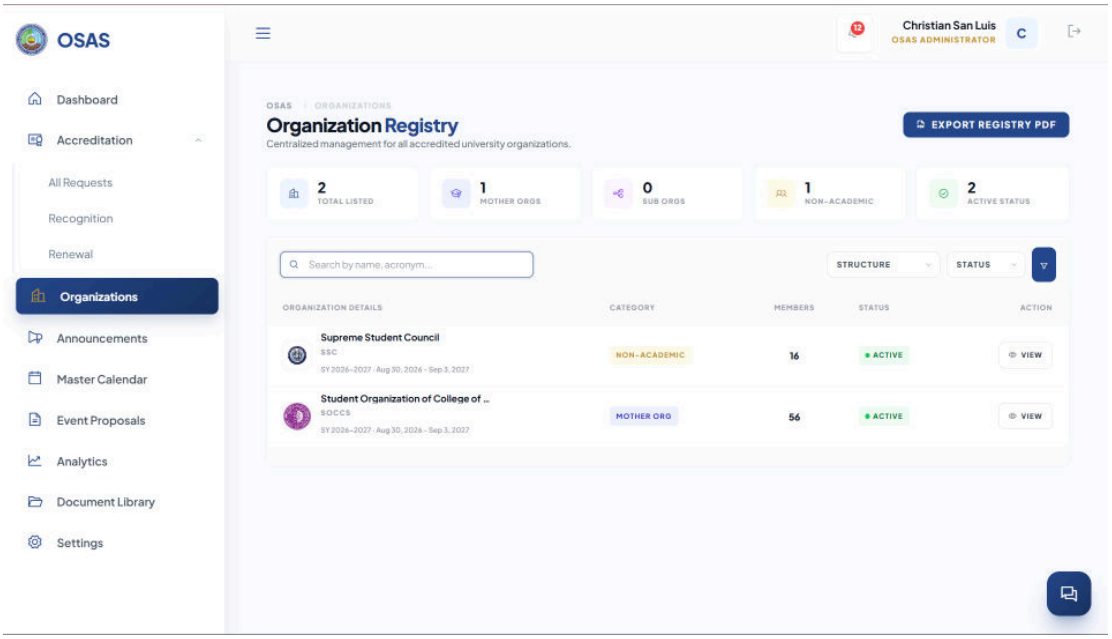


Figure 4.13. Organization registry list (OSAS).

Figure 4.13 is the OSAS organization registry list. Every student organization and its status are seen in one table. This is the central monitoring view required by Objective 1.

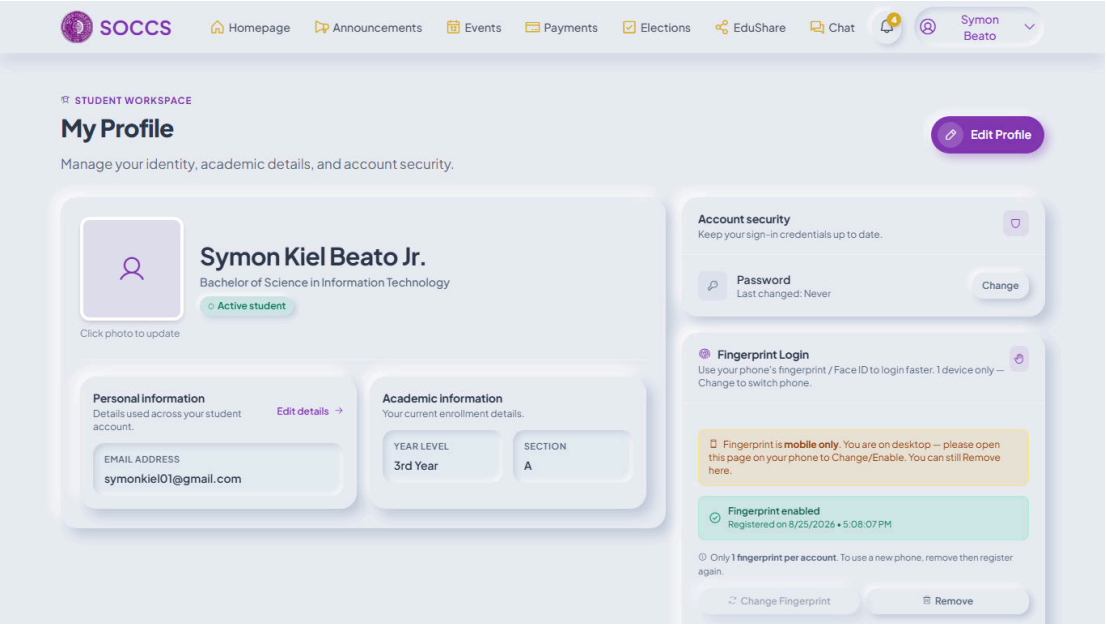


Figure 4.14. Student profile.

Figure 4.14 shows the student profile page. The personal details and organization memberships of the student are listed here. This is the profile side of the multi-org workspace in Objective 2.1.

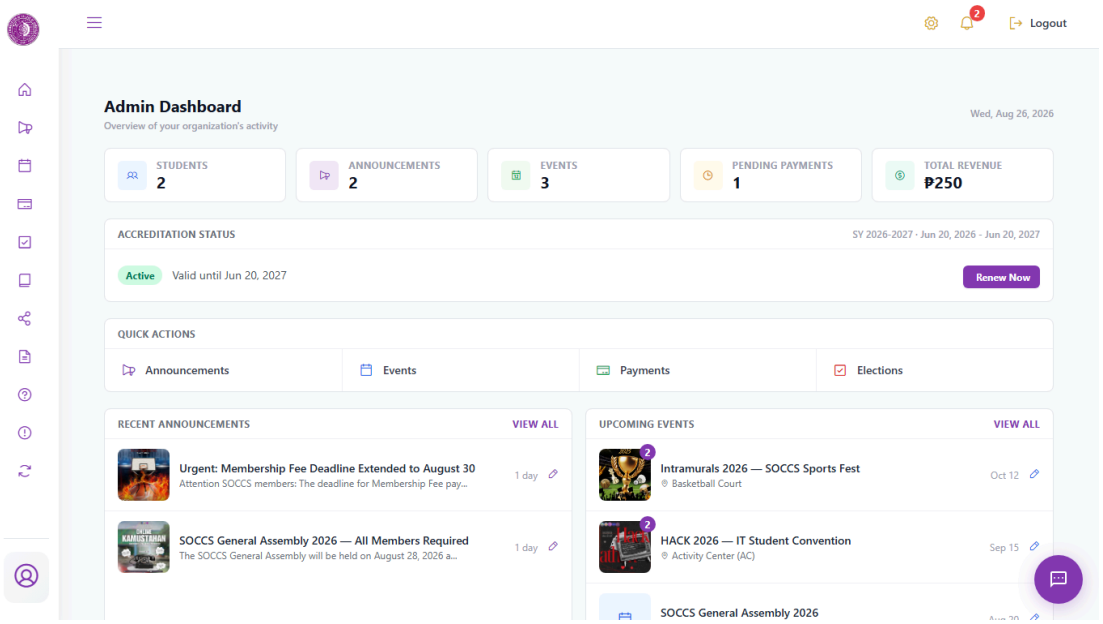


Figure 4.15. Organization admin dashboard.

Figure 4.15 is the organization admin dashboard. The officer manages announcements, events, payments, and elections for the owned organization from here. This is the per-organization workspace in Objective 2.1.

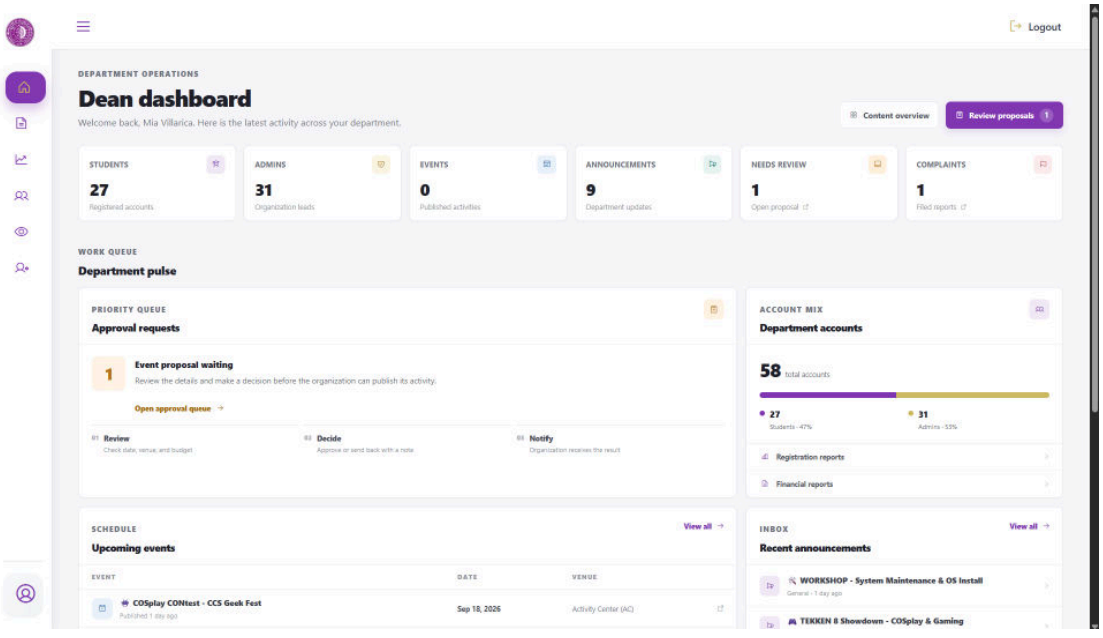


Figure 4.16. Dean dashboard.

Figure 4.16 shows the dean dashboard home. The dean sees pending proposals and content overviews for the handled colleges. This gives the dean a separate workspace scoped by college in Objective 2.1.

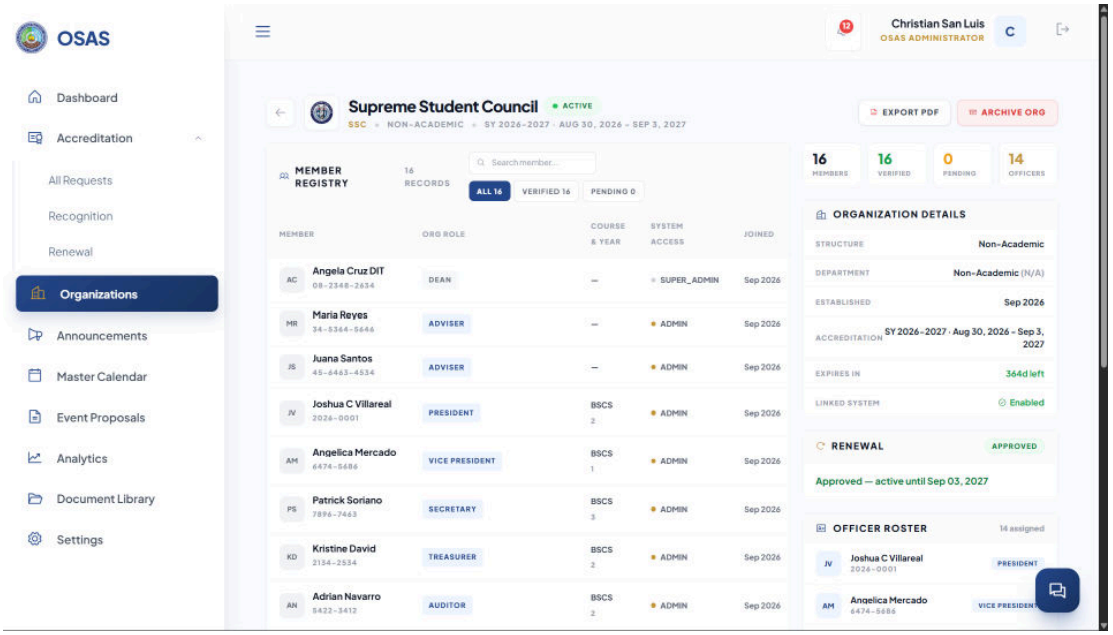


Figure 4.17. Organization details (OSAS).

Figure 4.17 opens one organization record from the registry. Its profile, status, and history are shown in full. This proves OSAS can profile every organization in Objective 2.1.

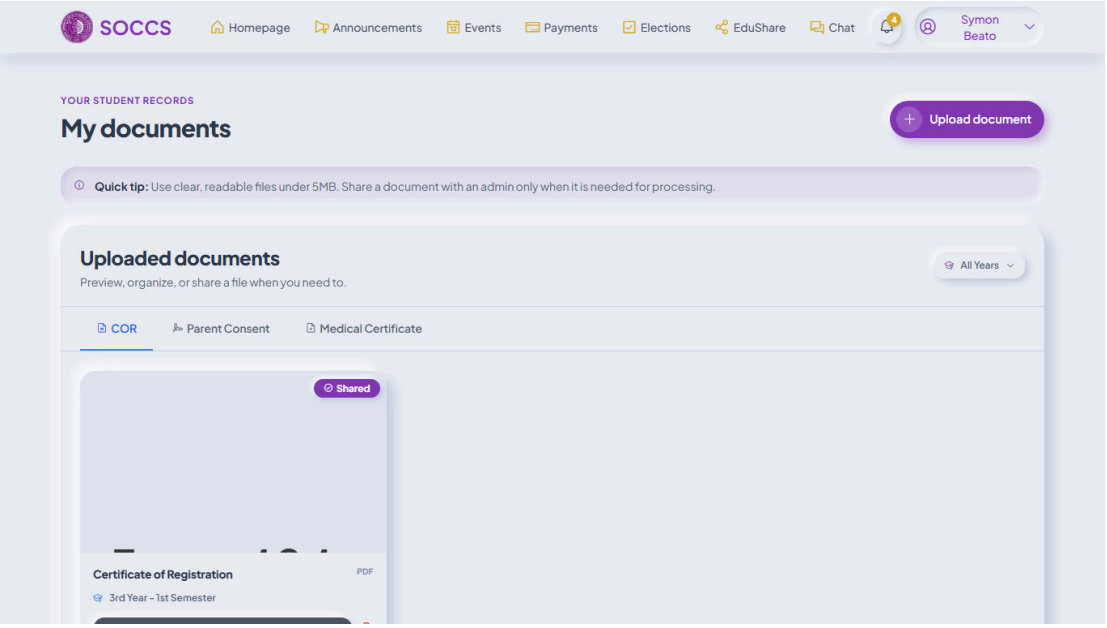


Figure 4.18. Student documents (MyDocs).

Figure 4.18 shows the student documents page. The student uploads and tracks personal submissions here. These files land in the same repository used for accreditation in Objective 2.2.

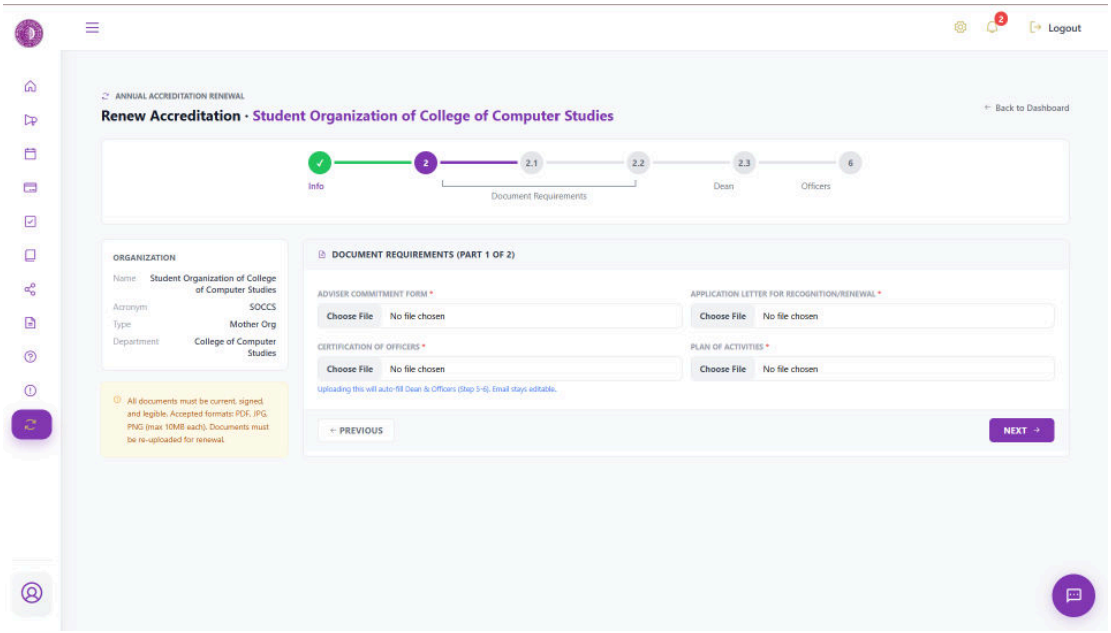


Figure 4.19. Accreditation documents upload, part 1 (organization admin).

Figure 4.19 is the first batch of accreditation attachments. The officer uploads the required papers per checklist item. This is where the tracking record in Objective

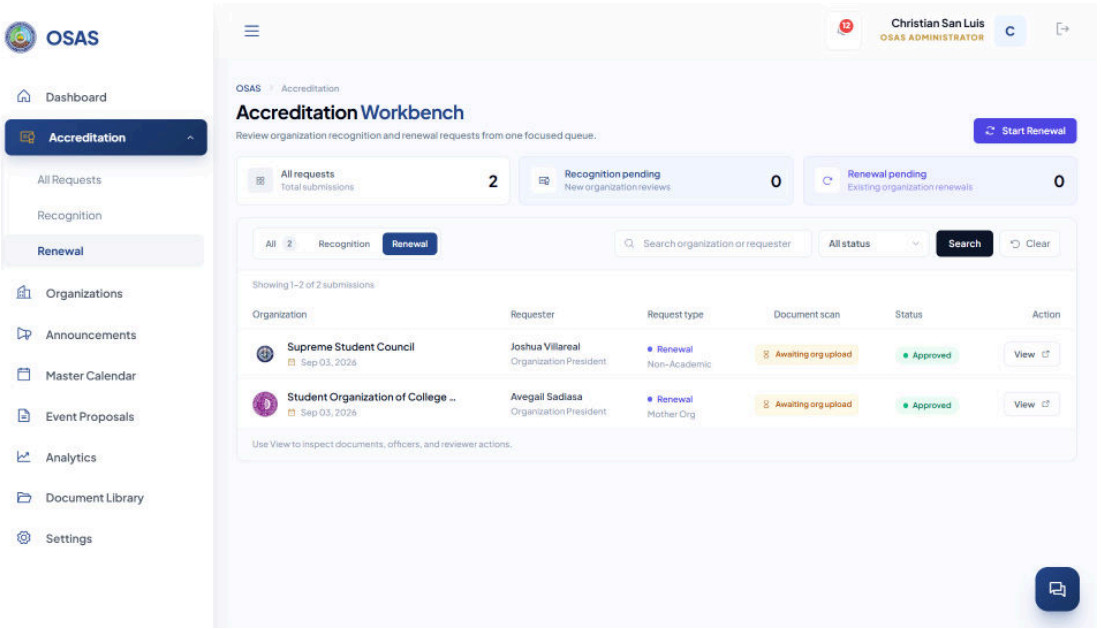


Figure 4.20. Accreditation renewal list (OSAS).

Figure 4.20 is the OSAS accreditation renewal list. All renewal requests and their progress are monitored in one queue. This is the tracking side of Objective 2.2. The dean has no separate accreditation screen, since approval happens in the same review flow.

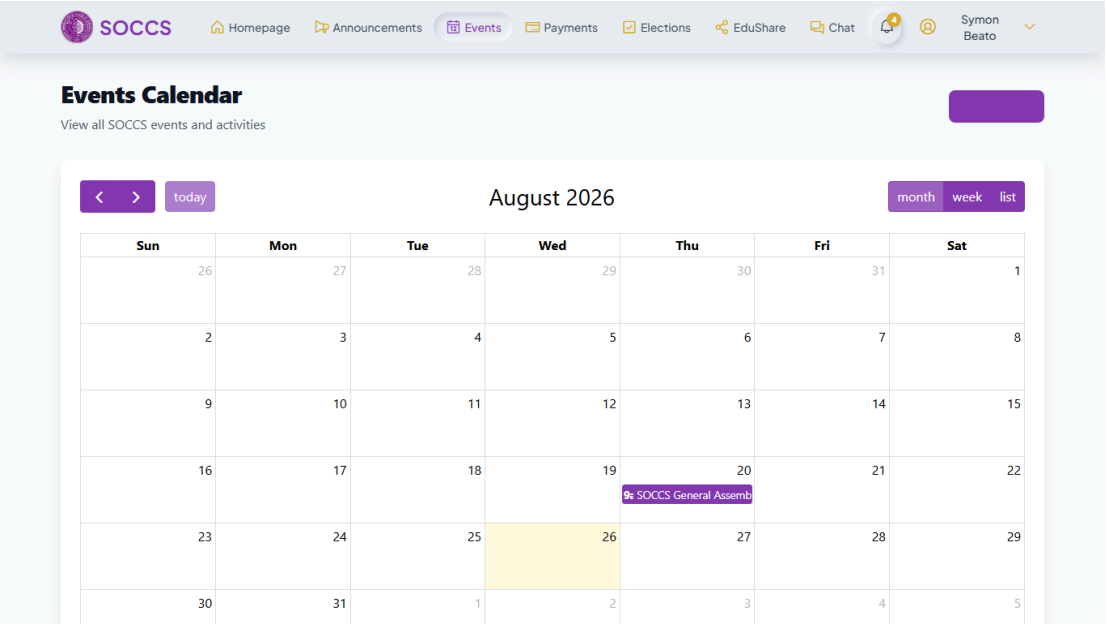


Figure 4.21. Student events calendar.

Figure 4.21 shows the student events calendar. Approved activities are plotted by date so the student can plan ahead. This is the student view of the master calendar in Objective 2.3.

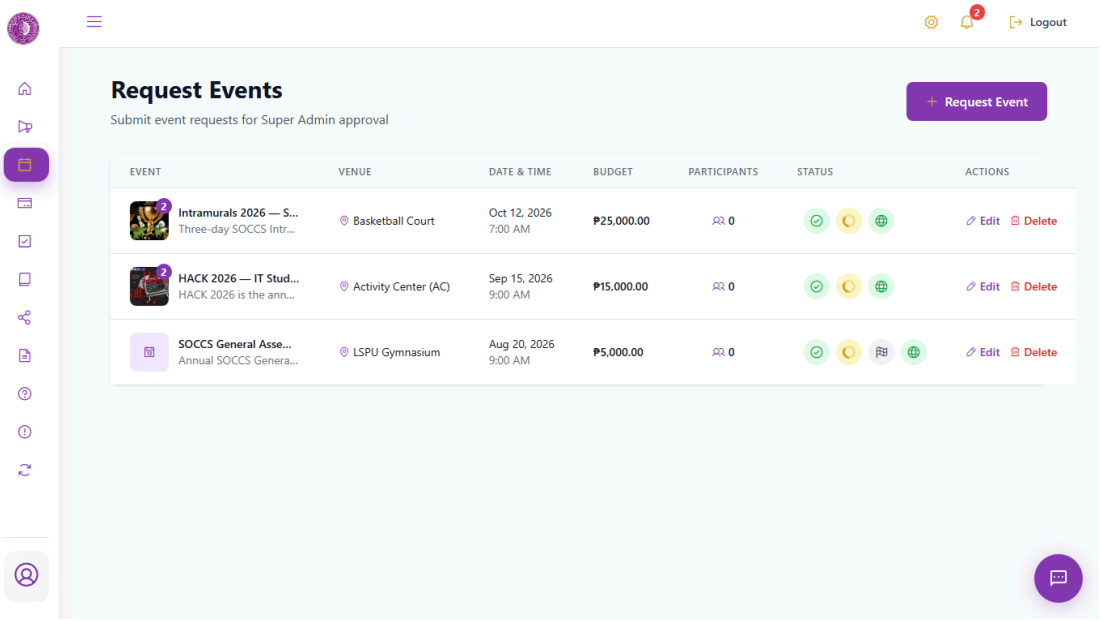


Figure 4.22. Events list (organization admin).

Figure 4.22 is the organization events list. The officer creates activities and watches their approval status here. This feeds the scheduling pipeline in Objective 2.3.

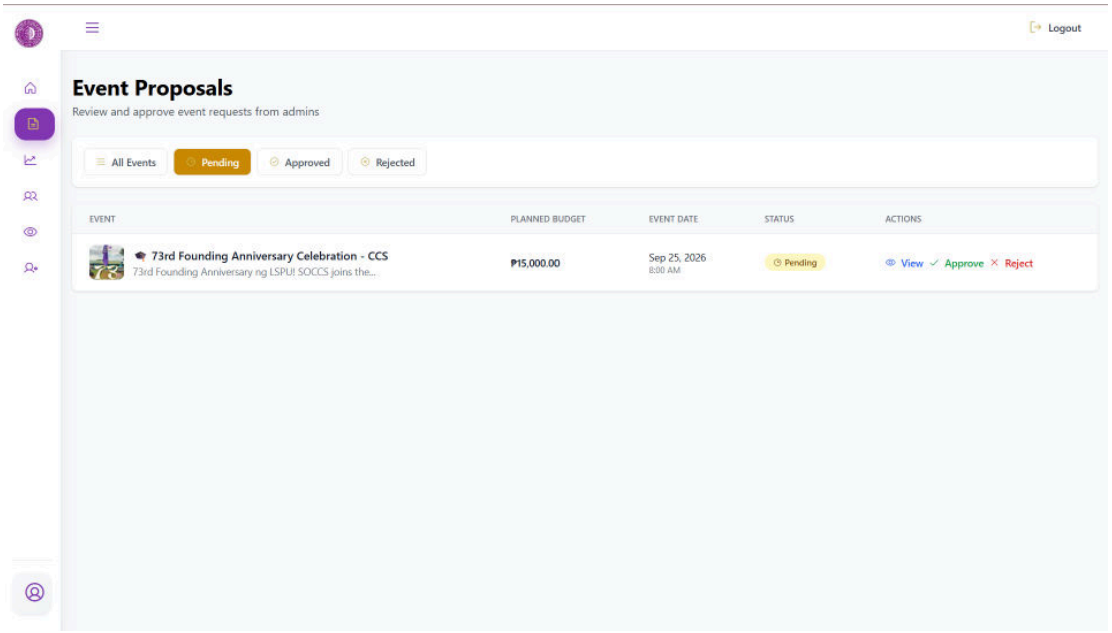


Figure 4.23. Pending event proposals (dean).

Figure 4.23 lists the event proposals waiting for dean action. The dean approves or rejects each proposal from this queue. This is the approval gate 2.3

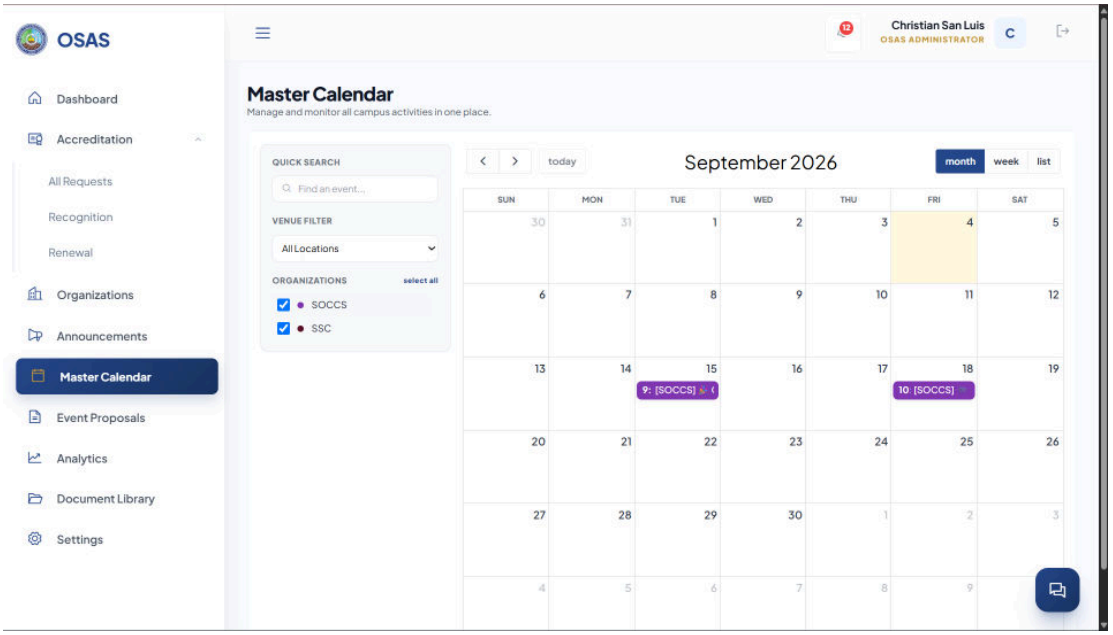


Figure 4.24. Master calendar month view (OSAS).

Figure 4.24 is the university-wide master calendar. All approved events sit in one month view so date clashes are easy to spot. This is the main output of Objective 2.3.

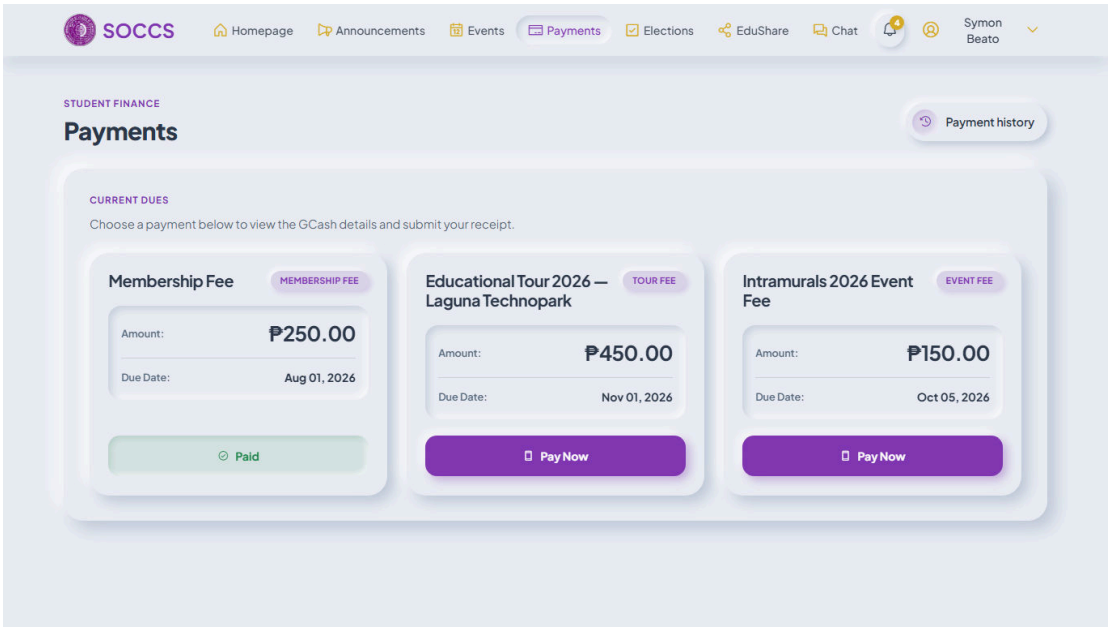


Figure 4.25. Student payments list.

Figure 4.25 shows the student payments list. Payables from different organizations appear under one account. The student opens the GCash, QR, history, or receipt modal from here in Objective 2.4.

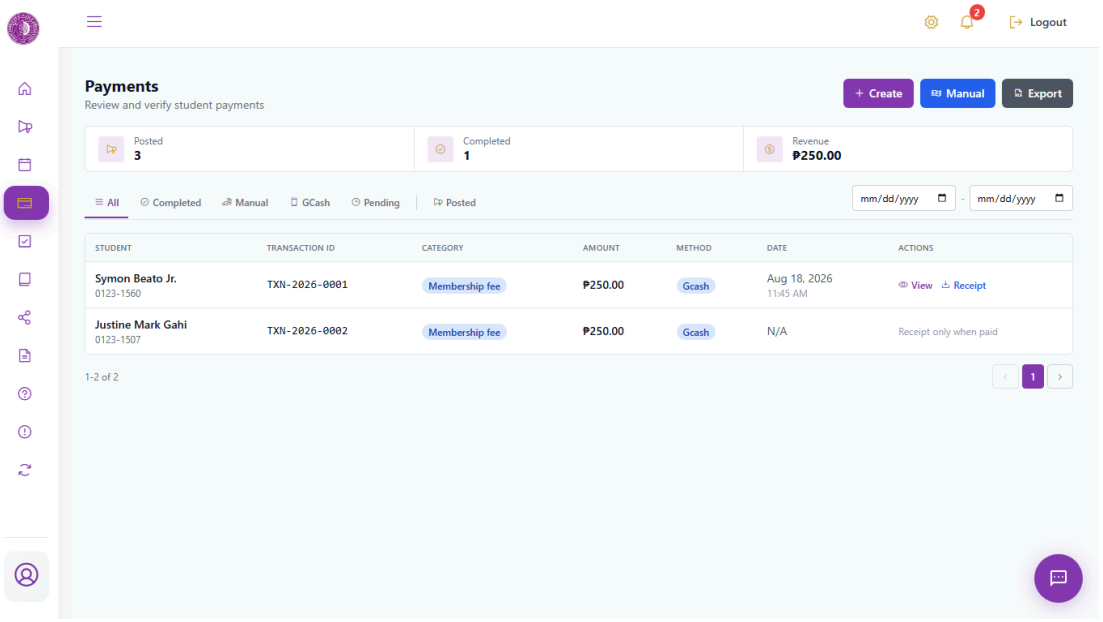


Figure 4.26. Payments list (organization admin).

Figure 4.26 is the officer payments list. Published payables and student submissions are tracked here for verification. This is the admin side of the centralized

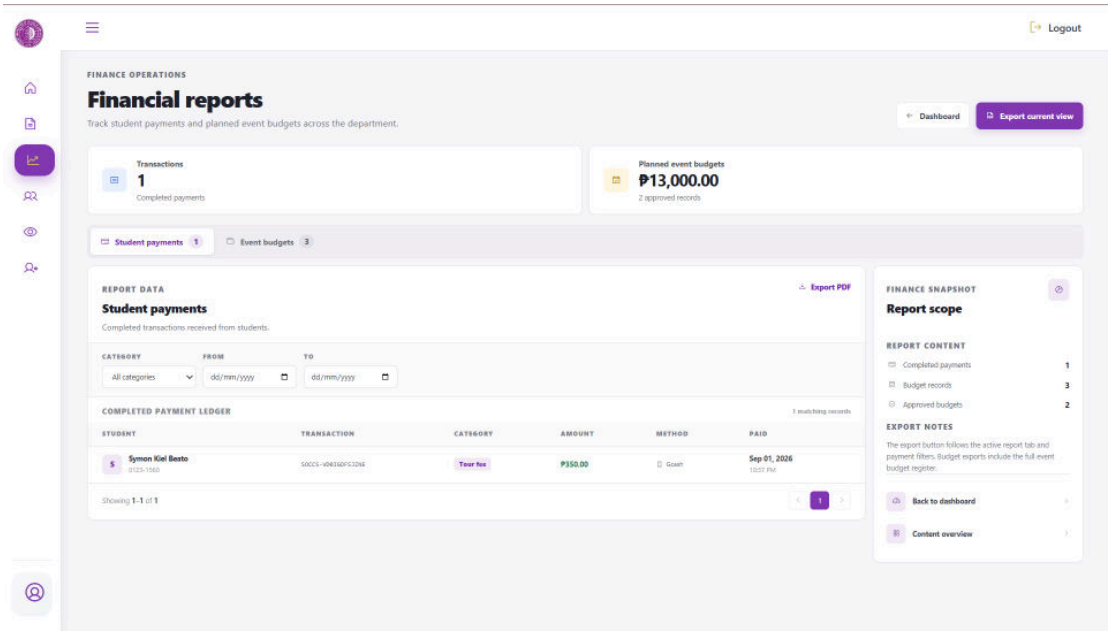


Figure 4.27. Finance reports, payments tab (dean).

Figure 4.27 shows the dean finance reports under the payments tab. Collected dues and payment records are summarized here for monitoring. OSAS has no separate payment screen, so this dean view plus the two lists above complete Objective 2.4.

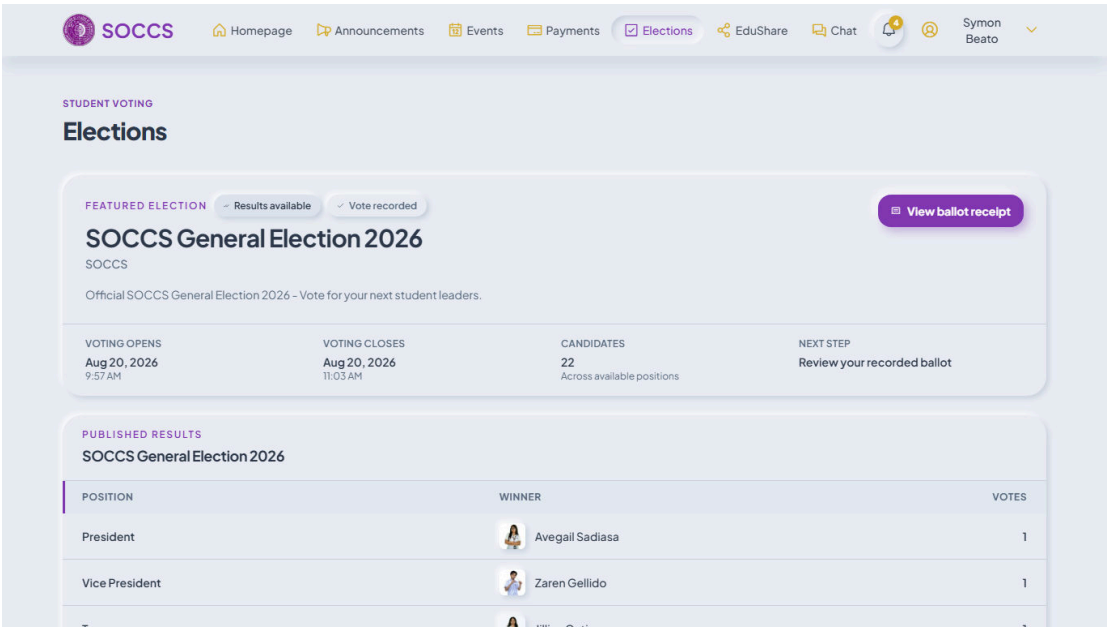


Figure 4.28. Student elections list.

Figure 4.28 shows the student elections list. Open and closed elections are listed with their schedules. The student enters the election details and casts a vote from here in Objective 2.5.

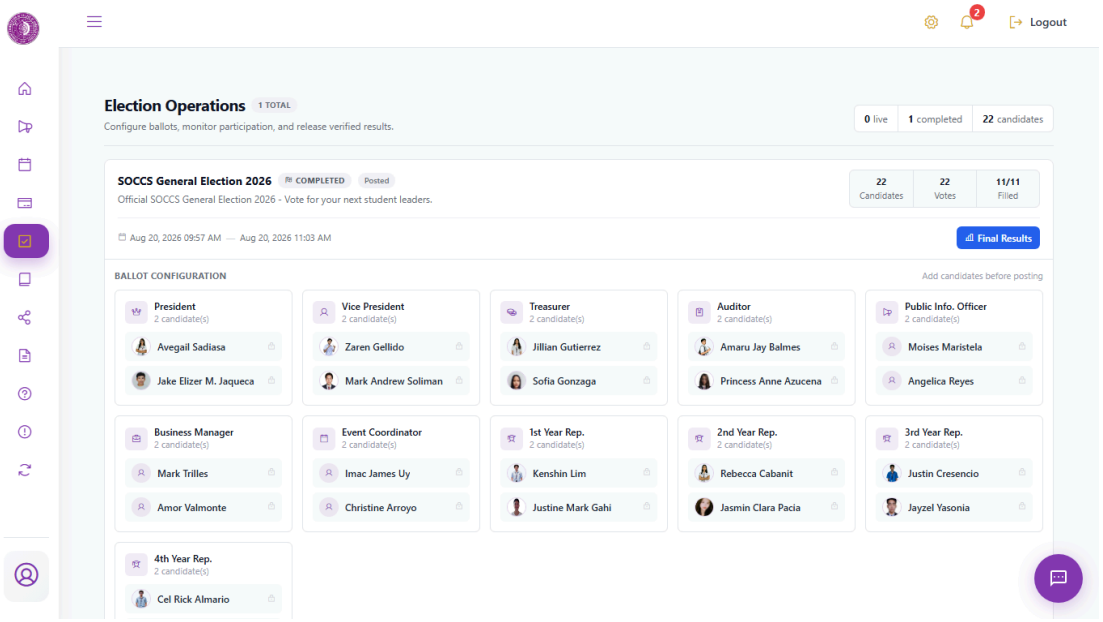


Figure 4.29. Elections list (organization admin).

Figure 4.29 is the officer elections list. The admin creates the election, manages candidates, and watches the live count. Deans and OSAS have no election screens since elections run between officers and students in Objective 2.5.

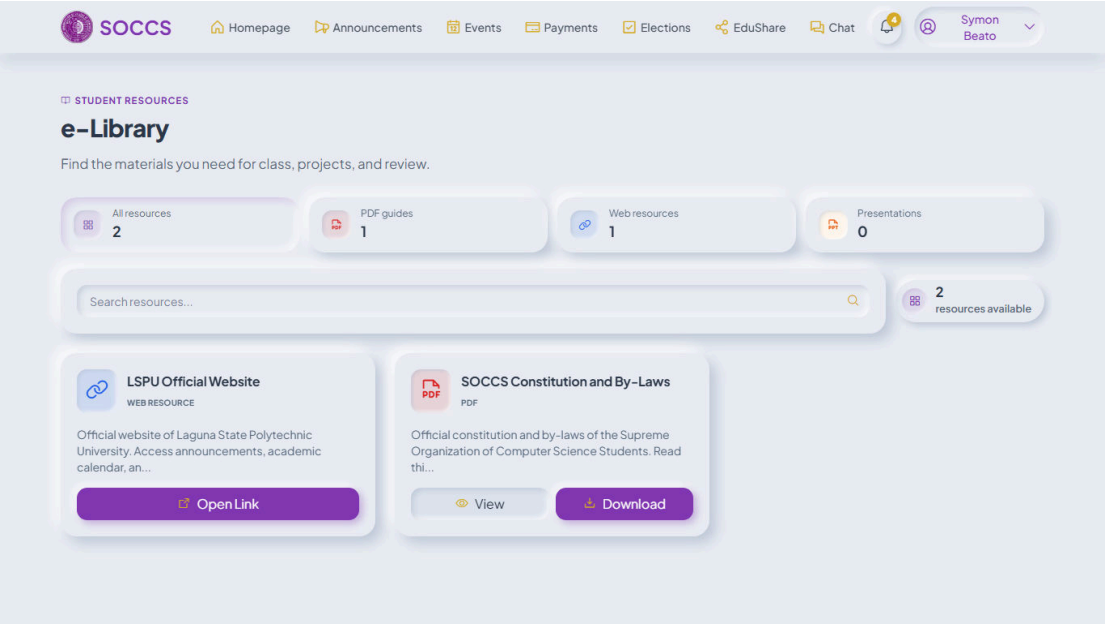


Figure 4.30. E-library resource list (student).

Figure 4.30 shows the student e-library list. Study materials posted by organizations are browsed and opened here. This is the student side of Objective 2.6.

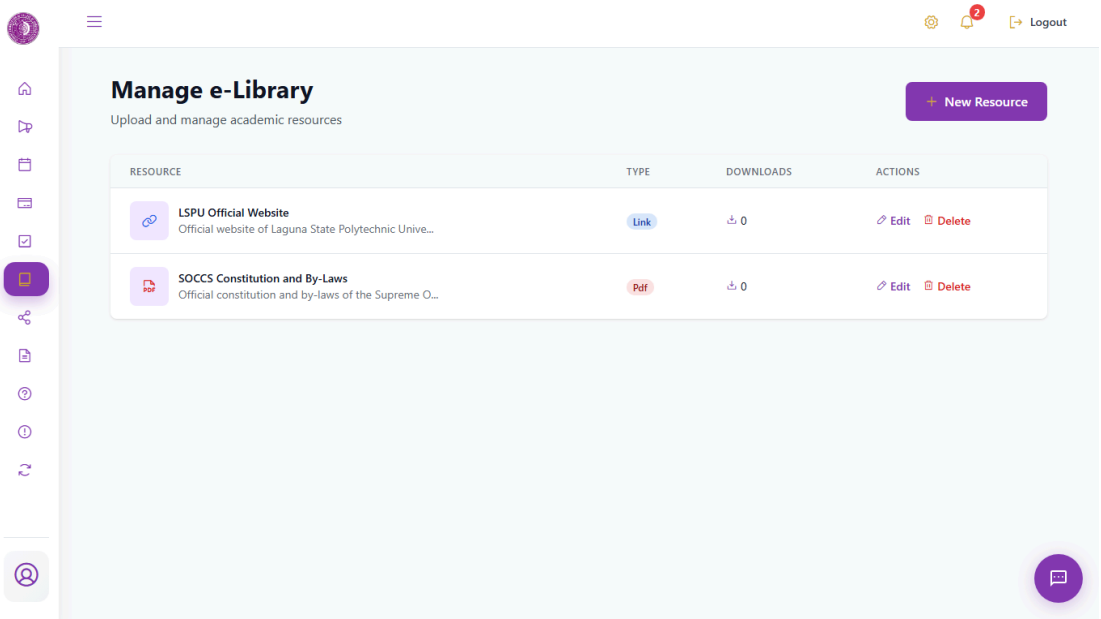


Figure 4.31. E-library (organization admin).

Figure 4.31 is the officer e-library view. The admin uploads and organizes resources for members, with EduShare folders for other file types. This is the management side of Objective 2.6.

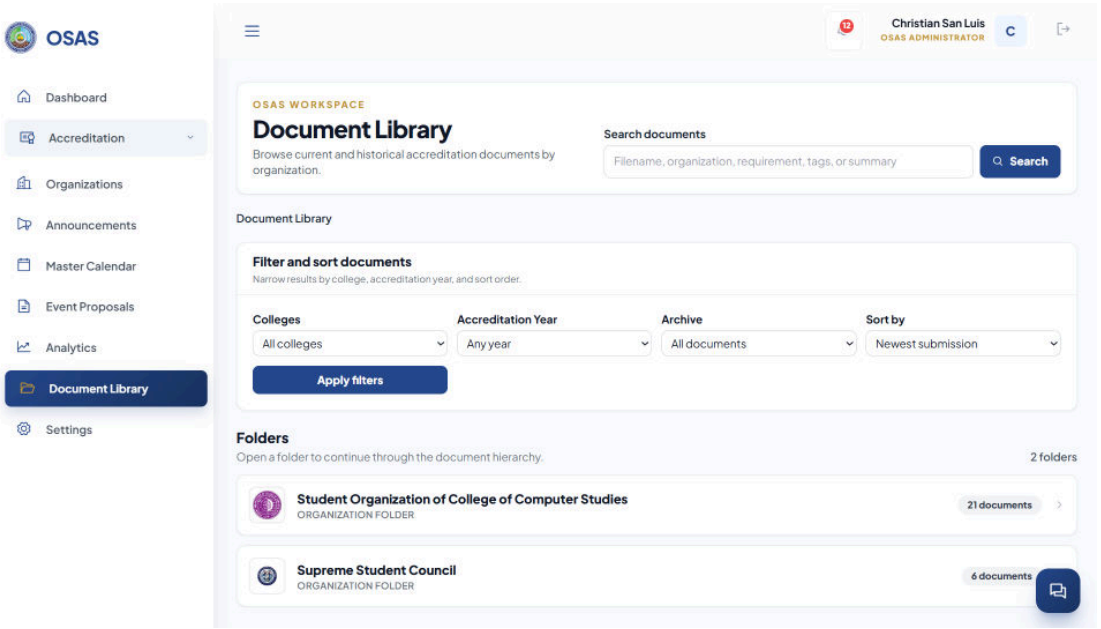


Figure 4.32. Document library folders (OSAS).

Figure 4.32 shows the OSAS document library folders. Submitted and processed files are grouped per organization for central keeping. The dean has no library screen, so these three views complete Objective 2.6.

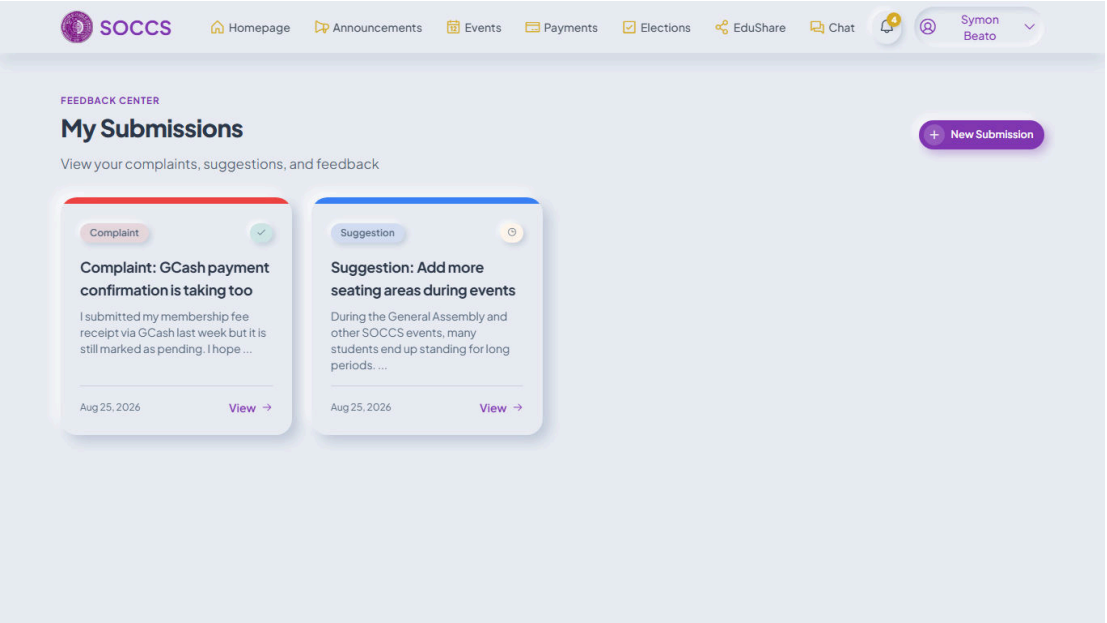


Figure 4.33. Complaints list (student).

Figure 4.33 shows the student complaints list. The student files a concern and follows the admin reply in the same thread. This is the feedback side of Objective 2.7.

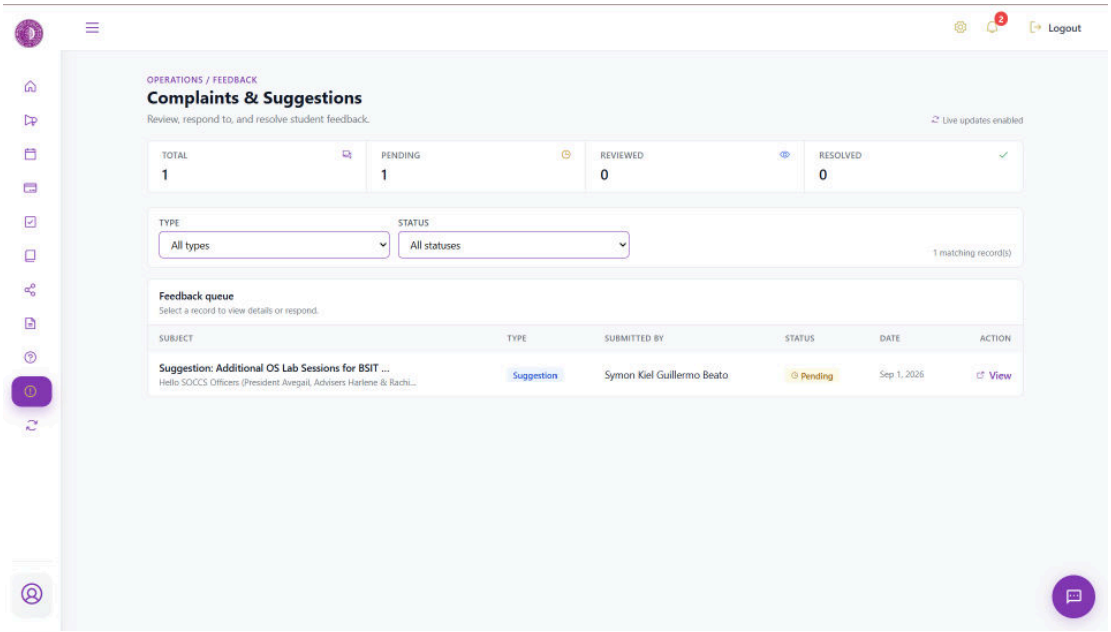


Figure 4.34. Complaints queue (organization admin).

Figure 4.34 is the officer complaints queue. Incoming concerns are listed for action and response. This closes the feedback loop in Objective 2.7.

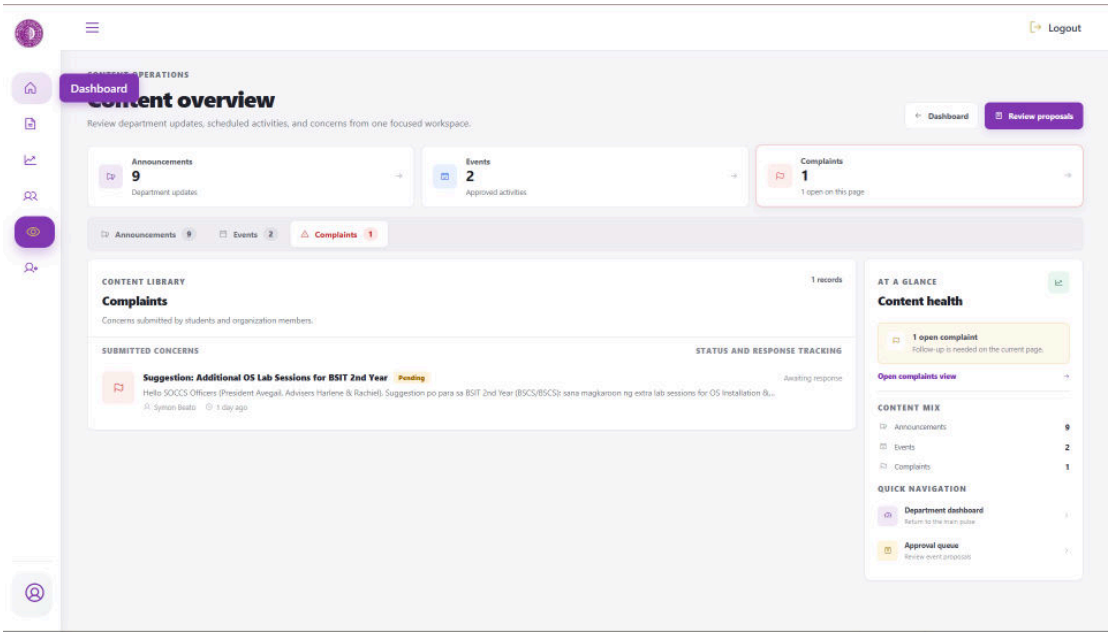


Figure 4.35. Content overview, complaints (dean).

Figure 4.35 shows the dean content overview for complaints. Reports under the handled colleges are monitored here. This gives the dean oversight in Objective 2.7.

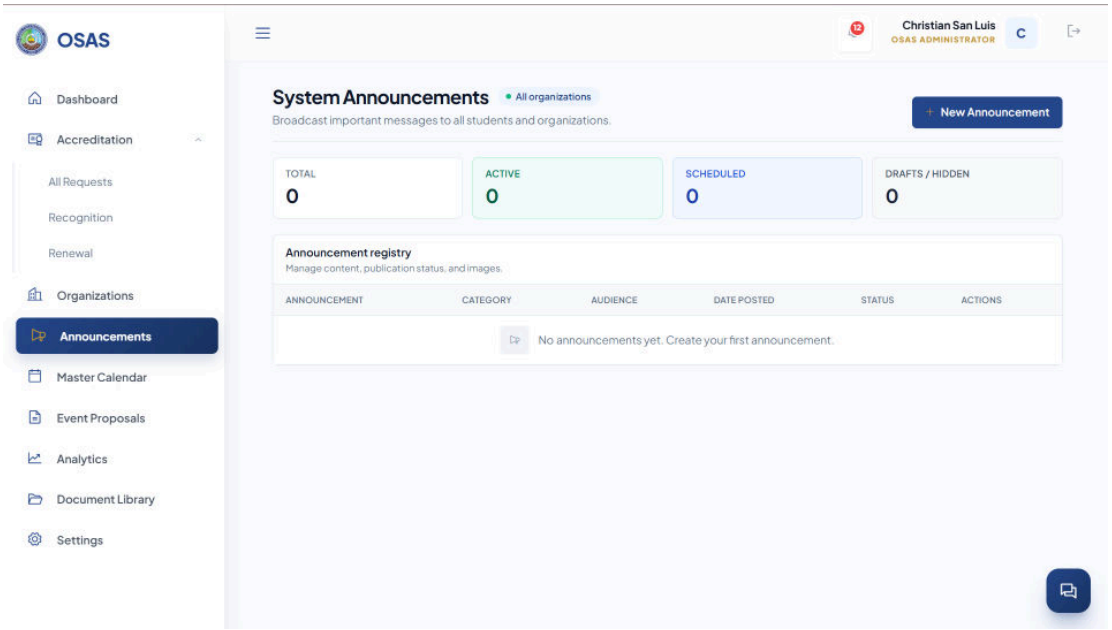


Figure 4.36. Announcements list (OSAS).

Figure 4.36 is the OSAS announcements list. University-wide notices are posted from here for all roles. Posting, complaints, and the chatbot FAQ together form Objective 2.7.

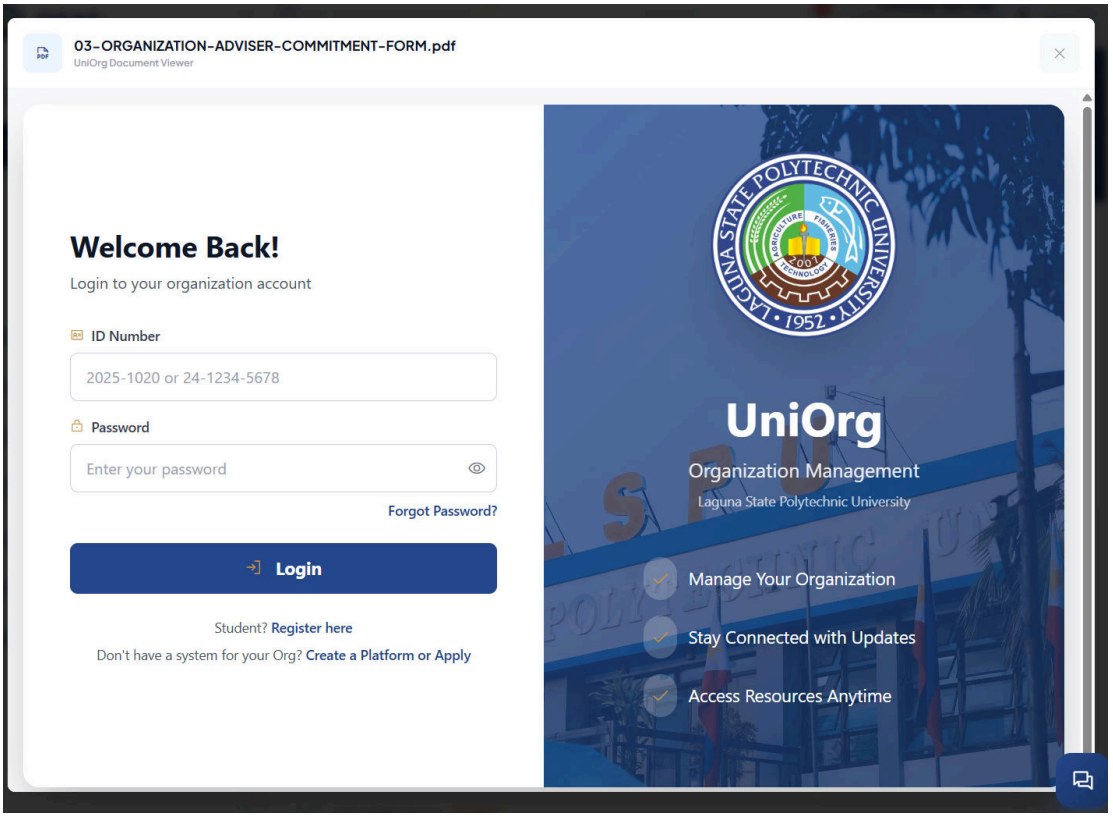


Figure 4.37. Login page of UniOrg.

Figure 4.37 is the shared login page. All users pass through it, while students register with their COR for identity checking. OSAS, dean, and admin accounts are pre-made with fixed roles in Objective 2.8.

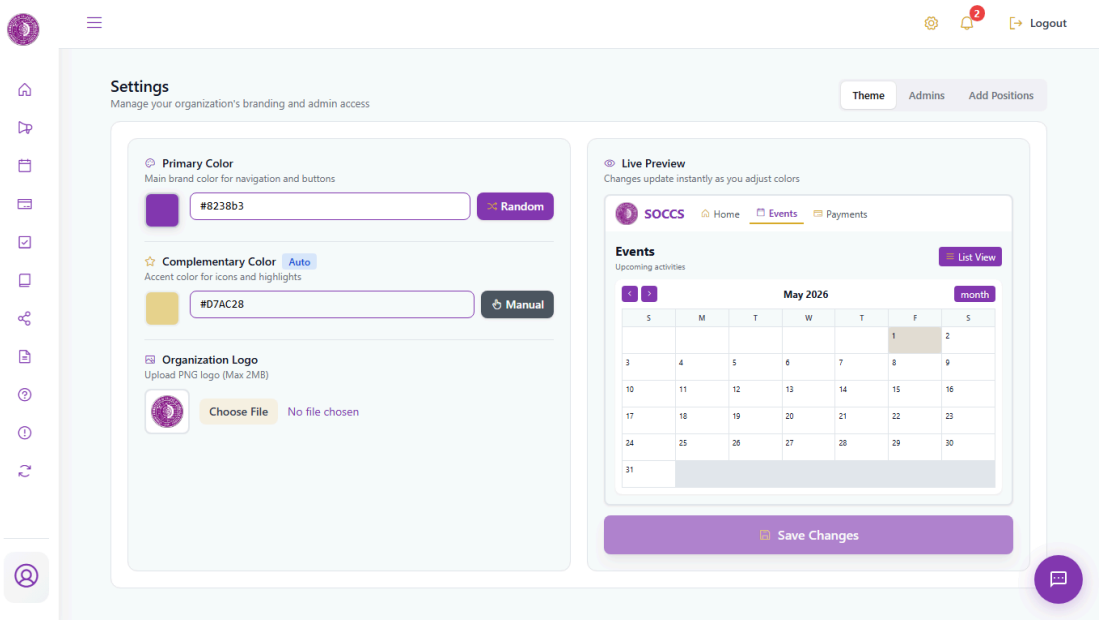


Figure 4.38. Settings (organization admin).

Figure 4.38 shows the organization admin settings. The officer manages profile and password settings within the owned workspace. This shows the limited scope of an officer account in Objective 2.8.

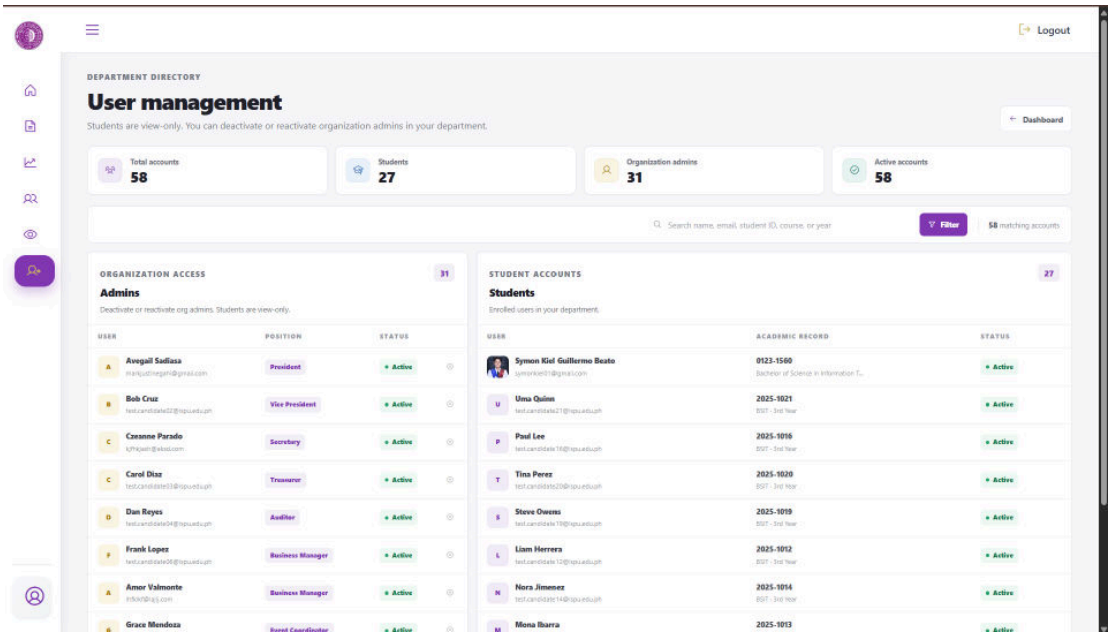


Figure 4.39. User management list (dean).

Figure 4.39 is the dean user management list. Accounts under the handled colleges are viewed and maintained here. Each account only opens modules for its role in Objective 2.8.

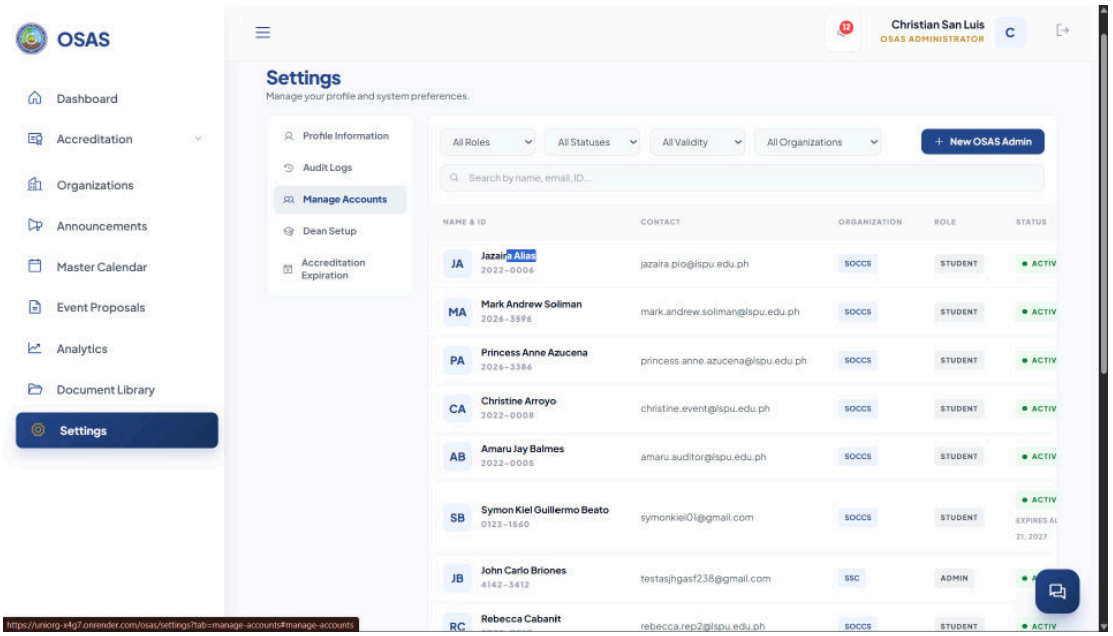


Figure 4.40. Account management list (OSAS).

Figure 4.40 shows the OSAS account management list. All roles and organizations are administered from this central page. This is the highest access level in Objective 2.8.

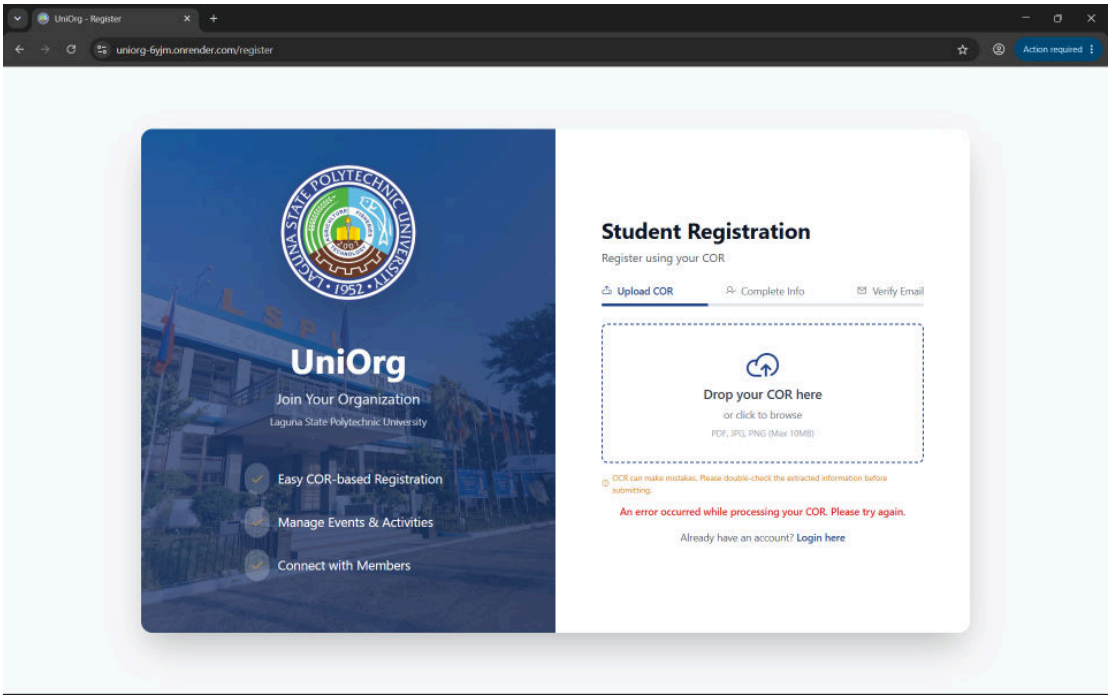


Figure 4.41. COR upload during registration.

Figure 4.41 shows the COR upload step in student registration. The uploaded COR image is the main file that OCR reads to verify identity. This is the first extraction point in Objective 2.9.

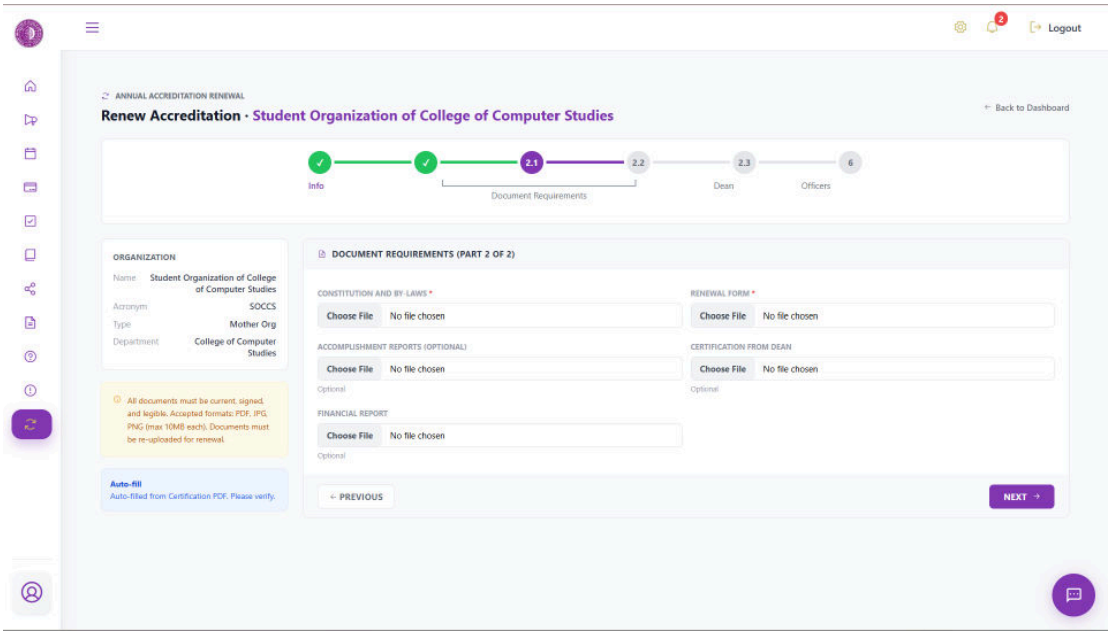


Figure 4.42. Accreditation documents upload, part 2 (organization admin).

Figure 4.42 is the second batch of accreditation attachments. Scanned papers uploaded here go through Tesseract OCR and Vision API reading. This is the heaviest extraction workload in Objective 2.9.

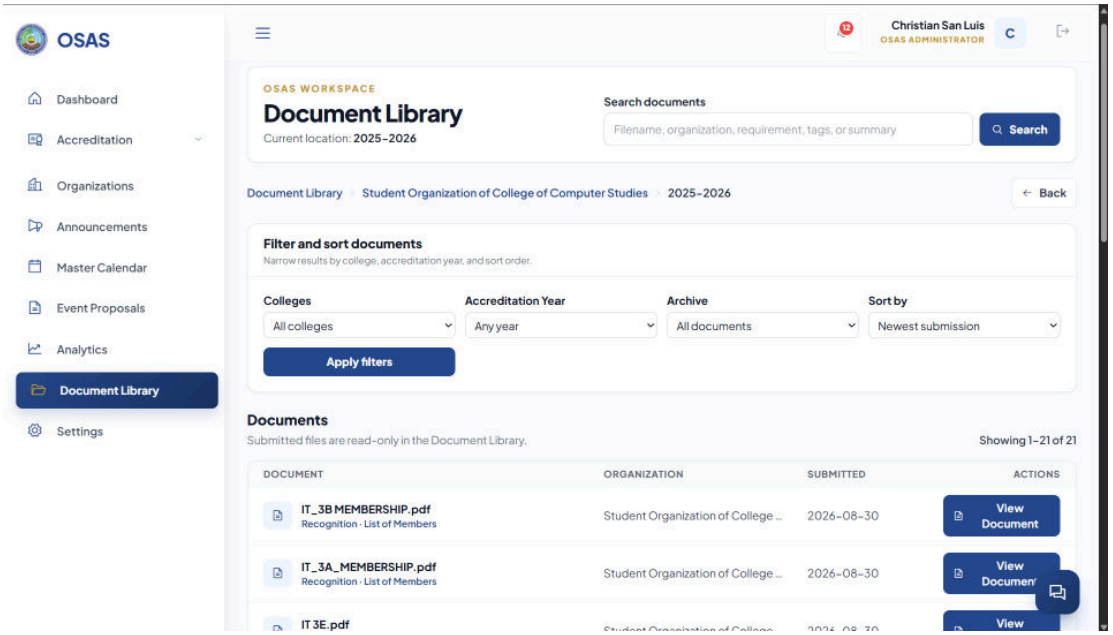


Figure 4.43. Organization documents view (OSAS).

Figure 4.43 shows processed organization documents at the OSAS side. Extracted text from the uploads is kept with each file for review. The dean has no OCR screen since extraction happens on upload in Objective 2.9.

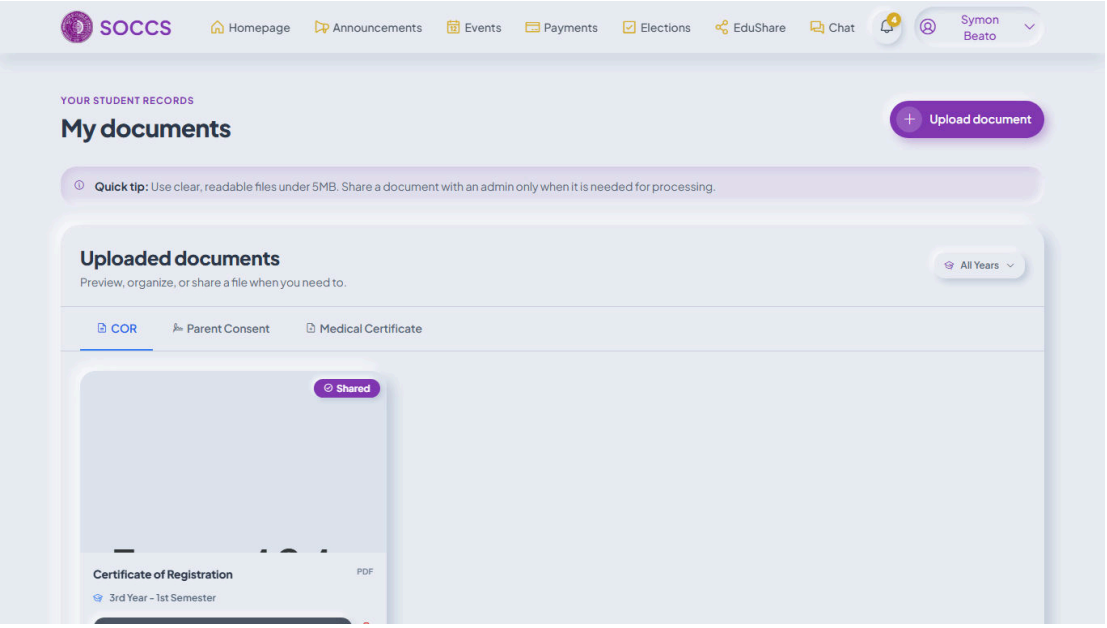


Figure 4.44. Student documents with auto-tags.

Figure 4.44 returns to the student documents page. Tags from the NLP pipeline describe each file for easier searching. This is the student view of auto-tagging in Objective 2.10.

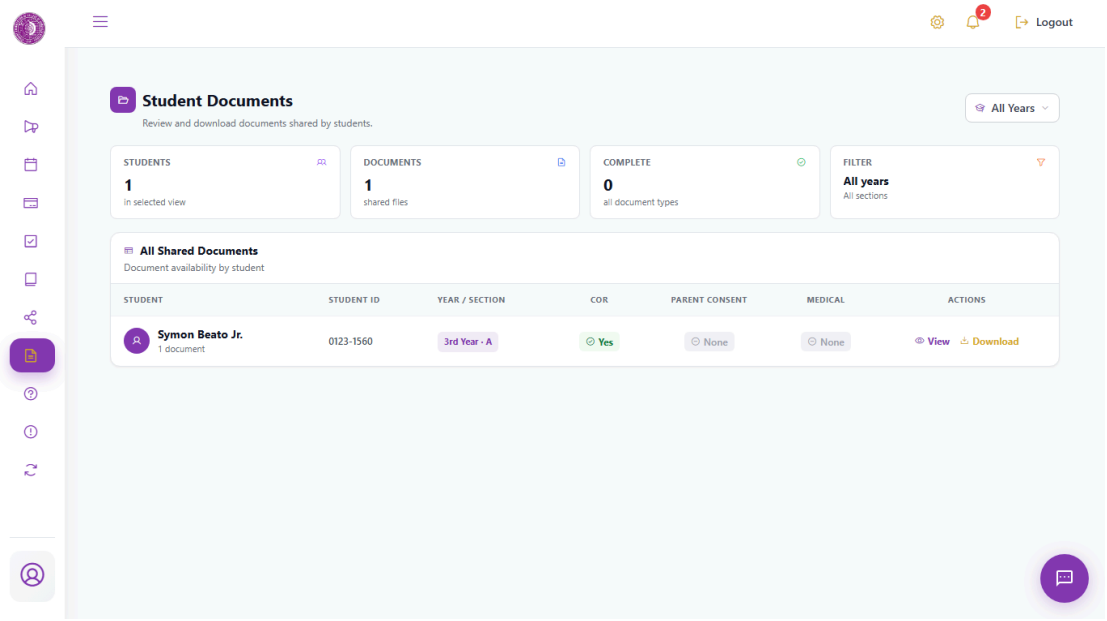


Figure 4.45. Submitted documents list (organization admin).

Figure 4.45 is the officer view of submitted student documents. Summaries and tags from Gemini and the Natural Language API guide the review. This is the admin side of Objective 2.10.

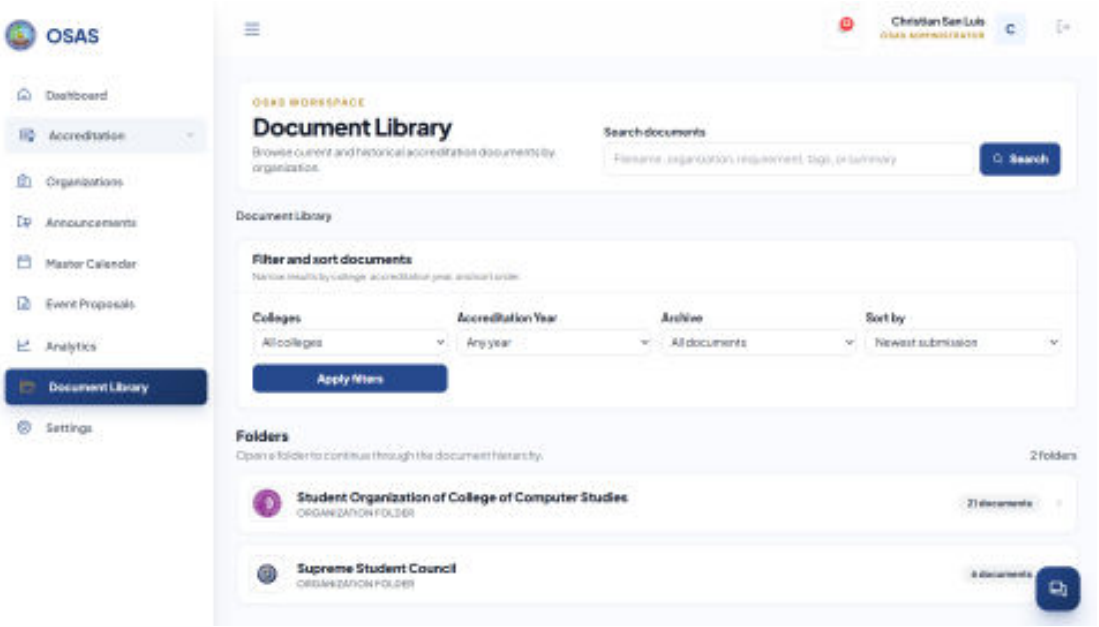


Figure 4.46. Document library folders with tagged files (OSAS).

Figure 4.46 shows the OSAS document library folders again. Grouping by organization, year, and auto-tag keeps the repository searchable. The dean has no NLP screen, so these three views close Objective 2.10.

4.2.6 Functional and Non-Functional Requirements

The developed UniOrg platform successfully satisfies all the functional and non-functional requirements established during the system analysis phase. Functional requirements pertaining to core operations such as

COR-upload user registration, accreditation and document submission with OCR text extraction and NLP auto-tagging, master calendar scheduling, centralized payments, e-voting, and EduShare and e-library sharing were fully operationalized through the Laravel web workspaces and the mobile-responsive browser interface. Behind the scenes, the application layer leverages Gemini 2.5 Flash for automated summarization and classification with Google Cloud Vision and Natural Language processing alongside Supabas.

On the non-functional side, the platform stays responsive on desktop and mobile browsers since all pages are mobile-responsive. Access control runs on two layers. The COR upload checks identity at registration, and role-based pages make sure students, officers, deans, and OSAS only open their own modules. Password settings complete the protection for day-to-day use.

For maintainability and growth, all 39 organizations share one Laravel codebase while Supabase handles structured records and Cloudflare R2 keeps the uploaded files. This separation lets OSAS onboard more organizations without new servers. The table 4.42 maps every requirement to its module and figure, so each claim in this section points to visible proof realtime to handle data flow.

In terms of non-functional requirements, the system delivers high performance, reliability, and security. Cross-platform compatibility was achieved for Android mobile browser users and standard web browser clients. Data security and structural integrity are maintained through strict PostgreSQL database constraints and Supabase backend management. Crucially, security parameters such as Role-Based Access Control (RBAC) configurations are isolated exclusively to the System Administrator dashboard to prevent unauthorized alterations by standard organization officers.

To explicitly demonstrate alignment between system design and empirical validation, The table 4.42 outlines the requirements-to-results matrix, detailing how each functional scope was verified through formal Black-Box testing phases yielding a 100% success rate.

Table 4.42. Requirement-to-Results Matrix

Requirement Category	System Requirement Description	Implemented Feature	Testing Evidence	Result
Functional	User registration with COR verification	COR upload with enrolled-student matching and OTP email verification	Test Case 02 (Registration & Login)	Passed (100%)
Functional	Automated accreditation and document processing	Accreditation submission with OCR text extraction, Gemini classification, and NLP auto-tagging	Test Case 03 (Accreditation & Documents)	Passed (100%)
Functional	Real-time event tracking and calendar mapping	Master calendar scheduling with organization registry and analytics dashboard	Test Case 04 (Events & Calendar)	Passed (100%)

Table 4.42—continued

Requirement Category	System Requirement Description	Implemented Feature	Testing Evidence	Result
Functional	Student participation and resource sharing modules	EduShare folders, e-library resources, and organization announcements	Test Case 05 (EduShare & Library)	Passed (100%)
Functional	Centralized payments and financial management	GCash-routed collections with receipt verification and financial reports	Test Case 06 (Payments & Finance)	Passed (100%)
Functional	E-voting and officer transition	System-managed elections with live counts, published results, and officer handover	Test Case 08 (Elections & Transition)	Passed (100%)
Non-Functional	Secure administrative control and RBAC	Restricted system configurations exclusive to the System Administrator account	Test Case 09 (Admin Login & Security)	Passed (100%)
Non-Functional	Real-time cross-platform responsiveness	Supabase realtime synchronization across web and mobile browser interfaces	System-wide Black-Box Phase 1 & 2 evaluation (670 step-level verifications)	Passed (100% Success Rate)

4.3 System Testing Results

This section presents the empirical findings obtained from the systematic functional testing procedures described in Chapter III, addressing Objective 2: To test the functionality of the developed UniOrg system across its core modules, including the Dashboard, Registration and Login, Accreditation and Document Management, Master Calendar and Events, Centralized Payments, E-Voting and Officer Transition,

EduShare and E-Library, Feedback and Complaints, and Administrator Login features.

Table 4.43. Black-Box Testing Results for TC-01 - User Authentication and RBAC

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The registration form opens properly with input fields: Student ID, Full Name, Email, Department/Course, Year Level, Section, Password, and COR Upload.	10	10	0	100%
2	The system validates input, verifies email format, creates the account, and sends an OTP/verification notice or success confirmation.	10	10	0	100%
3	System triggers client and server-side validation, shows specific error prompts, and blocks form submission.	10	10	0	100%
4	User logs in successfully and is redirected to the Student Dashboard (/dashboard) with student-level navigation and features.	10	10	0	100%
5	Officer dashboard loads with administrative controls. Switching between Admin View and Student View updates navigation options dynamically.	10	10	0	100%
6	System grants access to university-wide or college-level oversight dashboards with administrative authority.	10	10	0	100%
7	Session terminates immediately; direct protected URL access is intercepted and redirected to Login with an unauthorized access prompt.	10	10	0	100%

Table 4.44. Black-Box Testing Results for TC-02 - Multi-Org Workspace and Organization Profiling

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The organization portal loads displaying organization name, logo, banner, mission/vision, and quick statistics without broken assets.	10	10	0	100%
2	The settings panel opens with tabs for General Information, Branding/Themes, Role Permissions, and Officer Positions.	10	10	0	100%
3	Changes are saved to the database and updated across the student-facing organization landing page.	10	10	0	100%
4	The new position and assigned member appear in the officer roster with corresponding permissions updated.	10	10	0	100%
5	The workspace context, permissions, and displayed announcements switch instantly to the selected organization.	10	10	0	100%
6	Profile photo uploads securely (stored in Cloudflare R2), password updates after old password confirmation, and feedback toast is displayed.	10	10	0	100%

Table 4.45. Black-Box Testing Results for TC-03 - Accreditation Tracking and Application/Renewal Workflow

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The multi-step application form opens displaying fields for Organization Name, Department, Type (Academic/Non-Academic), and required document upload sections.	10	10	0	100%
2	The files are uploaded, validated for allowed file formats (PDF/images), and preview cards are displayed.	10	10	0	100%
3	System assigns a unique tracking identifier, sets status to 'Pending Review', and sends notification to OSAS administrators.	10	10	0	100%
4	The submitted application appears in the pending review list with organization summary, submitted documents, and timestamp.	10	10	0	100%
5	Documents open in the built-in viewer without download errors; feedback and review notes are recorded.	10	10	0	100%
6	Status changes to 'Approved' (activating the organization system) or 'Rejected' (with revision instructions sent to applicant).	10	10	0	100%
7	Pre-populates historical data, accepts updated officer and activity documents, and logs the renewal under OSAS renewal queue.	10	10	0	100%

Table 4.46. Black-Box Testing Results for TC-04 - OCR Document Data Extraction

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The document upload dropzone loads with format hints and allowed extensions (.pdf, .png, .jpg).	10	10	0	100%
2	The upload progresses with a loading indicator and initiates the OCR extraction pipeline.	10	10	0	100%
3	The OCR engine scans the image/PDF, recognizes printed text, and returns extracted raw text with confidence scores.	10	10	0	100%
4	The system accurately parses officer names, positions (President, VP, Secretary, etc.), and school year into structured form fields.	10	10	0	100%
5	Only validated tokens present in the source document are merged into empty slots; unreadable text prompts manual verification.	10	10	0	100%
6	System gracefully handles the error, flags unreadable content, and presents a clear error toast without system crash.	10	10	0	100%

Table 4.47. Black-Box Testing Results for TC-05 - NLP Summarization and Auto-Tagging

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The document is queued for natural language processing and entity extraction.	10	10	0	100%

Table 4.47—continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	System extracts recognized entities (Organizations, Dates, Locations, Events, Persons) with salience threshold > 0.1.	10	10	0	100%
3	The system generates a concise 2-3 sentence executive summary capturing the core objectives and outcomes of the document.	10	10	0	100%
4	Generates 3 to 10 context-aware tags (Document Type, Organization Name, Academic Year, Term, Key Topics) without duplicates.	10	10	0	100%
5	The smart search indexes the tags and entities; querying keywords returns the exact matching document instantly.	10	10	0	100%

Table 4.48. Black-Box Testing Results for TC-06 - Event Proposals and Master Calendar

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The Create Event form opens with fields for Event Title, Description, Venue, Start/End DateTime, Target Audience, Budget, and Request Letter upload.	10	10	0	100%
2	System checks venue availability in real-time and displays warning: 'This venue is already booked for the selected schedule'.	10	10	0	100%

Table 4.48—continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	Event is saved with status 'Pending Dean Approval' and forwards notification to the College Dean.	10	10	0	100%
4	Dean approval is recorded; event status automatically updates to 'Pending OSAS Approval' and alerts OSAS.	10	10	0	100%
5	OSAS approval is finalized; event status changes to 'Approved / Posted'.	10	10	0	100%
6	Approved event appears in FullCalendar with organization badge [SOCCS], color-coded category, hover tooltip, and registration button.	10	10	0	100%
7	Org requests new date; Dean and OSAS re-verify schedule and update calendar seamlessly upon approval.	10	10	0	100%

Table 4.49. Black-Box Testing Results for TC-07 - Centralized Payment and GCash Verification

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Form loads; admin creates fee announcement with Title, Amount (PHP), GCash QR code image, Account Number, and Deadline.	10	10	0	100%
2	Payment modal displays fee details, QR code, GCash account details, Reference Number input, and Receipt Upload field.	10	10	0	100%

Table 4.49—-continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	Payment record is created with status 'Pending Verification'; student receives confirmation prompt.	10	10	0	100%
4	Rejection reason is saved; status changes to 'Rejected'; student receives notification with remarks.	10	10	0	100%
5	Record re-enters pending verification queue with updated proof without creating duplicate rows.	10	10	0	100%
6	Status updates to 'Paid', revenue total increments in real-time, and student receives official payment status.	10	10	0	100%
7	System renders and downloads standardized PDF receipt containing Transaction ID, Student Info, Amount, and Org Stamp.	10	10	0	100%
8	Generates downloadable .xlsx spreadsheet with accurate payment logs, reference numbers, and total collections.	10	10	0	100%

Table 4.50. Black-Box Testing Results for TC-08 - E-Voting and Election System

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Election creation form opens with Election Title, Academic Year, Positions (President, VP, etc.), and Voting Window (Start/End DateTime).	10	10	0	100%

Table 4.50–-continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Candidate cards populate under respective position categories and save successfully.	10	10	0	100%
3	Election status changes to 'Active/Voting Open'; voting broadcast notification is dispatched to eligible student voters.	10	10	0	100%
4	Digital ballot loads showing positions, candidate profiles, platforms, and selection radio buttons/checkboxes.	10	10	0	100%
5	System securely encrypts and casts the ballot, flags student as 'Voted', and displays submission confirmation.	10	10	0	100%
6	System blocks submission with error message: 'You have already cast your vote for this election.'	10	10	0	100%
7	Real-time vote tallies, percentage breakdowns, and winning candidates are calculated and published on the results page (/elections/{id}/results).	10	10	0	100%

Table 4.51. Black-Box Testing Results for TC-09 - EduShare and E-Library Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Folder is created in directory tree and appears in the workspace.	10	10	0	100%

Table 4.51—continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Files upload to Cloudflare R2 storage; preview thumbnails and file size badges display properly.	10	10	0	100%
3	System validates upload client and server-side, blocks the file, and displays 'File too large / Invalid file type' toast.	10	10	0	100%
4	Sharing permissions update; authorized students can view the resource under their 'Shared with Me' tab.	10	10	0	100%
5	PDF renders in embedded viewer; download counter increments; direct unauthenticated access is blocked by canAccessItem().	10	10	0	100%
6	Live search filters resources instantly without page reload; clicking links redirects safely to educational resources.	10	10	0	100%
7	Real-time Pusher 'library' channel broadcasts update; resource appears/disappears on student view immediately.	10	10	0	100%

Table 4.52. Black-Box Testing Results for TC-10 - Feedback Hub and Real-Time Chat

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Complaint submission form opens with Category selector, Subject, Description, Anonymous toggle, and Evidence upload.	10	10	0	100%

Table 4.52–-continued.

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Ticket is generated with UUID tracking code; student identity is masked; ticket appears in Admin review queue (/admin/complaints).	10	10	0	100%
3	Response is logged, ticket status updates (In Progress / Resolved), and student receives resolution notification.	10	10	0	100%
4	Chatbot parses user query and returns instant matching answers from the pre-configured FAQ database.	10	10	0	100%
5	Message and image bubble render instantly on Admin floating chat widget (#admin-chat-widget) via Supabase Realtime without page refresh.	10	10	0	100%
6	Student receives instant message bubble and download tile; unread counter updates in real-time.	10	10	0	100%
7	Org Admin receives real-time OSAS communication banner and toast notification with full attachment support.	10	10	0	100%

4.3.1 Test Results by Requirements

To rigorously evaluate whether UniOrg fulfilled its intended operational scope, the system was subjected to black-box testing across its primary modules. Each test case was measured against predefined criteria to contrast system behavior with expected outcomes. The table 4.53 outlines the

detailed test results by requirement, detailing the expected output, actual system behavior, and final test status.

Table 4.53. Test Results by Requirements

Test Case / Module	Requirement Description	Expected Result	Actual Result	Test Status
TC-01: Authentication and RBAC	Register, authenticate, switch views, route users to role dashboards, block unauthorized URLs.	Accounts created with COR and OTP verification; RBAC enforced per role.	System validated inputs, created accounts, and blocked unauthorized URL access.	Passed
TC-02: Multi-Org Workspace	Isolate organization workspaces, custom branding, multi-org association, profile updates.	Branding, permissions, and announcements switch per selected organization.	Workspace context and permissions switched instantly without data leakage.	Passed
TC-03: Accreditation and Renewal	Submit accreditation packages, track milestones, OSAS review, approve, or reject.	Tracking ID assigned, pending list updated, approval activates the system.	Applications tracked, reviewed, and decided with notifications sent.	Passed
TC-04: OCR Data Extraction	Extract text, officer rosters, student numbers, and academic periods from scans and PDFs.	Raw text with confidence scores parsed into structured form fields.	Engine extracted and parsed entities with confidence scores.	Passed
TC-05: NLP and Auto-Tagging	Extract entities, summarize narrative reports, and assign categorization tags.	Two to three sentence summary and 3 to 10 tags generated; smart search finds documents.	Summaries, tags, and smart search returned the exact documents.	Passed

Table 4.53–-continued.

Test Case / Module	Requirement Description	Expected Result	Actual Result	Test Status
TC-06: Events and Master Calendar	Submit event proposals with venue booking through Dean and OSAS approval to calendar posting.	Conflict warnings shown, approval chain completed, event posted on FullCalendar.	Proposals approved through the chain and posted on the calendar.	Passed
TC-07: Payments and GCash Verification	Create fees, submit GCash proofs, verify or reject, generate receipts, export reports.	Pending, Paid, and Rejected lifecycle with PDF receipts and Excel export.	Payments verified, receipts generated, and reports exported.	Passed
TC-08: E-Voting and Elections	Configure elections, register candidates, secure one-time voting, automatic tallying.	Encrypted ballots cast once, double voting blocked, results published.	Votes cast once, tallied, and published correctly.	Passed
TC-09: EduShare and E-Library	Upload, organize, share, preview, and download resources with validation and access control.	R2 storage, file validation, live search, and realtime library updates.	Resources shared, searched, and synced in real time.	Passed
TC-10: Feedback Hub and Chat	Submit complaints securely, query FAQ chatbot, realtime messaging with attachments.	UUID tickets resolved, instant chatbot answers, Supabase realtime delivery.	Tickets resolved, chatbot answered, and messages delivered instantly.	Passed

4.3.2 Testing Summary

To provide a holistic view of the system evaluation outcomes, the quantitative data gathered across all black-box testing phases were consolidated. This statistical summary aggregates the performance of the UniOrg platform across multiple user roles and operational scenarios, measuring system reliability against the core objectives established for the Office of Student Affairs and Services (OSAS) of LSPU Santa Cruz Campus.

The evaluation involved ten (10) independent evaluators comprising OSAS personnel and student organization stakeholders, who executed ten (10) comprehensive test cases spanni670 step-level verificationsons. The table 4.54 outlines the overall statistical outcomes of the testing process

Table 4.54. Overall Statistical Outcomes of Black-Box Testing

Evaluation Metric	Quantitative Value	Percentage / Status
Total Number of Evaluators	10 Personnel / Stakeholders	100% Participation
Total Test Cases Executed	10 Modules / Core Functions	100% Coverage
Total Step-Level Evaluations	670 Step-Level Verifications	100% Executed
Passed Evaluations	670 Steps	100.00%
Failed Evaluations	0 Steps	0.00%
Overall Functional Success Rate	100% Pass Rate	Operationally Ready

As reflected in the summary table, the UniOrg platform achieved a 100% functional success rate with zero failed test steps. This consolidated

outcome demonstrates that every integrated feature, ranging from COR-based registration with OTP verification and OCR-driven document processing with NLP auto-tagging to master calendar coordination and strict System Administrator RBAC enforcement operates with high stability and precision. These outcomes show functional compliance under the tested conditions, with ten test cases and 670 step-level verifications executed in a controlled black-box setting. Wider deployment readiness would still depend on pilot runs and live-environment validation.

4.3.3 Discussion of Testing Results

The empirical findings gathered from the black-box testing phases. The achievement of a 100% functional success rate across all ten test cases and 670 step-level verifications demonstrates that the platform core components spanning the Laravel web workspaces and the mobile-responsive browser interface. The findings offer evidence of functional compliance with the documented requirements under the tested conditions.operated as intended in the tested scenarios.

The testing results confirm that the system successfully bridges the operational gaps identified during preliminary stakeholder interviews with the Office of Student Affairs and Services (OSAS) of LSPU Santa Cruz Campus. Specifically, the successful validation of Test Case 04 and Test Case 05 proves that the integration of Google Cloud Vision and Tesseract OCR for document extraction with Gemini AI and Cloud Natural Language for summarization and auto-tagging performs accurately under real-world testing conditions.

Furthermore, the flawless execution of Supabase realtime updates verifies that the backend maintains low-latency data streams between student browsers and the OSAS oversight dashboards. Security boundaries are equally well-enforced, as shown by the strict validation of Test Case 01, which confirms that Role-Based Access Control configurations remain exclusively isolated to the System Administrator account to prevent unauthorized privilege escalation by standard organization officers.

In adherence to rigorous evaluation standards, the testing process also accounted for iterative refinements. While no formal test cases failed during the final evaluation phase with the university evaluators, preliminary developmental reviews conducted with technical specialists yielded critical insights regarding OCR confidence handling on low-quality scanned documents and realtime reconnection behavior during concurrent payment and election submissions. These initial observations were addressed prior to formal testing by tightening validation prompts with manual verification fallbacks and optimizing realtime reconnection protocols, thereby eliminating potential points of failure and directly contributing to the flawless execution recorded during the final black-box testing phase. Met the functional criteria set for the tested scenarios, which supports further stages toward university-wide deployment.

4.4 System User Evaluation Results

The System User Evaluation Results present the findings obtained from the intended users and stakeholders who evaluated the completed UniOrg system, the

results are presented according to the evaluation criteria and correspond to the instrument and procedures described in Chapter III.

4.4.1 Respondent Profile

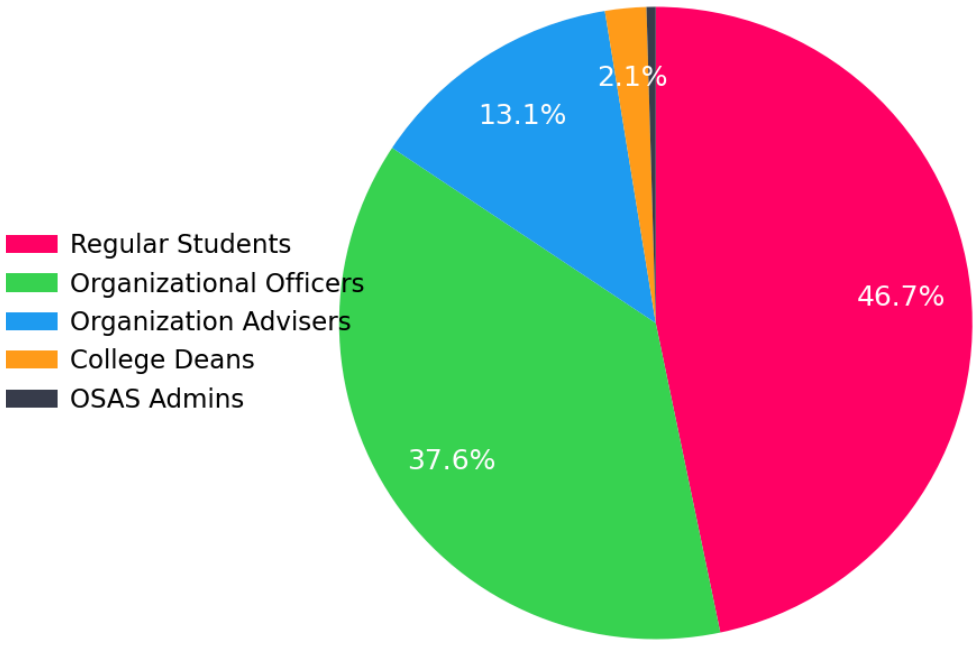


Figure 4.47. Distribution of respondents by role.

Figure 4.47 shows that the majority of the respondents were Regular Students, comprising 200 out of the 428 total respondents (46.7%), followed by Organizational Officers at 161 (37.6%), Organization Advisers at 56 (13.1%), College Deans at 9 (2.1%), and OSAS Admins at 2 (0.5%). This distribution reflects the sampling design established in Chapter III, wherein the OSAS Admins were included through total enumeration given their limited number, while students, officers, advisers, and deans were drawn through stratified sampling using Slovin Formula to represent the population

of LSPU Santa Cruz Campus. The predominance of student respondents is consistent with the objective of evaluating UniOrg primarily from the perspective of the general student body it was built to serve, while still capturing the perceptions of the officers, advisers, deans, and OSAS personnel responsible for its day-to-day operation.

4.4.2 Evaluation Results by Criteria

Table 4.55. Level of Acceptability of the Respondents in Terms of Quality Factors

Indicator	WM	SD	Interpretation
1. The platform's layout allows users to move through its features without confusion.	4.71	0.56	Strongly Agree
2. The platform provides the necessary tools for managing organization activities and information.	4.65	0.59	Strongly Agree
3. The platform's color scheme and icons are visually appealing and professionally designed.	4.64	0.61	Strongly Agree
Overall Weighted Mean	4.66	0.59	Strongly Agree

The Quality Factors results show a very high level of acceptance, with an overall weighted mean of 4.66 and a standard deviation of 0.59, interpreted as Strongly Agree. The three indicators were consistently rated highly, with the clarity of the platform layout for moving through features without confusion receiving the highest weighted mean of 4.71. This rank is numerical only, since all three indicators fall within the same Strongly Agree band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were relatively concentrated

around the mean, suggesting a strong and consistent perception of the system quality and interface.

These results align with prior findings on system quality and user experience. Zardari et al. (2021) confirmed that clear navigation and layout are the strongest drivers of ease of use in e-learning acceptance, while visual appeal remains significant but secondary. Similarly, Mailizar et al. (2021) found that system quality in terms of access and flexibility significantly raises usefulness and attitude. Al-Rahmi et al. (2021) also showed that ease of navigation predicts attitude and intention more than hedonic factors, which explains why layout clarity scored highest at 4.71 and icons lowest at 4.64 yet both Strongly Agree.

Table 4.56. Level of Acceptability of the Respondents in Terms of Perceived Usefulness

Indicator	WM	SD	Interpretation
1. Using the UniOrg platform enables me to complete student organization-related tasks more quickly.	4.66	0.59	Strongly Agree
2. The platform helps me easily search, access, and view organization records and announcements.	4.60	0.60	Strongly Agree
3. The centralized system helps me stay informed and updated about student organization activities and events.	4.68	0.57	Strongly Agree
Overall Weighted Mean	4.64	0.59	Strongly Agree

The Perceived Usefulness results show a very high level of acceptance, with an overall weighted mean of 4.64 and a standard deviation of 0.59, interpreted as Strongly Agree. The three indicators were consistently rated highly, with the centralized system keeping users informed and updated about activities and events receiving the highest weighted mean of 4.68. This rank is numerical only, since all three indicators fall within the same Strongly Agree band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were relatively concentrated around the mean, suggesting a shared recognition of the platform usefulness in daily organization transactions.

The pattern supports earlier work on university portals and usefulness. Alkhateeb and Abdalla (2023) described the portal as the gateway to information and proved that information quality drives usefulness and satisfaction, which matches the highest item on staying informed at 4.68. Alfadda and Mahdi (2021) likewise reported that usefulness for continuous connection and self-efficacy in searching both correlate with attitude and use, so the lowest item on search and access at 4.60 is still very high. Al-Adwan et al. (2023) further confirmed usefulness as the key enabler of intention while enjoyment is secondary.

Table 4.57. Level of Acceptability of the Respondents in Terms of Perceived Ease of Use

Indicator	WM	SD	Interpretation
1. Learning how to use the UniOrg platform is easy for me.	4.60	0.61	Strongly Agree
2. Navigating through the platform to view organization details and schedules is clear and understandable.	4.67	0.54	Strongly Agree
3. Becoming skillful in using the features of the UniOrg platform would be easy for me.	4.60	0.63	Strongly Agree
Overall Weighted Mean	4.62	0.60	Strongly Agree

The Perceived Ease of Use results show a very high level of acceptance, with an overall weighted mean of 4.62 and a standard deviation of 0.60, interpreted as Strongly Agree. The three indicators were consistently rated highly, with the clarity of navigating through organization details and schedules receiving the highest weighted mean of 4.67. This rank is numerical only, since all three indicators fall within the same Strongly Agree band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were relatively concentrated around the mean, suggesting that users across roles found the platform easy to learn and operate.

Prior studies point to the same order where clarity leads and initial learning trails slightly. Lin and Yu (2023) showed that ease for collaborative learning and self-efficacy significantly predict ease of use and attitude, which fits the highest item on clear navigation at 4.67. Alyoussef (2021) found with

432 students that ease of use affects usefulness and task fit, and Alyoussef (2023) added that information and system quality drive ease of use. This explains why learning at 4.60 and becoming skillful at 4.60 are a bit lower, users still need brief familiarization.

Table 4.58. Level of Acceptability of the Respondents in Terms of Attitude Toward Using

Indicator	WM	SD	Interpretation
1. I feel comfortable using a digital platform for student organization activities.	4.64	0.60	Strongly Agree
2. I feel confident when using the UniOrg platform.	4.63	0.59	Strongly Agree
3. I have a positive attitude toward using UniOrg for university-wide student organization engagement.	4.65	0.59	Strongly Agree
Overall Weighted Mean	4.64	0.59	Strongly Agree

The Attitude Toward Using results show a very high level of acceptance, with an overall weighted mean of 4.64 and a standard deviation of 0.59, interpreted as Strongly Agree. The three indicators were consistently rated highly, with the positive attitude toward using UniOrg for university-wide engagement receiving the highest weighted mean of 4.65. This rank is numerical only, since all three indicators fall within the same Strongly Agree band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were relatively concentrated around the mean, suggesting a uniform confidence and comfort in adopting the digital platform.

The attitude scores are consistent with technology acceptance literature. Rosli and Saleh (2023) proved with 303 students that self-efficacy and motivation shape ease of use and usefulness, so confidence at 4.63 can lag a little behind general positive attitude at 4.65. Barz et al. (2024) also found that affinity for technology raises usefulness and ease, and that attitude strongly predicts intention. Zhou et al. (2022) extended TAM for online platforms and showed ease affects usefulness which then drives intention, which supports the university-wide engagement result.

Table 4.59. Level of Acceptability of the Respondents in Terms of Behavioral Intention to Use

Indicator	WM	SD	Interpretation
1. I intend to use the UniOrg platform for student organization transactions and activities in the future.	4.64	0.58	Strongly Agree
2. I would use this platform whenever I need to access organization services or information.	4.66	0.54	Strongly Agree
3. I would recommend the UniOrg platform to other students and organizations in the university.	4.61	0.61	Strongly Agree
Overall Weighted Mean	4.64	0.58	Strongly Agree

The Behavioral Intention to Use results show a very high level of acceptance, with an overall weighted mean of 4.64 and a standard deviation of 0.58, interpreted as Strongly Agree. The three indicators were consistently rated highly, with the willingness to use the platform whenever organization services are needed receiving the highest weighted mean of 4.66. This rank is numerical only, since all three indicators fall within the same Strongly Agree

band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were elatively concentrated around the mean, suggesting a firm intention to keep using and recommending the system.

The intention pattern mirrors continuance literature. Kala et al. (2024) showed satisfaction significantly predicts continuance intention, which supports the highest item on using whenever services are needed at 4.66. Xin et al. (2022) found attitude and expectancy predict intention to adopt e-government services. Srivastava and Singh (2023) explained the lowest item, since recommendation at 4.61 follows from continuance intention rather than preceding it, so direct use is naturally a bit higher than recommending to others.

Table 4.60. Level of Acceptability of the Respondents in Terms of Experience

Indicator	WM	SD	Interpretation
1. My previous experience with online platforms helps me use UniOrg effectively.	4.68	0.60	Strongly Agree
2. I find it easy to adapt to the UniOrg system based on my general familiarity with digital tools.	4.61	0.60	Strongly Agree
3. Transitioning from traditional manual processes to UniOrg's digital platform improves my overall user experience.	4.69	0.56	Strongly Agree
Overall Weighted Mean	4.66	0.59	Strongly Agree

The Experience results show a very high level of acceptance, with an overall weighted mean of 4.66 and a standard deviation of 0.59, interpreted as

Strongly Agree. The three indicators were consistently rated highly, with the improvement from transitioning manual processes to the digital platform receiving the highest weighted mean of 4.69. This rank is numerical only, since all three indicators fall within the same Strongly Agree band, so the gap reflects ordering rather than a substantive difference. The low standard deviation indicates that respondents ratings were relatively concentrated around the mean, suggesting that prior familiarity with online platforms translated evenly into effective UniOrg use.

Experience effects reported before match this table. Akritidi et al. (2022) found with 858 users that familiarity with e-services significantly determines intention, which explains why adapting based on familiarity at 4.61 is lowest but still high. Alketbi and Masuri (2026) showed with 419 experienced users that system quality raises satisfaction and that usefulness and satisfaction raise continuance, which fits the highest item on manual to digital transition at 4.69. Shahzad et al. (2022) likewise confirmed attitude strongly predicts intention with a large coefficient, so improved experience carries over to use.

Table 4.61. Summary of Technology Acceptance Ratings Across Criteria

Criterion	WM	SD	Interpretation	Rank
Quality Factors	4.66	0.59	Strongly Agree	1
Experience	4.66	0.59	Strongly Agree	2

Table 4.61—continued.

Criterion	WM	SD	Interpretation	Rank
Perceived Usefulness	4.64	0.59	Strongly Agree	3
Attitude Toward Using	4.64	0.59	Strongly Agree	4
Behavioral Intention to Use	4.64	0.58	Strongly Agree	5
Perceived Ease of Use	4.62	0.60	Strongly Agree	6

The table 4.61 presents the overall summary of technology acceptance ratings across six key criteria for the UniOrg platform: Quality Factors, Experience, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, and Perceived Ease of Use.

Overall, the findings demonstrate an exceptionally high level of acceptance among respondents, with all criteria falling consistently within the Strongly Agree band. The weighted means range narrowly from 4.62 to 4.66, and the low standard deviations (ranging from 0.58 to 0.60) indicate a high degree of consensus and minimal variability in the respondents' answers.

The ranking agrees with integrated TAM and IS success work. Prasetyo et al. (2021) found with 500 students that system quality and interface drive ease of use toward intention and actual use, which supports Quality at top with 4.66. Castiblanco Jimenez et al. (2021) reported experience and self-efficacy as strongest predictors of ease and content quality as main driver of usefulness. Alkhawaja et al. (2022) proved system quality

strongly predicts usefulness and intention, so Ease of Use at 4.62 being lowest yet Strongly Agree is expected.

4.4.3 Statistical Results

In accordance with the statistical treatment specified in Chapter III, descriptive statistics, specifically the weighted mean and standard deviation were applied to quantify the ratings provided by the 428 respondents. This statistical treatment measures the central tendency and dispersion of the evaluations across the 18 indicators grouped into six acceptance criteria, determining the overall acceptability and performance of the UniOrg platform. Tables 4.55 to 4.60 present the statistical results, detailing the weighted means, standard deviations, and corresponding verbal interpretations for each measured criterion

Table 4.62. Overall Grand Mean of Technology Acceptance Criteria

Evaluation Criteria	Weighted Mean	Standard Deviation	Verbal Interpretation
1. Quality Factors	4.66	0.59	Strongly Agree
2. Perceived Usefulness	4.64	0.59	Strongly Agree
3. Perceived Ease of Use	4.62	0.60	Strongly Agree
4. Attitude Toward Using	4.64	0.59	Strongly Agree
5. Behavioral Intention to Use	4.64	0.58	Strongly Agree
6. Experience	4.66	0.59	Strongly Agree
Overall Grand Mean	4.64	0.59	Strongly Agree

As shown in the statistical summary, the computed weighted means across all evaluated criteria reflect exceptionally high ratings from the respondents. The low standard deviation values indicate a general agreement among the evaluators, comprising OSAS personnel, college deans, organization advisers, officers, and regular students, regarding the effectiveness and usability of the system.ean confirms that UniOrg successfully meets its technical and operational benchmarks. The statistical evidence proves that the integration of cross-platform accessibility (Laravel web and mobile browsers), automated OCR and NLP document processing, master calendar coordination with Supabase realtime, and secure administrative controls provides a reliable, efficient, and user-friendly solution that completely addresses the organization management requirements of LSPU Santa Cruz Campus.

The grand mean of 4.64 reflects the high acceptance seen in recent TAM studies. Li and Zhu (2022) showed system and information satisfaction drive ease and usefulness in both online and blended groups without significant difference, which matches the low SD and uniform Strongly Agree. Belo et al. (2024) found with 503 students that system quality leads ease, usefulness outweighs ease on attitude, and attitude strongly drives intention. Al-Nuaimi and Al-Emran (2021) systematically reviewed LMS acceptance and confirmed the same hierarchy, so the overall result is well supported.

4.4.4 Discussion of User Evaluation

The user evaluation results, culminating in an overall grand mean of 4.64 with a comparatively low standard deviation of 0.59, demonstrate an highly positive assessment of the UniOrg platform by its intended users. In alignment with the research objective to measure system acceptability through the Technology Acceptance Model (TAM), these figures indicate a general agreement among OSAS personnel, college deans, organization advisers, officers, and regular students. The limited variance in the respondents ratings confirms that the operational benefits are consistently recognized across all administrative and student-facing roles.

Furthermore, the high ratings in Perceived Ease of Use and Experience validate the project user-interface design strategy. Organization management applications demand intuitive navigation, as users range from first-year students to senior administrators with varied technical backgrounds. The evaluation proves that the workspace architecture featuring role-specific Laravel dashboards for officers with a mobile-responsive interface for students minimizes the learning curve. Users can effortlessly submit accreditation documents, register for events, and settle payments without requiring extensive technical training.

Finally, the exceptional scores in Attitude Toward Using and Behavioral Intention to Use represent the most critical indicator of the project ultimate success. These results demonstrate that the students and university personnel do not merely appreciate the software as an academic prototype; they are actively willing and prepared to adopt UniOrg into their daily

organizational routines. Ultimately, the evaluation confirms that the system has achieved both technological stability and deep end-user trust, proving its complete readiness for deployment across LSPU Santa Cruz Campus.

4.5 Discussion and Interpretation of Findings

The development, testing, and evaluation of the UniOrg system yielded highly favorable results that directly address the core research objectives and the identified operational problems within the Office of Student Affairs and Services. The primary issue motivating this project was the reliance on fragmented, manual, and paper-bound workflows for organization accreditation and document management. The empirical data gathered from both the technical black-box testing and the user evaluation demonstrate that UniOrg successfully mitigates these bottlenecks by introducing an integrated, AI-powered digital infrastructure.

Addressing the first objective of system development, the findings show that the implemented architecture utilizing a Laravel monolith with Supabase and Cloudflare R2 successfully bridges the technological gap identified in Chapter II. Related literature and reviews of existing local systems frequently highlight a lack of cohesive university organization platforms capable of handling document intelligence and realtime coordination simultaneously. By integrating Gemini 2.5 Flash for summarization with Google Cloud Vision OCR and Cloud Natural Language auto-tagging, alongside Supabase realtime for live updates, UniOrg provides an enterprise-grade solution that surpasses legacy spreadsheet and messenger-based coordination. Furthermore, isolating Role-Based Access Control (RBAC) configurations exclusively to the System Administrator addresses the critical security

gaps often found in decentralized student applications, ensuring that standard officers cannot compromise system integrity.

In relation to the second objective regarding system functionality, the black-box testing results revealed a 100% success rate across 670 step-level verifications. This result is critically important; in the context of university-wide deployment, system reliability is non-negotiable. The flawless execution of modules such as accreditation tracking, OCR and NLP document processing, event approvals, centralized payments, and secure e-voting proves that the logic performs exactly as intended under simulated real-world conditions. The results suggest that the logic performs as intended under the simulated conditions used in testing. These findings align with recent studies in educational information systems, which argue that minimizing administrative latency and automating document categorization significantly reduces the workload on student affairs offices during peak accreditation periods.

Finally, addressing the third objective regarding user acceptability, the Technology Acceptance Model (TAM) evaluation yielded a remarkable overall grand mean of 4.64 with a standard deviation of 0.59. This statistical outcome demonstrates that the intended users, including OSAS personnel, college deans, organization advisers, officers, and regular students, find the system exceptionally useful and easy to navigate. The results strongly support existing TAM literature, which consistently reports that high Perceived Ease of Use and Perceived Usefulness directly drive Behavioral Intention to Use in academic management tools. The low standard deviation implies a unified consensus among stakeholders, confirming that the web plus mobile browser approach successfully lowers the technical barrier to entry for ordinary students while providing robust oversight tools for OSAS officers

shows favorable potential for university-wide deployment, and it indicates that the objective of transitioning LSPU Santa Cruz Campus from a manual accreditation model to an intelligent, automated, and student-inclusive digital platform was successfully achieved.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary, conclusions, and recommendations of the study based on the results gathered from the testing and evaluation of the developed UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging. It includes the analysis and interpretation of the data obtained from system testing and user acceptance evaluation conducted among OSAS administrators, Deans, organization advisers and officers, and students of LSPU Santa Cruz Campus.

5.1 Summary

This study aimed to design, develop, test, and evaluate a system for university-wide organization accreditation, document intelligence, and student-OSAS coordination through OCR processing, NLP auto-tagging, and realtime synchronization. The major findings of the study, aligned with the specific research objectives, are summarized as follows:

- System Development (Objective 1): The study successfully designed and developed UniOrg as a web-based system featuring a Laravel monolith with Supabase and Cloudflare R2, serving role-specific workspaces for OSAS administrators, Deans, organization officers and advisers, and students. The developed core modules include COR-based registration with OTP verification and RBAC, multi-organization workspaces, accreditation and renewal tracking, OCR text extraction, NLP summarization and auto-tagging, event proposals with Dean and OSAS approvals plus a master calendar, centralized GCash payments with receipt

verification, secure e-voting with officer transition, EduShare and e-library resources, and a feedback hub with chatbot.

● System Testing and Functionality (Objective 2): The developed system underwent rigorous Black-Box testing across ten (10) distinct test cases covering all major features. The testing resulted in a 100% success rate, with all 670 recorded step-level verifications marked as passed and zero (0) failed results. Testing was run under controlled black-box conditions with ten IT testers working from an end-user view.

● User Acceptance Evaluation (Objective 3): The acceptance was evaluated using the Technology Acceptance Model (TAM) by 428 respondents, predominantly comprising Regular Students at 200 (46.7%), along with Organizational Officers, Organization Advisers, College Deans, and OSAS Admins. The evaluation yielded an overall grand mean of 4.64, interpreted as Strongly Agree, across all six constructs: Quality Factors (4.66), Experience (4.66), Perceived Usefulness (4.64), Attitude Toward Using (4.64), Behavioral Intention to Use (4.64), and Perceived Ease of Use (4.62).

5.2 Conclusions

Based on the summary of findings and the specific research objectives of the study, the following conclusions are drawn:

● Resolution of Operational Gaps: The development of UniOrg successfully transitions the Office of Student Affairs and Services of LSPU Santa Cruz Campus from a manual, paper-bound workflow to a structured digital infrastructure. The integration of COR-based registration, OCR extraction with NLP auto-tagging, and

master calendar coordination directly resolves previously identified operational gaps in accreditation processing, document management, and event oversight.

● Functional Reliability and Readiness: The system is functionally robust and operationally ready for university-wide deployment. Shows functional compliance across the tested scenarios. The flawless 100% pass rate across 670 step-level Black-Box verifications proves thatsuggests that the application logic, AI processing, realtime database synchronization, and role-based access controls operate exactly as intended without technical failure.operated as intended in the tested scenarios.

5.3 Recommendations

In light of the findings and conclusions of this study, the researchers propose the following recommendations:

● System Improvement: The university may consider enhancing the system by adding payment anomaly detection to flag suspicious receipt submissions ahead of enrollment peaks. Additionally, refining OCR classification for low-quality scans could further improve document processing accuracy. Developers should also look into refining user interface elements related to Perceived Ease of Use, as it was the lowest-scoring construct at 4.62 despite its positive rating.

● Platform Expansion: To maximize reach and ensure inclusivity, future development should explore building a dedicated mobile application, since the system is currently accessed through mobile browsers. This would improve accessibility and offline resilience for students with limited connectivity on campus.

- Integration with Other University Systems: LSPU should explore integrating UniOrg database and reporting outputs with the registrar enrollment records and other campus information systems. This would facilitate seamless verification of student credentials and coordinated scheduling across academic units.

- Future Research: Future researchers are encouraged to evaluate the system using an extended TAM or UTAUT model that incorporates additional constructs such as trust and facilitating conditions. Furthermore, subsequent studies should investigate the long-term operational impact of automated accreditation and AI-assisted document processing on the actual cycle time of organization recognition at the campus.

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APPENDICES

(A. TECHNICAL BACKGROUND)

Transcribed Interview



Figure 27. **TRANSCRIBED INTERVIEW 1.**

The researchers conducted their interview to Dr. Christian P. San Luis, LPT, PhD, OSAS Chairperson. The interview happened at the Office of Student Affairs and Services (OSAS) in Laguna State Polytechnic University – Sta. Cruz Campus last March 16, 2026, Monday. The researchers are Justine Mark Gahi (Researcher 1) and Symon Kiel Beato (Researcher 2) of LSPU-SCC.

Researchers	*...para ma-recap na din po...*
Client	*...most probably 'yan ay monitoring do'n sa mga activities ng mga orgs.*
Researchers	*Nabanggit din po namin 'yung accreditation po.*
Client	*Okay.*

Researchers *Paano po ba nagwu-work 'yung accreditation?*

Client *Accreditation kasi tinatanong nila kung ilan 'yung mga student organizations, ano 'yung mga activities. So pinakamaganda kung halimbawa 'yung isang org merong account diyan, tapos lahat ng activities nila pwede nilang i-up. So parang database siya, parang kumbaga sa learning management system, lahat ng mga activities nila: letter request, mga sabihin natin 'yung mga documents ng organization, nandiyan. 'No, tulad ng mga ganyan. So 'yan ay nandiyan 'yung registration ng mga students, tapos 'yung bylaws nila, 'yung POA nila—'yung plan of activities—so kung nandiyan, tapos 'pagka may mga activities sila, 'yung mga highlight, 'yung after no'ng activity siyempre meron diyang mga evaluation, tapos mga narrative reports, tapos 'yun, 'yun 'yung mga basically pictures. So ayon.*

Researchers *'Yun po, 'yun po 'yung goal nitong Capstone po namin.*

Client *Okay.*

Researchers *Kumbaga bawat student org po ng bawat department 'yan mag...*

Client *Pwede silang mag-create no'ng sarili nila.*

Researchers *Opo. Kumbaga parang mag-a-ask pa po ng permission sa OSAS po siguro kapag gagawa pa po ng bago, 'yung bagong simula po.*

Client *Okay. Tapos ang isa pa, halimbawa 'yung mga calendar of activities, kung halimbawa may sampung organization, may mga activities sila, makikita na do'n sa calendar nila lahat.*

Researchers *Ayon, opo, opo. Possible po 'yun.*

Client *'Yun, isa. Maganda 'yun. Pwede-pwede 'yun. Ano pa?*

Researchers *Halos 'yun po...*

Client *Pero 'yan ipapagamit niyo ha? Baka naman mamaya pagka natapos ninyo, kasi sawang-sawa na ako do'n sa magsasabi dito na gagawa po kami sa inyo ng system, after matapos 'yung thesis nila, wala na. Sayang 'yung pinaghirapan ninyo.*

Researchers *Opo nga po. Ang mahirap lang po kasi kapag parang sasakupin po lahat namin no'ng ginagawa po ng OSAS kaya ang focus lang po nito sa student organization.*

Client *Ah hindi, oo. Student organization, okay, goods! Okay. So 'yun, most probably dapat meron silang ano, may sariling account 'yung mga orgs. Makapag-create sila ng sariling account for approval ng OSAS. So siyempre may mga accreditation kasi 'yan eh. Tapos do'n nila ilalagay 'yung kanilang POA, nando'n 'yung ano, nando'n 'yung ilan pa mga documents nila, mga activities, tapos 'yung mga request. 'Pagka may certain activity sila, halimbawa for this month meron silang naging activity, nandiyan na 'yung evaluation. Magsisimula sa request letter hanggang sa approved letter, tapos evaluation, tapos 'yung mga pictures, narrative reports, financial reports. 'Yun.*

Researchers *'Yung halos requirements po ba for accreditation, 'yung nasa citizen charter natin?*

Client *Yeah. Ang 'yung sa Citizen's Charter natin, 'yan ay accreditation lang. Ibig sabihin kung gaano katagal dumadaan 'yung papeles 'pagka tsine-check 'no, and 'yung accreditation process 'no. After no'n, wala na, tapos na 'yon. Pero 'pagka diyan, pwede naman... hindi, ang ang tingin ko kasi diyan baka 'yung parang may individual account ang mga organization.*

Researchers *Opo, opo, gano'n po.*

Client *So pupwede din na halimbawa for this year, kasi ang ang expiration ng isang organization ay one year. So halimbawa for 2025 to 2026 academic year, ilan 'yung registered na ilan 'yung registered na organization. Tapos makikita, "Uy may 38. Ano 'yung mga activities?" Tapos do'n sa mga activities nila, halimbawa nagkaroon ng search bar, sabihin natin, outreach program. Lahat ng may mga activities across all organization, maso-sort out niya. So halimbawa 'yung activities nila bago nila i-up, meron tayong categories 'no. Halimbawa, Days, okay? College Days, organization, o kung ano man 'no. So pupwedeng mag-suggest ng mga gano'n. Kasi 'di ba ang pinaka-importante diyan, not merely on parang storage lang. Siyempre ang gusto natin diyan, pagka kinlick natin, "Uy, ilan ba 'yung organization ninyo na nakapag-conduct ng outreach program na hinahanap sa accreditation?" "O, heto po." Ngayon pagka kinlick 'yon, 'di lalabas, "O heto, sort, mga lima ang lumabas." Pagka kinlick 'yon, 'yung program niya, "O heto 'yung narrative, heto 'yung simula do'n sa request letter, approved letter, narrative reports." Ngayon kung may mga resolution kasi hinahanap din 'yun, resolutions, tapos 'yung 'yung outcome, 'tas financial.*

Researchers *Parang madami din po talagang process po siguro.*

Client *Oo, oo. Pero madali lang naman kasi ang pinaka-ano naman diyan ay kasi isa lang naman 'yung format na hahanapin. Kasi mag-a-upload lang siya eh. So ang pinaka... kasi kung uploading lang napakasimple lang naman. Maglagay ka ng database, upload ka, parang Google Drive lang siya.*

Researchers *Opo.*

Client *Pero pagka halimbawa 'yun tulad no'n, maghahanap ka ng mga activities o kaya sa ganitong calendar magpo-post ako o che-check-in ko 'yung calendar, ilan 'yung mga may activities ngayon.*

Researchers *Kapag po ba kukuha kami kunwari po sample data, pwede po bang makahingi no'n?*

Client *Pwede naman ang sample...*

Researchers *Siguro kahit po blank paper lang po.*

Client *Oo. Pwede naman, pwede naman. We can provide naman do'n sa do'n sa mga ano.*

Researchers *Sige po.*

Client *Kasi isa lang naman eh, isa lang namang data 'yun eh. Ulitin niyo lang, paltan-paltan niyo lang ng mga details 'yun 'no. So isa 'yun, siguro nakita ko do'n 'yung una 'yung calendar, pangalawa 'yung pagka may mga hahanapin na mga ano mga activities, pangalawa 'yung ano nga din 'yung for accreditation 'no na pagka sinort niya o heto, 'tas lahat makikita na 'yung mga documents related to that. Tapos maso-sort mo siya 'no based on academic year kasi tiyak 'yan 'yung iba halimbawa 'yung organization present lang siya for academic year 2025-2026, next year hindi, o next academic year wala sila. So 'yun.*

Researchers *Parang lahat po talaga matatatak po.*

Client *Oo, oo. Ibig sabihin pagka... kasi ilan 'yan, tulad ngayon we have 39 organizations.*

Researchers *39?*

Client *39 organizations. Magkakaiba 'yan. Dalawa lang naman category niyan: acad... academic at non-acad.*

Researchers *Pero halos tulad-tulad din po sila ng process?*

Client *Parehas lang din naman sila. Ang 'yung sa... ay hindi pala academic, sorry. College-based tsaka non-college-based. Kasi 'yung iba halimbawa tulad no'ng sa inyo, 'yung E2, 'yung isa pa, tapos may mother org 'no. 'Yung E2 ay sub-org 'yan ng college ng ano. Pero 'yung Red Cross, wala naman siyang ano kinabibilangan na college kasi multi-college sila, 'yung mga members nila.*

Researchers *Ayun, ask ko lang din po do'n sa about sa mother org tsaka po do'n sa mga sub-org. Alam niyo din po ba 'yung process no'ng hatian kapag sa budget po?*

Client *Yes. 150 sa mother org, 100 sa ano... sa sub-org. Baliktad... Nicole, baliktad ba ako? 100 ang mother org 'no? 150 pala. 100 ang ano kasi marami... Mas mataas ang sub-org kaysa sa mother org, kukunin lang niya 'yun. Basta total niyan 250. And dahil sa mother org, lahat ng members ng sub-org, member ng mother org. So 'yun ang ano do'n.*

Researchers *'Yun lang po. Kasama din po 'yun sa mga ilalagay namin.*

Client *Yeah.*

Researchers *May note na po.*

Client *So tingnan ninyo, tapos siguro 'yung highlights ng mga organizations 'no.*

Researchers *Opo, kasi parang do'n din makikita na po... May nasabi po kasi sa amin na parang binibigyan niyo po ng...*

Client *Recognition?*

Researchers *Opo, opo.*

Client *Yes. 'Pagka 'yun nga depende do'n sa mga activities na na-conduct nila.*

Researchers *Parang 'di na lang po muna i-close.*

Client *Sige, okay.*

Researchers *Pero pwede naman po sa ata... bumalik po kami dito?*

Client *Oo naman, wala namang problema. Ay kasi siyempre pagkaano 'di pag-isipan ninyo kung paano ninyo aanahan 'no 'yung mga*

nabanggit ko kung may mga kulang naman eh. Halimbawa mag-sample kayo, sa inyo na lang, ano kayo sa E2 ba o sa ano?

Researchers *Ang ia-ano na po namin is 'yung sa SOCS po.*

Client *Ay sa SOCS? O 'yun, kunin niyo sa SOCS.*

Researchers *'Yun nga po, 'yung the former pinapakita po naming system sa inyo. Ito po 'yun.*

Client *Ah, 'yung sa SOCS lang 'yan?*

Researchers *SOCS lang po 'yan mismo. Pero 'di pa po siya integrated na pa-OSAS. 'Yun lang po 'yung gusto namin.*

Researchers *Bale papalawakin lang po 'yung sakop.*

Client *Okay, sige. Ibig sabihin 'yung SOCS tapos kunwari 'yung iba naman ay halimbawa CAS, o kaya sabihin natin 'yung iba ay SUPREME, mga gano'n. Pwede naman 'yun ano. O gusto niyo lang isa, SOCS lang?*

Researchers *Hindi po, sa lahat na po. Kasi kapag ganto na po 'yung sa OSAS ang gagamit, kailangan po is iba't ibang...*

Client *Oo, oo.*

Researchers *Kaya ayun po, 'yun din po 'yung gusto siguro po ng mga panel po.*

Client *'Yung na hindi lang siya limited sa SOCS 'no.*

Researchers *Opo, opo.*

Client *Okay. Anong gamit ninyo na platform? Java o...*

Researchers *Um, PHP pa din po.*

Researchers *PHP.*

Client *PHP.*

Researchers *Pero siguro po mabago din, madagdagan din po ng mga gagamitin naming platform kasi malawak nga po 'yung...*

Client *Oo, oo. Eh 'yung 'yung storage na lang eh.*

Researchers *'Yun po, opo.*

Researchers *Malaki nga po.*

Client *Oo, kasi siyempre pagka nag-upload ng mga pictures 'yan hindi pupwede na 'yung 15GB lang na ano kasi mga minsan ang dami nila, may video pa, may...*

Researchers *Opo.*

Client *Pero limit niyo na lang 'yung ano...*

Researchers *Opo, maglalagay po kami.*

Client *...may mga cap lang. 'Yun.*

Researchers *'Yun lang po.*

Client *Wala naman akong kailangang ano sa inyo 'no?*

Researchers *Pero pumayag naman po kayo na ikuha din namin kayo as client po?*

Client *Oo naman.*

Researchers *Ay sige po. Pwede po bang magpa-picture po?*

Client *Picture? Oo naman.*

Researchers	*Sige po.*
Client	*1, 2, 3... Okay. Sige, dito din.*
Researchers	*1, 2, 3... Isa pa po. 1, 2, 3...*
Client	*Okay, salamat! Salamat.*
Researchers	*Hintayin ko na lang...*
Client	*Sa pagbalik niyo na lang... Justin tsaka...*
Researchers	*Symon po.*
Client	*Symon, okay. Pagka sa susunod eh 'di balikan ninyo kung ano man 'yung pagka may mga questions pa kayo.*
Researchers	*Opo. Thank you po!*
Client	*Sige, salamat, salamat.*
Researchers	*Thank you po sa oras.*
Client	*Pagbutihin ninyo ha. Opo, opo. Sige, sige.*

TRANSCRIBED INTERVIEW 2.

The researchers conducted their interview to Ms. Nicole D. Bartolome, OSAS Secretary. The interview happened at the Office of Student Affairs and Services (OSAS) in Laguna State Polytechnic University – Sta. Cruz Campus last March 31, 2026, Tuesday. The researchers are Justine Mark Gahi (Researcher 1) and Symon Kiel Beato (Researcher 2) of LSPU-SCC.

Client *Dito sa 1, basahin ko na lang ha. "Who are the user... users in the OSAS admin part?" So, dalawa lang naman kami dito: staff tapos 'yung chairperson natin which is si Sir Ian. "Itong accounts of each departments have one year expiration?" Specifically, ano 'to?*

Researchers *'Di ba po 'pag 'yung mga... 'pag sa mga org po may... 'pag nagawa po ng system magkakaroon po sila ng mga accounts. Tapos sa accreditation po meron po silang one year na expiration po.*

Client *Ah, so 'yung accreditation naman nila... per ano kasi siya eh, per semester siya noon, 'di ba?*

Researchers *Ah, per semester.*

Client *Ay hindi, isang... isang taon siya. So tama, isang taon. Not unless may madagdag na organization.*

Researchers *Ah may ano po, parang nagre-request po, nagbubuo po ng new org?*

Client *Mhm, may gano'n. So dito sa atin recently may dalawa na nagpa—nagpapa-accredit. So nadagdag siya do'n sa bilang natin na 38 na na-accredit natin no'ng last time no'ng first sem. 'Tas merong nag-follow up na madaragdag sila na dalawa. So possible 'yun, may mga madadagdag pa.*

Researchers *Paano po kaya 'yun? Parang from scratch po 'yung org nila po?*

Client *Mhm, oo. Mag-i-start sila sa una.*

Researchers *Paano po 'yun? 'Yung 'pag magpapasa po ng mga reports?*

Client *Oo, meron naman kaming ano, set ng documents na kailangan nilang i-comply na dapat nilang maipasa for them to be accredited. Mhm. So 'yung sa accreditation, "Ano-ano lahat ng requirements na kailangan ng OSAS per department?" So 'yun nga, meron tayong anim na forms na ginagamit for accreditation. So anim na forms 'yun. 'Yun 'yung kailangan nilang kumpletuhin tapos maipasa dito sa amin. So right after na maibigay nila sa amin 'yun, um, 'yun na 'yung magiging indication natin... ayon 'yung example. 'Yun 'yung mga forms na kailangan nilang i-comply sa amin, dapat kumpleto 'yun. Na andu'n 'yung certification nila ng bawat officers, list of members, tapos plan of activities, accreditation form, renewal form.*

Client *So 'yun 'yung mga kailangan nilang i-comply. So kapagka okay na 'yan, um pipirmahan 'yan ng ating Student Organization Units which is si Sir Lava and then kay Sir Ian. So after no'n, magka-conduct na ng accreditation. So sino 'yung mga nakapag-comply within do'n sa binigay na time ng completion no'ng accreditation form, 'yun lang 'yung mapapasama sa bilang ng accreditation.*

Researchers *'Yung pipirma po, dito din po ba sa office ng OSAS 'yung isang pipirma po o sa ibang office na po 'yun?*

Client *Um, minsan napunta dito 'yun si Sir Lava. Minsan naman pinapadala na lang do'n sa CBAA office, faculty, 'yung pipirmahan diyan. So mas madalas siyang naando'n compare na naandito siya.*

Researchers *Okay. May ano po ba dito, parang kapag certain access lang po no'ng bawat staff? Kunwari kayo po.*

Client *Paanong certain access?*

Researchers *Kunwari po ah ang hina-handle niyo lang pong task ay 'yung mga sa documents lang po...*

Client *Um, so dito kasi since isa nga lang 'yung staff, um...*

Researchers *Lahat na? Lahat na po pala.*

Client *...may access ako sa lahat no'n.*

Researchers *Pati po 'yung chairperson, parang same lang po kayo?*

Client *Oo, same lang kami.*

Researchers *Ah, same lang pala. Okay.*

Researchers *Pati po 'yung pag-post ng announcements?*

Client *Oo, pagpo-post ng announcement do'n sa page or sa kahit sa GC ng mga student leaders. Pwedeng si Sir, pwede ako.*

Researchers *Kapag po ano ah nagre-request po ba muna 'yung mga bawat 'yung bawat student org po kapag may activity po silang gagawin?*

Client *Oo, 'yun. 'Yun ang pinakaunang ginagawa nila. Magpapasa sila ng request letter sa amin upon approval pa ng OSAS hanggang kay CD, then sunod sa VPAA hanggang kay President. So upon approval pa din 'yun.*

Researchers *May mga ano po 'yun? Ano po 'yung mga ano? May ibang ibang papel na po 'yung ano no'n inaasikaso po?*

Client *Depende do'n sa activity na ika-conduct nila. So kung inside dito sa atin, ang signatories na lang 'yan ay hanggang kay CD. So iba lang kapagka 'yung activity na gagawin sa labas. So kapagka sa labas, may compliance na sila na dapat ipasa.*

Researchers *Ah, katulad katulad po ng tour?*

Client *Oo, halimbawa tour, tree planting, or outreach program, or any activity na gagawin nila sa labas o may mga attendance sila na seminar. Iba ang letter and compliance no'n compare sa dito niyo lang gagawin 'yung activity.*

Researchers *'Di dapat po pala... may account din sa... kahit dito na lang. Dito na lang.*

Researchers *May pagkakasunod-sunod po ba 'yung mga ano?*

Client *Signatories? 'Yung mga pipirma? Oo, meron 'yung pagkakasunod-sunod. Sa una, manggagaling sa kanila 'yung letter. So of course, requestor 'yung dean. Sunod no'n, after mapirmahan na ng dean nila, aakyat na 'yun dito sa amin. So upon approval pa din namin 'yun. So kung hindi kung hindi ma-approve sa amin, uulit-ulit sila no'n. After sa amin, 'pag na-approve, diretso na 'yun sa Records Office. For signature naman no'ng Campus Director. After no'n, kay Vice President for Academic Affairs, kay VPAA, bago pa siya kay President. So hindi pwedeng mauna si CD bago kami. So kami ang mauuna, CD, VP, then Pres.*

Researchers *'Di ano naman po pala talaga 'yun, dapat 'yung pirma po talaga hindi naman po pwedeng e-sig 'yun ngayon 'no? 'Di okay lang pala, okay lang.*

Researchers *Next po, ano po 'yung ano mga non-college org po?*

Client *Non-college org meron apat tayo. Hindi lang ako sure kung lima. So meron tayo diyang Red Cross Youth... ngayon 'yung isang humabol na magpapa-accredit is 'yung Campus Ministry, Theater Arts Guild, Supreme Student Council... Kapilyo.*

Researchers *Kapilyo?*

Client *Mhm. Non-acad siya.*

Researchers *May list po ba kayo no'n? Lahat po no'ng mga non-college based tsaka college based?*

Client *Oo, meron naman kami.*

Researchers *Pati po 'yung mga sub-orgs, pwede po kaya makahingi?*

Client *Oo, makahingi. Bigyan ko kayo ng copy later.*

Researchers *'Di ba po may mga non-college based po, kunyari po Red Cross, ano po 'yun, 'yung mga members lang po nila 'yung makaka-access po do'n or pwede pong makita ng lahat ng students?*

Client *Um, 'yung members lang nila.*

Researchers *Ah, 'yung members lang po mismo?*

Client *Kasi masyadong malawak kapagka non-college based kasi halo-halo siya. Halimbawa Red Cross kayo, or sabihin na nating Supreme, 'yung mga officers na naando'n manggagaling siya sa iba't ibang department. So may manggagaling sa CONA, representative ng CONA, CAS, CBAA, CTE, CCS. So iba-iba sila, medyo malawak siya macompare do'n sa college-based na kapagka ang program mo ay halimbawa CCS lang, kayong students lang na CCS ang naando'n.*

Researchers *Pero ano po 'di ba, kapag nag-a-announce po sila, visible po 'yun lahat ng students?*

Client *Oo. Visible 'yun sa all. Halimbawa 'yung ano nila...*

Researchers *'Pag may ipo-post daw pong announcement?*

Client *Ng announcement, oo, visible siya sa lahat. May kanya-kanya lang din silang page ang mga non-college based. Oo, tapos ang makakakita no'n of course 'yung lahat, para sa lahat po.*

Researchers *'Yung mga ano po ba, 'yung mga sub-org po sa bawat department, nagre-request po ba muna sila do'n sa mother org bago po dito sa OSAS or diretso na po dito sa OSAS?*

Client *Diretso na 'yun dito.*

Researchers *Ah, diretso na?*

Client *Diretso na. Halimbawa sa CBAA, ilang program meron CBAA dito? Ang mother org nila 'yun 'yung CBAA. Ang sub-org do'n um 'yung JFINEX, JPIA, JMAP, DEPASO, YES-O... Madami, kasi pito sila. Pitong program tapos isang mother org tapos anim na sub-org. So 'yung letter na ginagawa nila, diretso na 'yun sa amin. Hindi na siya upon approval ng mother org.*

Researchers *Pero 'yung process po daw no'ng ano non-college based ay same lang po sa mga college based?*

Client *Oo, same lang siya. Lahat ng letter, college-based, non-college based, same process lang din siya. Kung merong activity sa labas, gano'n din, may compliance din silang need i-comply. Pero do'n sa paggamit ng system, 'pag non-college based, so hindi na siya requestor ng dean kasi wala naman silang dean. Wala silang dean so sila advisor lang. Advisor lang nila. Or 'yung mismong president din ng non-college based 'yung magre-request ng letter upon approval ng advisor, Sir, CD, VPAA, and the Pres. 'Yun lang 'yung kaibahan ng non-college sa college-based.*

Researchers *Pero iisa lang naman 'yung format po no'n talaga ng lahat?*

Client *Oo.*

Researchers *Kailangan po kasi namin no'ng mga type ng documents na napasok po sa inyo lahat.*

Client *Type ng documents? Sige. Tapos 'yung kasama na do'n 'yung attachment. So bibigyan ko kayo ng sample ng request letter ng college-based sa non-college based. For conduct ng activity lang naman, ano?*

Researchers *Kahit blank lang po 'yun, kahit blank.*

Client *Oo, bibigyan ko kayo ng kopya para ma-determine niyo 'yung difference. Tapos 'yung isa, 'yung list ng organizations, non-college tsaka college. Sige.*

Researchers *Tingin niyo po ba mapapabilis 'yung buhay niyo 'pag 'yung naglagay po kami ng auto-tagging? 'Yung auto-tagging po, heto po 'yung babasahin niya po 'yung type 'yung document and ika-categorize niya na po kung ano po 'yung document na 'yun.*

Client *Mas mabilis siyang ma-categorize. Kasi kapagka 'yung halimbawa 'yung ginagamit kasi naming system dito ay Nexus. So lahat ng letter hindi namin ma-distinguish kung anong letter siya. So kung ganyan, kumbaga naka-separate na 'yung request letter for gantong event, para sa gantong event, kung anong department siya, kung anong organization siya, mas mabilis.*

Researchers *Kaya po ano, parang hihingi po kami ng endorsement letter po 'pag kakausapin po namin 'yung iba't ibang organization.*

Client *Kayang-kaya niyo namang kausapin 'yung 40 organizations?*

Researchers *Siguro po sa ngayon 'yung mga mother orgs muna.*

Client *Mother org, oo. Sige, tingnan natin kung makakagawa tayo ng endorsement. Possible naman 'yun. Para mahingian niyo lang din sila ng data.*

Researchers *Sa part po kasi ng system namin, 'yung mga orgs po ay pwede silang mag-request ng system dito po. Tapos ano po kaya 'yung pwedeng requirements na pwede niyong makita para mabigyan ng proof na org talaga sila?*

Client *Org talaga sila? Ayun, 'yung Certificate of Accreditation nila.*

Researchers *Ay 'yun lang po? 'Yun na po 'yung pinaka...*

Client *Mhm.*

Researchers *Heto pa lang naman sa ngayon.*

TRANSCRIBED INTERVIEW 3.

The researchers conducted their interview to Dr. Rachiel R. Rivano, PhD. The interview happened at the Faculty Office of College of Computer Studies (CCS) in Laguna State Polytechnic University – Sta. Cruz Campus last April 27, 2026, Monday. The researchers are Justine Mark Gahi (Researcher 1) and Symon Kiel Beato (Researcher 2) of LSPU-SCC.

Client	*What about your thesis?*
Researchers	*Do'n po sa system, 'di ba ma'am, lahat po ba no'ng mga mother org kailangan meron na po agad do'n sa system na gagawin po namin?*
Client	*Lahat ng ano?*
Researchers	*Mother orgs po.*
Client	*Ah, ilalagay niyo na? Since kukuha kayo ng data sa ano sa OSAS, lahat ng naka—lahat ng na-accredit na org, ilagay niyo na.*
Researchers	*'Di lahat po talaga no'n ma'am? Kasi may nilagay din po kasi kaming parang application ng nagagawa po ng system. Parang isa-submit po nila do'n 'yung requirements 'tas saka pa po titingnan ng OSAS kung ia-approve po nila 'yung application no'ng org na 'yon.*
Client	*Ah, magse-sample kayo ng ano, ng mag-a-apply na bago or... eh 'di ang ilagay niyo na lang muna 'yung lahat ng... Kasi 'yung mga org kasi sa OSAS ay hindi lahat ay student organization. Meron do'ng mga clubs, parang gano'n.*
Researchers	*Opo, opo, non-academic po.*
Client	*Eh 'di ang ilagay niyo na lang 'yung mga academic. 'Yung mga non-academic... O kaya naman pwedeng ano, para dalawa na 'yung makuha natin, mag-sampling na lang kayo. Like halimbawa*

kung 10 'yung academic tapos 10 din 'yung non-academic, kuha kayo do'n no'ng ano, konti lang. Para seen meron.

Researchers *Pero paano po ba 'yung naisip namin, 'di ba po may request ng system po kami? 'Yung mga current orgs po pwede silang mag-request sa OSAS na mag-create po ng system tapos 'yung mga 'pag na-approve po nila...*

Client *Anong system?*

Researchers *'Di ba po 'yung base po namin ay 'yung SocSync? 'Pag nag-request po sila, ayon po 'yung magiging configurable kung alin po 'yung features na gusto nila. May announcement, events...*

Client *Oo. 'Pag in-interview kayo do'n, tatatanungin pa kung meron pa silang gustong ipadagdag aside do'n sa features niyo na una na nilagay? Kasi ang ano natin, 'yung SOCS...*

Researchers *Sino po ma'am? 'Yung bawat org po o si 'yung OSAS po mismo?*

Client *OSAS.*

Researchers *OSAS po.*

Client *Kasi 'yung bawat org naman, susunod lang 'yan do'n sa ano... kung ano 'yung guidelines ni OSAS eh.*

Researchers *Pero kayo po ba ma'am, parang okay lang po sa inyo kahit parang walang unique na functionalities 'yung bawat org? Parang same-same lang po sila ng mga nagagawa?*

Client *Oo, tapos kung may idadagdag sila...*

Researchers *Ah, kung may idadagdag.*

Researchers *Ah, kunyari po 'pag ano, CHMT may specific feature po sila...*

Client *Halimbawa 'yung CHMT may process sila ng ganito na wala naman do'n sa ibang org, pero nando'n sa system, pwede niyang i-add 'yun. Kumbaga pwede niyang ma... 'pag ginamit niya 'yung system na dinevelop niyo para sa OSAS na may kinalaman sa mga student organization, heto 'yung mga pwede niyang gawin, pipiliin niya do'n. 'Yun lang 'yung mapupunta do'n sa... parang naka-LMS type siya.*

Researchers *Kaya po ba 'yun sa ano, limited time po? Kung bawat org po may specific... Ang balak po kasi namin general, 'yung kailangan lang po talaga. 'Tas do'n na po sila pipili.*

Client *General, I mean magiging generic 'yung ano niya.*

Researchers *'Yung mga kailangan: announcement, events, EduShare... 'Yung mga current po naming system din.*

Client *Ay nando'n sa SocSync? 'Yun na mismo 'yung ilalagay?*

Researchers *Opo. Tapos pwede na nilang i-configure kung ano po 'yung gustong feature na gusto lang nila. Ayun lang po 'yung kukunin nila.*

Client *Pero usually kasi kung ano 'yung nasa SocSync, 'yun din 'yun. Kung maidagdag man 'yung ibang org, minimal, very minimal lang. Kasi halos same lang naman din 'yung ano 'yan sila eh, 'yung common processes: 'yung botohan, mga koleksyon sa membership, mga koleksyon sa payments sa mga activities, ano pa ba... submission sa requirements sa OSAS. Same-same lang naman 'yan sila. Kung may madadagdag man na process, siguro ano lang, it's very internal lang sa college na 'yon. It's very minimal lang.*

Researchers *Ayon po ma'am. Ano po bang nangyayari kapag na-expire po 'yung accreditation sa org? Na-di-disable po ba 'yung org tapos...*

Client *Pag... teka lang ha. Sige. Actually hindi siya na-e-expire as in totally mawawala, ma-e-expire. Every year renewable siya. 'Pag hindi pa siya accredited, halimbawa every start ng sem kasi every first sem ang accreditation eh, anong tawag dito, hindi lang kami pwedeng magpa-conduct ng mga activities hangga't hindi pa accredited. Tapos hindi pa rin kami pwedeng mag-collect ng membership 'pag hindi pa accredited.*

Researchers *'Yun lang po 'yung parang... pero hindi po nawawala 'yung org na 'yon?*

Client *Hindi. Not unless 'di mo siya papa-accredit. Hindi siya ano, walang walang operations.*

Researchers *Eh 'di kung halimbawa po nangyari nga po 'yun, hindi nakapag-accredit 'yung org na 'yon, ang maganda po ba i-di-disable na po namin 'yung org sa system?*

Client *I-disable lang. Kasi pwede naman siyang ano eh, pwede naman siyang i-renew. Like for example kung ngayong taon na 'to hindi siya na-renew, so ano siya kumbaga lahat ng operations ng org ay naka-halt, stop siya. Tapos next sem naisipan nilang ipa-accredit, pwede natin siyang ulit i-activate. Pero nasa log. Para at least si OSAS may monitoring pa rin kung, "Ah ito palang org na 'to ay ano, hindi pala nagpa-accredit no'ng ganitong taon." Tapos...*

Researchers *Ay, may nangyayari pong gano'n?*

Client *Meron! Sa Engineering. 'Yung mother org nila. 'Di ba ilang taon silang walang mother org? Puro ano... kaya parang I think one or two years silang puro ano, EE Days, CE Days, pero wala silang COE Days. Ngayon na lang ulit. Ngayon na lang ulit sila nagkaroon ng COE Days, 'yung one week na simula Monday hanggang... Merong nangyayari.*

Researchers *Para tama po ba, renewal form lang po 'yung kailangan?*

Client *Renewal every year.*

Researchers *Renewal form.*

Client *Pero parang 'di renewal nga 'yun eh. Hindi, parang renewal nga kasi existing na siya eh. Tapos nire-renew lang 'yung accred—kumbaga nagre-re... nagpapa-re-accredit lang sila sa OSAS. 'Yung E2 tsaka 'yung CSS, 'yun renew 'yun. Kasi ano naman siya eh, ilang two years na ba o ilang taon na?*

Researchers *'Yun pala ma'am, do'n sa E2 tsaka sa CSS, parang may naging officer din po ba ng SOCS na napapunta sa E2 or CSS or magkaiba po talaga sila?*

Client *Magkaiba talaga sila. Walang officer ang SOCS na officer ng E2.*

Researchers *Ah, bawal po 'yun. 'Tas iba rin po 'yung advisor?*

Client *Oo, iba din 'yung advisor. Hanggang sa representatives. Walang officer ng E2 na officer din ng SOCS. Kumbaga sa election kasi sa filing ng candidacy ay ano na 'yun, ano siya, individual. Hindi ka pwedeng like mag-a-apply ka sa SOCS 'tas mag-a-apply ka din do'n sa isa.*

Researchers *Nilagyan po kasi namin siya ng role-based access sa bawat org. Eh 'di po namin ma-sure kung tama po ba 'yung sa 1st to 4th year representative. Eto lang po 'yung ginawa namin: E-Library, EduShare, tsaka Complaints. Ayon lang po 'yung sa kanila, do'n sa 1st to 4th year representative. Ang may full access po ayan: Senior, Junior Adviser, President, Vice President.*

Client *Tingnan niyo sa bylaws. Best i-base niyo 'yung roles nila sa bylaws kasi nasa bylaws 'yung rules.*

Researchers *Ah, hihingi na lang din po kami.*

Client *Sa SOCS meron. Kausapin niyo si... actually meron ako kaya lang siyempre 'di ko pwedeng ibigay sa inyo. Kailangan niyo manghingi ng formal letter. May bylaws 'yung SOCS.*

Researchers *Anong pong letter 'yun, may format naman po or...*

Client *Format? Basta 'yung sa letter, wala namang format. Ang nakalagay lang do'n is you are getting ano parang approval or permit na makahingi ng mga documents. Halimbawa kung ayaw niyong per org kayo magkukuha, pwedeng sa OSAS na kayo dumiretso kasi lahat ng bylaws ng lahat ng orgs, meron silang kopya.*

Researchers *'Yung bylaws po ba ano, kayo pong gumawa or bigay po 'yun no'ng...*

Client *May format 'yung pinaka-bylaw tapos ine-edit. Pinapasadahan sa OSAS kung approve nila 'yung content.*

Researchers *Sila na po ba 'yung gagawa para sa bylaws? Sino po 'yung lalagay namin do'n?*

Client *Kayo, tsaka ako, kasama ako sa letter. 'Tas approve ni Sir Ian. Para lahat ng bylaws pwede ninyong makita kasi for sure may iba-ibang bylaws 'yon.*

Researchers *Ay, parang 'yung gagawin po pala namin ay endorsement letter? Or pang-SOCS lang?*

Client *Paanong endorsement?*

Researchers *Kapag po pupunta po kami sa ibang org...*

Client *Hindi, 'yun na 'yon! Tapos ino-note na lang siguro ni Sir Ian na ano, kumbaga in-approve niya 'yon tapos bibigyan niyo ng kopya or ipapakita niyo 'yun do'n sa kada org na meron kayong permit from OSAS to conduct data gathering do'n sa mga orgs. Pili na lang kayo, tanungin niyo na lang sa stat niyo kung... halimbawa pumunta kayo sa OSAS 'di ba tatanungin niyo kung ilan 'yung academic tsaka non-academic sa org. Kung halimbawa pagpalagay na natin na 40: 20 na academic, 20 na non-academic. So tatanungin niyo sa stat niyo kung 'yung bang 40 na 20-20, lahat ba 'yun kukunin niyo or hihingan niyo ng... gagamitin niyo as respondents? Or do'n sa 40, do'n sa 20-20, pagpalagay na nating 10-10? Kasi pwede naman 'yun eh na hindi lahat. Kasi kumbaga*

halos same-same lang naman 'yung magiging ano niyan eh, magiging inputs niyan sa inyo.

Researchers *Kung gano'n po pala, sino po 'yung magiging... ano 'yung bibigyan po namin ng questionnaires? May questionnaires po pala?*

Client *'Yun! Do'n nga magbe-base 'yun. Kasi ang questionnaires niyo, since ang pinakagagamit no'ng system ay mga orgs, 'di mga officers. Tapos 'yung kada org, ilan do'n sa mga officers na 'yon 'yung target. Siyempre i-target niyo 'yung nandito sa atin, 'yung tatlo. Kasama na 'yun do'n sa pinaka... kung halimbawa do'n sa 20, so kasama na siya do'n sa tatlo. Tapos kukuha pa kayo ng ibang org from different colleges. Halimbawa tatlo na dito sa atin, matik, tapos ang natitirang college, lahat lang ng mother orgs, pwedeng gano'n. Tapos do'n sa non-academics, eh 'di kumuha lang kayo ng mga major organizations kasi 'yung iba diyan mga maliliit lang.*

Researchers *'Di naman pala kailangan lahat eh.*

Client *Pipili lang kayo do'n kasi kung kukunin niyo naman kasi 'yung... kung halimbawa 40, pare-parehas lang naman ang magiging sagot no'n eh. So kukuha lang kayo ng sample. Kaya tatanungin niyo sa stat niyo, kunin niyo muna 'yung data kung ilan 'yung... kung ilan 'yung total number ng organizations tapos ilan do'n sa total na 'yon 'yung kukunin niyo. Tapos 'yung respondents, eh 'di do'n tatanungin niyo rin sa ano... sa stat niyo kung lahat ba ng officers no'ng kinuha niyong sample or 'yung major officers lang like from the President up to the Secretary or...*

Researchers *Paano po 'yung mga students, kasama rin po?*

Client *Meron pa rin kasi 'yung operations ng org, may estudyante pa rin do'n eh. Pwede sa OSAS mismo, si Sir Ian tapos 'yung staff. Tapos 'yung per org, advisers din. Basta lahat ng involved sa operations ng organization dapat kasama as respondents.*

Researchers *Pero may limit lang pala?*

Client *Oo, iba-iba lang sila ng access patulod no'ng diniscuss natin kanina. Iba-iba lang sila ng access doon sa system. Halimbawa si Dean, 'di ba involved naman siya kasi pina-approve niya 'yung mga activities, 'di ba? So dapat siya meron din. Tapos 'yung mga advisers ng org, dalawa 'yun: co-adviser tsaka adviser. Ang tawag lang sa atin is Senior, Junior, pero laging dalawa 'yun.*

Researchers *'Yung mga ano, 1st year to 4th year na representative...*

Client *'Yung bylaws sa OSAS na lang kayo humingi, meron naman silang copy no'n.*

Researchers *Hihingi pala po kami do'n ng type ng document tapos list ng orgs.*

Client *Willing naman sila.*

Researchers *Opo, nabigyan na kami.*

Client *Anong sabi ni Sir Ian?*

Researchers *Pumunta daw po kami kung may kailangan pa raw po.*

Client *'Yun pala eh. Alam niya na adviser niyo ako.*

Researchers *Ayon, hindi pa po namin nasabi.*

Client *Sabihin niyo na adviser niyo ako para...*

Researchers *'Yung ano po ba... lahat po ng mga org may COMELEC po?*

Client *Ha?*

Researchers *Lahat po ng org, organization, may COMELEC?*

Client *May COMELEC? Oo, lahat. Pati Supreme meron din.*

Researchers *Kahit po 'yung mga sub-org meron?*

Client *Ay hindi, isa lang 'yun sa atin, isang COMELEC.*

Researchers *Ah per ano po, per college?*

Client *Per college. Except 'yung Supreme kasi may sarili sila.*

Researchers *Ilan po ba 'yung mga tao do'n sa COMELEC?*

Client *Si Ma'am Lyn kasi ang COMELEC dito sa atin eh. 'Di ko alam kung ilan.*

Researchers *'Pag may election po, nag-uusap po 'yung SOCS tapos COMELEC kung paano po 'yun?*

Client *Ah hindi, 'yung operations ng election, sila lang 'yun. Labas kami do'n. Kasi hindi namin pwedeng pakialaman ang COMELEC sa election. Sila talaga 'yung bahala do'n.*

Researchers *'Di election po lagay lang po namin sa COMELEC?*

Client *'Yun lang talaga 'yung function nila: election. Do'n lang sila magfa-function. Si Ma'am Lyn 'yun eh sa atin, 'di ba? Ewan ko lang sa ibang college kung gano'n din.*

Researchers *Ma'am, paano po kapag ano... 'yung sa ngayon po, kapag po ba nagpapasa kayo ng mga documents katulad po ng 'pag magre-renew ng accreditation, physically po ba ipinapasa 'yung papel or nagpapasa po ng mga soft copy?*

Client *Hindi, hard copy. Pinapasa do'n 'yung mga requirements na kailangan. Halimbawa 'yung bylaws, tapos 'yung pinapapirmahan sa inyo 'di ba kung officer ka ng ng section, 'di ba may pinapa-fill out sa inyo na naglalagay kayo ng picture 'tas pirma? Pini-print nila 'yon together with pictures tapos 'yung bylaws, 'yun, isa-submit 'yun.*

Researchers *Sino po 'yung ano 'yun, 'yung nagpapasa no'n? Kung sino lang po ma-assign?*

Client *Depende kung sinong available pero usually naman secretary 'yun. Kasama si President. Kasi siya gumagawa no'n eh, secretary.*

Researchers *Pero 'yung mga adviser, president, kaya rin po nilang magpasa?*

Client *Oo naman. Pero uswal naman, one process lang. Halimbawa hindi pa napapasa tapos kami lang ng adviser 'yung available, kami magpapasa. Pero mas madalas, officers... sa officers lang talaga 'yun. Sila 'yung nag-aasikaso no'n.*

Researchers *Balak po kasi namin soft copy tapos naka-record na po do'n sa...*

Client *Pwede naman 'yun kasi gagawa naman natin ng system eh. So since OSAS naman ay may access do'n sa system, pwedeng do'n na mismo iche-check. Kaya kung gusto niyo tulungan niyo si OSAS na ano lagyan niyo ng ng AI.*

Researchers *Meron po kaming AI, 'yung sa summarize tapos auto-tagging.*

Client *Tutulong ni AI si OSAS... tutulungan ni OSAS si AI para mag-judge if lahat ng documentary requirements no'ng org para maka-accredit ay submitted na.*

Researchers *Submitted na.*

Client *Pwede naman, patulong ni OSAS si AI para ma-check 'yung bylaws kung tama ba 'yung wordings, may kulang ba, parang gano'n. Para may iba lang do'n sa...*

Researchers *Balak po kasi namin 'yung mga pinapasa pong documents isa-summarize po ng AI tapos makikita na po 'yung mismong details no'ng mga bawat doc. Tapos 'yung isa po auto-tagging, partner po sila, ita-tag po kung ano po siyang type na document, kung request...*

Client *Para malaman kung complete 'yung requirements. Pwede 'yun, maganda 'yun.*

Researchers *Kapag po kasi gagawan ng account 'yung mga adviser, ang mangyayari po do'n sa system ay... dadaan po do'n sa pag-request ng account tapos ilalagay niyo po do'n 'yung Gmail po tapos do'n niyo po mare-receive 'yung parang temporary na account na ibibigay sa inyo ng OSAS. Pero may format po ba 'yung ID ng mga adviser? Katulad po sa mga students, number lang?*

Client *Wala kaming ID.*

Researchers *Kahit mga number po, wala?*

Client *'Pag adviser? Wala, adviser lang kami.*

Researchers *'Di temporary na lang po ano?*

Client *Ah, magbibigay kayo?*

Researchers *Kasi po ang pang-login namin ay 'yung student ID and 'yung password.*

Client *Ah, 'di gawan niyo na lang kami ng ano... as adviser. Pero wala kaming ID.*

Researchers *'Di unique po talaga 'yung...*

Client *Meron kaming ID, 'yung employee ID. 'Yun, pwede namin 'yung gamitin as pang-log in, 'yung employee ID.*

Researchers *Ilang number po 'yun?*

Client *2, 4, 6, 8, 10... 10.*

Researchers *'Yung 01 po sa unahan, ano po 'yun? Para iba-iba po kayo?*

Client *'Di ko alam kung anong ibig sabihin no'ng 01 eh. Pero ito, ito lang kasi 'yung number ko dati. Tapos nadagdag 'to kasi ito ay year kung kailan ako na-regular. Ay, month at saka day... ay hindi pala, month at year. Basta 10 digits.*

Researchers *Sige po.*

Client *Not unless gagawa kayo ng sarili ninyong ano, ID. Pero kung existing, meron na kaming existing na ID. Tapos email, 'yung mga institutional email naman, so pwedeng 'yun na lang gamitin, 'yung lspu.edu.ph.*

Researchers *Para verified na po agad.*

Client *Oo, para sa email na lang kami magbe-verify kung existing talaga 'yung account kaysa sa ID number.*

Story Board



Figure 28. This Figure shows the student login/register.



Figure 29. This Figure shows the creation of the Organization Platform.

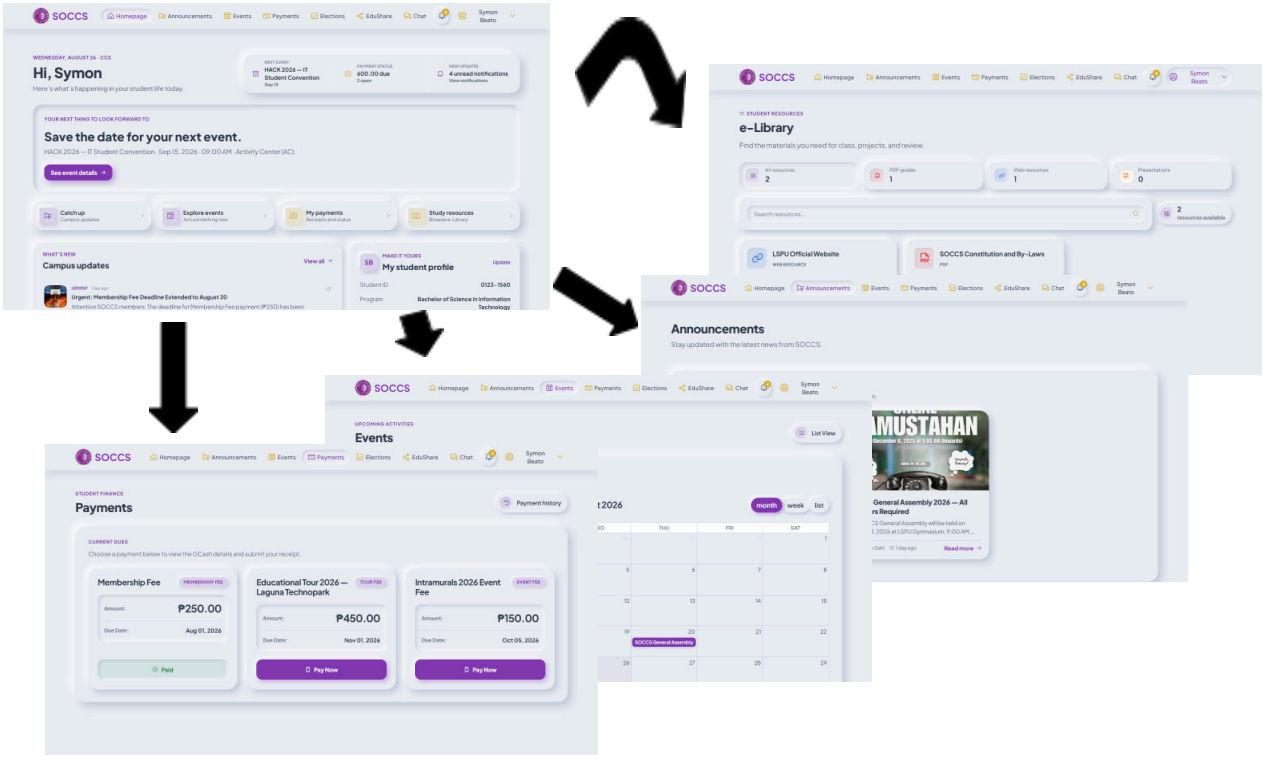


Figure 30. This Figure shows that the homepage can navigate to announcements/events/payments/study resources.

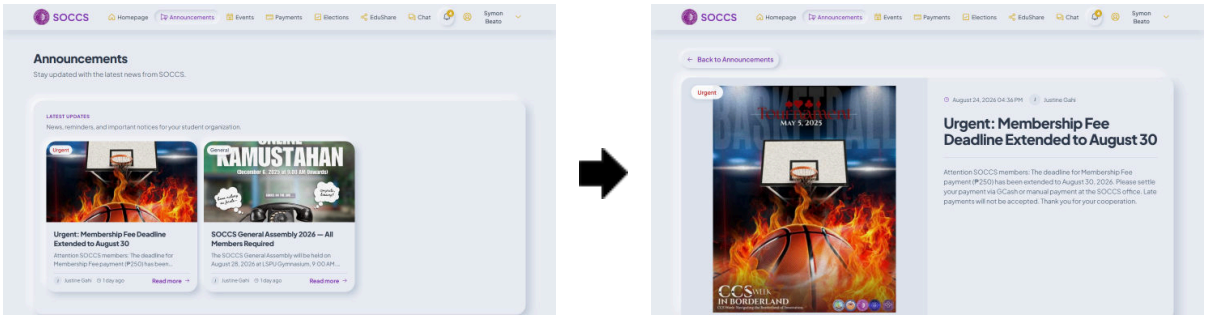


Figure 31. This figure shows the announcement list and the announcement view.

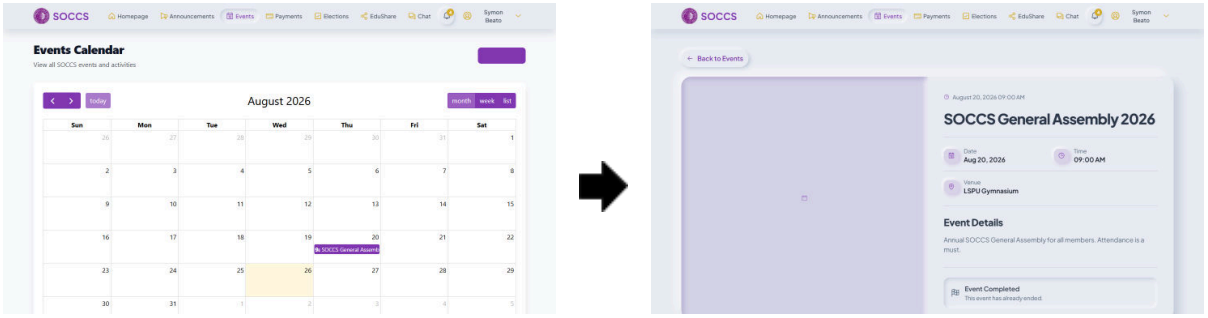


Figure 32. This figure shows the event calendar and the event details.

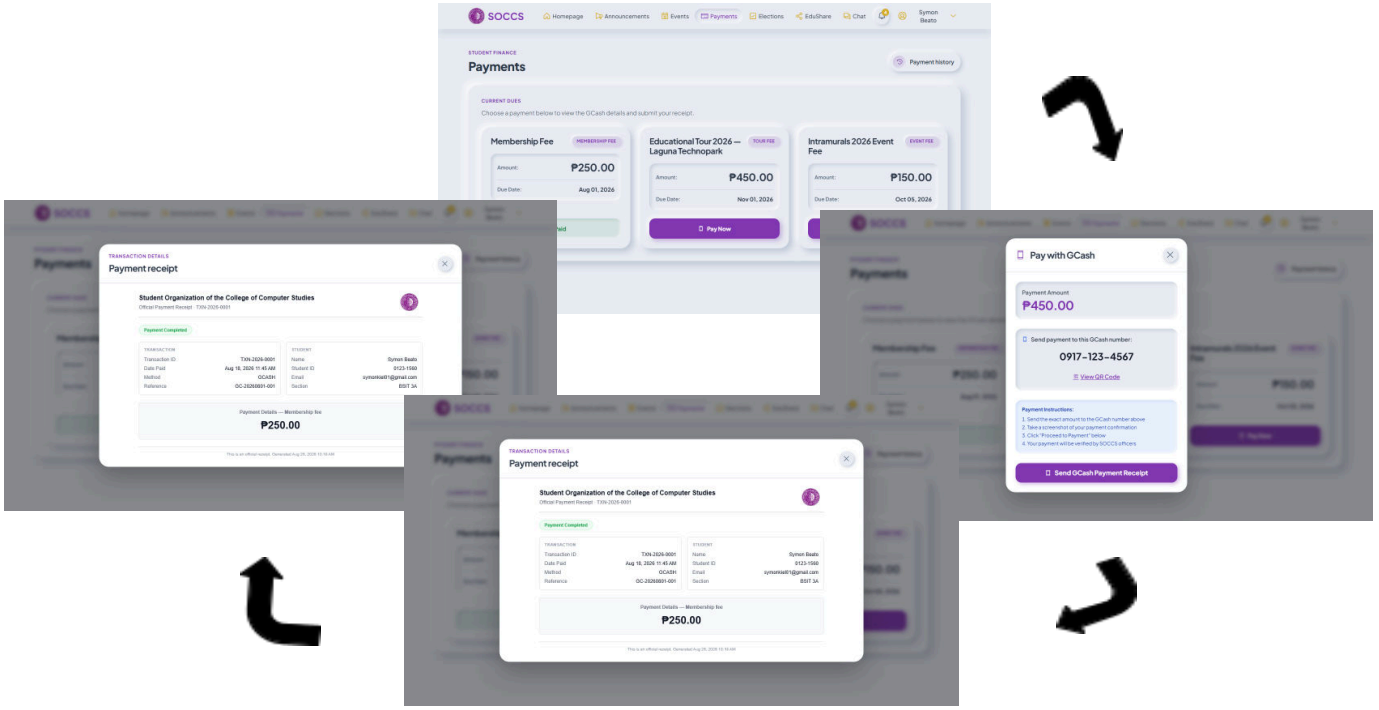


Figure # This Figure shows the payment list/payment with gcash/ receipt/payment history.

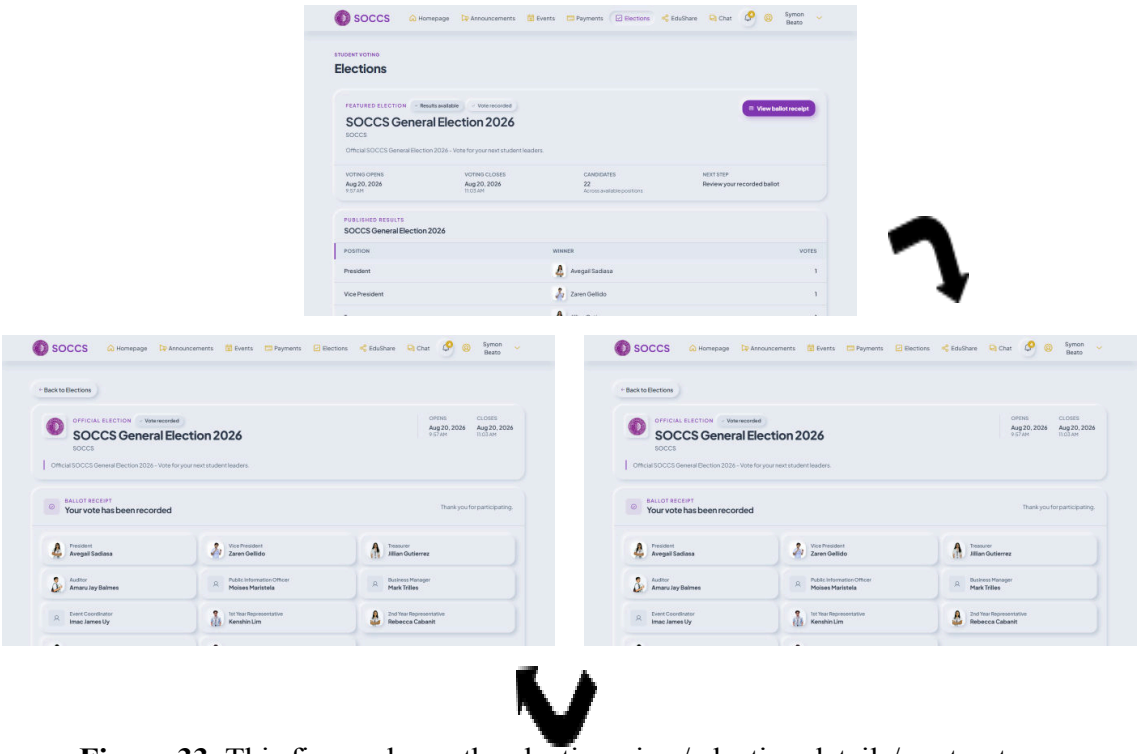


Figure 33. This figure shows the election view/ election details/ cast vote.

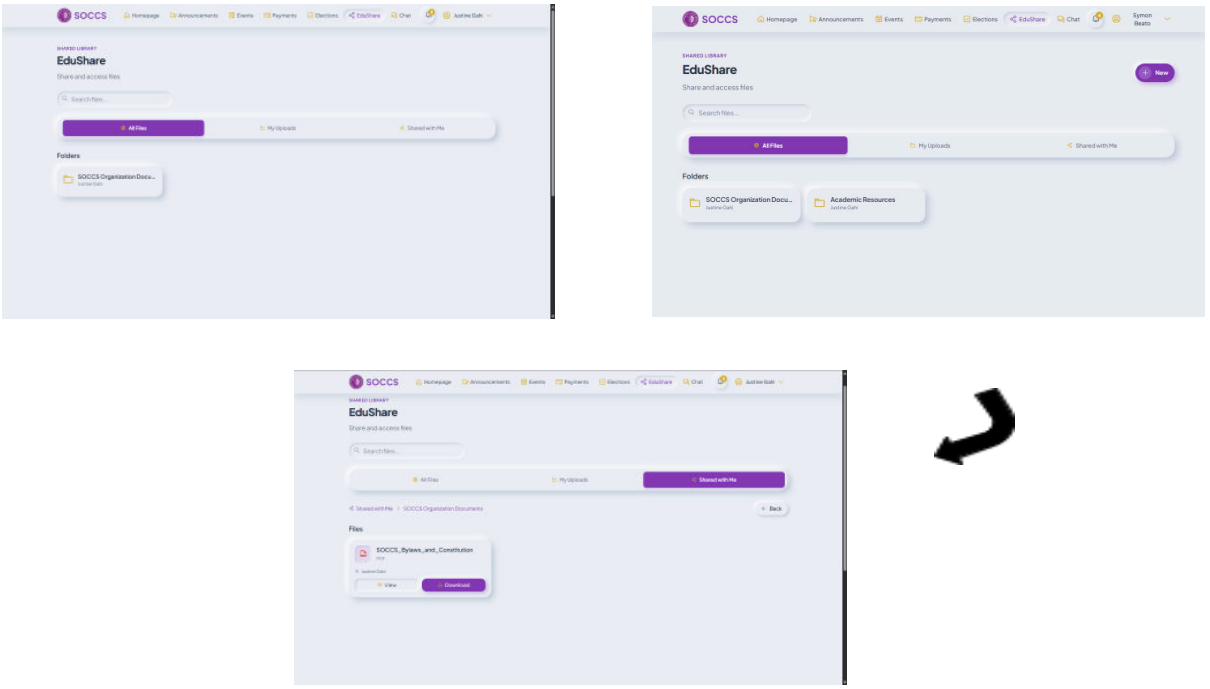


Figure 34. This Figure shows the edushare files/ my uploads/ shared with me files.

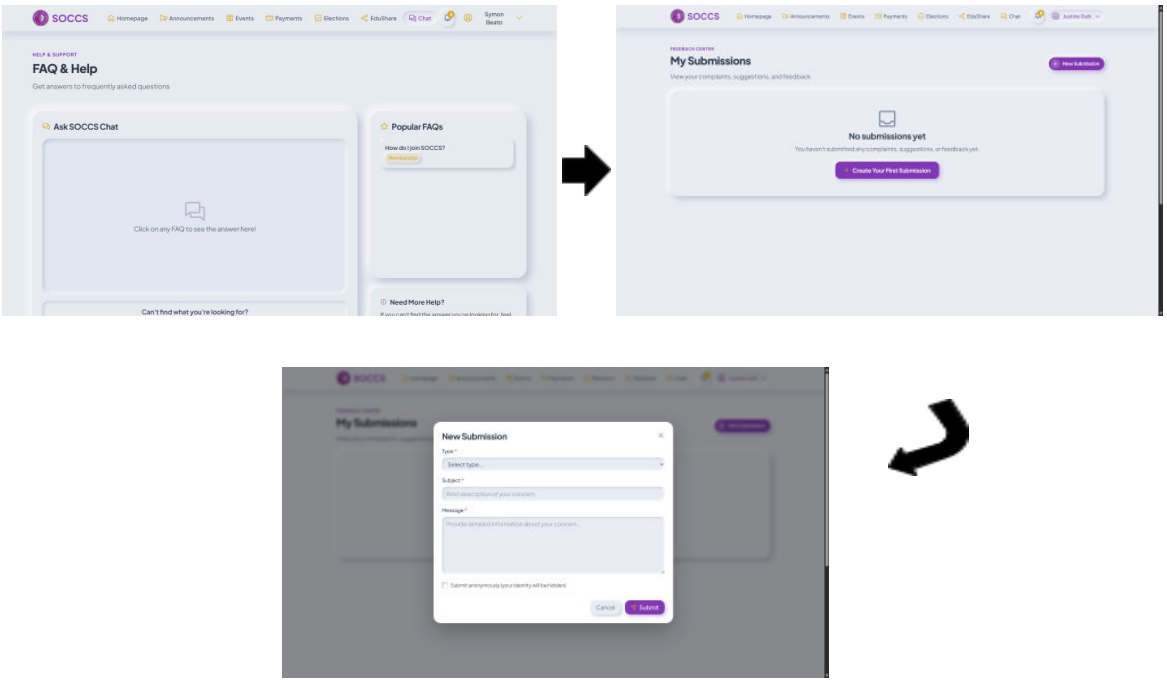


Figure 35. This figure shows the chat and complaints/ complaints modal.

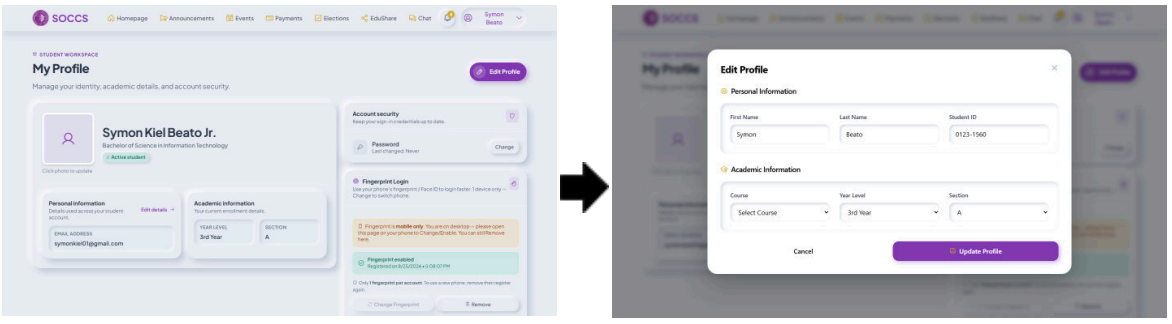


Figure 36. This figure shows the student profile and edit modal of the profile.

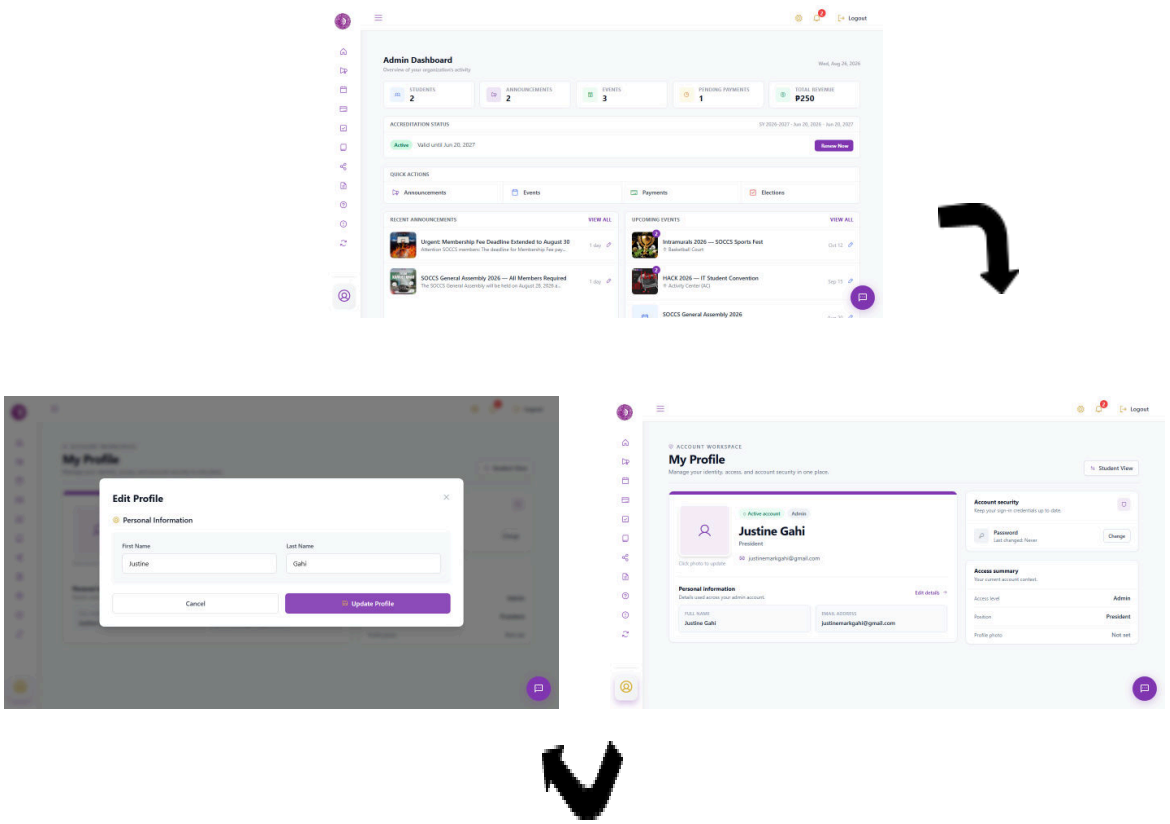


Figure 37. This Figure shows the how admin profile customization.

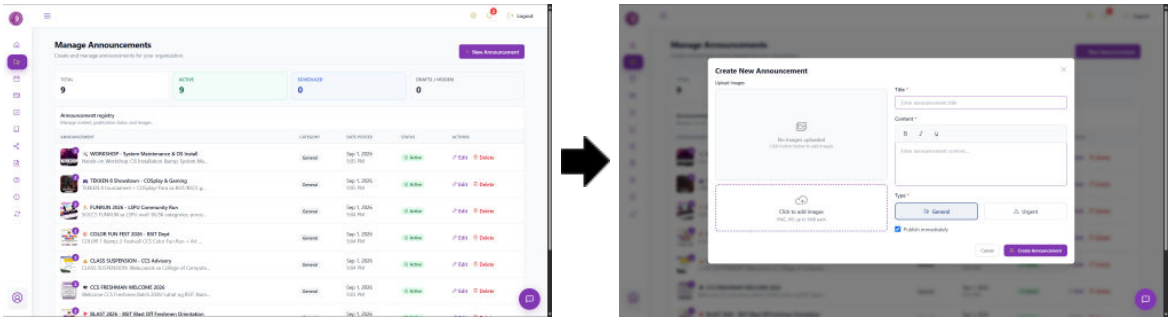


Figure 38. This Figure shows the creation of announcement.

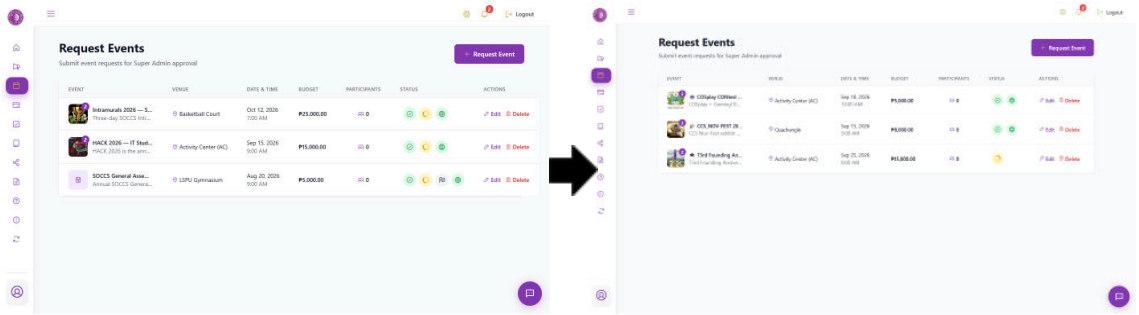


Figure 39. This Figure shows the request and publish of events.

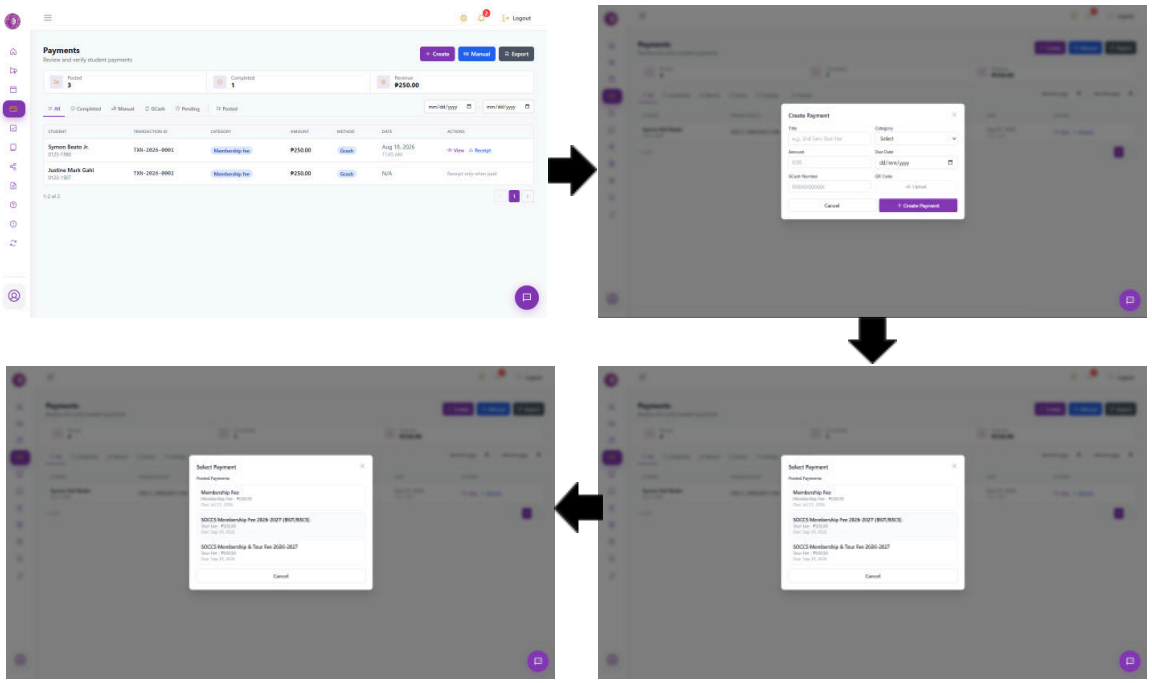


Figure 40. This Figure shows the payment creation and scheduling.

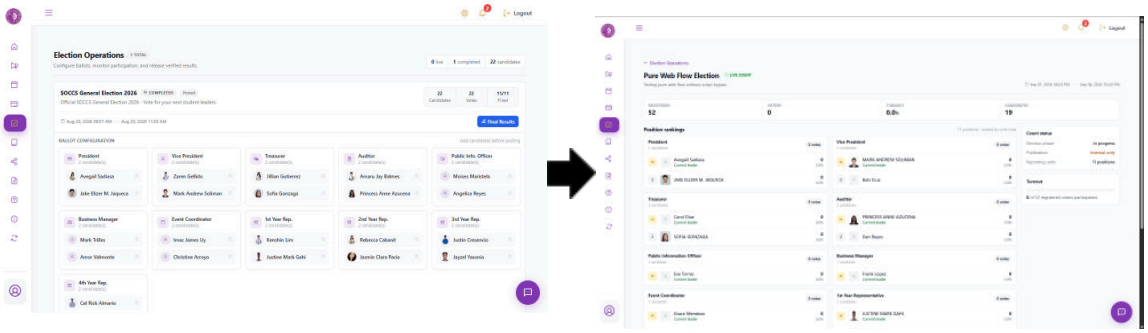


Figure 41. This Figure shows the election listing and winning of the candidates.

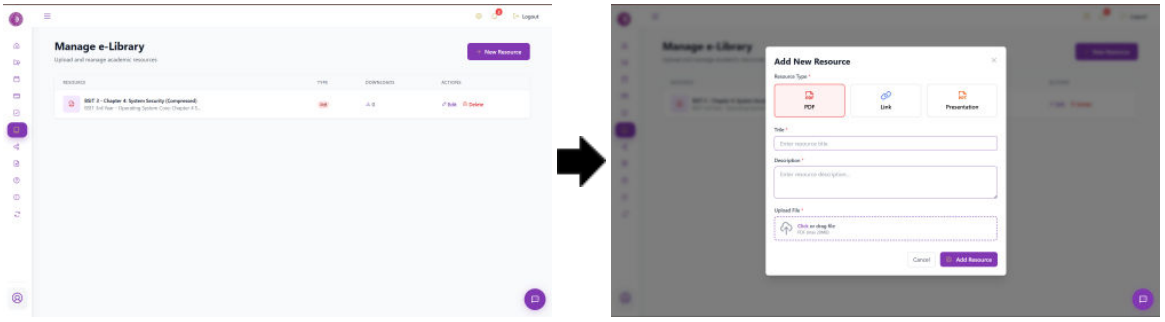


Figure 42. This Figure shows how to make Library Material.

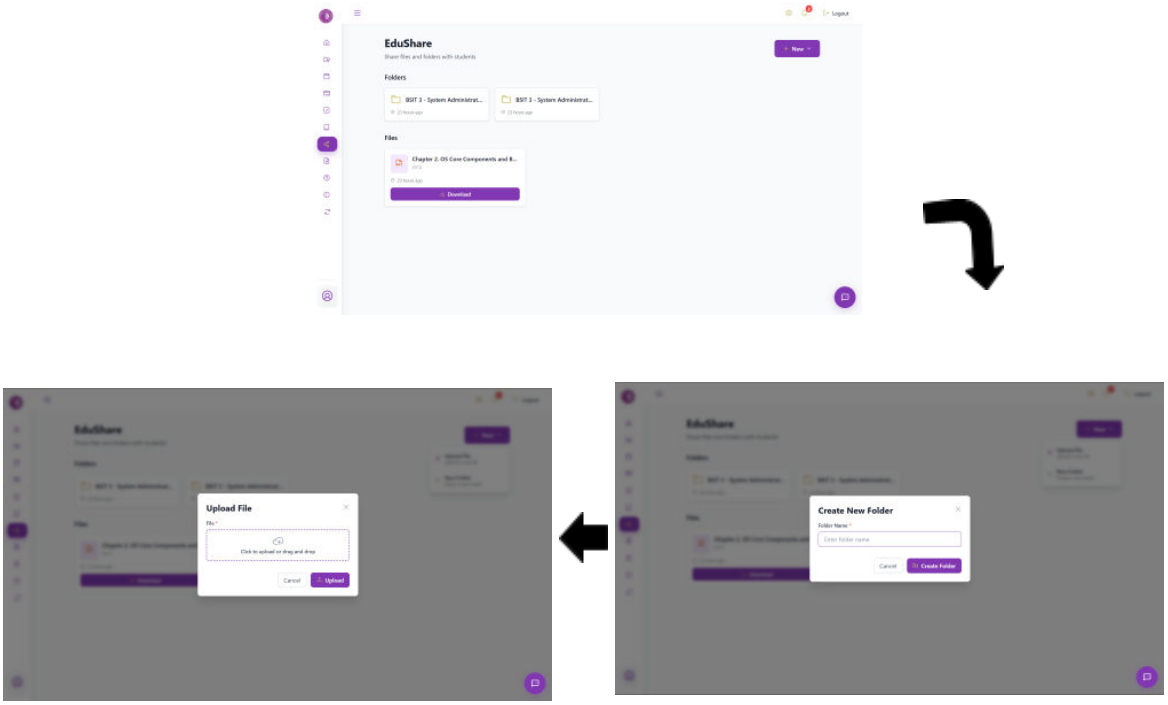


Figure 43. This Figure shows how to create Edu Share files as an admin.

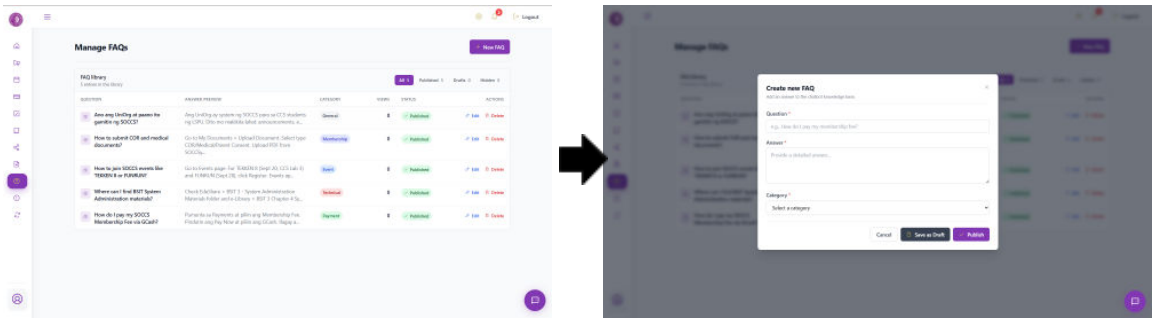


Figure 44. This Figure shows how to set the frequently asked questions answer.

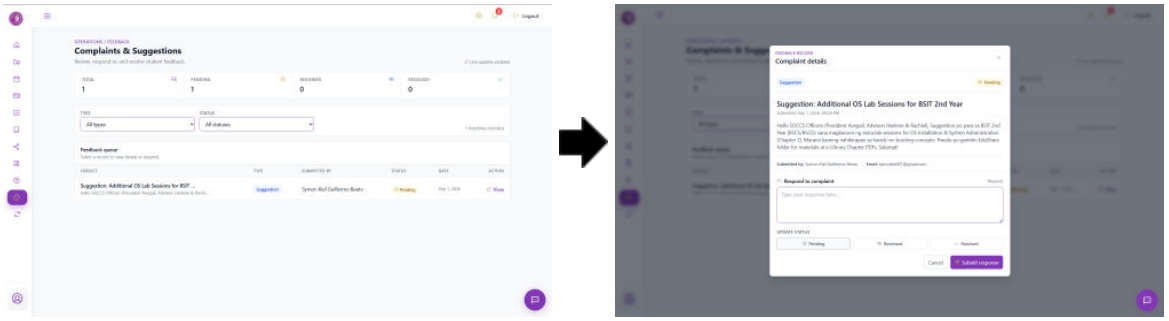


Figure 45. This Figure shows how to reply and view complaints and suggestions.

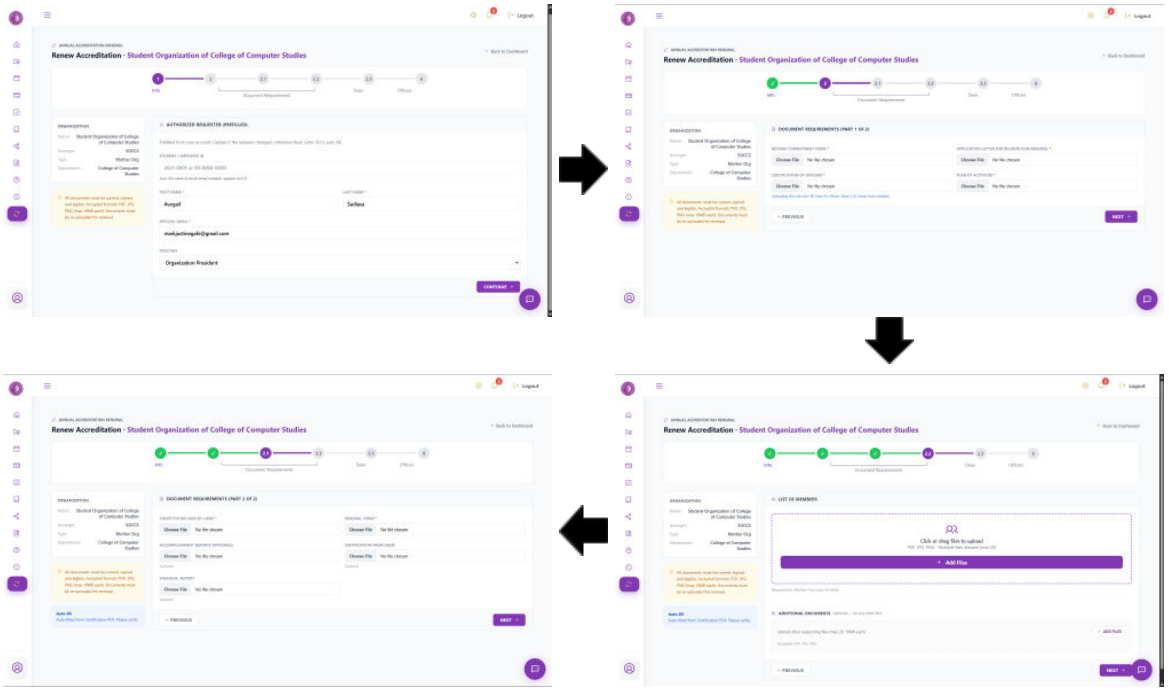


Figure 46. This Figure shows how to renew the accreditation on an org part 1.1.

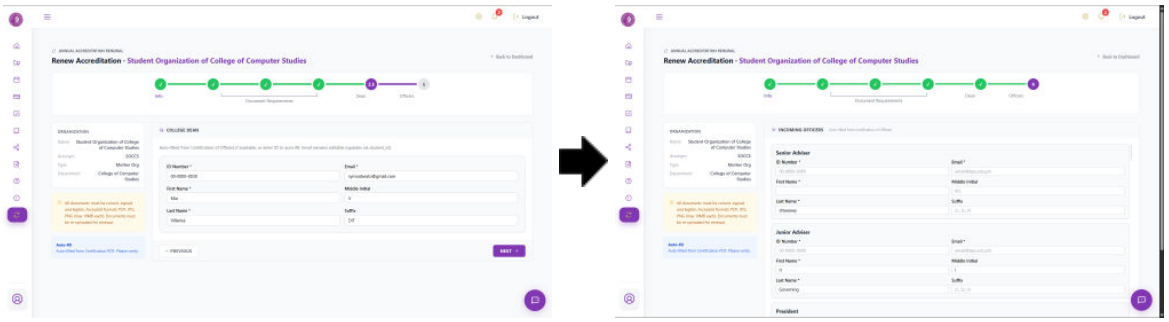


Figure 47. This Figure shows how to renew the accreditation on an org part 1.2 .

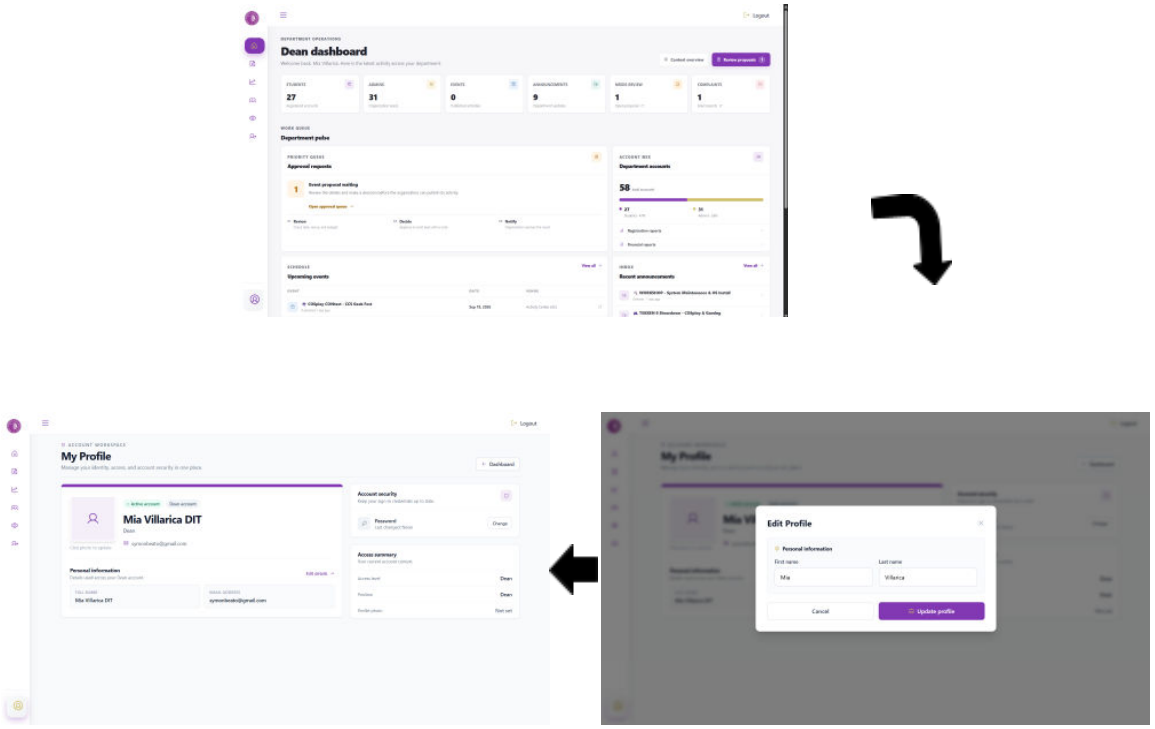


Figure 48. This Figure shows the dashboard and details of the dean account .

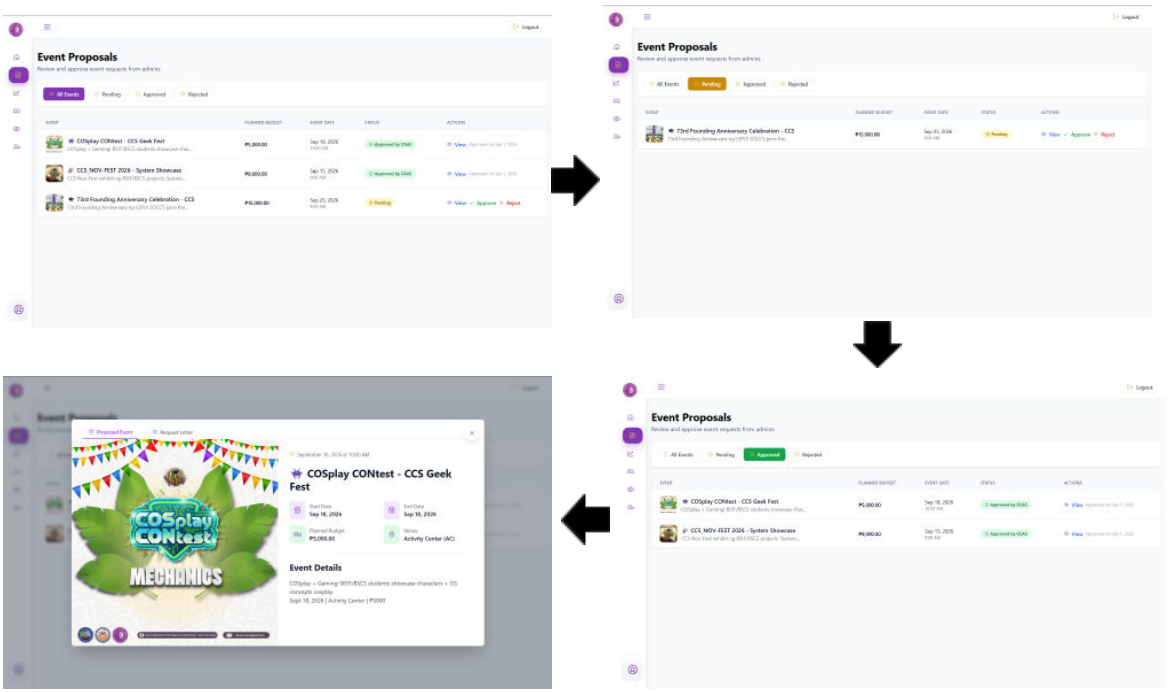


Figure 49. This Figure shows the dean view and approve events.

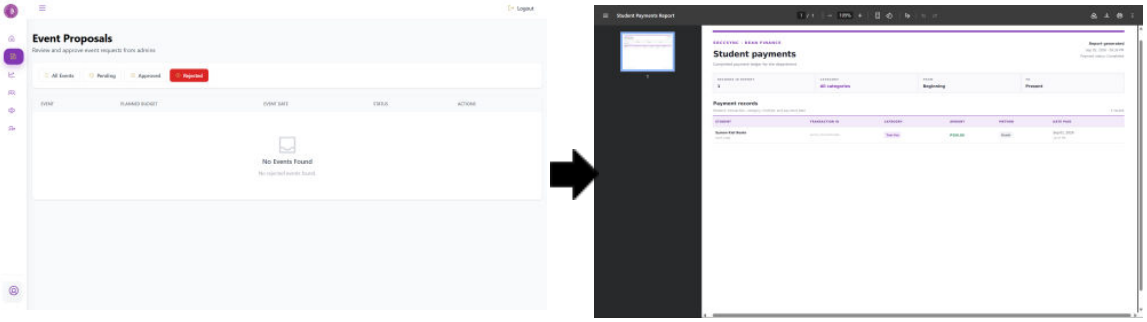


Figure 50. This Figure shows the dean view and review payment summary.

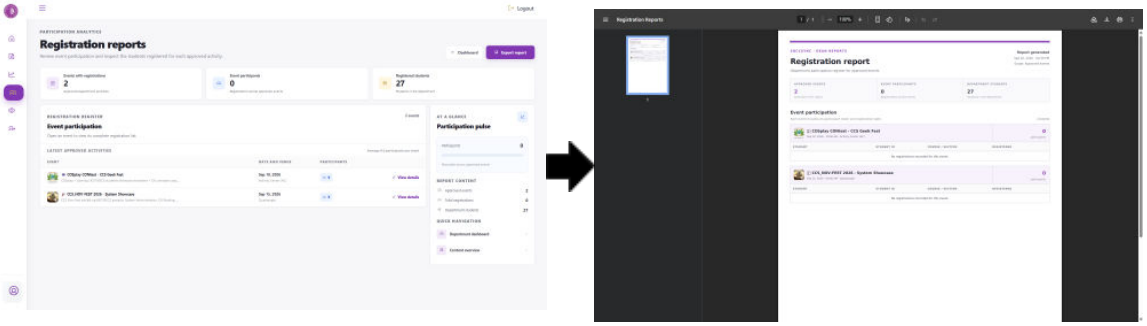


Figure 51. This Figure shows the dean view and review registration report summary.

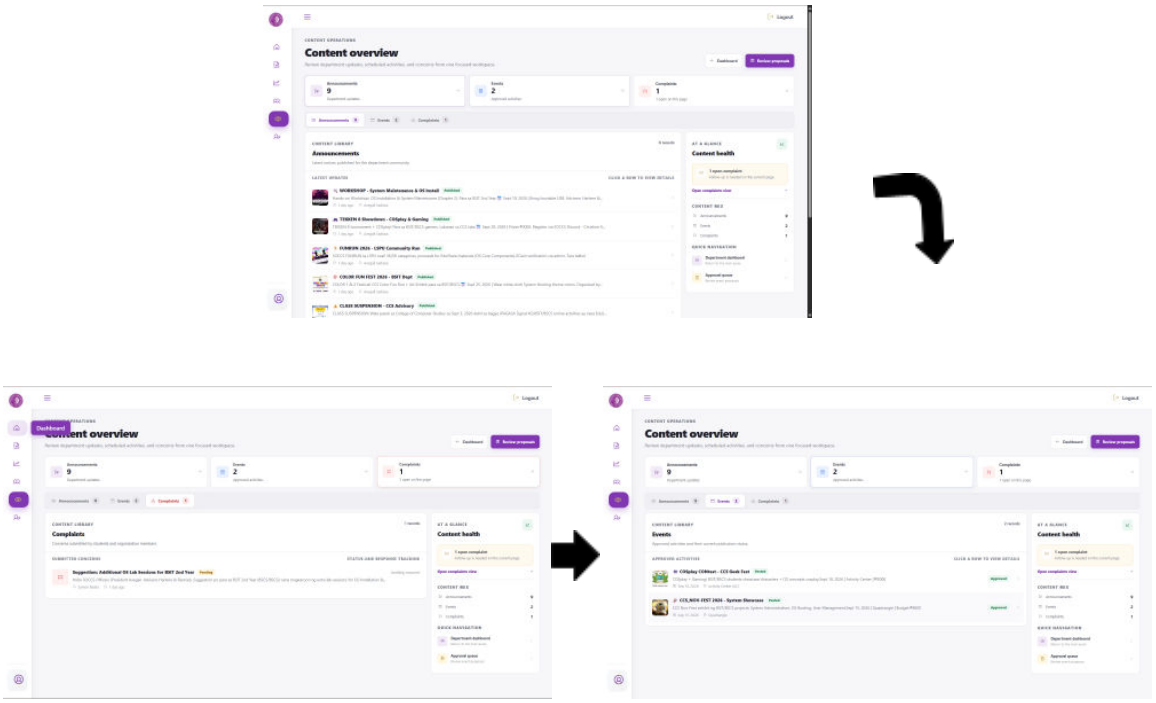


Figure 52. This Figure shows the dean view and review content overview.

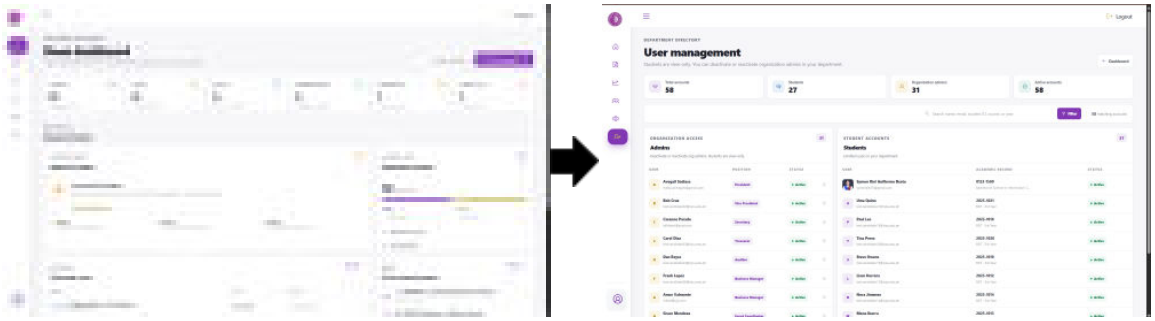


Figure 53. This Figure shows the dean view and review user management.

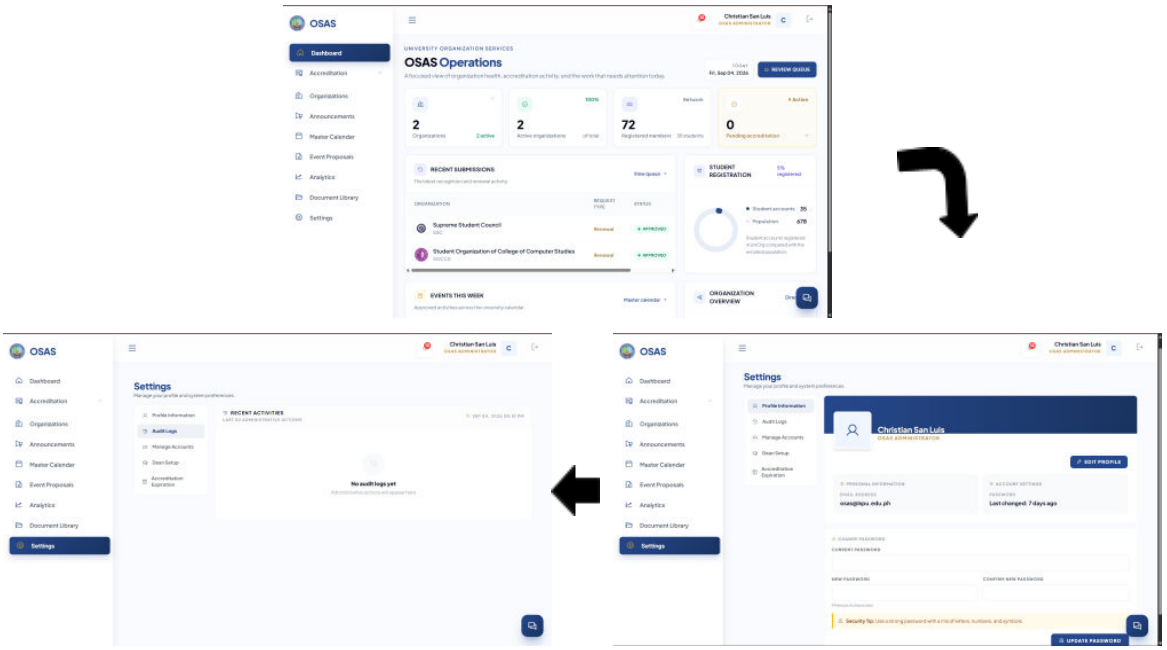


Figure 54. This Figure shows the OSAS view and updates their profile.

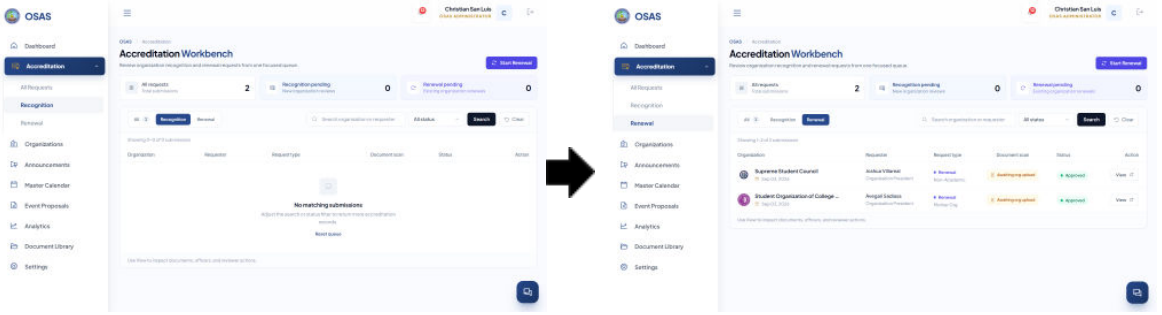


Figure 55. This Figure shows the OSAS view, review and approval of the accreditation for orgs.

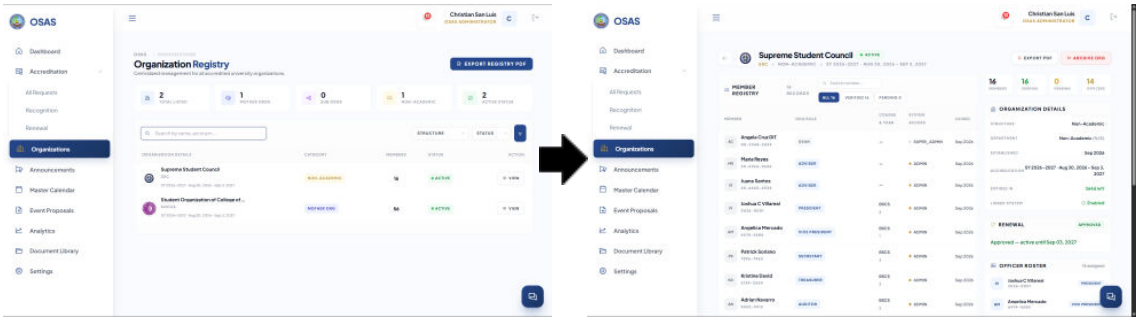


Figure 56. This Figure shows the OSAS view, review and archive orgs.

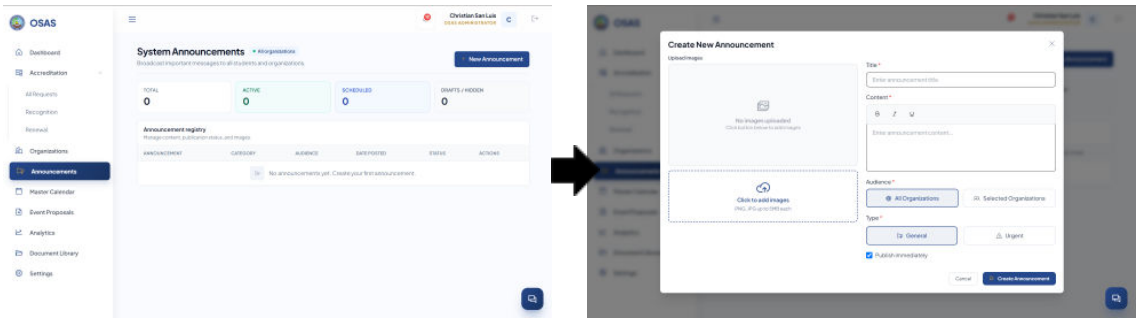


Figure 57. This Figure shows the OSAS create an update for all the org users.

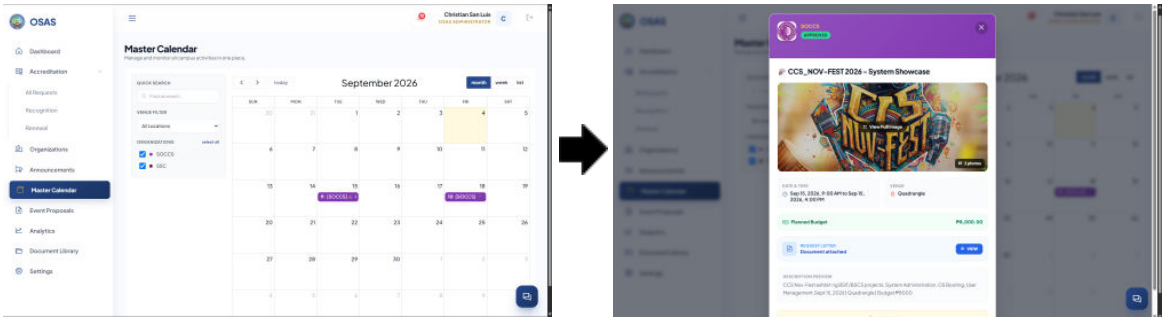


Figure 58. This Figure shows the OSAS create an update for all the org users.

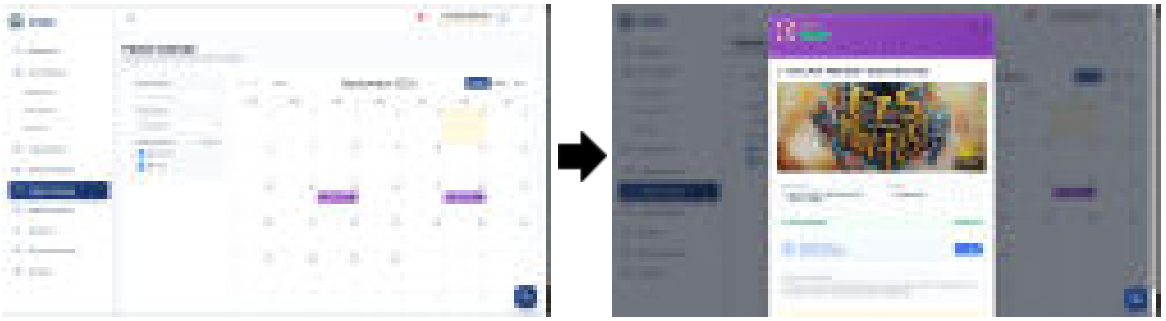


Figure 59. This Figure shows the OSAS view of the calendar of the events.

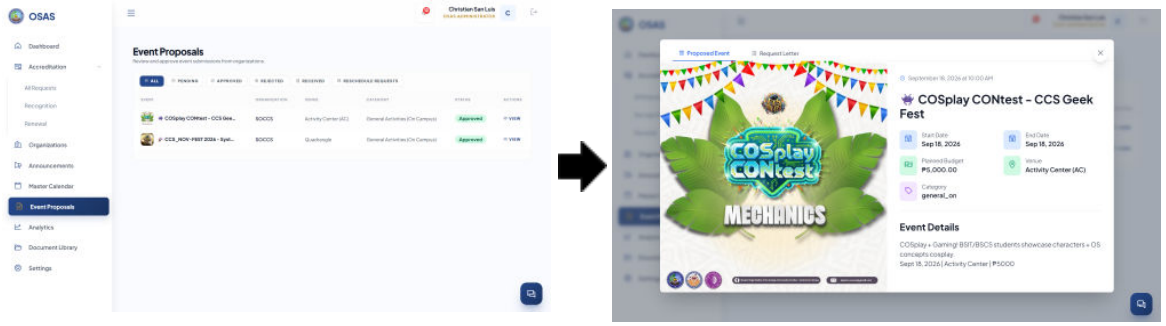


Figure 60. This Figure shows the OSAShow to approve events proposed by the orgs.

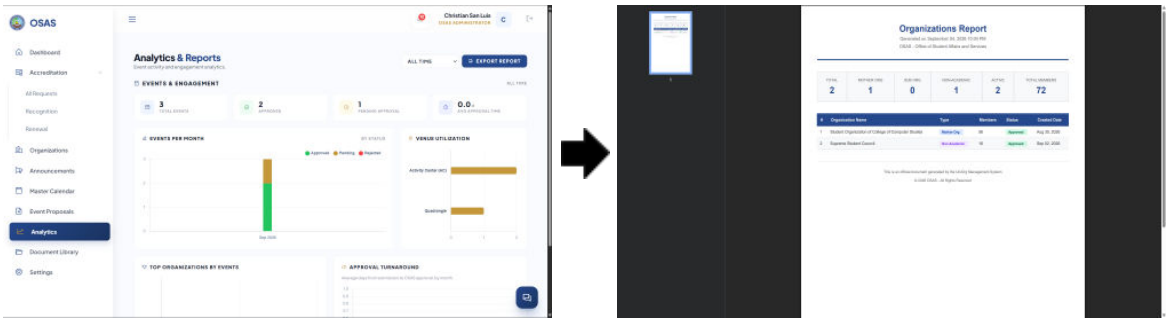


Figure 61. This Figure shows the OSAShow to review the Organization status.

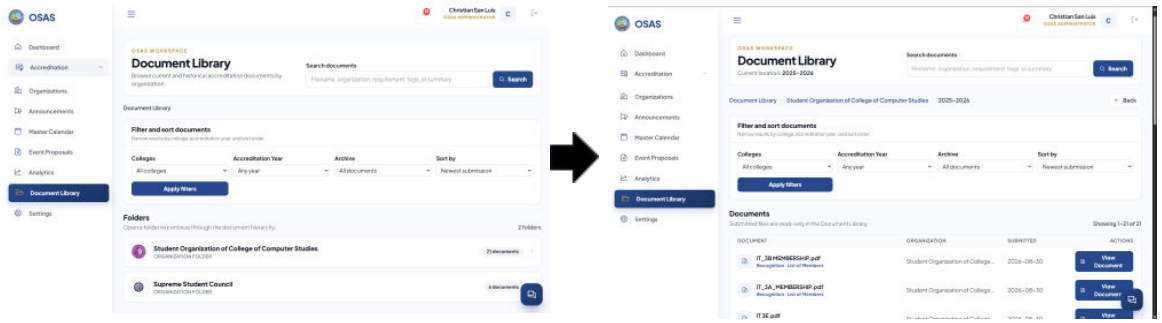


Figure 62. This Figure shows the OSAS view and reviews the archived documents of the past accreditation years.

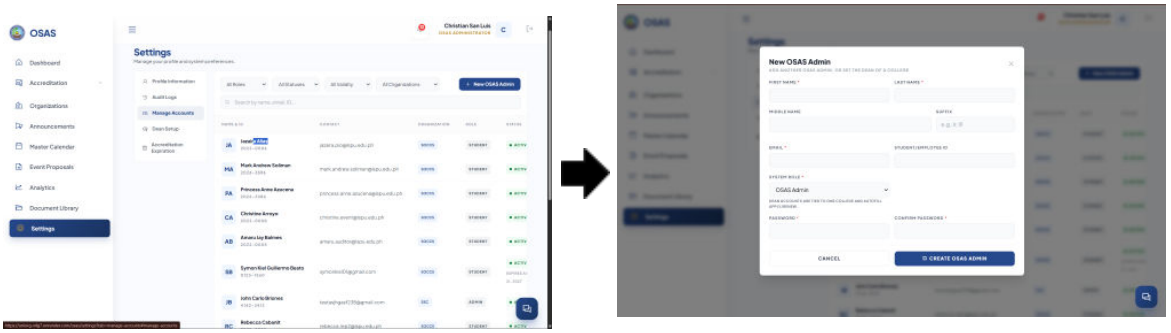


Figure 63. This Figure shows the OSAS account creation.

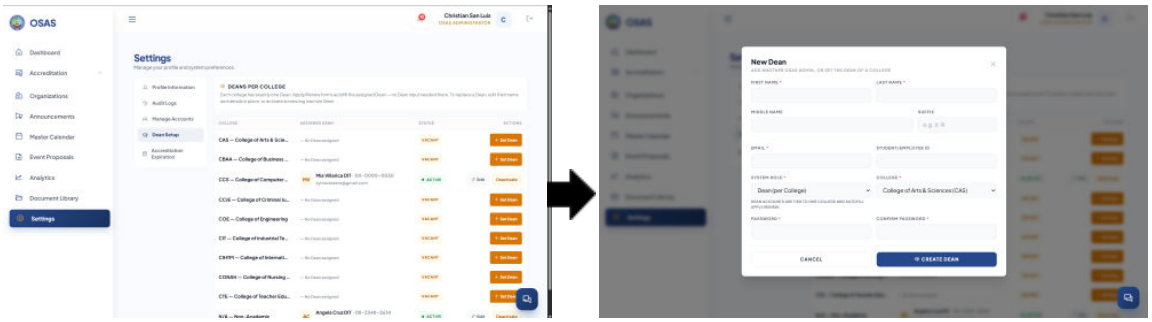


Figure 64. This Figure shows the OSAS how to create and set the deans per college.

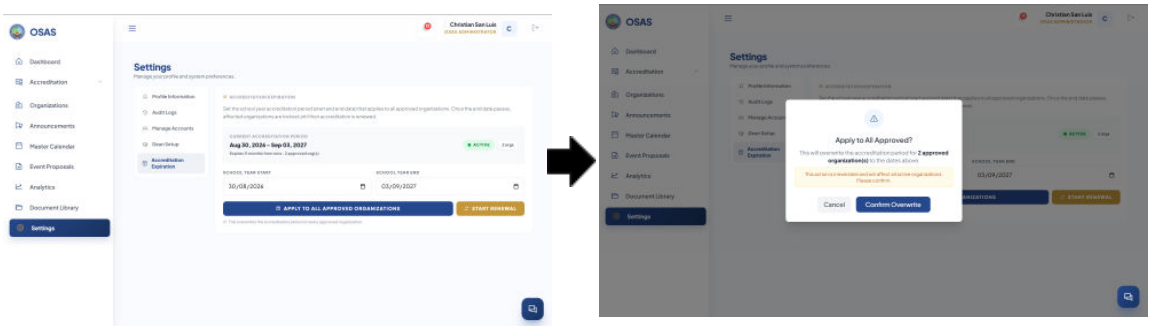


Figure 65. This Figure shows the OSAS how to set the accreditation year.

Current Process Flowchart

This shows the current process followed by the Office of Student Affairs and Services (OSAS) of LSPU-SCC for student organization accreditation.

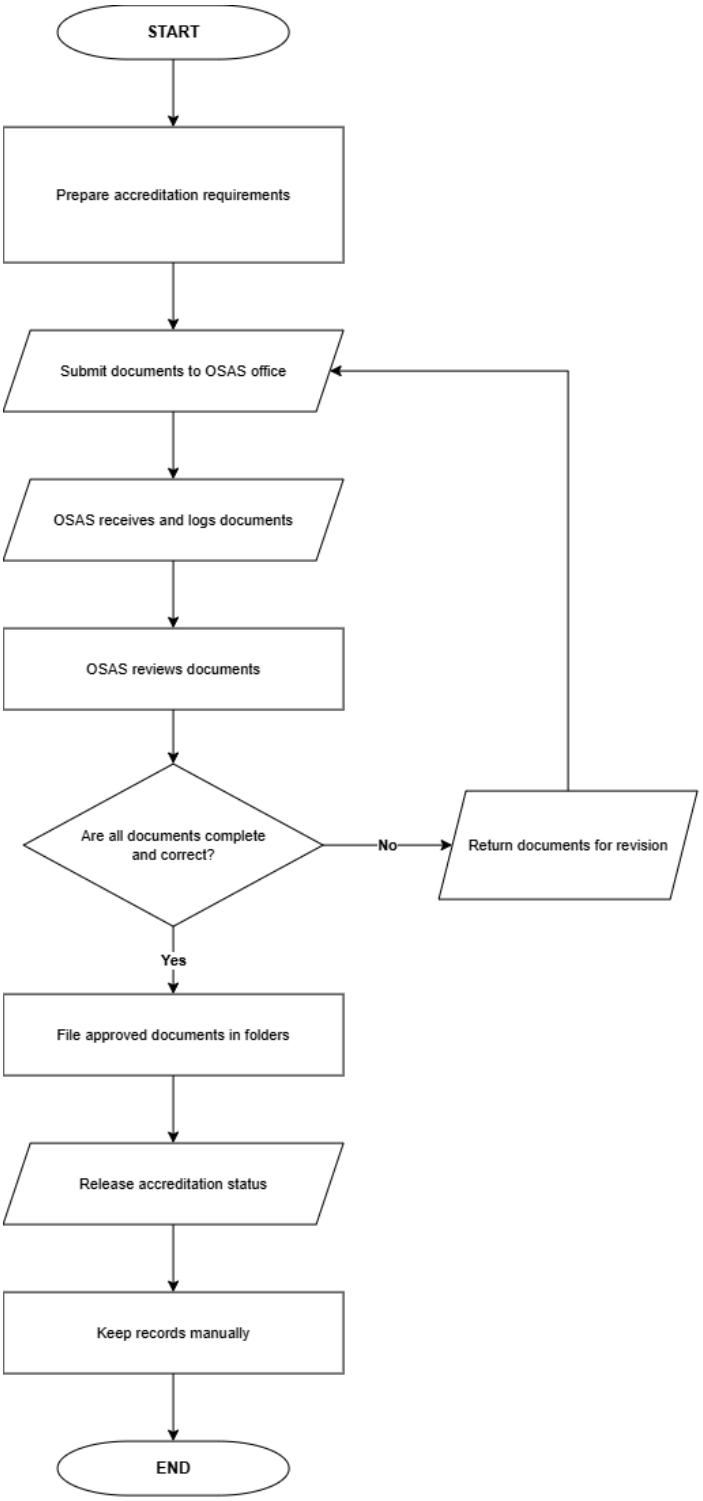


Figure 66. This figure shows the current process flowchart of the OSAS at LSPU-SCC.

Proposed Process Flowchart

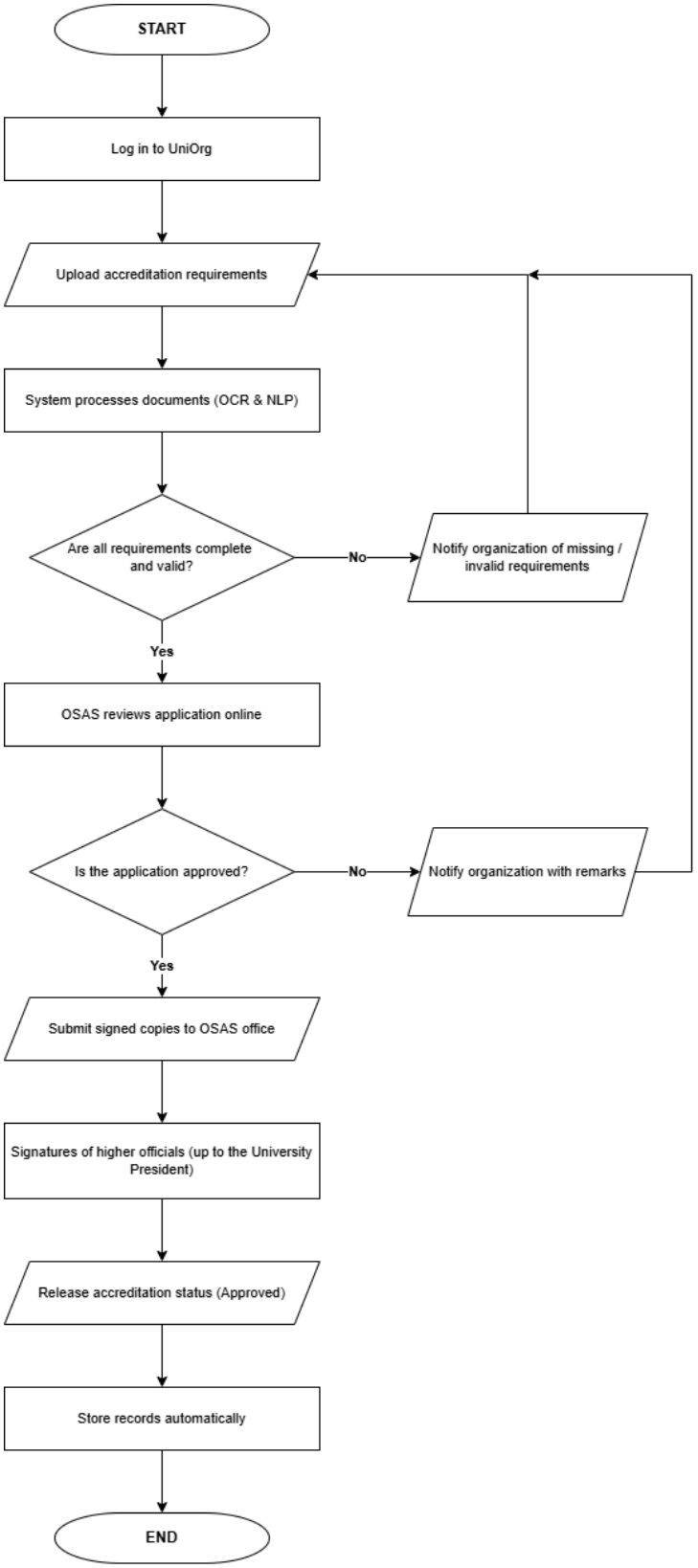
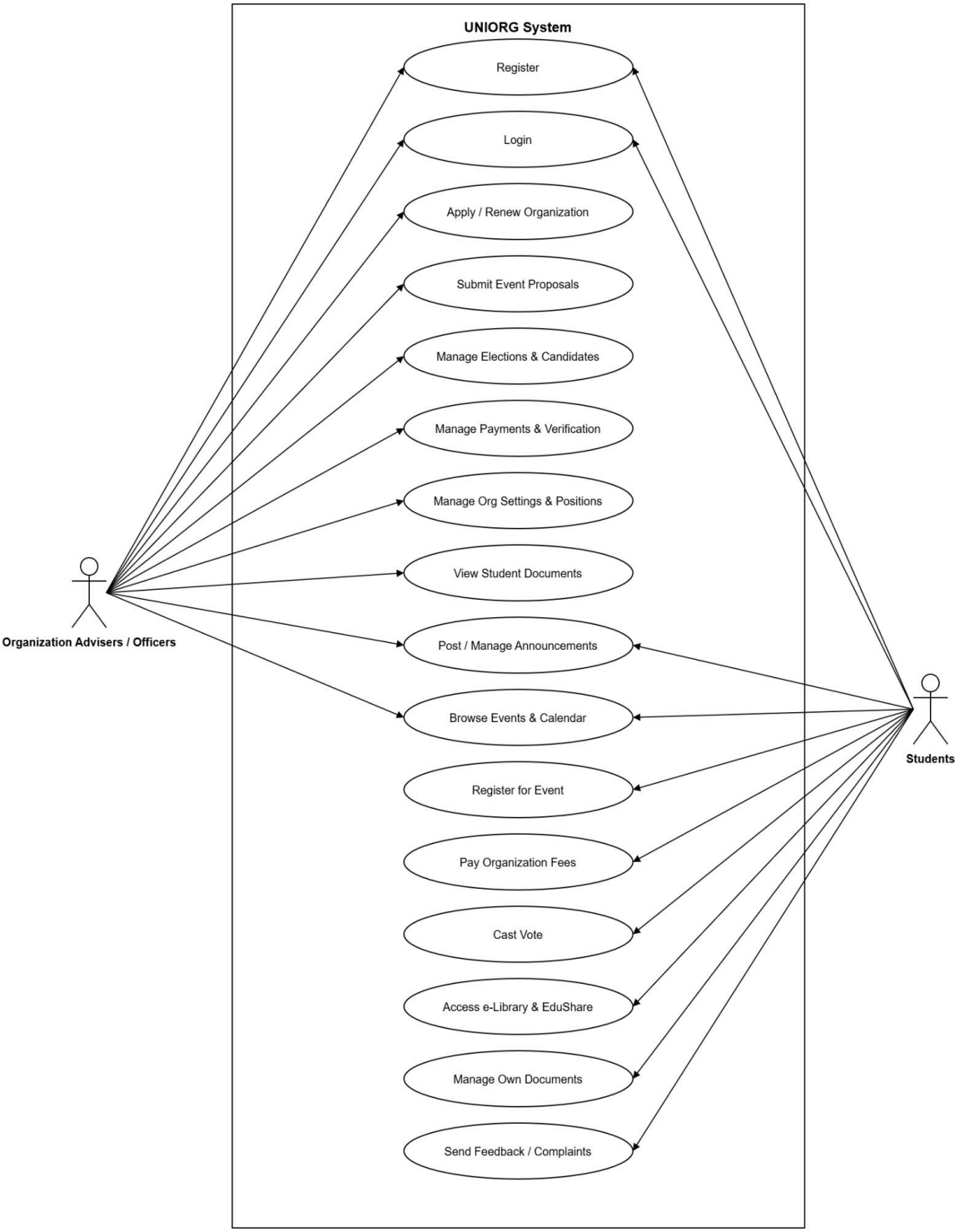


Figure 67. This figure shows the proposed process flowchart for the Organization accreditation of the OSAS, LSPU-SCC.

Use Case Diagram

The diagram shows the use case diagram of the system and demonstrates the process and who manage its process.



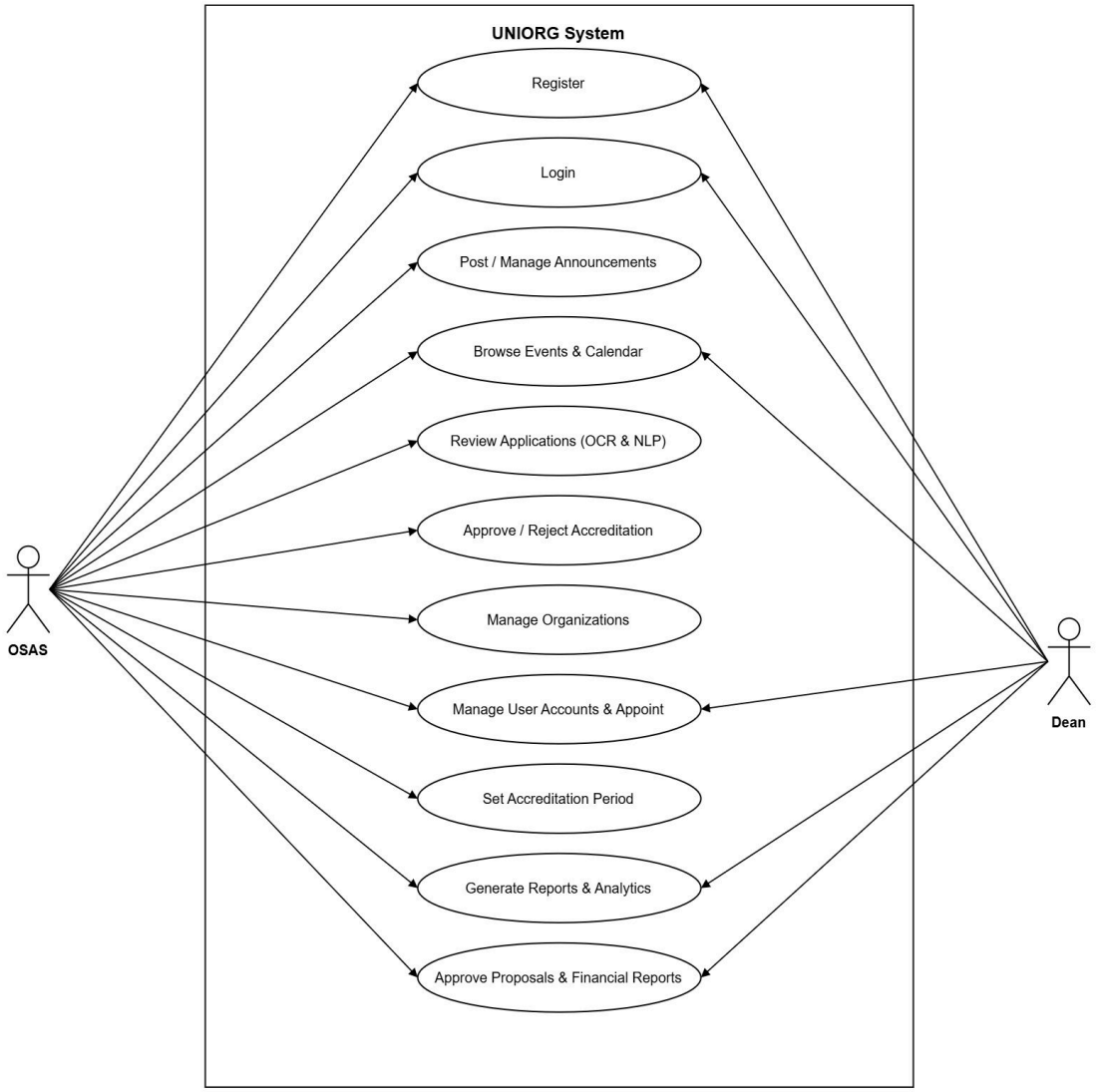


Figure 68. This figure shows the proposed use case diagram for the UniOrg Platform of LSPU-SCC

System Context Diagram

The diagram shows the context of the UNIORG Platform and the users who interact with the system.

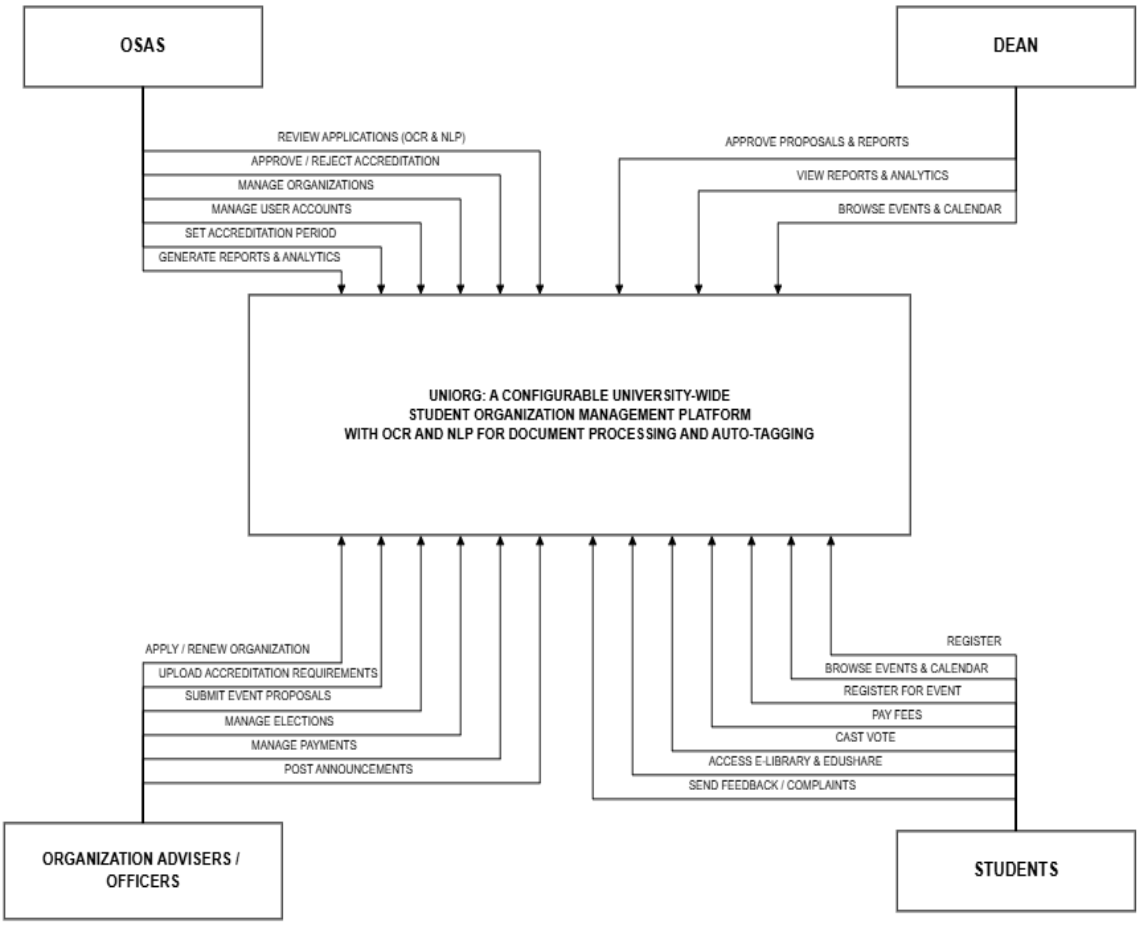


Figure 69. This figure shows the system context diagram of the UNIORG Platform showing its main users and their interactions.

Data Flow Diagram

1.0 User Account & Profile Management

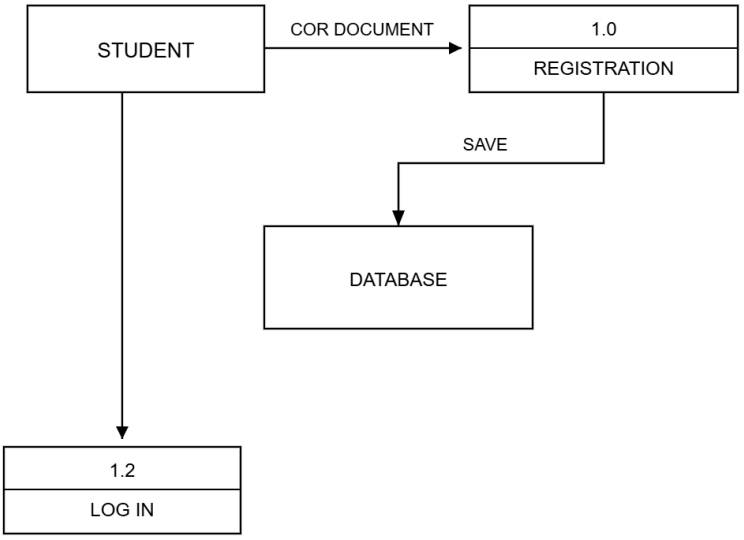


Figure 70. 2.0 Announcements System

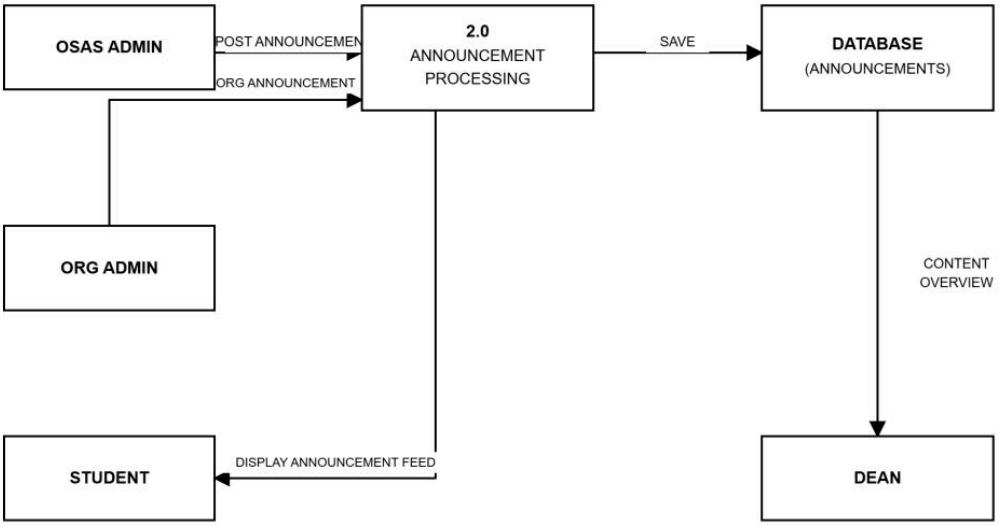


Figure 71. 3.0 Events & Master Calendar

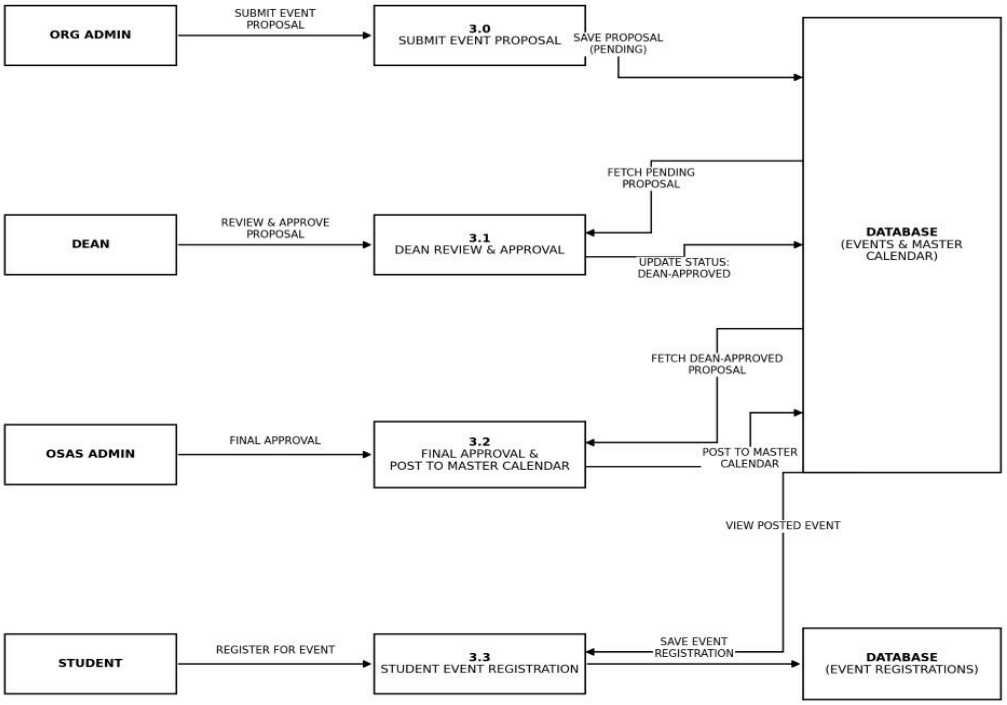
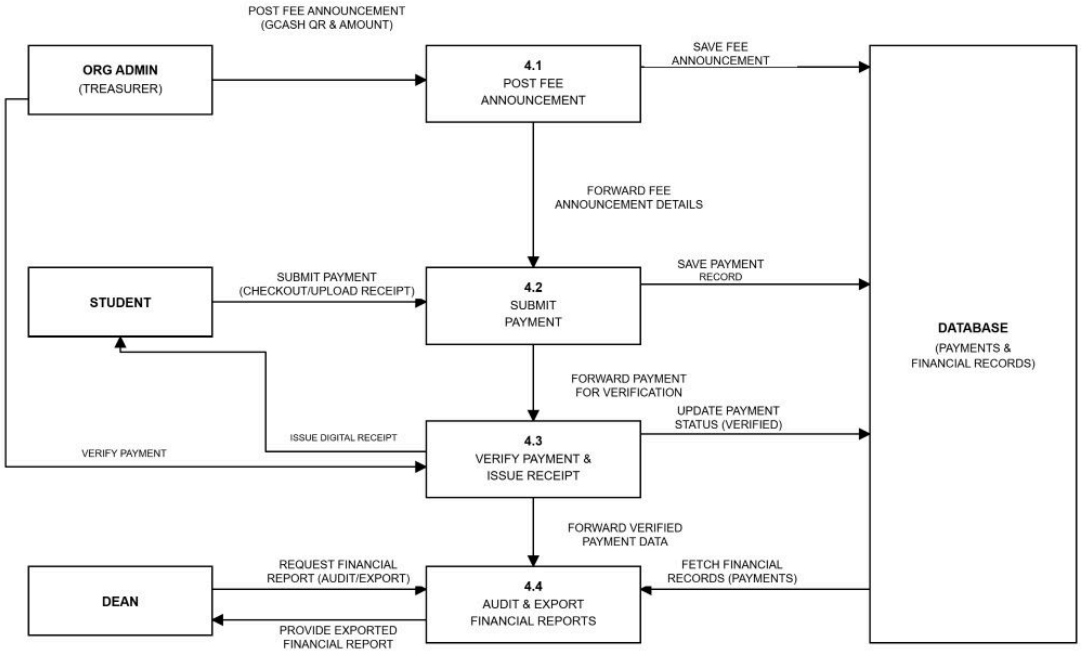


Figure 72. 4.0 Payments & Financial Management



5.0 Elections & Automated Voting

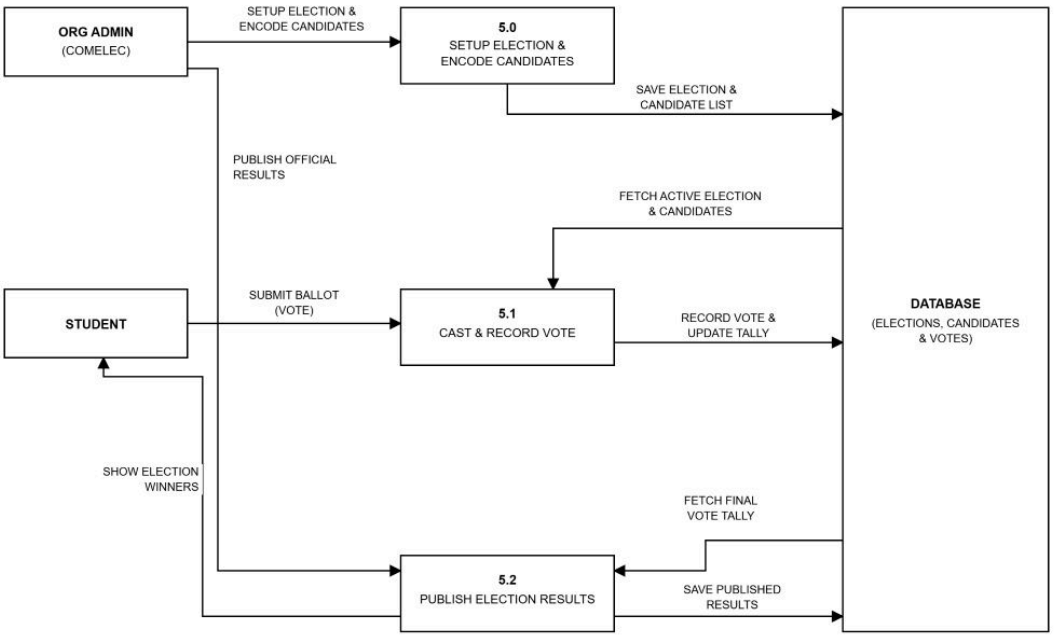
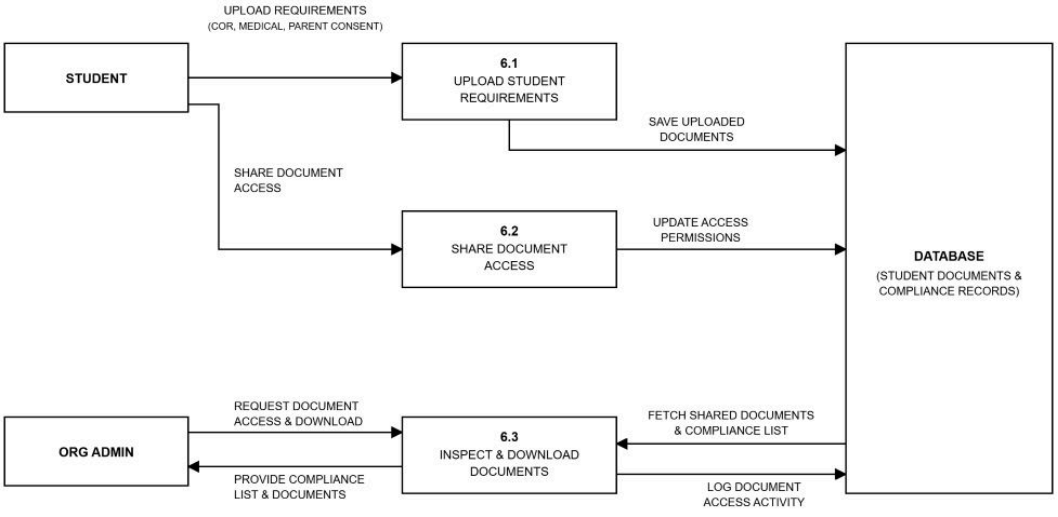
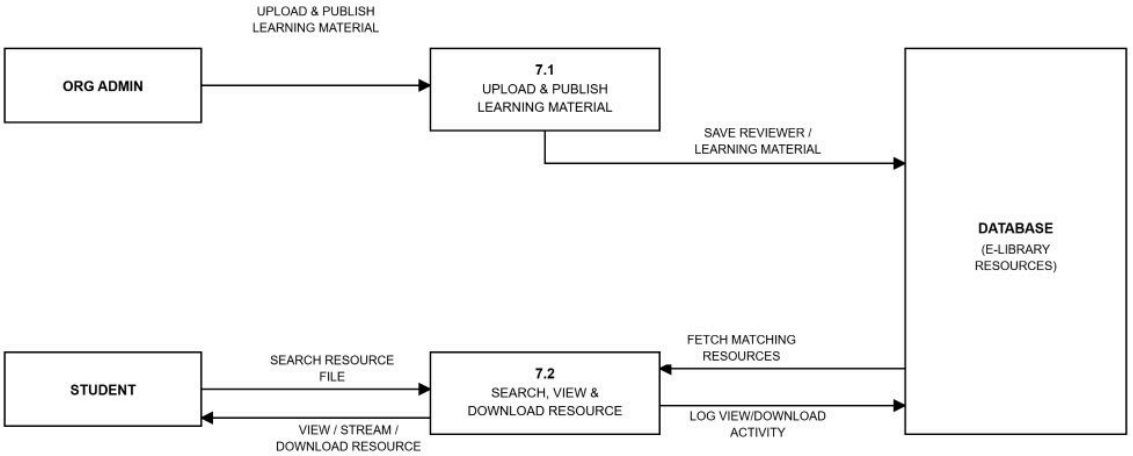


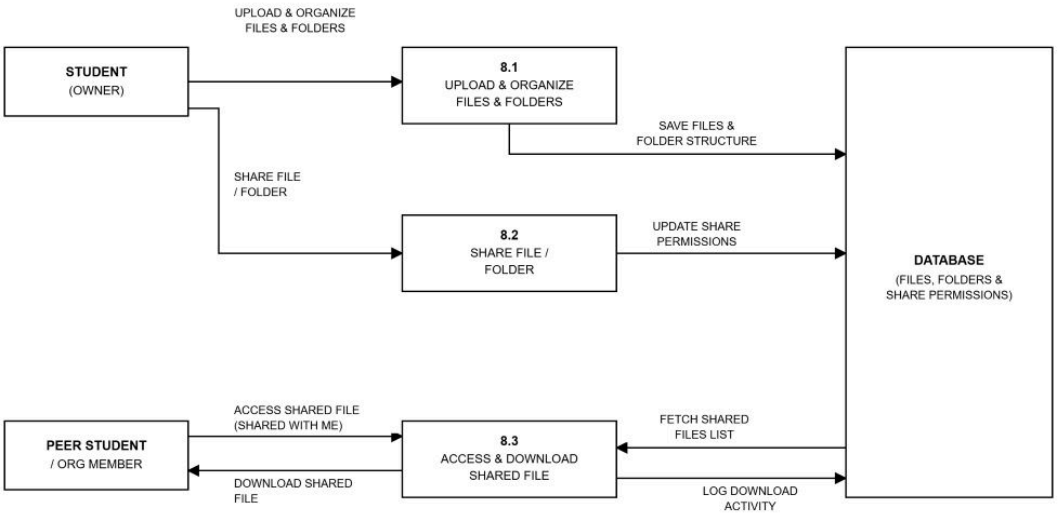
Figure 73. 6.0 Student Document Vault & Compliance



7.0 e-Library Repository



8.0 EduShare



9.0 Complaints

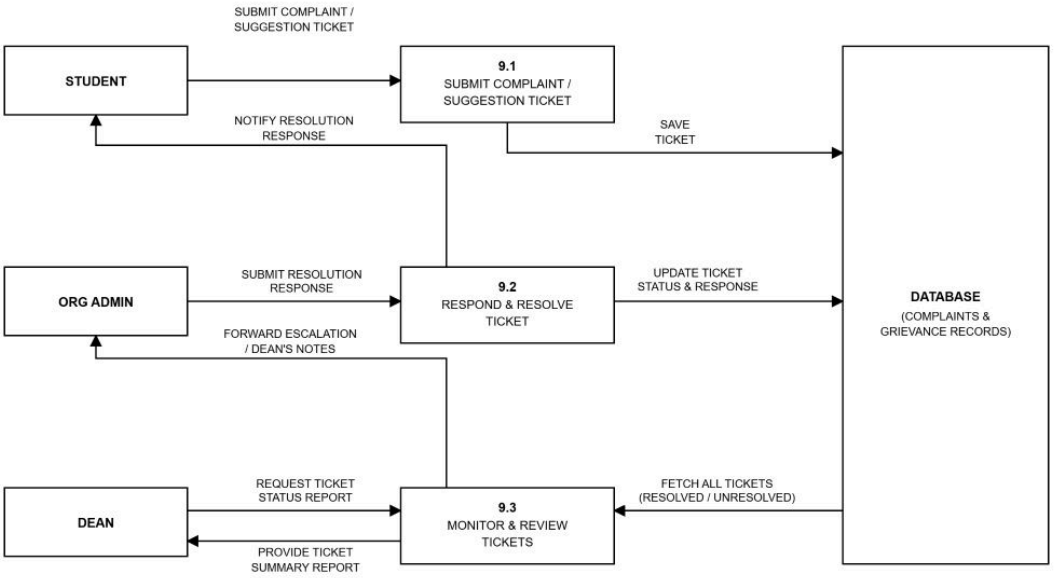
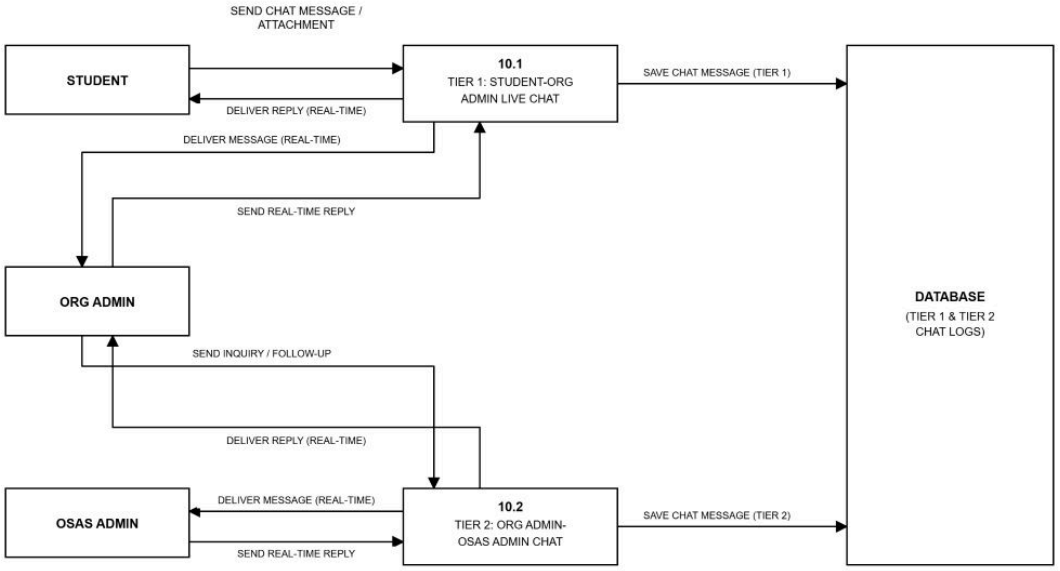
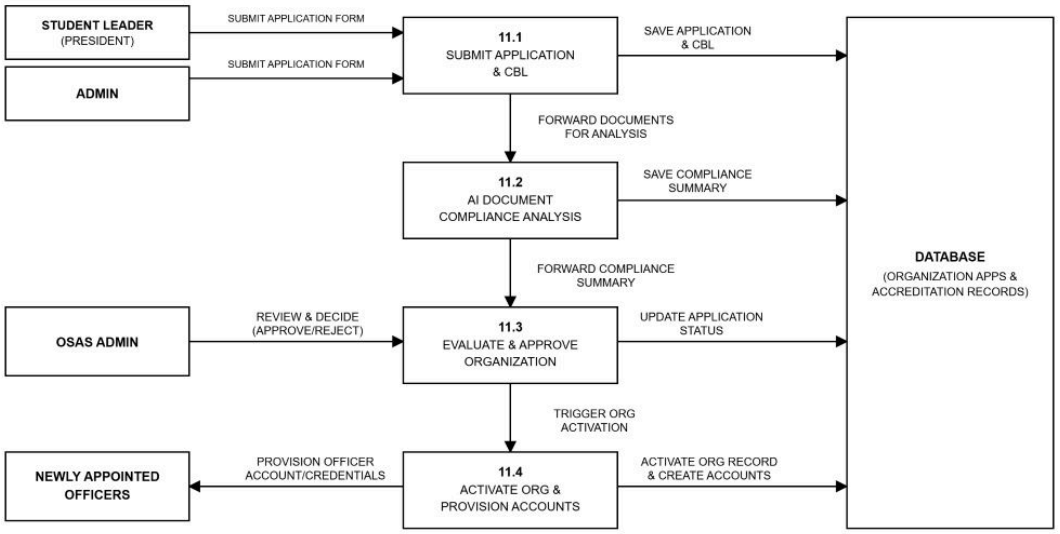


Figure 74. 10.0 Chat



11.0 Organization Accreditation & Renewal (Apply Org)



Hierarchical Input Process Output Model

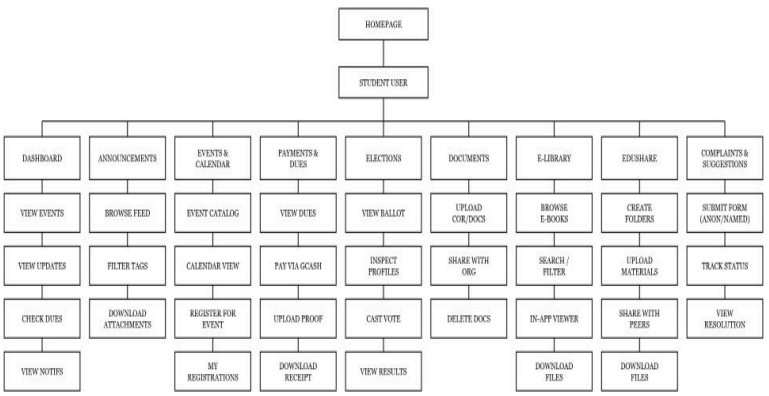


Figure 75. Hierarchical input-process-output model for the Student user.

This figure illustrates the hierarchy of system modules, functional processes, and input-output interactions available to enrolled students within the UniOrg Web Application.

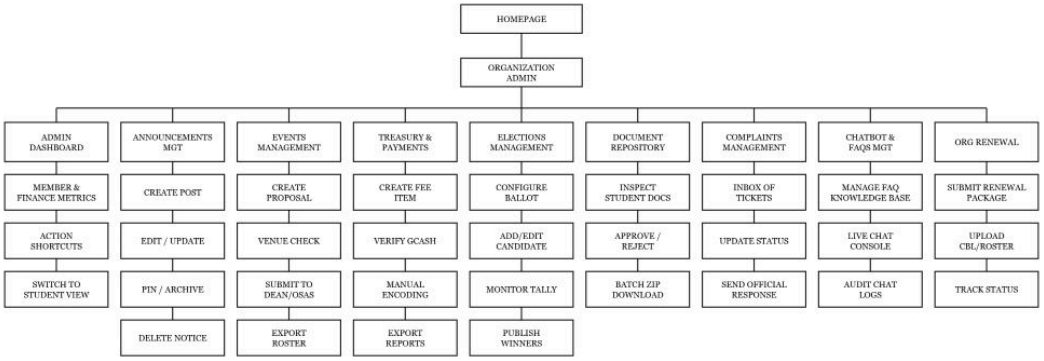


Figure 76. Hierarchical input-process-output model for the Organization Admin user.

This figure illustrates the hierarchy of administrative modules, management processes, and input-output operations available to student leaders and organization officers in the UniOrg Web Application.

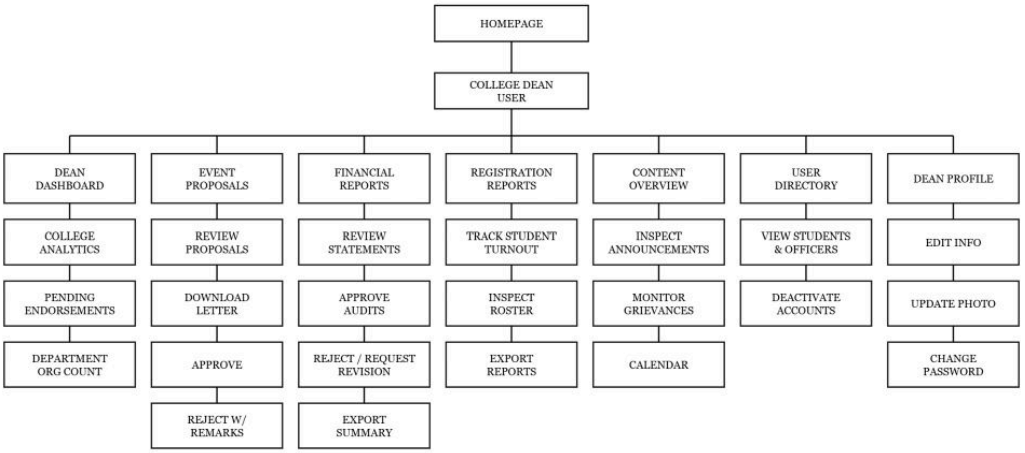


Figure 77. Hierarchical input-process-output model for the Dean user.

This figure illustrates the hierarchy of supervisory modules, approval mechanisms, and input-output operations available to College Deans and Academic Heads in the UniOrg Web Application.

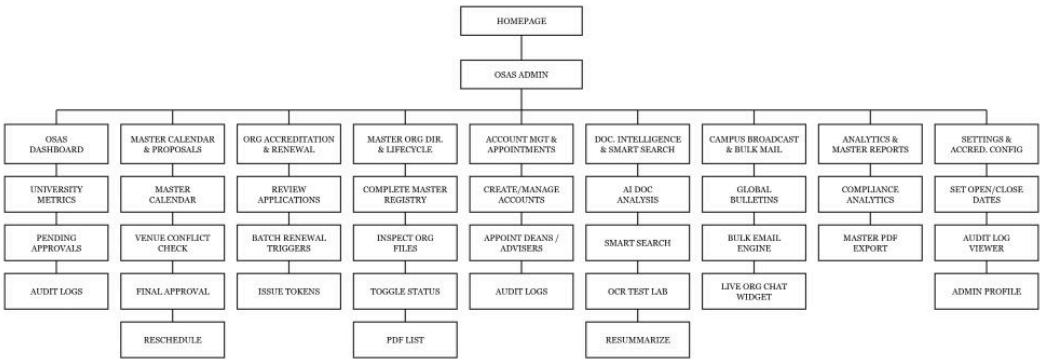


Figure 78. Hierarchical input-process-output model for the OSAS Administrator user.

This figure illustrates the hierarchy of university-level administrative modules, governance processes, and input-output operations available to the Office of Student Affairs and Services in the UniOrg Web Application.

Project Schedule

This shows the Gantt chart which presents the project schedule. It helps plan, coordinate and track the tasks and phases in developing the system.

2026									
Activity	March	April	May	June	July	Aug	Sept	Oct	Nov
Preliminaries									
Coordination with OSAS									
Project Planning									
Data Gathering									
Documentation									
System Development									
Designing System Interfaces									
Coding									
Testing									
System Testing									
Error Detection									
User Evaluation									

Figure 79. Gantt Chart

User Testing and Feedback									
Assessment of Results									
Bug Fixing and Final Revision									

Figure shows the project schedule created by the researchers to guide them about the next steps they took and to help track the tasks from the preliminary phase up to system development, testing and user evaluation.

Hardware and Software Resources

This part presents the hardware and software resources used in the development of the UniOrg Platform.

Software Resources

A. UNIORG Platform Software Resources

- Hyper Text Markup Language (HTML)
- Cascading Style Sheets (CSS)
- JavaScript
- PHP 8.4
- Laravel Framework (11.0)
- Supabase PostgreSQL

- AJAX
- Vite
- Node.js / NPM
- Composer
- Visual Studio Code
- Windows OS
- Git / GitHub

Application Programming Interface (API)

- Google Cloud Vision API
- Google Cloud Natural Language API
- Groq API
- Pusher API
- Supabase API
- Gmail API

External Libraries

- Supabase
- Cloudflare R2 Storage
- Laravel DomPDF
- PHPSpreadsheet
- Tesseract OCR
- Smalot PDF Parser
- Spatie PDF to Image

- FullCalendar
- Phosphor Icons
- Laravel Echo with Pusher.js

Hardware Resources

- Desktop / Laptop/ Mobile Phone
- Processor Intel Core i3 or higher
- RAM 4GB or higher
- Storage 256GB or higher
- Stable Internet Connection

Input/Output

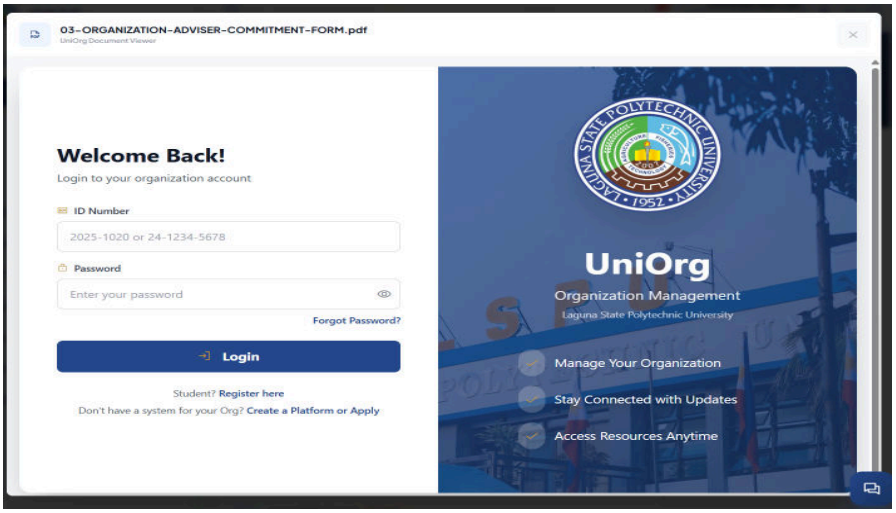


Figure 80. This figure shows the login page of UniOrg.

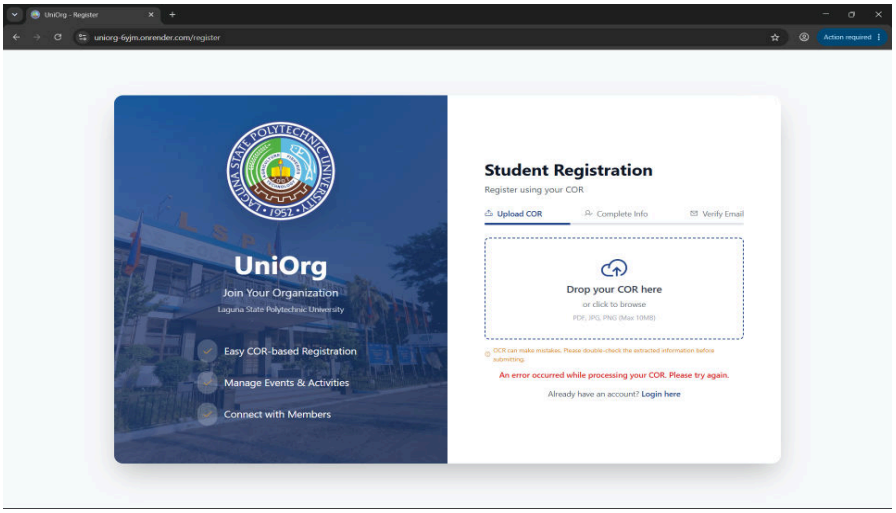


Figure 81. This figure shows the Upload COR step of student registration in UniOrg.

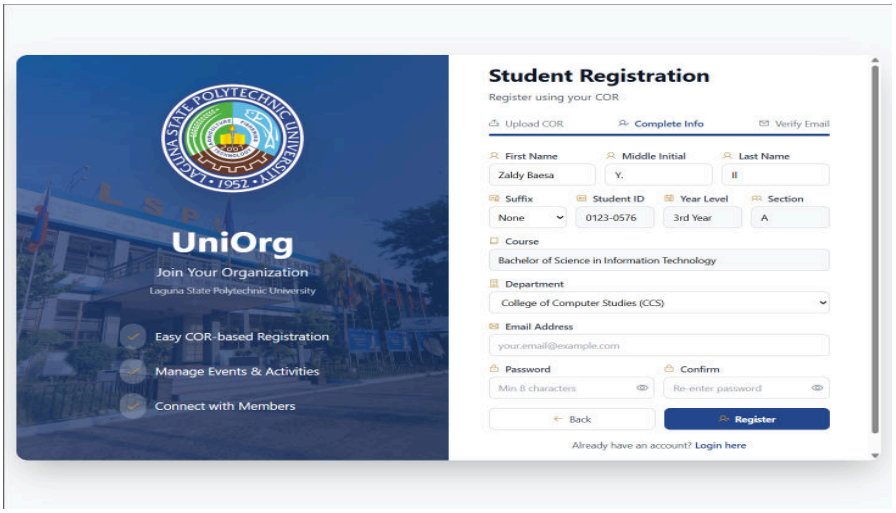


Figure 82. This figure shows the Complete Info step of student registration in UniOrg.

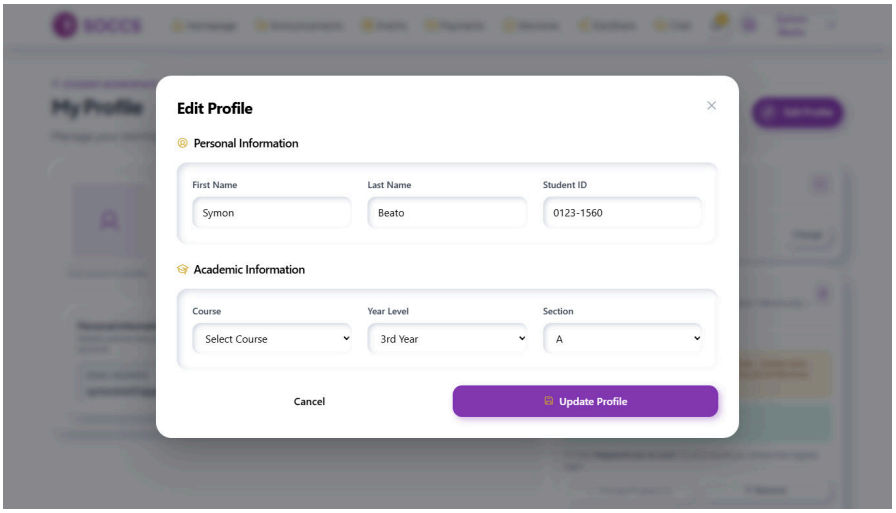


Figure 83. This figure shows the student profile edit modal of UniOrg.

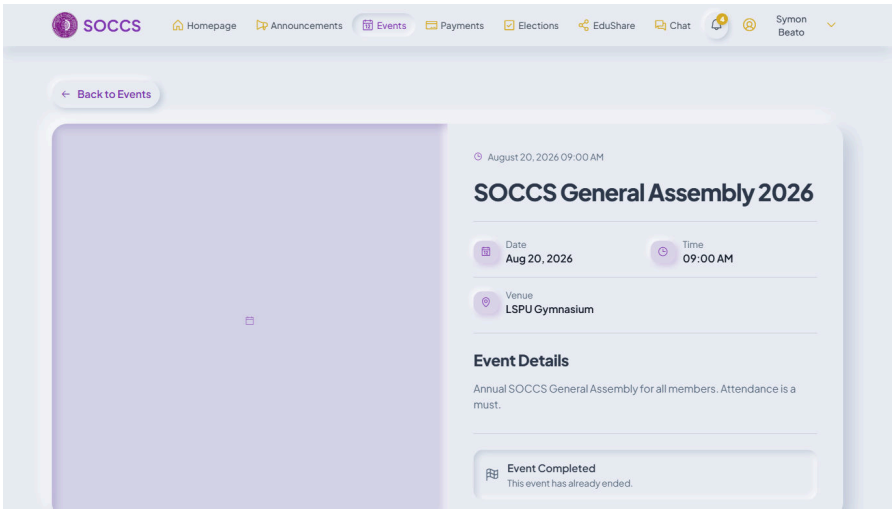


Figure 84. This figure shows the event registration modal of UniOrg.

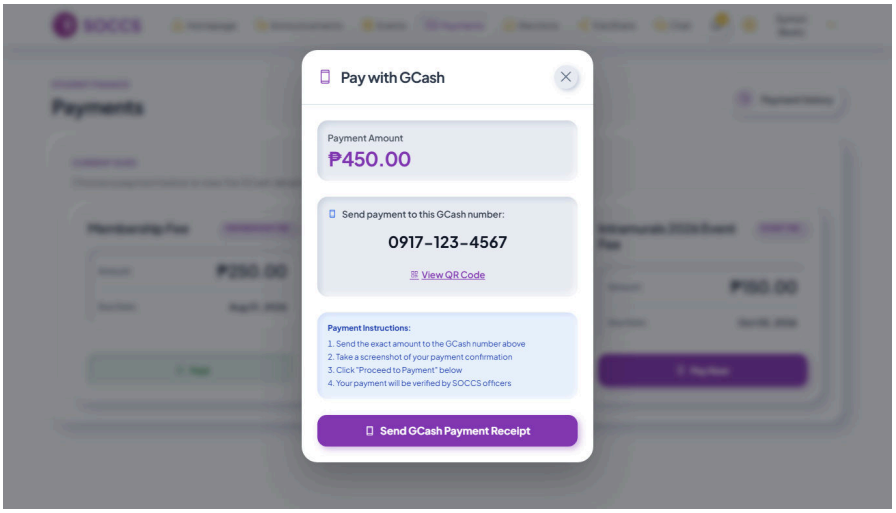


Figure 85. This figure shows the GCash payment submission of UniOrg.

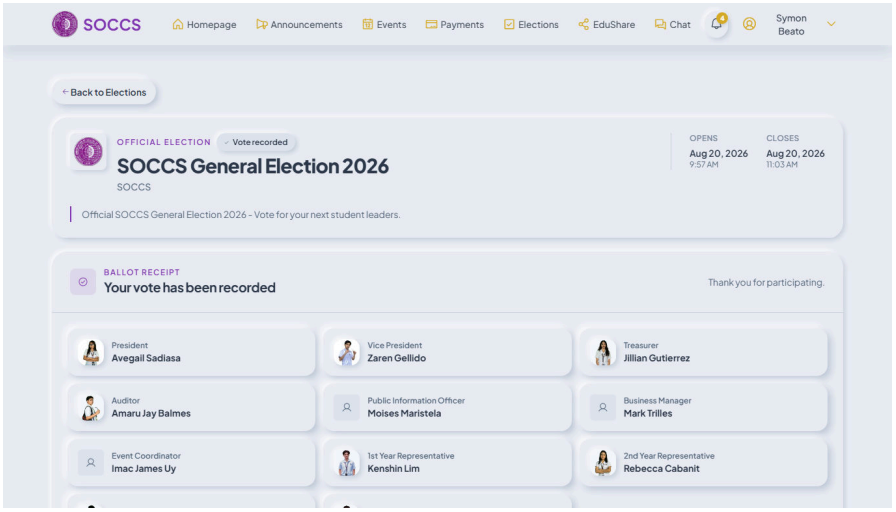


Figure 86. This figure shows the election vote casting modal of UniOrg.

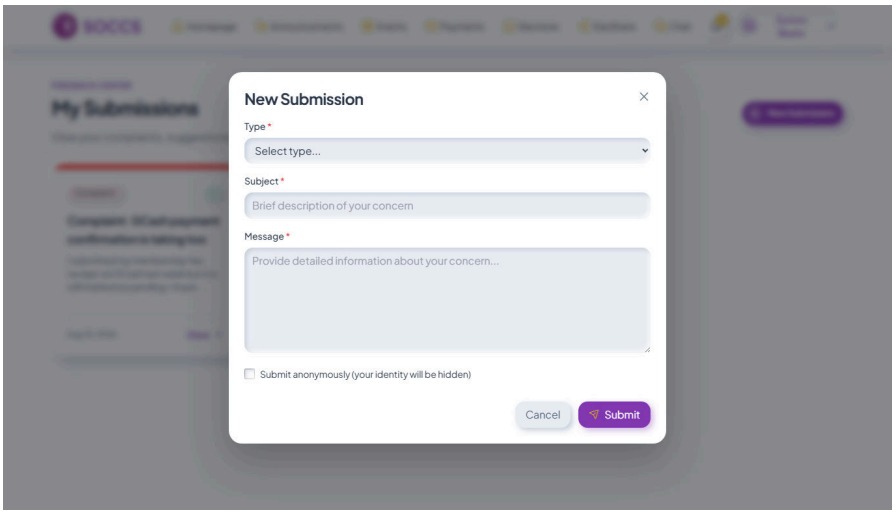


Figure 87. This figure shows the new complaint submission modal of UniOrg.

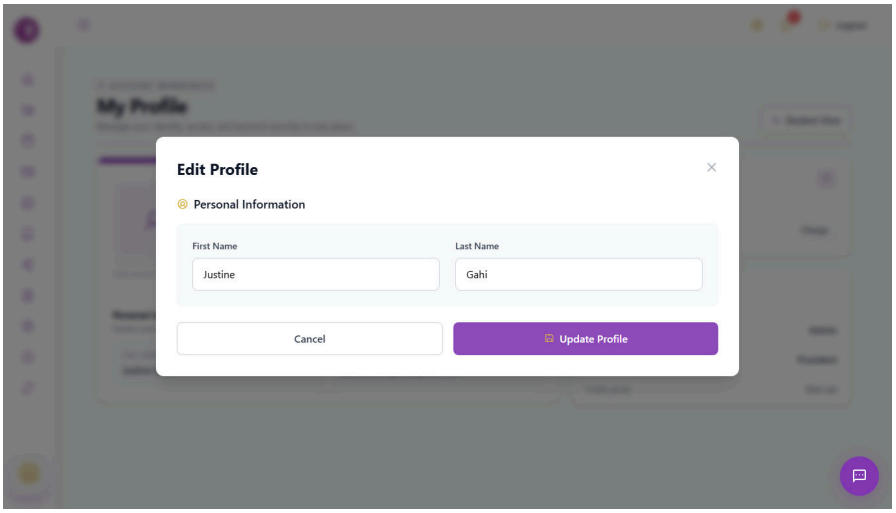


Figure 88. This figure shows the organization admin profile edit modal of UniOrg.

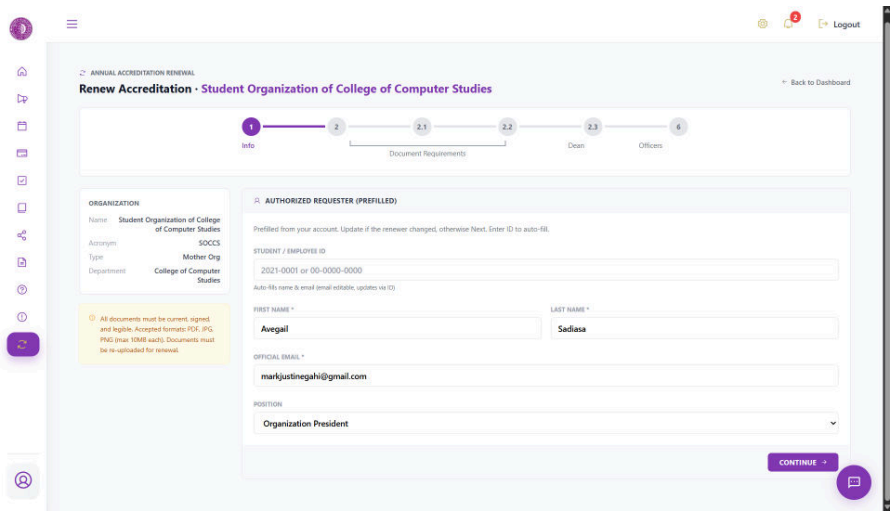


Figure 89. This figure shows the accreditation renewal requester information form of UniOrg.

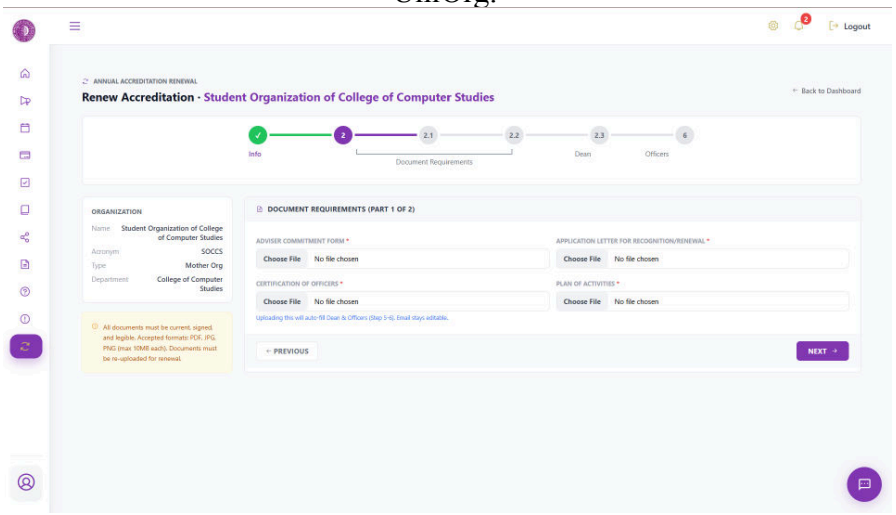


Figure 90. This figure shows the accreditation document requirements part 1 of UniOrg.

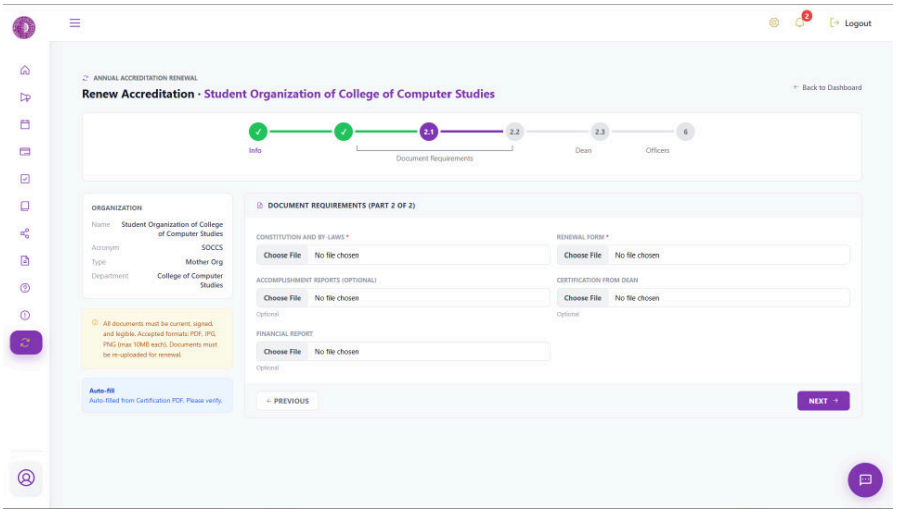


Figure 91. This figure shows the accreditation document requirements part 2 of UniOrg.

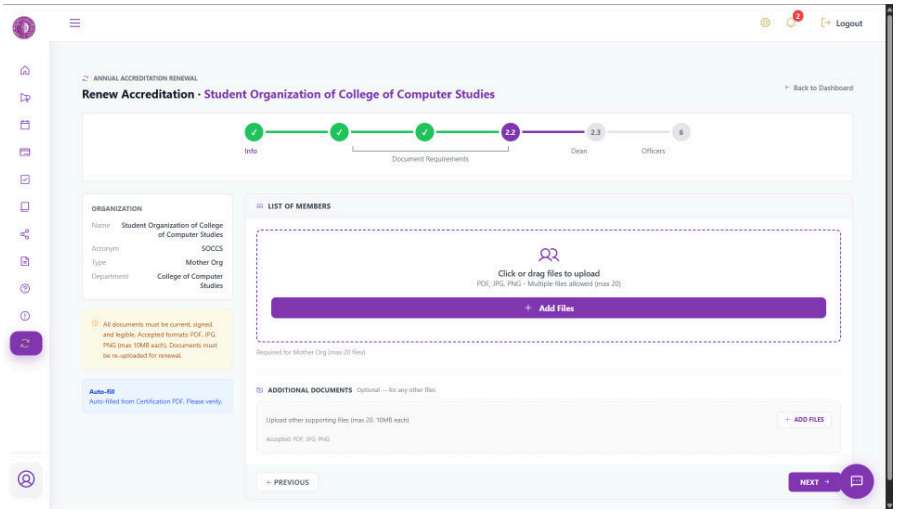


Figure 92. This figure shows the members list upload step of accreditation renewal in UniOrg.

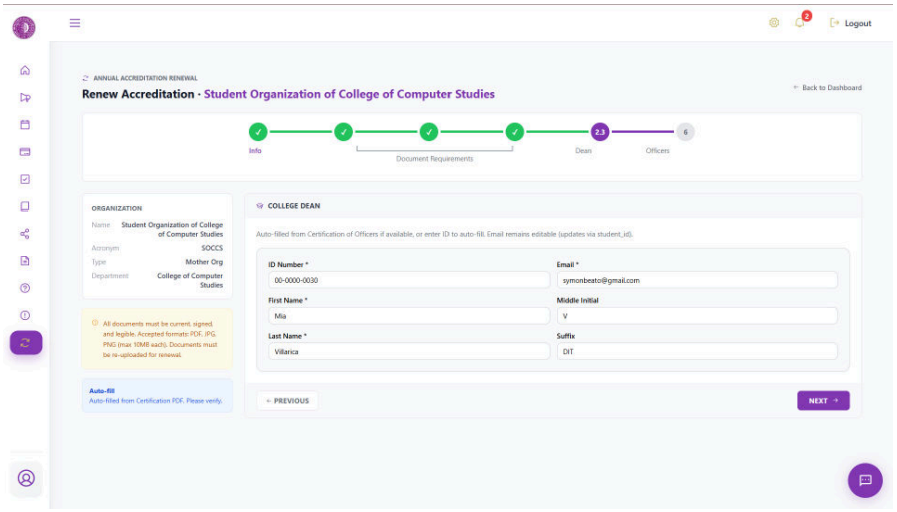


Figure 93. This figure shows the college dean form step of accreditation renewal in UniOrg.

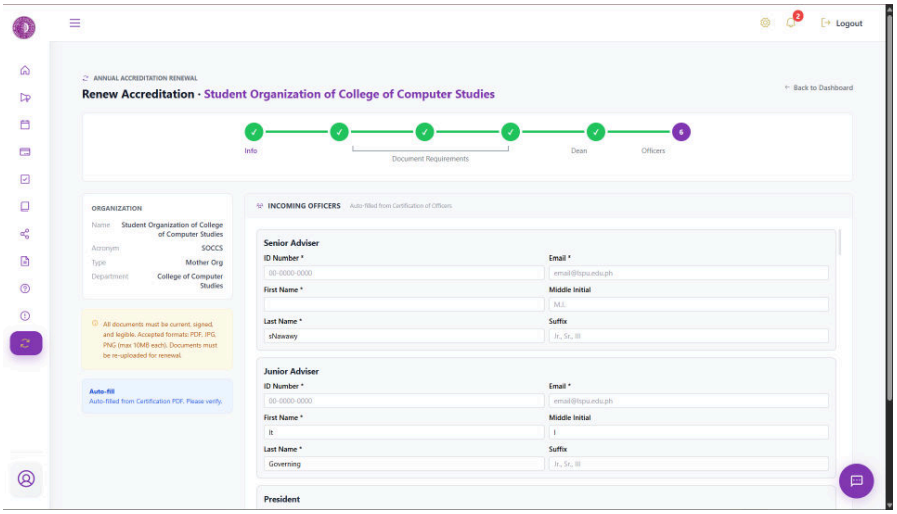


Figure 94. This figure shows the incoming officers step of accreditation renewal in UniOrg.

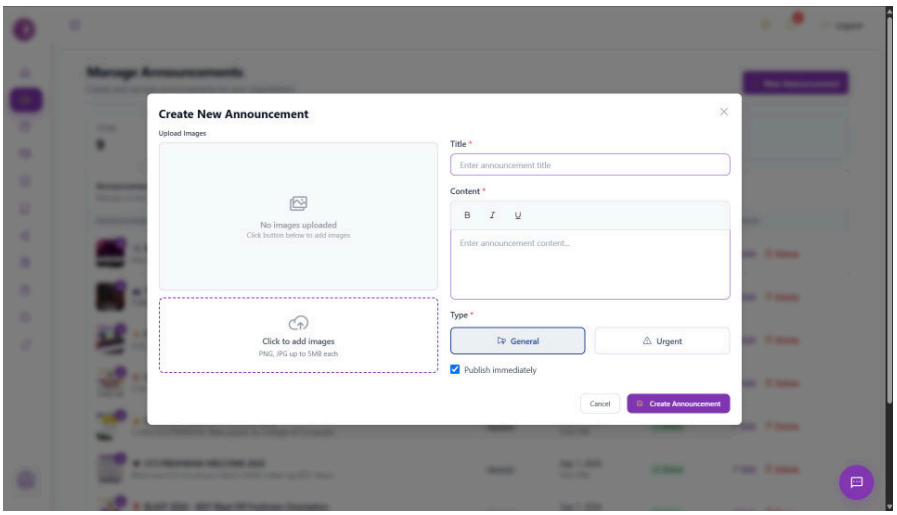


Figure 95. This figure shows the announcement creation modal of UniOrg.

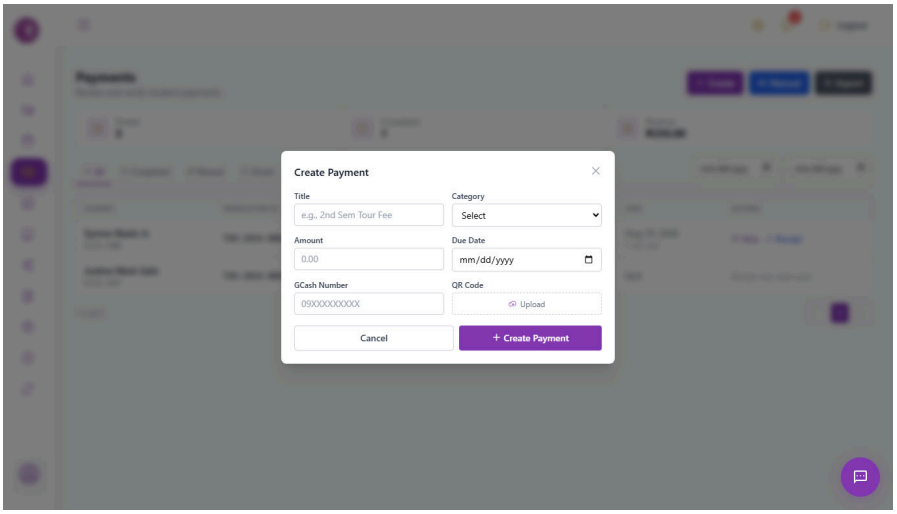


Figure 96. This figure shows the payment announcement creation of UniOrg.

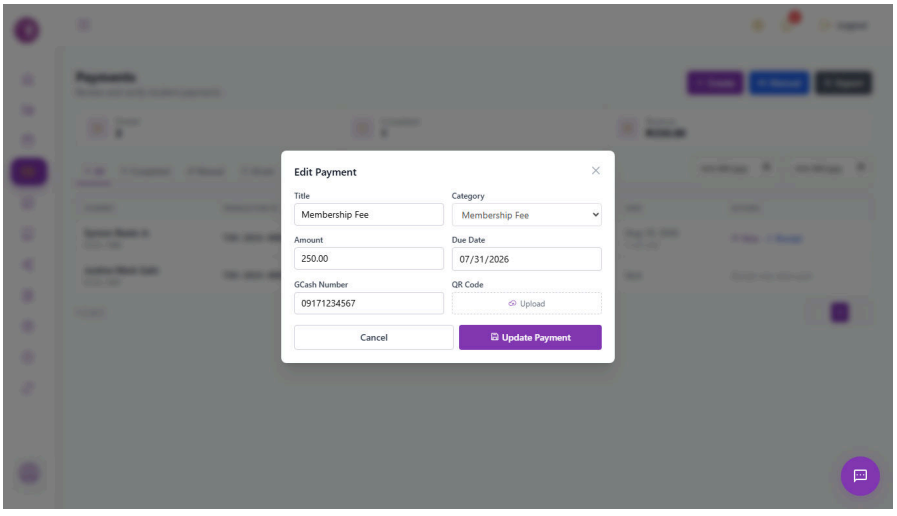


Figure 97. This figure shows the payment announcement edit of UniOrg.

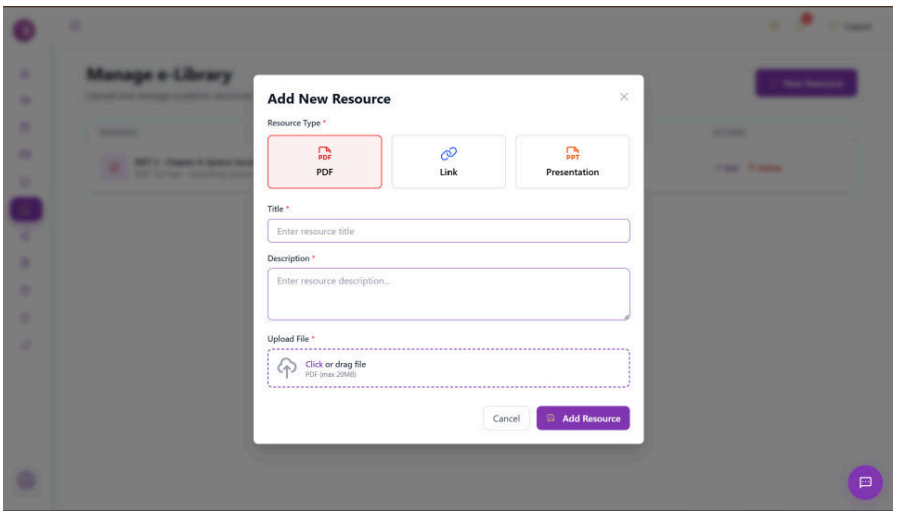


Figure 98. This figure shows the e-library resource upload modal of UniOrg.

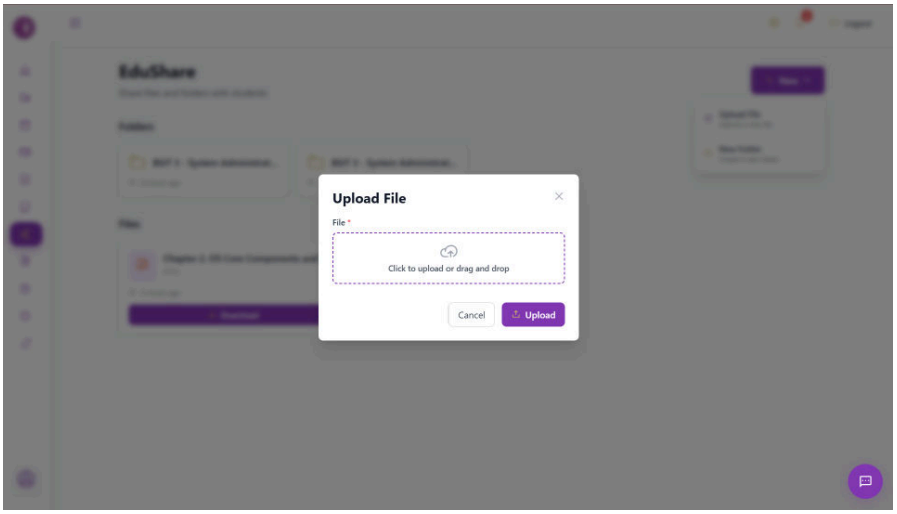


Figure 99. This figure shows the EduShare file upload modal of UniOrg.

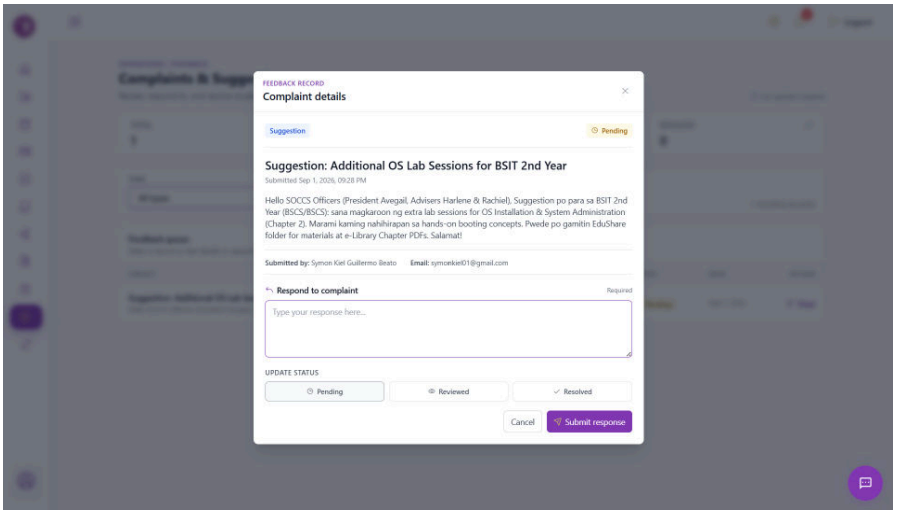


Figure 100. This figure shows the complaint response modal of UniOrg.

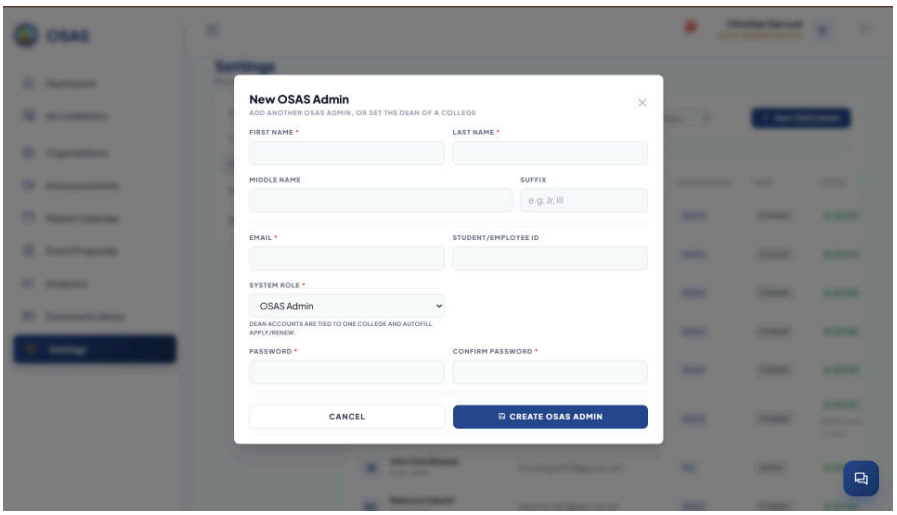


Figure 101. This figure shows the OSAS admin creation modal of UniOrg.

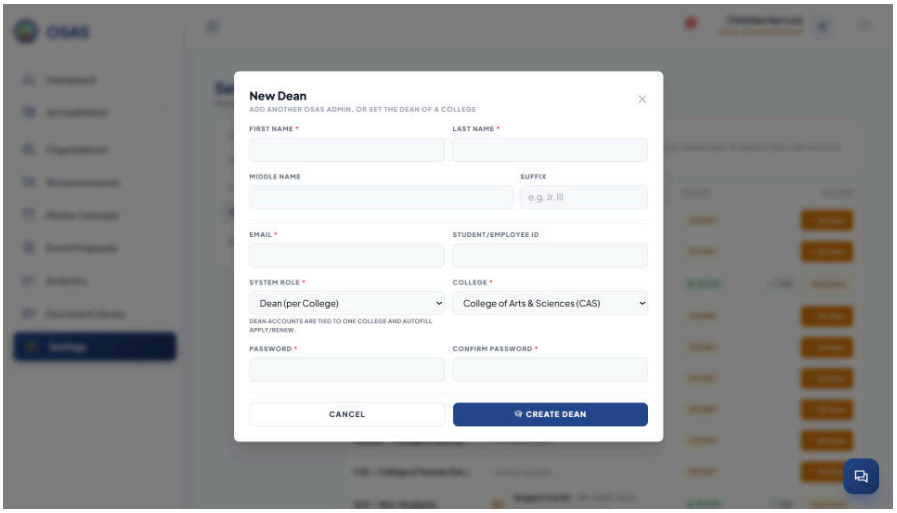


Figure 102. This figure shows the dean account creation modal of UniOrg.

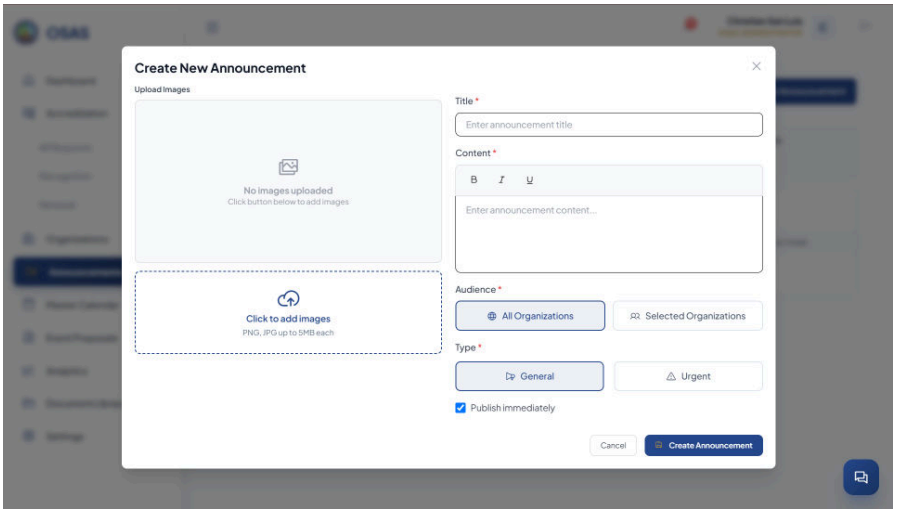


Figure 103. This figure shows the OSAS announcement creation modal of UniOrg.

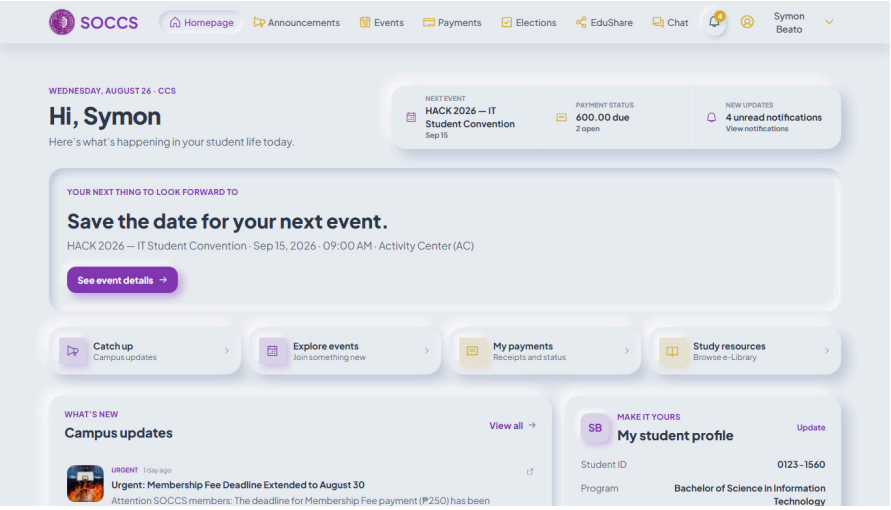


Figure 104. This figure shows the student dashboard of UniOrg.

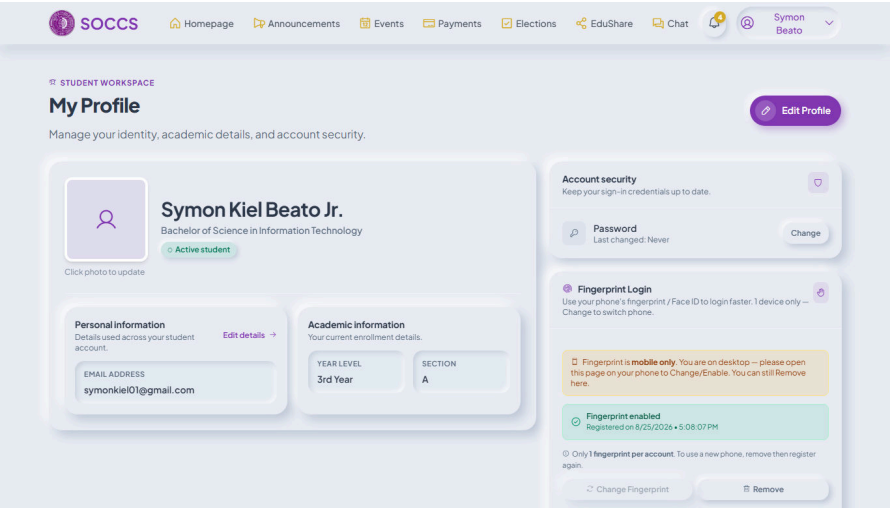


Figure 105. This figure shows the student profile page of UniOrg.

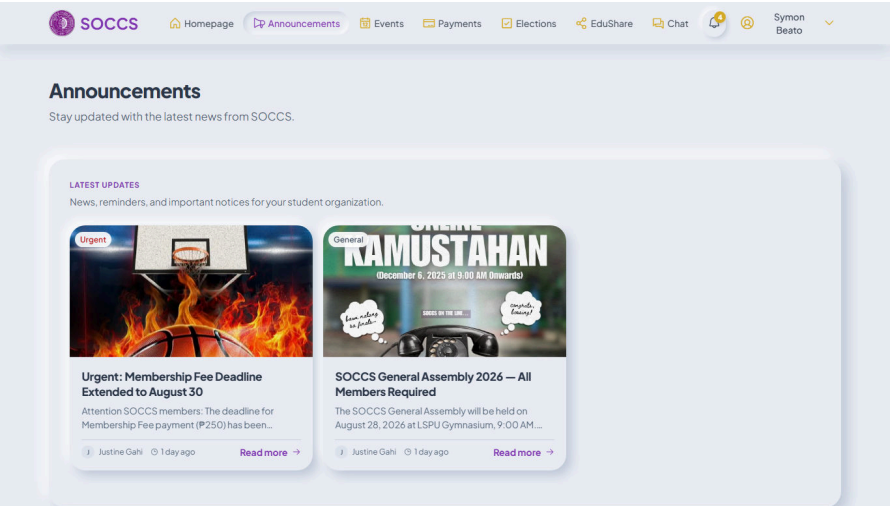


Figure 106. This figure shows the student announcements list of UniOrg.

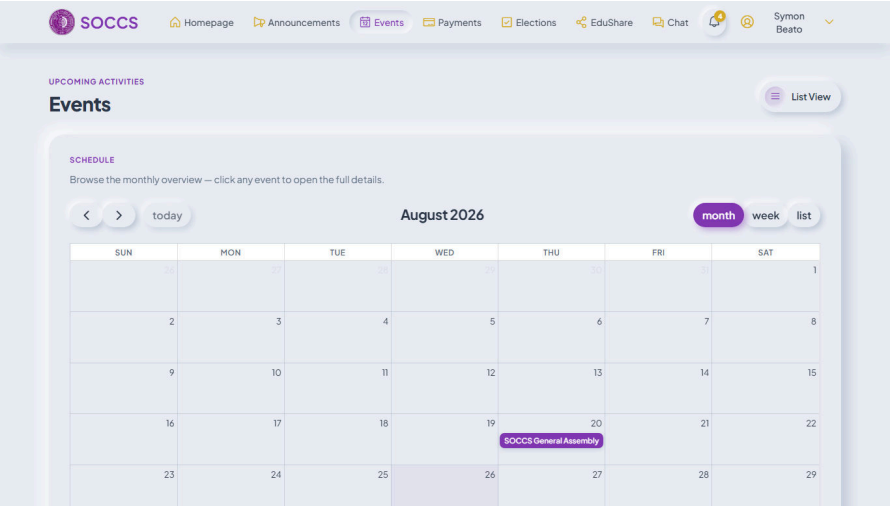


Figure 107. This figure shows the student events list of UniOrg.

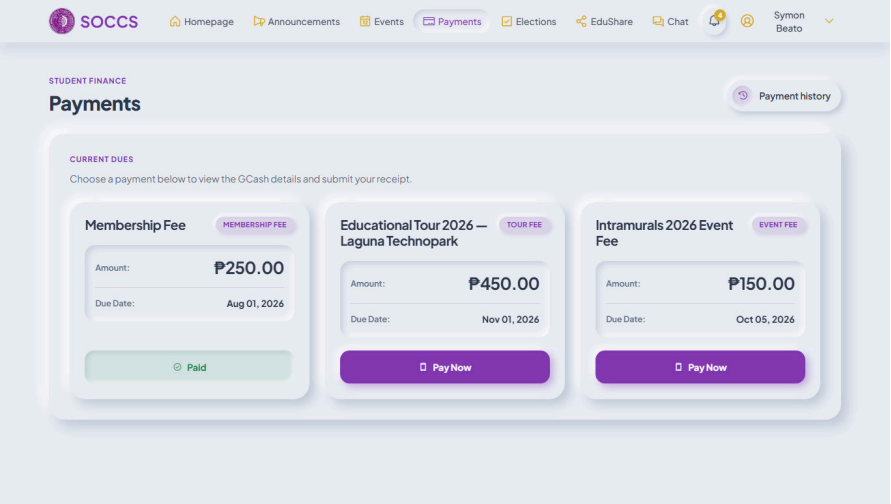


Figure 108. This figure shows the student payments list of UniOrg.

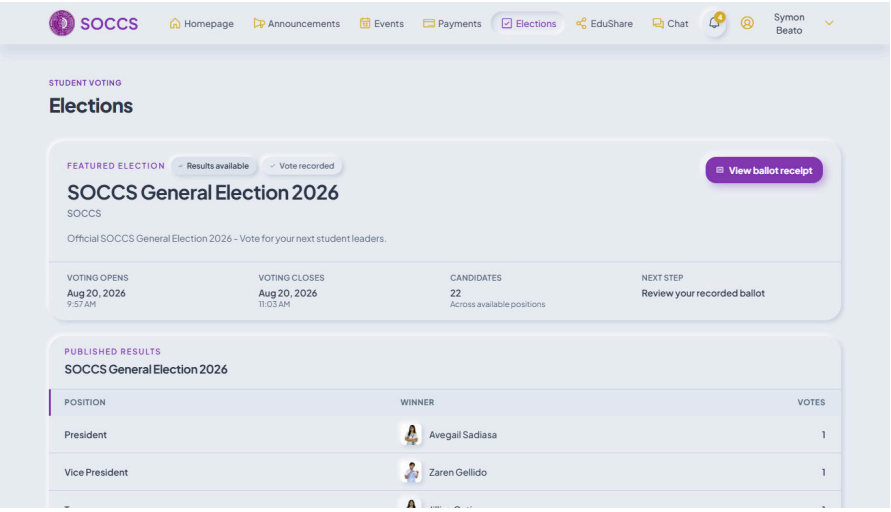


Figure 109. This figure shows the student elections list of UniOrg.

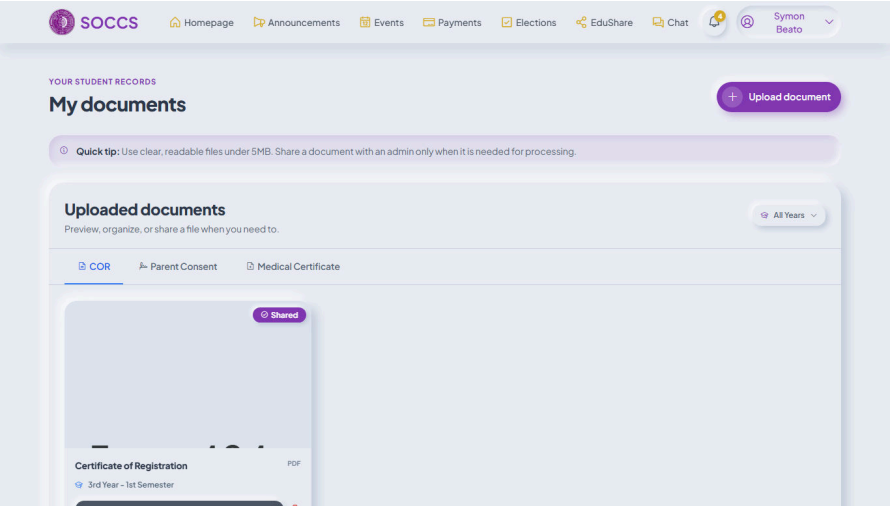


Figure 110. This figure shows the student documents page of UniOrg.

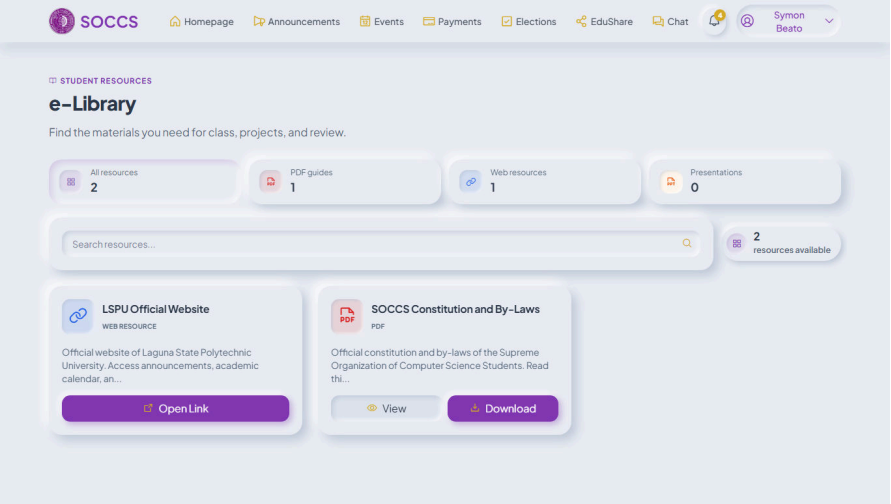


Figure 111. This figure shows the student e-library list of UniOrg.

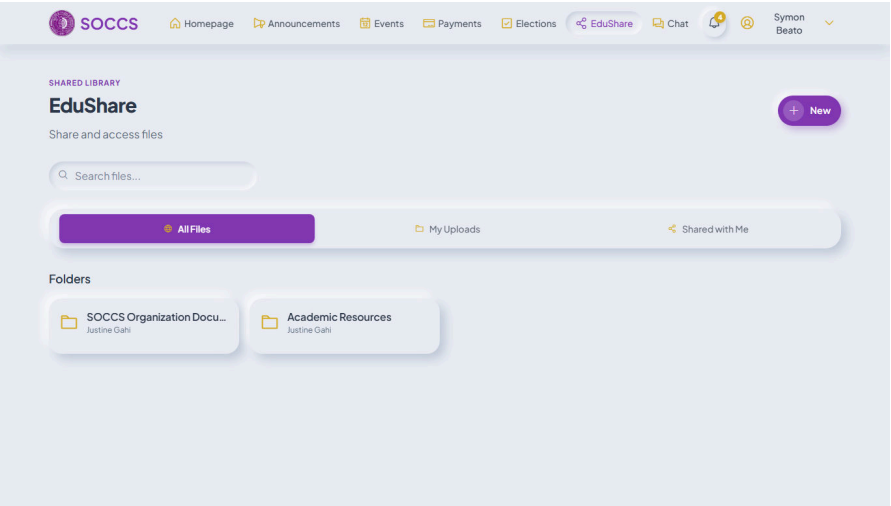


Figure 112. This figure shows the student EduShare list of UniOrg.

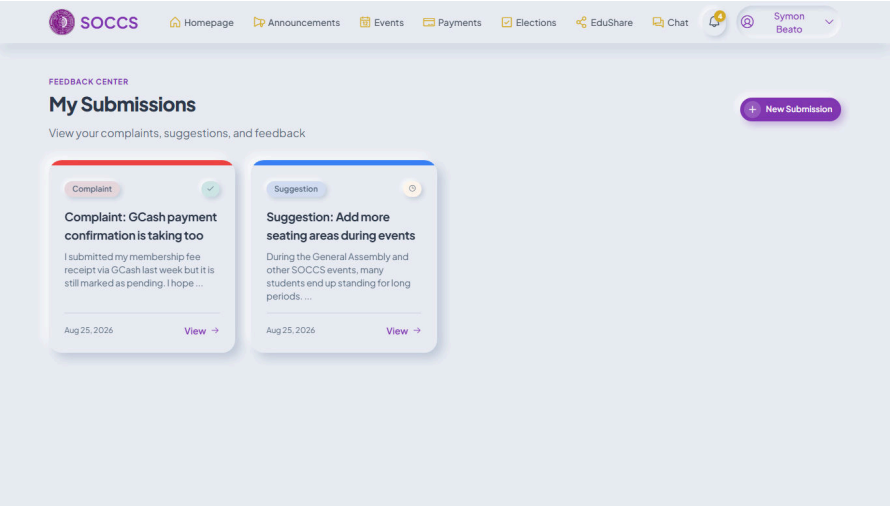


Figure 113. This figure shows the student complaints list of UniOrg.

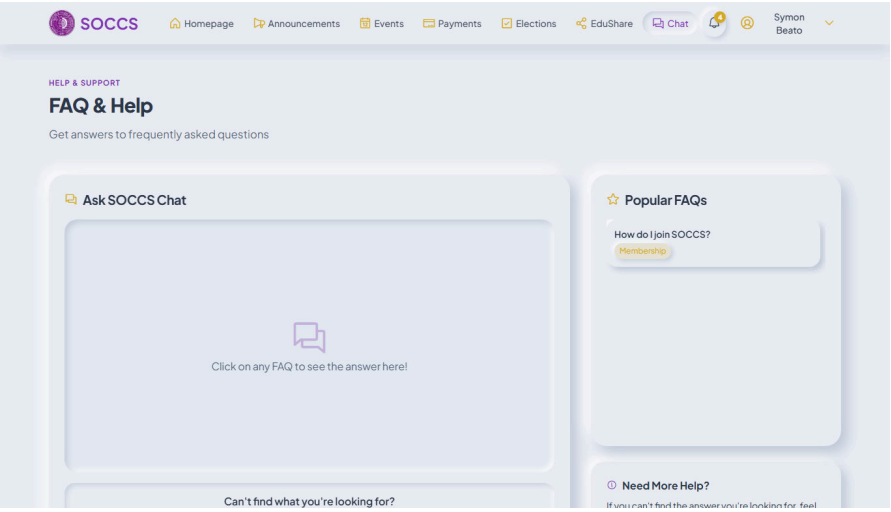


Figure 114. This figure shows the student chatbot of UniOrg.

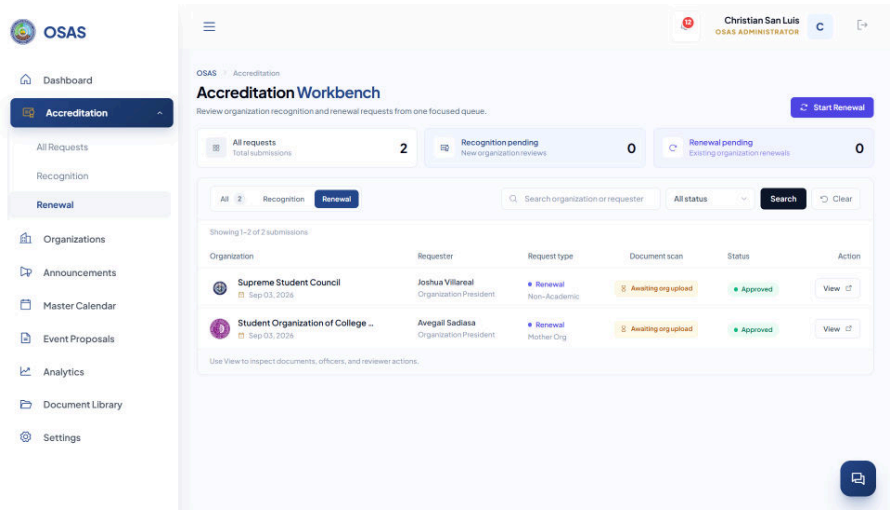


Figure 115. This figure shows the accreditation renewal list of UniOrg.

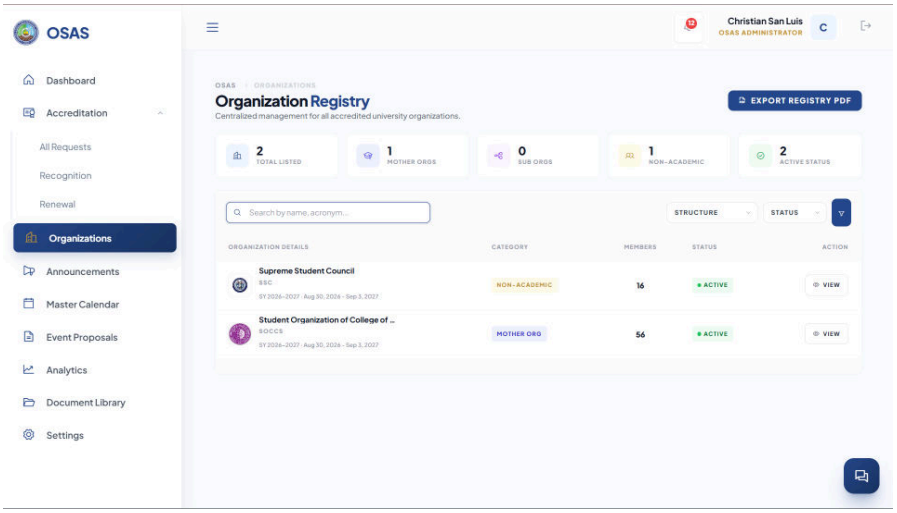


Figure 116. This figure shows the organization registry list of UniOrg.

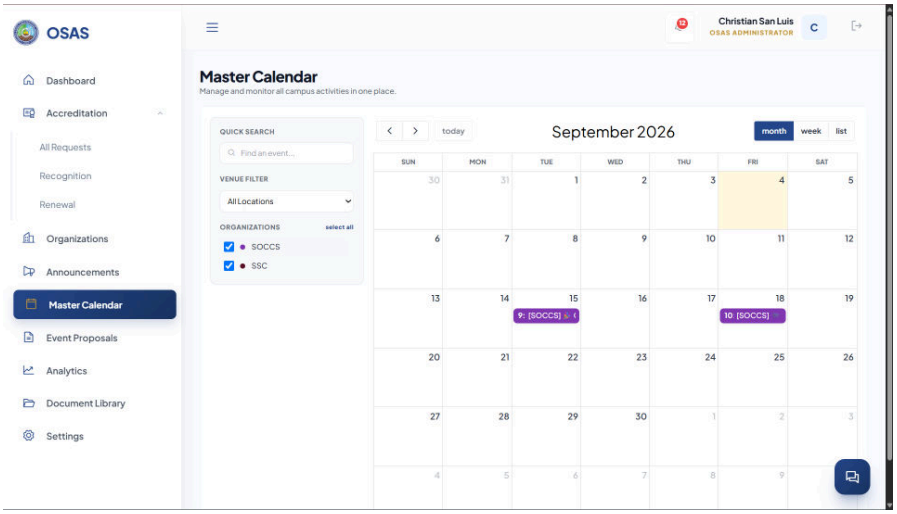


Figure 117. This figure shows the master calendar month view of UniOrg.

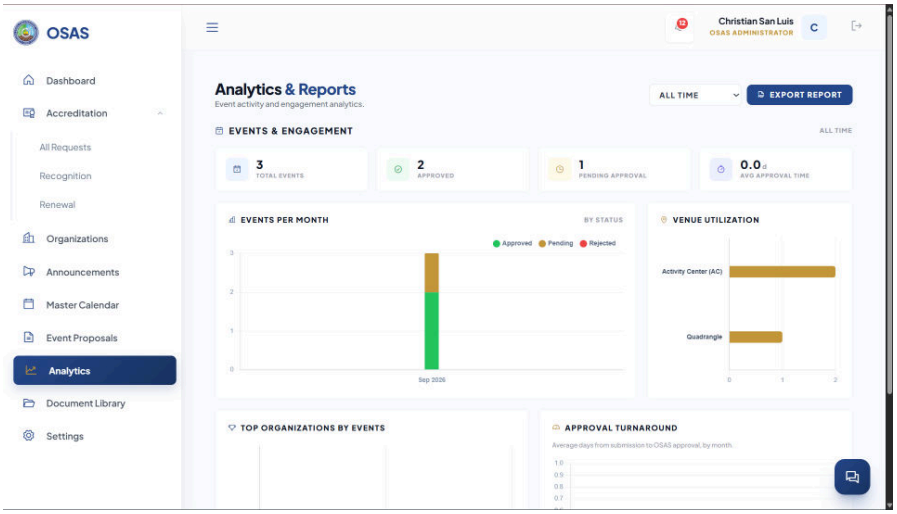


Figure 118. This figure shows the analytics and reports dashboard of UniOrg.

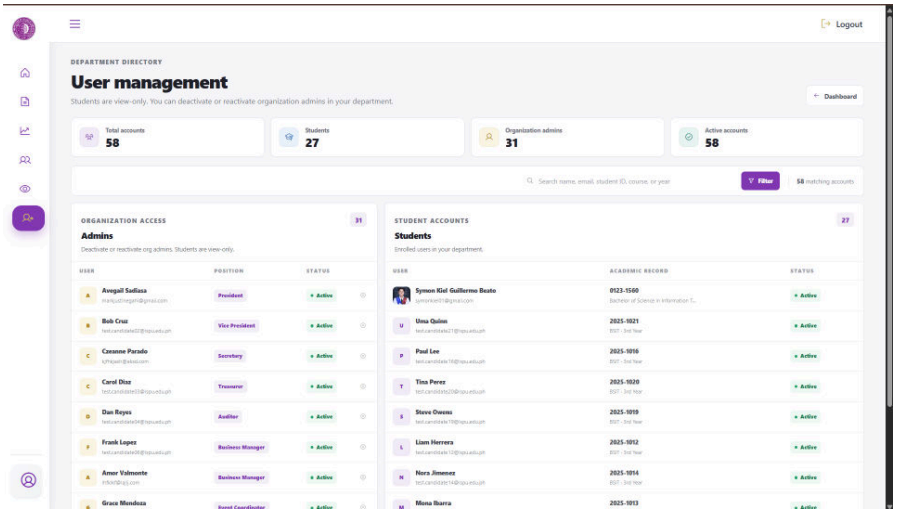


Figure 119. This figure shows the department user management list of UniOrg.

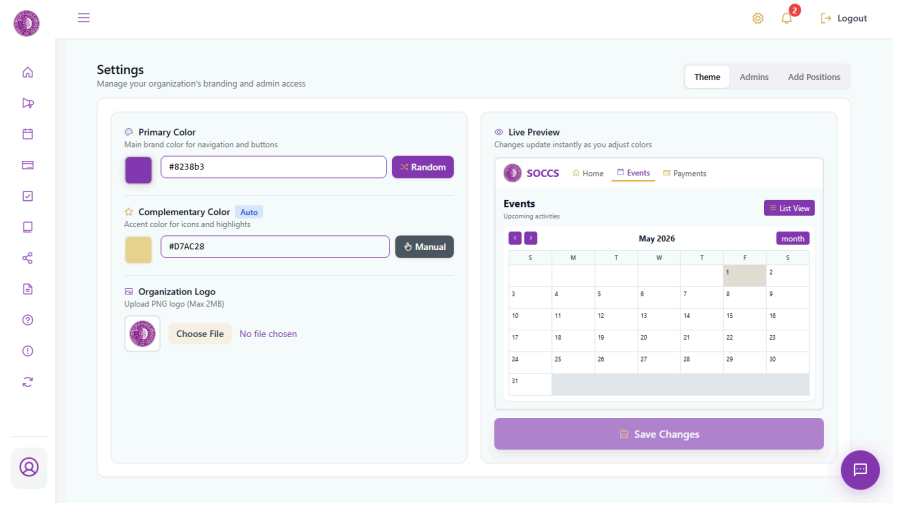


Figure 120. This figure shows the organization settings page of UniOrg.

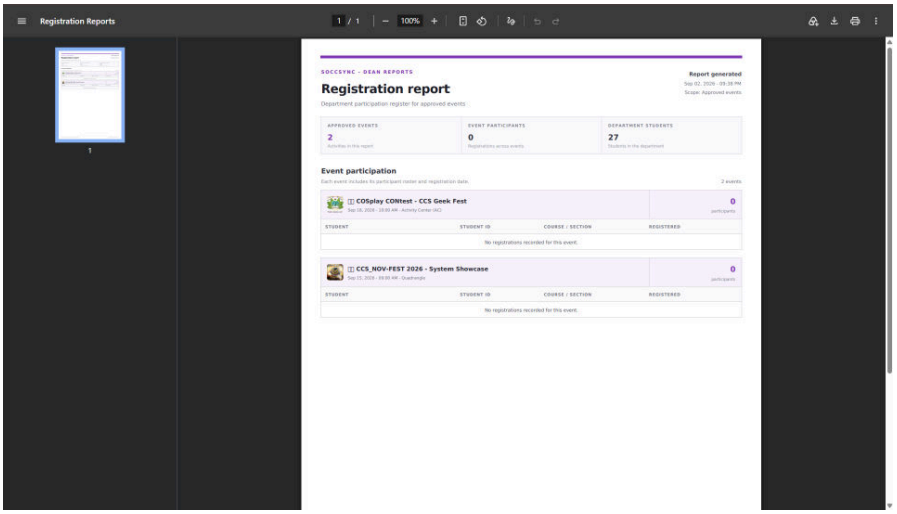


Figure 121. This figure shows the registration report PDF preview of UniOrg.

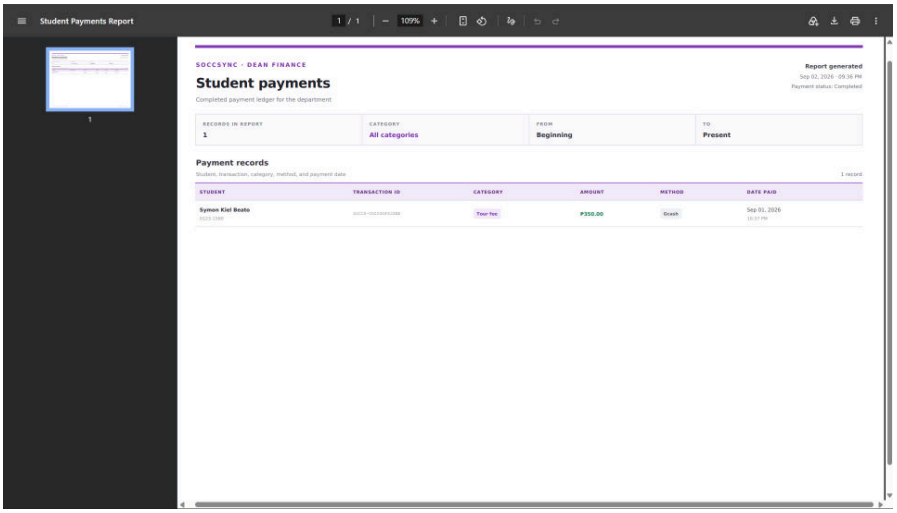


Figure 122. This figure shows the student payments PDF preview of UniOrg.

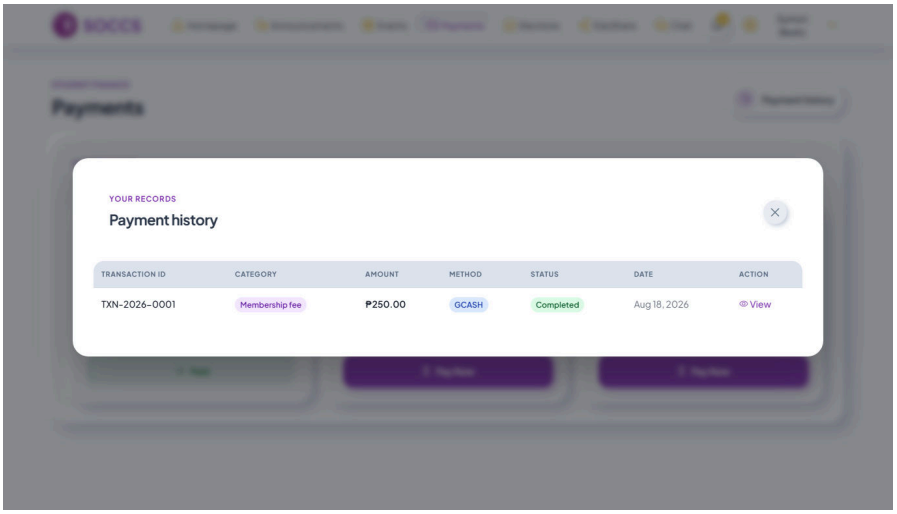


Figure 123. This figure shows the payment history modal of UniOrg.

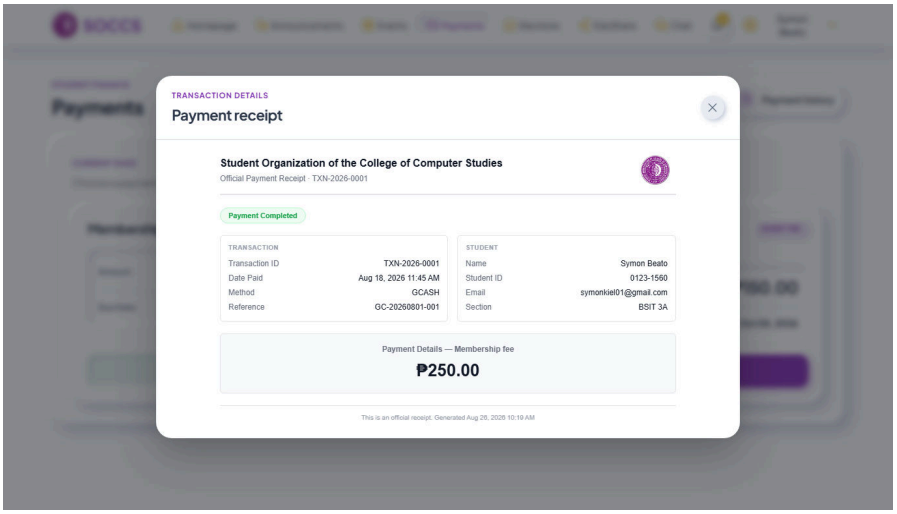


Figure 124. This figure shows the payment receipt modal of UniOrg.

Testing and Evaluation Instruments

This section shows the technology acceptance questionnaire used for the evaluation of the user acceptance towards the developed system.

TECHNOLOGY ACCEPTANCE QUESTIONNAIRE

Dear Respondents,

Greetings in the name of the Lord!

The purpose of the survey is to evaluate user satisfaction and usage intentions for the created system, **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR And NLP for Document Processing and Auto-Tagging"**. Your responses to this questionnaire will be kept completely confidential and will be very helpful to the outcome of this study. Please respond as completely as you can, as only the project researchers will have access to the data you submit. Additionally, no details that could be used to identify you will be shared.

Have a nice day and thank you so much for your help.

Sincerely,

BEATO, SYMON KIEL G.
GAHI, JUSTINE MARK R.

DATA PRIVACY STATEMENT

I hereby affirm my right to be informed, object to processing, access, and rectify, suspend or withdraw my personal data, and be indemnified in case of damages pursuant to the provisions of the Republic Act No. 10173 of the Philippines, Data Privacy of 2012 and its corresponding Implementing Rules and Regulations.

Do you consent with your personal data being processed as described above?
□ Agree

PART 1: RESPONDENTS DATA

Directions: Please fill in the following information:
Name of the Respondent: (Optional) _____ Date: _____

Age: _____ Sex: □ Male □ Female

Which of the following are you?
□ Student/Officer □ Organization Adviser □ Dean □ OSAS Administrator

PART 2: TECHNOLOGY ACCEPTANCE QUESTIONNAIRE

Directions: Kindly put a CHECK (✓) on the corresponding scale to indicate your level of acceptance relative to each question. Indicate your acceptability with the following scale:

1 - Unacceptable	2 - Slightly Unacceptable	3 - Undecided	4 - Acceptable	5 - Highly Acceptable

You may also indicate your comment and suggestions in the last part of the questionnaire.

QUALITY FACTORS	5	4	3	2	1
1. The platform's layout allows users to move through its features without confusion.					
2. The platform provides the necessary tools for managing organization activities and information.					
3. The platform's color scheme and icons are visually appealing and professionally designed.					

PERCEIVED USEFULNESS	5	4	3	2	1
1. Using the UniOrg platform enables me to complete student organization-related tasks more quickly.					
2. The platform helps me easily search, access, and view organization records and announcements.					
3. The centralized system helps me stay informed and updated about student organization activities and events.					
PERCEIVED EASE OF USE	**5**	**4**	**3**	**2**	**1**
1. Learning how to use the UniOrg platform is easy for me.					
2. Navigating through the platform to view organization details and schedules is clear and understandable.					
3. Becoming skillful in using the features of the UniOrg platform would be easy for me.					
ATTITUDE TOWARD USING	**5**	**4**	**3**	**2**	**1**
1. I feel comfortable using a digital platform for student organization activities.					
2. I feel confident when using the UniOrg platform.					
3. I have a positive attitude toward using UniOrg for university-wide student organization engagement.					
BEHAVIORAL INTENTION TO USE	**5**	**4**	**3**	**2**	**1**
1. I intend to use the UniOrg platform for student organization transactions and activities in the future.					
2. I would use this platform whenever I need to access organization services or information.					
3. I would recommend the UniOrg platform to other students and organizations in the university.					
EXPERIENCE	**5**	**4**	**3**	**2**	**1**
1. My previous experience with online platforms helps me use UniOrg effectively.					
2. I find it easy to adapt to the UniOrg system based on my general familiarity with digital tools.					
3. Transitioning from traditional manual processes to UniOrg's digital platform improves my overall user experience.					

Comment if you have any recommendations and suggestions.

Validated By:

Rachiel R. Rivano, PhD
Thesis Adviser

User's Manual

UNIORG: A CONFIGURABLE UNIVERSITY-WIDE STUDENT ORGANIZATION MANAGEMENT PLATFORM WITH OCR AND NLP FOR DOCUMENT PROCESSING AND AUTO-TAGGING

USER'S MANUAL

GENERAL INFORMATION

System Overview

UNIORG is a configurable university-wide student organization management platform developed for the Office of Student Affairs and Services (OSAS) of Laguna State Polytechnic University Santa Cruz Campus. It functions as the central workspace of 39 recognized student organizations and covers the full cycle of organization life, from accreditation and document management to events, payments, elections and communications. The platform provides role-based web portals for OSAS personnel, college deans, organization advisers and officers, as well as students. Its document pipeline combines OCR text extraction with NLP auto-tagging and AI summaries so accreditation papers and organizational records remain searchable and organized.

Each organization is given its own configurable workspace with announcements, events calendar, payment collections, elections, e-library and EduShare folders, feedback hub and member list. Transactions follow a clear approval chain. Event proposals pass through the dean and OSAS before publishing, accreditation applications move through a guided multi-step renewal flow, while manual e-wallet payments are verified by organization admins against uploaded receipts. This keeps every record traceable, from submission up to approval.

Authorized Use Permission

UNIORG is a restricted university-operated system. Access to its administrative, accreditation, financial and configuration modules is strictly limited to authorized personnel of OSAS, the college deans and the designated advisers and officers of recognized student organizations. Students may only access their own records and the public or member-level functions of organizations they belong to. Access privileges are enforced through Role-Based Access Control (RBAC), so each account only sees the data and functions needed for its duties. Security policies, role assignments and accreditation period settings are restricted to OSAS administrators, and field-level users cannot alter system hierarchies. All accreditation decisions, proposal approvals, payment verifications and access events are recorded in the system logs for transparency and accountability.

By accessing UNIORG, users agree to its terms of service and acknowledge that the system processes sensitive information, including student identification records, Certificates of Registration, accreditation documents and payment receipts. All collection and processing comply with Republic Act No. 10173, or the Data Privacy Act of 2012. Any unauthorized extraction, distribution or misuse of student data or organizational records is prohibited and subject to administrative and legal penalties.

Points of Contact (Help Desk)

For account issues, accreditation concerns and technical problems, users may reach the OSAS help desk through the official OSAS office of LSPU Santa Cruz Campus during office hours. Organization-level concerns such as membership, payments and event details should first be coordinated with the respective

organization adviser or officers through the in-system complaints and feedback hub or announcements channel. For quick answers, the built-in chatbot and FAQ library inside each workspace provide guidance on common procedures.

Authentication and Account Setup

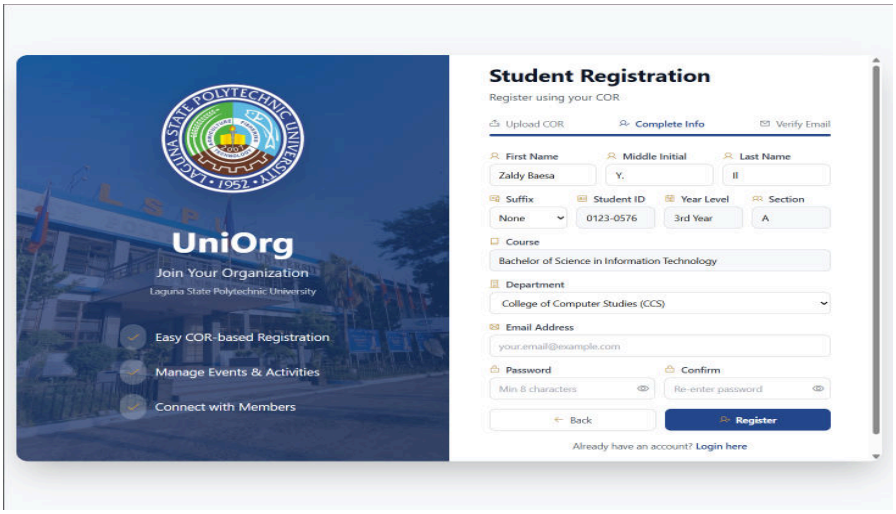


Figure 125. This figure shows the login page of UniOrg.

Step 1. Open the UniOrg web application in the browser.
Step 2. Enter your ID Number (format 2025-1020 or 24-1234-5678) and your password. Use the eye icon to show or hide the password.
Step 3. Click the Login button.
Step 4. The system verifies your credentials and redirects you to your role-based dashboard.

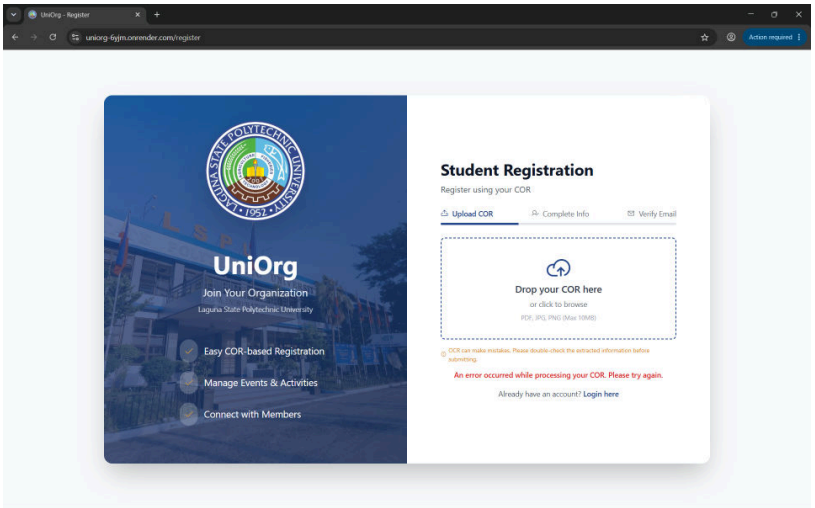


Figure 126. This figure shows the Upload COR step of student registration in UniOrg.

Step 1. Open the student registration page and fill out the basic account details first.
Step 2. Proceed to the verification step and ready a clear photo or scan of your Certificate of Registration.
Step 3. Upload the COR file in the upload slot provided.
Step 4. Click Submit and wait for the confirmation message.
Step 5. Your account moves to the Complete Info step once the COR is accepted.

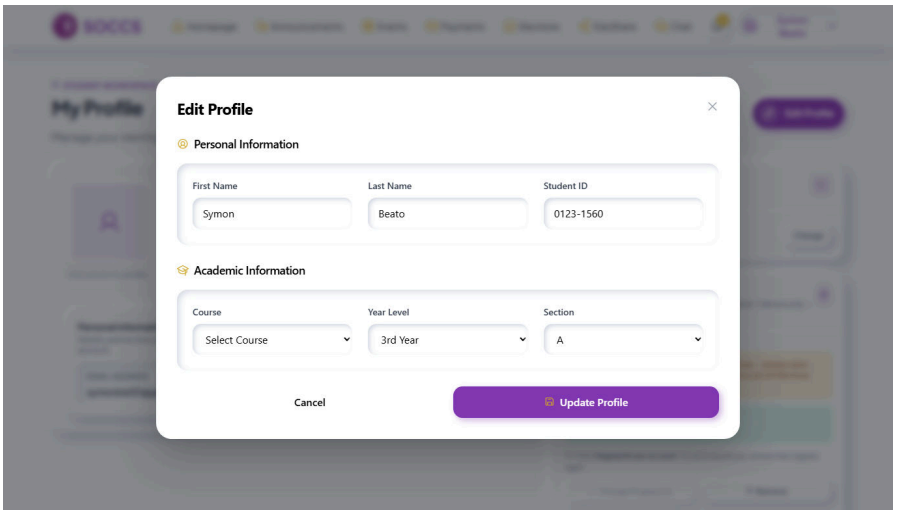


Figure 127. This figure shows the Complete Info step of student registration in UniOrg.

Step 1. From the login page, click the Register or Sign up button.
Step 2. Upload your Certificate of Registration (COR) for student verification.
Step 3. Fill out your personal details: First Name, Last Name, Student Number, Program, Year Level, Email Address, Contact Number, and Password.
Step 4. Review your entries and click the Submit button.
Step 5. A success message displays, and you can now log in to your account.

Student Users

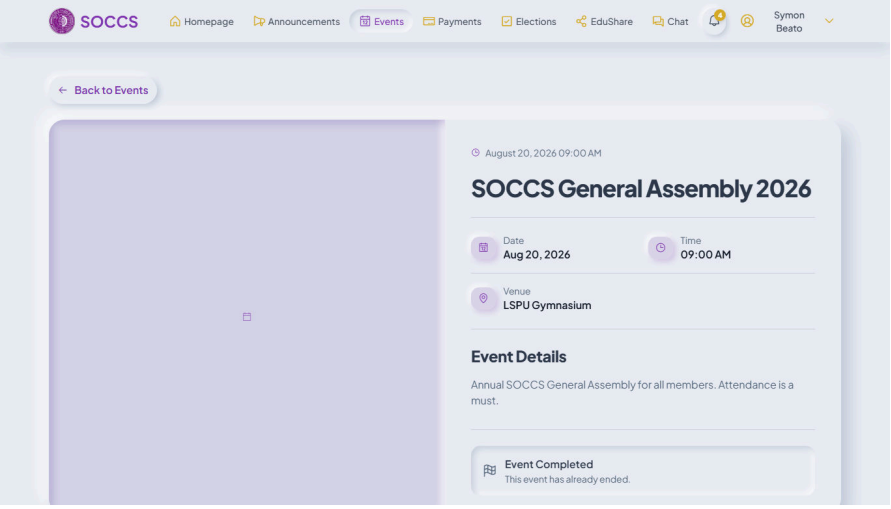


Figure 128. This figure shows the student profile edit modal of UniOrg.

Step 1. Log in and go to your Profile page.
Step 2. Click the Edit Profile button to open the edit modal.
Step 3. Update your profile photo, contact number, or other personal details.
Step 4. Click the Save button.
Step 5. A confirmation message displays and your updated details appear on your profile.

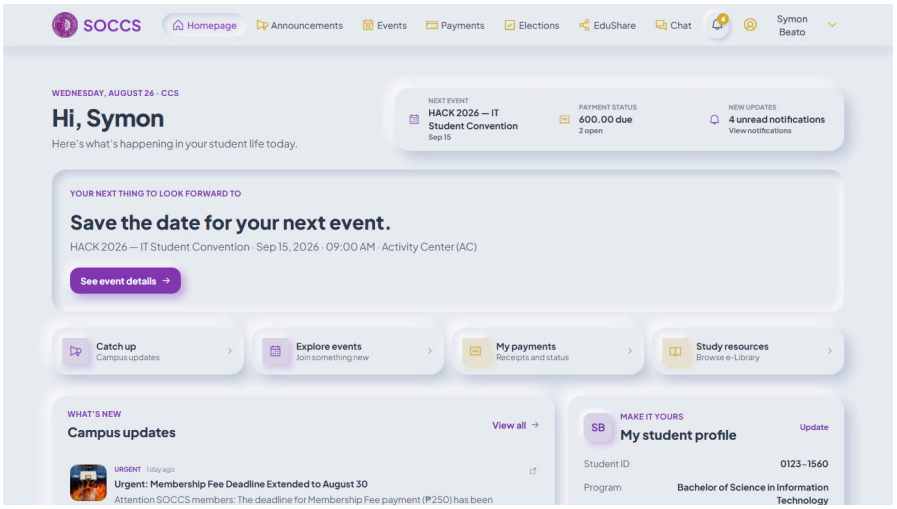


Figure 129. This figure shows the student dashboard of UniOrg.

Step 1. Log in with your student account to land on the dashboard.
Step 2. View the summary cards for announcements, events, payments, and elections.
Step 3. Use the sidebar to jump to any module you need.
Step 4. Click the notifications bell to check the latest updates.

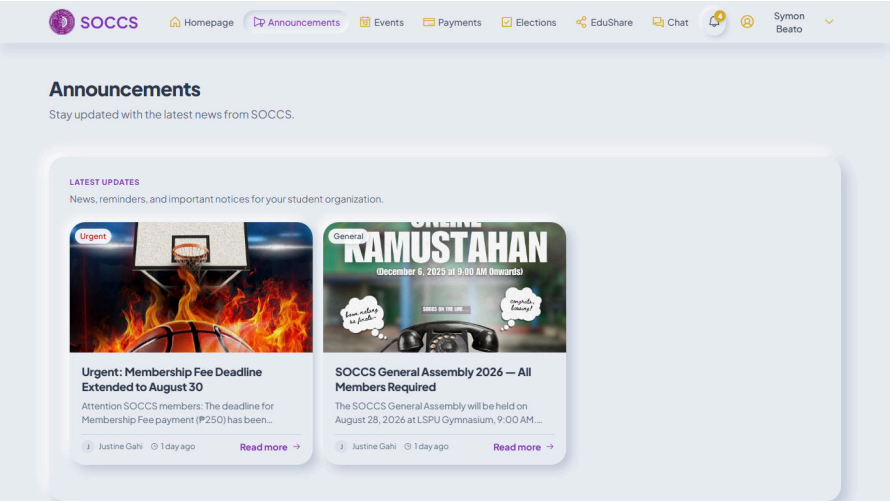


Figure 130. This figure shows the student announcements list of UniOrg.

Step 1. Go to the Announcements tab to view all posted announcements.
Step 2. Click an announcement to read its full details.
Step 3. Check the pinned items for the most important updates.
Step 4. Use the search or filter if you are looking for a specific post.

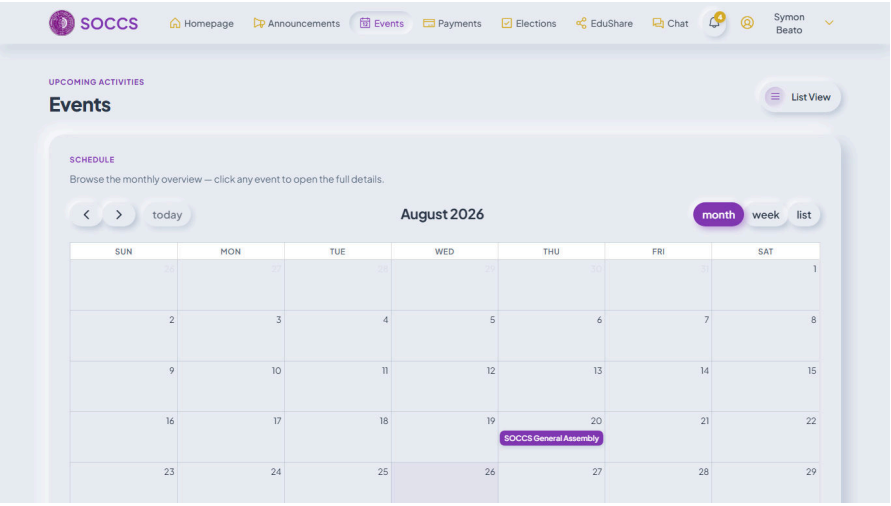


Figure 131. This figure shows the student events list of UniOrg.

Step 1. Go to the Events tab to browse the upcoming events.
Step 2. Use the search or filter to find an event you like.
Step 3. Click an event to view its details and schedule.
Step 4. Click Register to join the event.

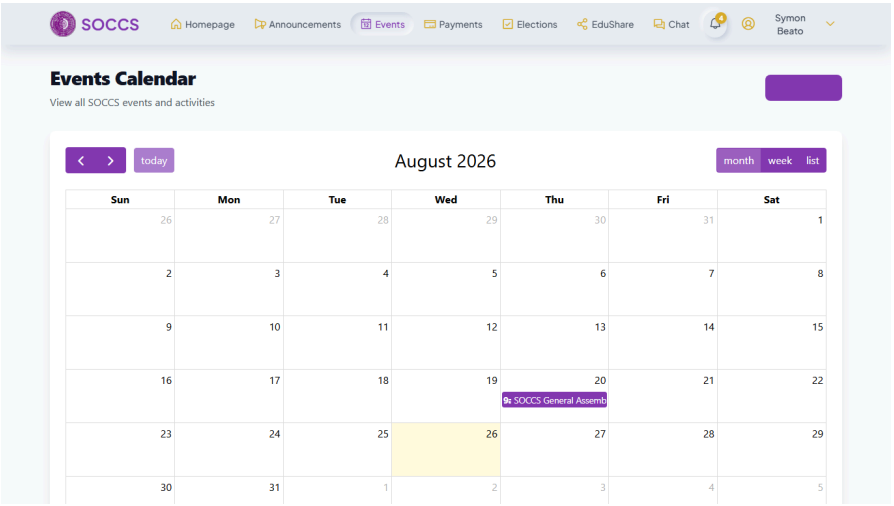


Figure 132. This figure shows the student events calendar view of UniOrg.

Step 1. Open the Events calendar view to see activities by date.
Step 2. Switch between month and week views as needed.
Step 3. Click an event chip on the calendar to view its details.
Step 4. Register directly from the details shown.

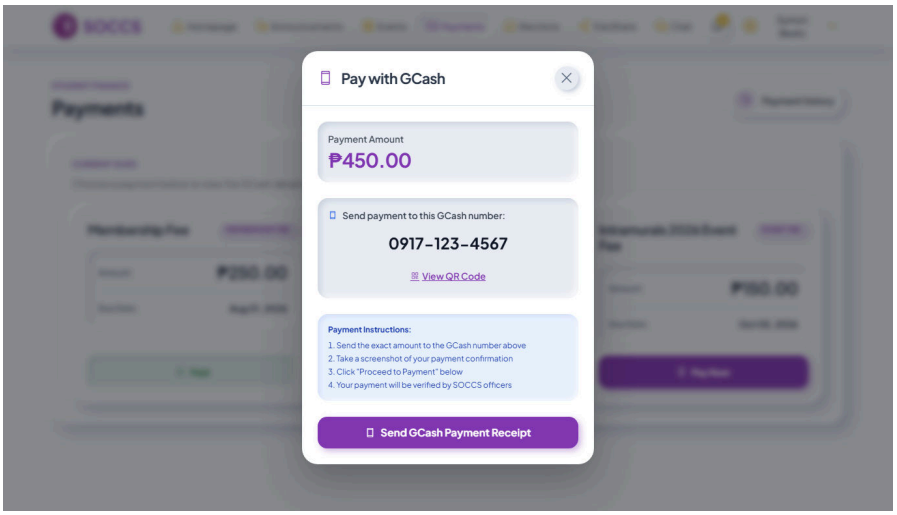


Figure 133. This figure shows the event registration modal of UniOrg.

Step 1. Go to the Events tab to view the list of upcoming events.
Step 2. Click the event you want to join to view its details.
Step 3. Click the Register button to open the registration modal.
Step 4. Confirm your registration in the modal and click Submit.
Step 5. Your registration is recorded and appears in your registered events list.

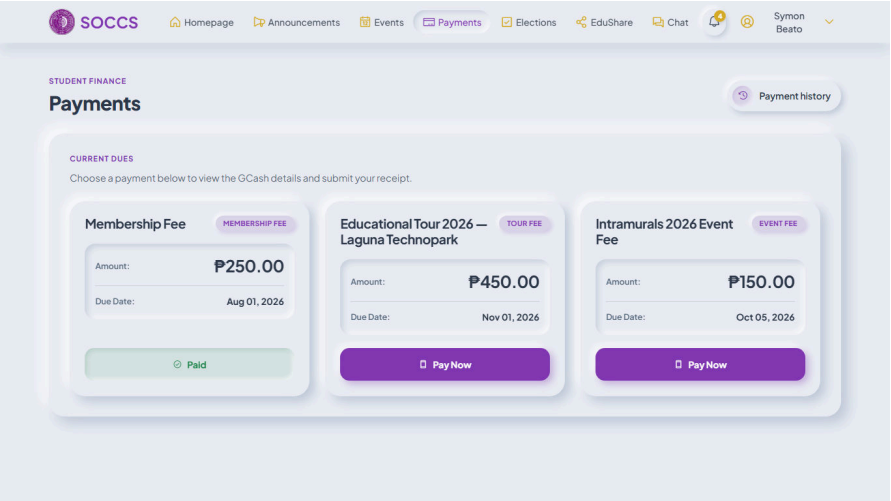


Figure 134. This figure shows the student payments list of UniOrg.

Step 1. Go to the Payments tab to view all payables.
Step 2. Check the amount, category, and due date of each item.
Step 3. Select the payment you want to settle.
Step 4. Submit your GCash payment and receipt for verification.

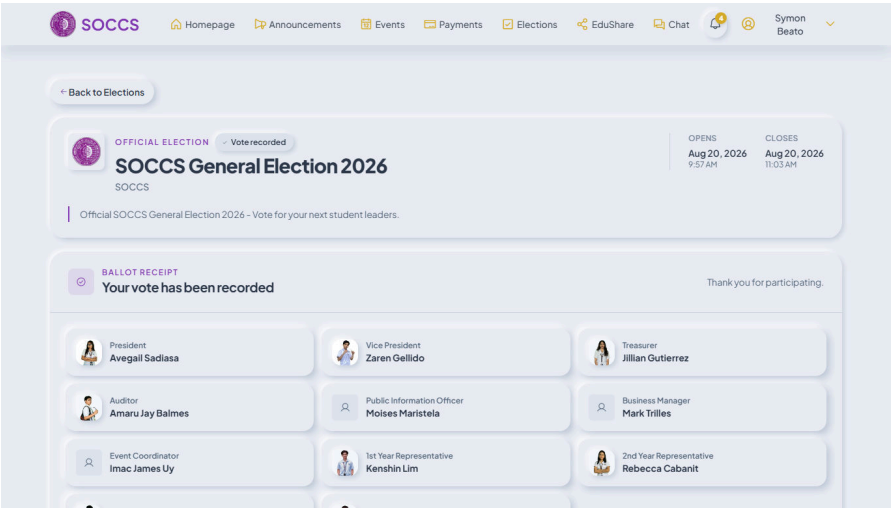


Figure 135. This figure shows the GCash payment submission of UniOrg.

Step 1. Go to the Payments tab to view the list of payables.
Step 2. Select the payment item and choose GCash as the payment method.
Step 3. Send the exact amount to the displayed GCash account number.
Step 4. Enter your GCash reference number and upload the screenshot of your receipt in the modal.
Step 5. Click the Submit button.
Step 6. Your payment is marked as pending until the organization admin verifies it, and you may view your receipt in the payment history.

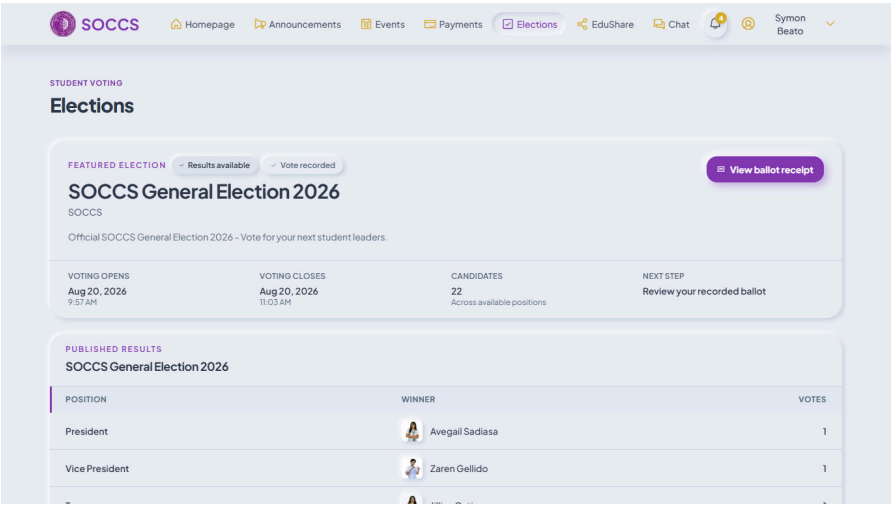


Figure 136. This figure shows the student elections list of UniOrg.

Step 1. Go to the Elections tab to view all elections.
Step 2. Check which elections are active and which are closed.
Step 3. Open an active election to view its positions and candidates.
Step 4. Cast your vote within the voting period.

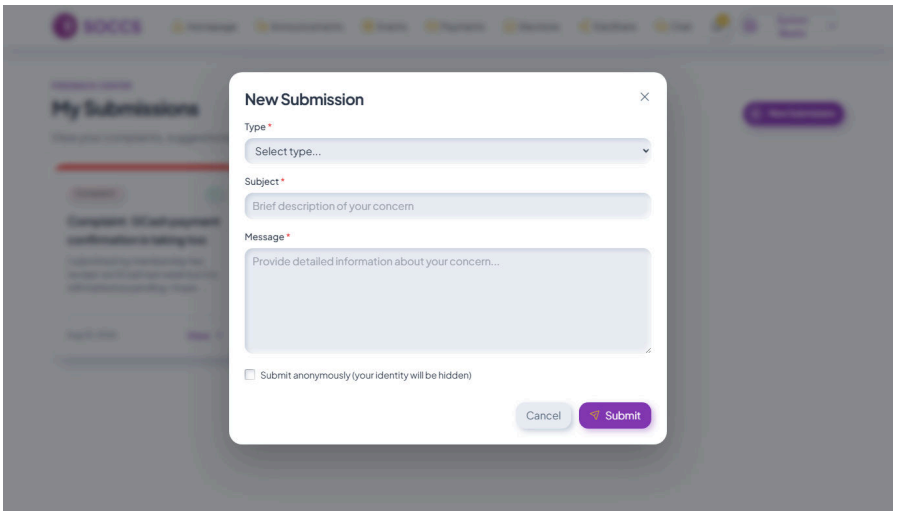


Figure 137. This figure shows the election vote casting modal of UniOrg.

Step 1. Go to the Elections tab during the voting period to view the active election.
Step 2. Select the position and the candidate you want to vote for.
Step 3. Click the Vote or Cast Vote button to open the confirmation modal.
Step 4. Review your selection and click Confirm.
Step 5. A success message displays, and your vote is recorded. Note that each student can only vote once per position.

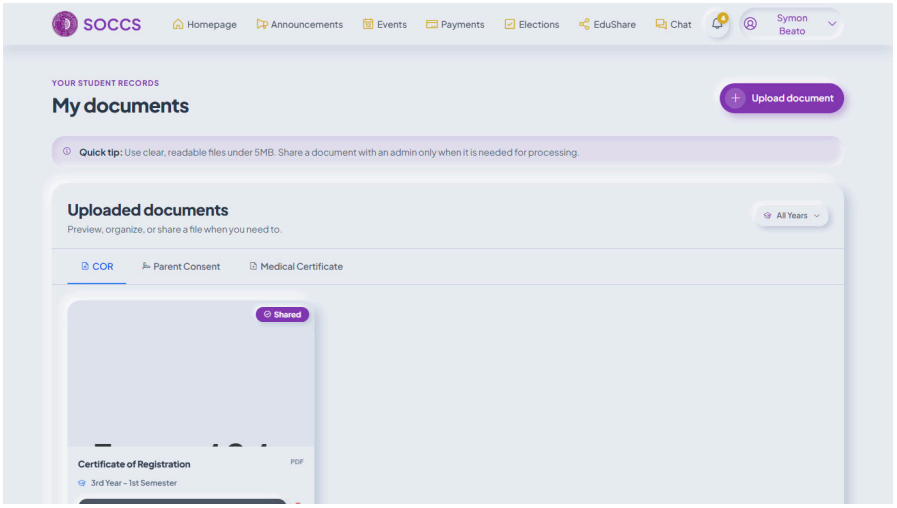


Figure 138. This figure shows the student documents page of UniOrg.

Step 1. Open the My Documents page to view your submitted documents.
Step 2. Check the verification status of each document.
Step 3. Upload a new document with its type and validity details.
Step 4. Track the status until an officer verifies it.

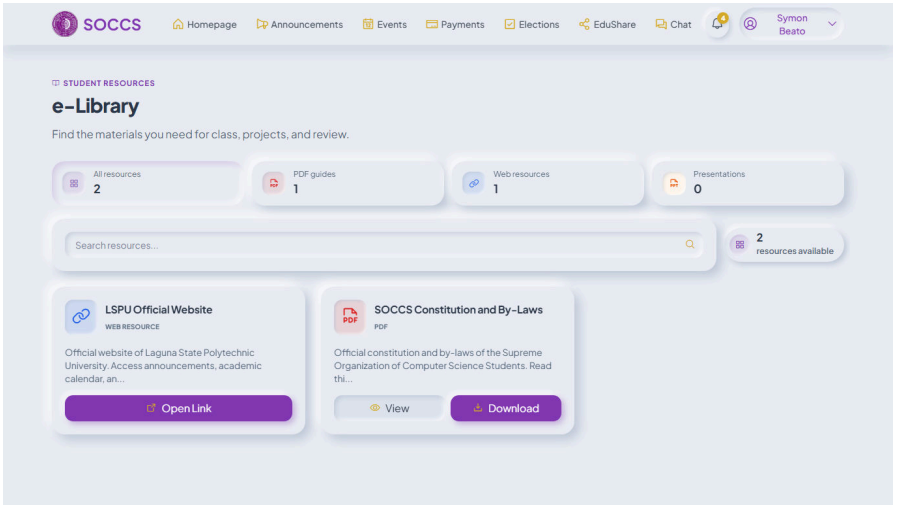


Figure 139. This figure shows the student e-library list of UniOrg.

Step 1. Go to the E-Library tab to browse shared resources.
Step 2. Use the search bar to find a title or topic.
Step 3. Click a resource to preview its details.
Step 4. Download the file if you need a copy.

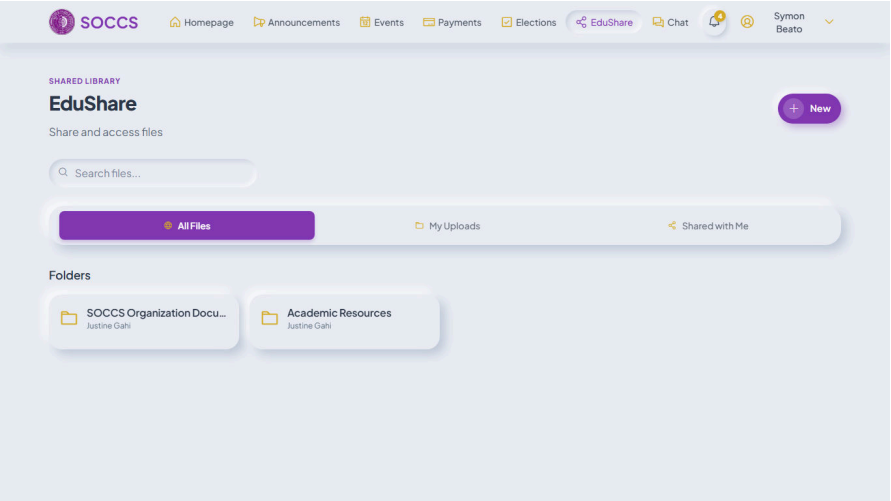


Figure 140. This figure shows the student EduShare list of UniOrg.

Step 1. Go to the EduShare tab to browse shared folders and files.
Step 2. Open a folder to view its contents.
Step 3. Use the search to locate a specific material.
Step 4. Download the file you need.

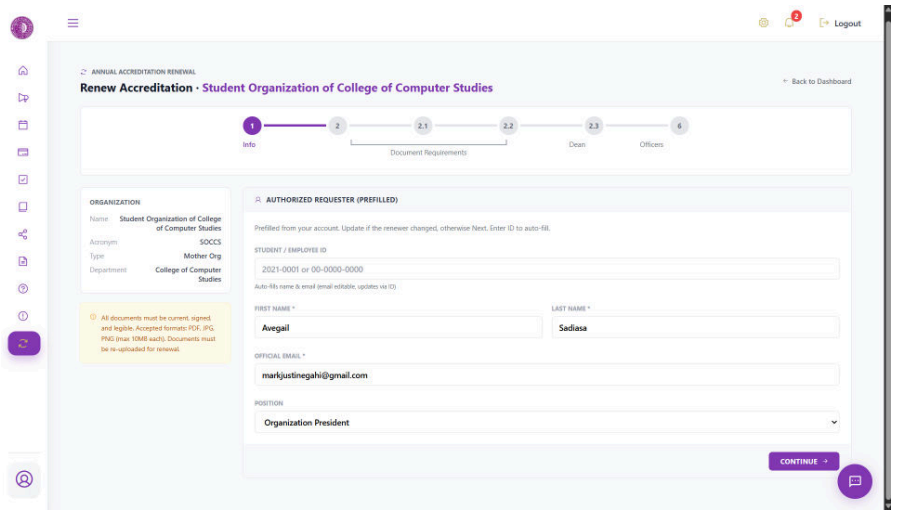


Figure 141. This figure shows the new complaint submission modal of UniOrg.

Step 1. Go to the Complaints or Feedback Hub tab.
Step 2. Click the New Complaint or File Complaint button.
Step 3. Fill out the subject, category, and the detailed description of your concern.
Step 4. Attach any supporting file if needed and click Submit.
Step 5. Your complaint is recorded and you can track its status in your complaints list.

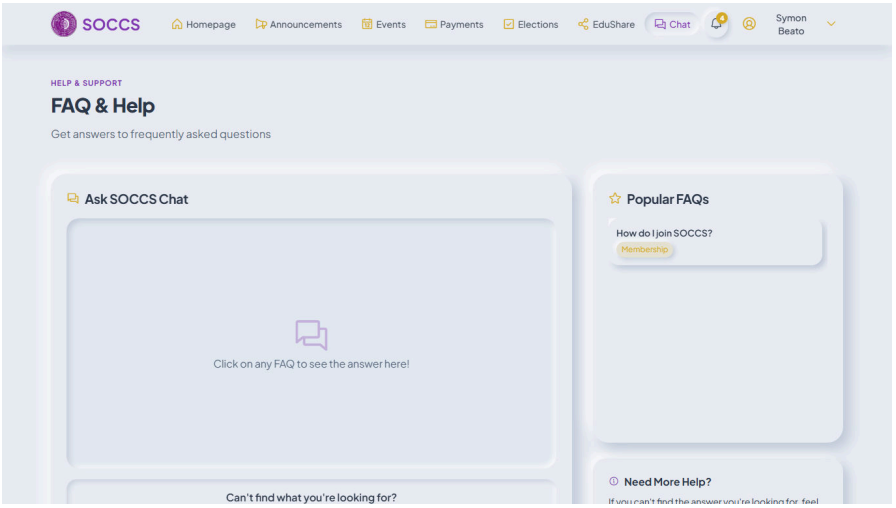


Figure 142. This figure shows the student chatbot of UniOrg.

Step 1. Open the chatbot panel from your dashboard.
Step 2. Type your question in the chat box and send it.
Step 3. Read the answer generated from the FAQ library.
Step 4. Ask a follow-up or contact an officer if needed.

Organization Admin and Officer Users (covers Adviser)

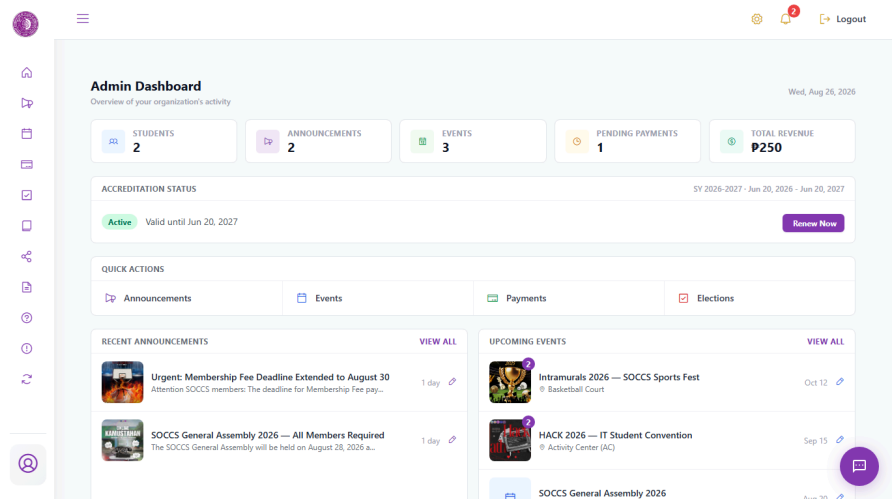


Figure 143. This figure shows the organization admin dashboard of UniOrg.

Step 1. Log in with your organization admin account.
Step 2. View the overview cards for members, events, payments, and pending items.
Step 3. Click any card or sidebar item to manage that module.
Step 4. Check notifications for accreditation and proposal updates.

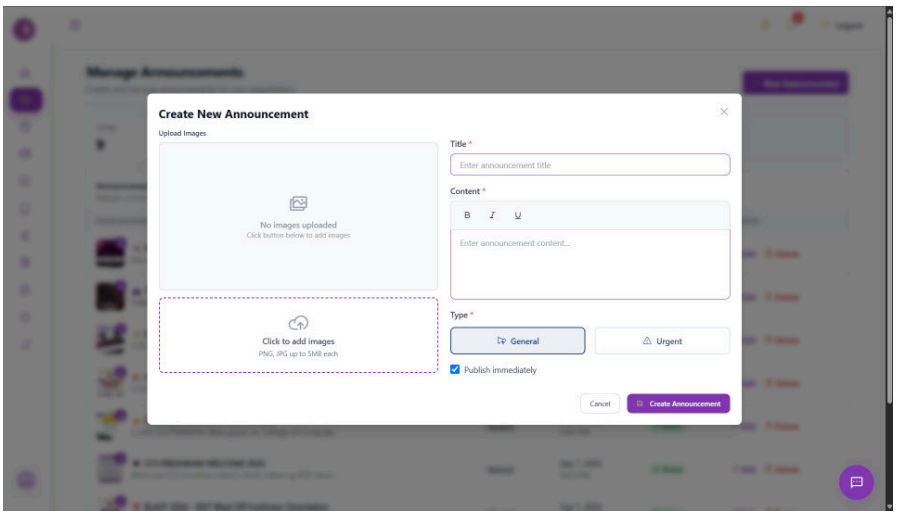


Figure 144. This figure shows the accreditation renewal requester information form of UniOrg.

Step 1. Log in as an organization admin and go to the Accreditation module.
Step 2. Click the Renew or Apply for Accreditation button.
Step 3. Fill out the requester information form with the organization name, school year, and contact person details.
Step 4. Click Next to proceed to the document requirements step.
Step 5. Complete the remaining steps: upload the required documents, upload the members list, fill out the college dean form, and encode the incoming officers.
Step 6. Submit the application. OSAS reviews it and updates its accreditation status.

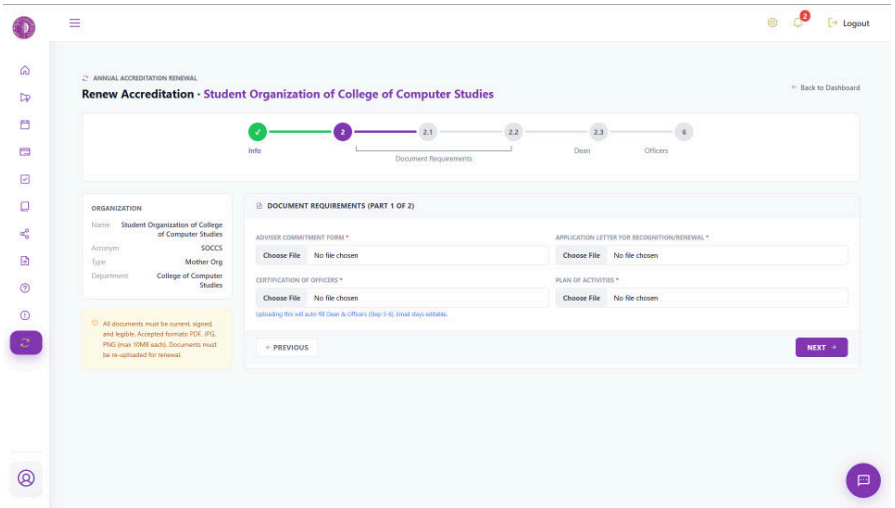


Figure 145. This figure shows the accreditation document requirements part 1 of UniOrg.

Step 1. After the requester info step, click Continue to reach Document Requirements.

Step 2. Upload each required document in its assigned slot.
Step 3. Accepted formats are PDF, JPG, and PNG with max 10MB each.
Step 4. Wait for the OCR and NLP processing status to complete per file.
Step 5. Proceed to the next part once all slots are filled.

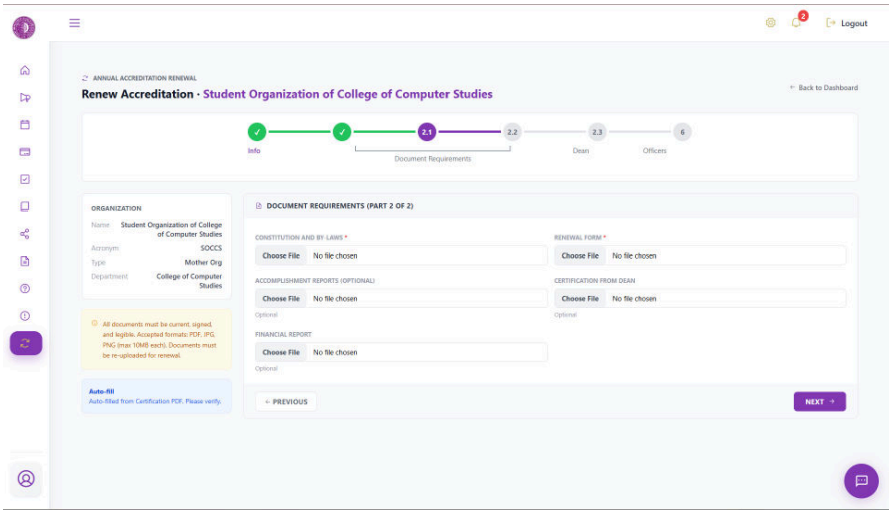


Figure 146. This figure shows the accreditation document requirements part 2 of UniOrg.

Step 1. Continue with the remaining requirement slots in this part.
Step 2. Replace any rejected file by uploading a clearer copy.
Step 3. Check the AI summary and tags per document if displayed.
Step 4. Click Next to move to the members list step.

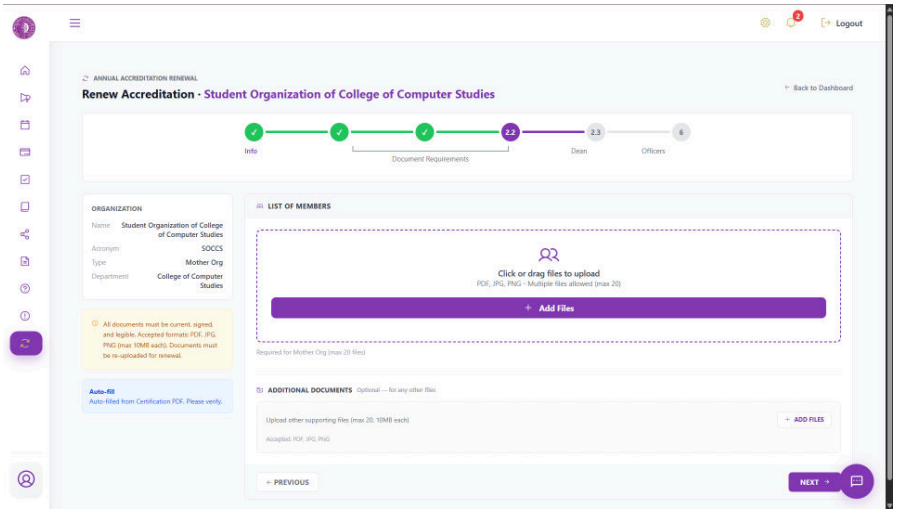


Figure 147. This figure shows the members list upload step of accreditation renewal in UniOrg.

Step 1. Upload the official members list file in the upload slot.

Step 2. Verify that the encoded count matches your expected members.
Step 3. Fix any errors in the file and re-upload if needed.
Step 4. Click Next to proceed to the dean form step.

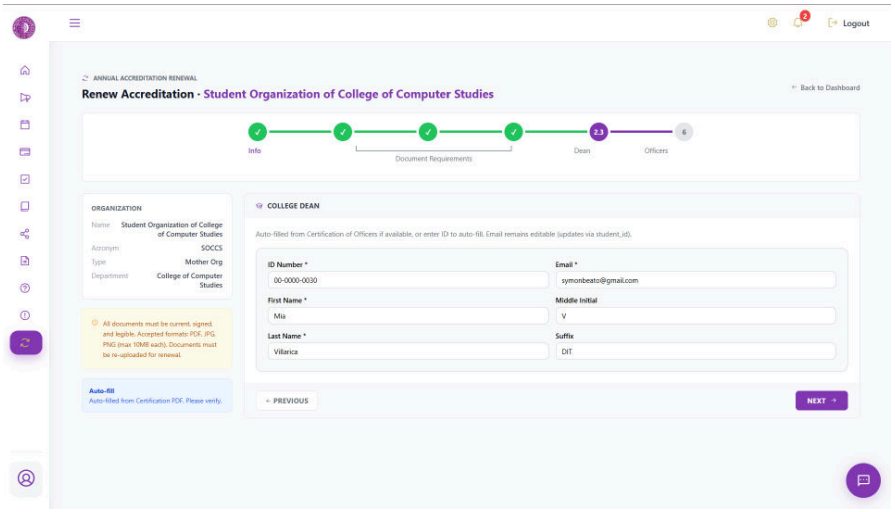


Figure 148. This figure shows the college dean form step of accreditation renewal in UniOrg.

Step 1. Fill out the college dean certification details required.
Step 2. Attach the signed certification file in its slot.
Step 3. Double-check the entries for correctness.
Step 4. Click Next to move to the officers step.

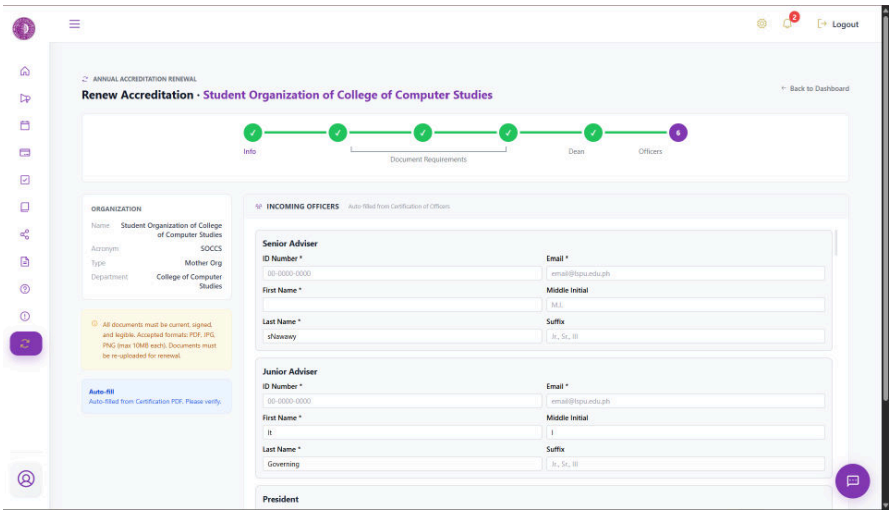


Figure 149. This figure shows the incoming officers step of accreditation renewal in UniOrg.

Step 1. Encode the incoming officers with their respective positions.
Step 2. Review the list to make sure it is complete and accurate.
Step 3. Remove or correct any wrong entry before submitting.
Step 4. Submit the application for OSAS review.

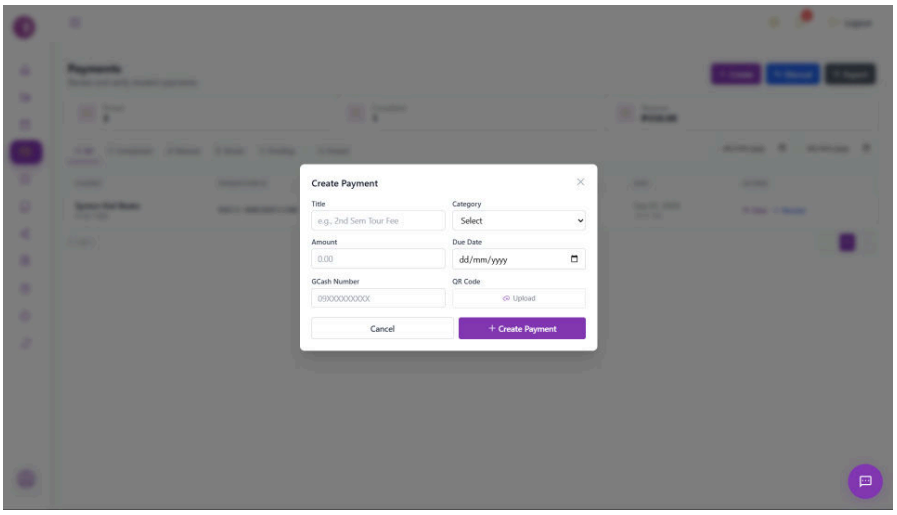


Figure 150. This figure shows the announcement creation modal of UniOrg.

Step 1. Go to the Announcements module.
Step 2. Click the Create or New Announcement button.
Step 3. Enter the announcement title, details, and target audience.
Step 4. Click the Post or Publish button.
Step 5. The announcement is posted and appears in the announcements list of the members.

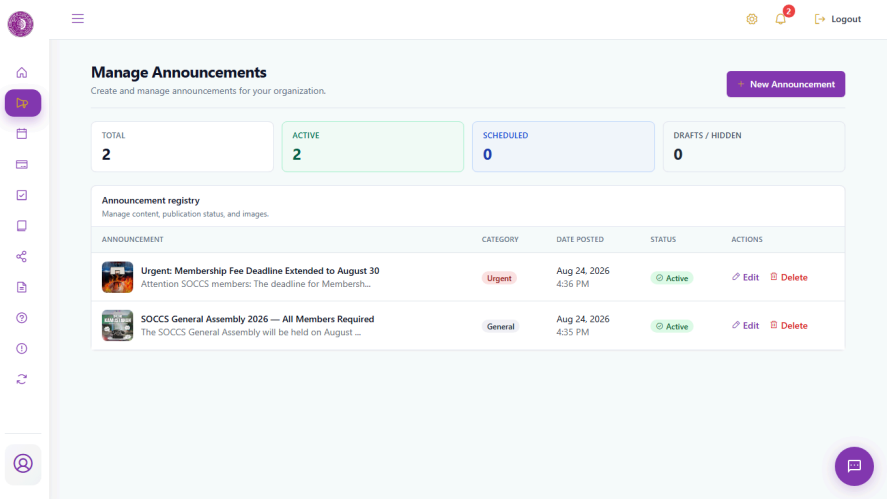


Figure 151. This figure shows the admin announcements list of UniOrg.

Step 1. Go to Announcements to view all posted announcements.
Step 2. Click an item to view or edit its contents.
Step 3. Delete outdated posts to keep the list clean.
Step 4. Click Create to post a new announcement.

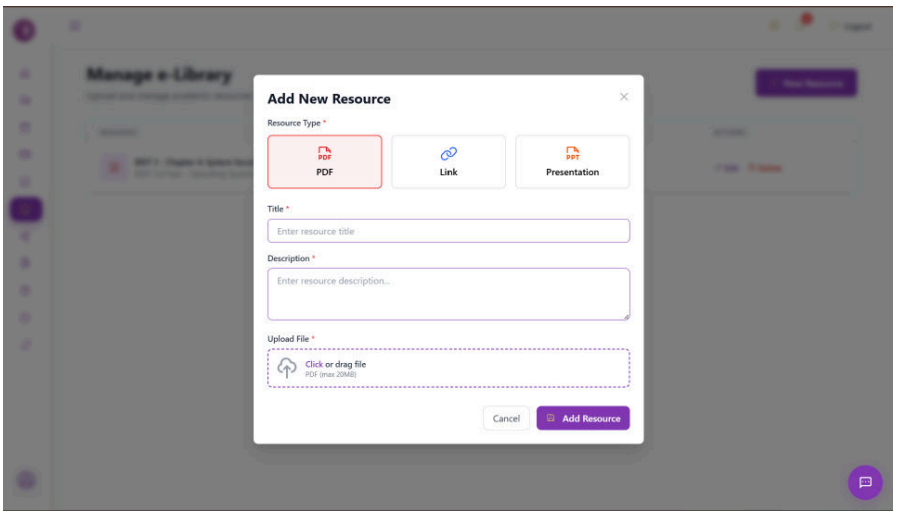


Figure 152. This figure shows the payment creation modal of UniOrg.

Step 1. Go to the Payments module.
Step 2. Click the Create or New Payment button.
Step 3. Enter the payment title, amount, deadline, and payment instructions.
Step 4. Click the Save or Publish button.
Step 5. The payable is published and members can now submit their payments for verification.

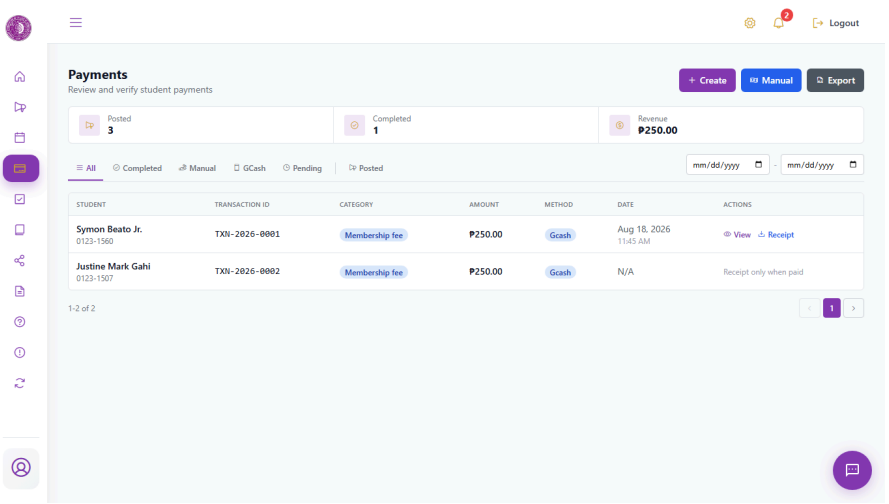


Figure 153. This figure shows the admin payments list of UniOrg.

Step 1. Go to Payments to view all payment collections.
Step 2. Track which members have paid and which are pending.
Step 3. Open a collection to verify submitted receipts.
Step 4. Click Create to publish a new collection.

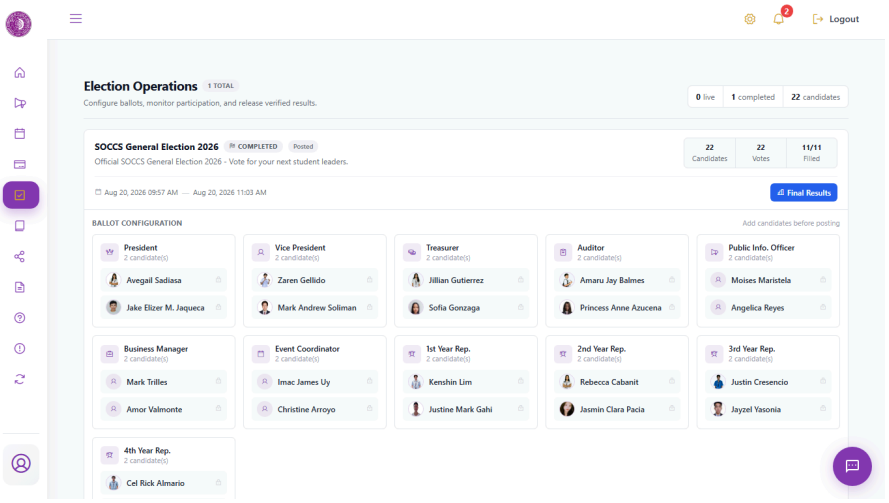


Figure 154. This figure shows the admin elections list of UniOrg.

Step 1. Go to Elections to view all elections of the organization.
Step 2. Open an election to manage its candidates and schedule.
Step 3. Monitor the voting progress while it is active.
Step 4. Publish the results once voting ends.

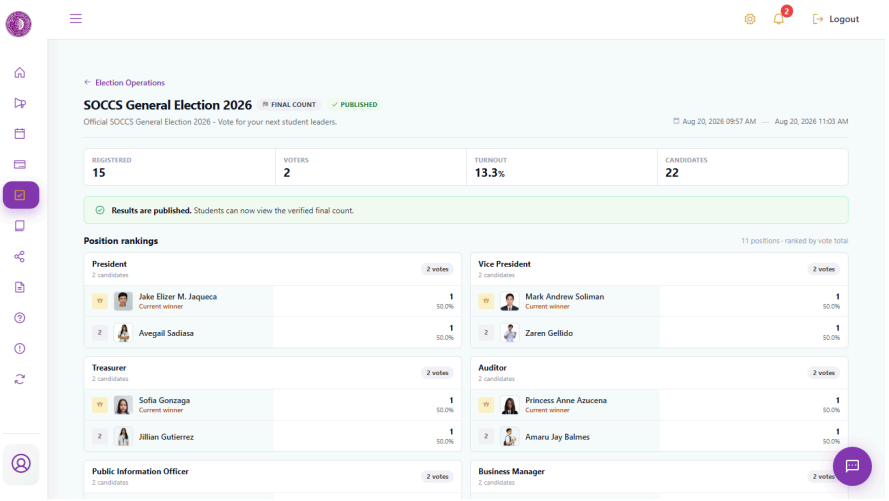


Figure 155. This figure shows the election results view of UniOrg.

Step 1. Open the finished election to view the tally.
Step 2. Check the vote count per candidate per position.
Step 3. Verify the winners before publishing.
Step 4. Click Publish so members can view the results.

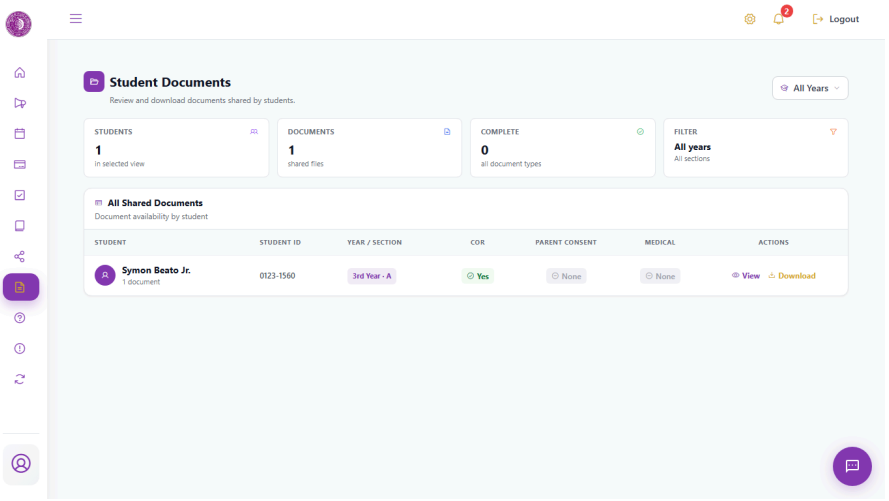


Figure 156. This figure shows the student documents management page of UniOrg.

Step 1. Go to Documents to review submitted student documents.
Step 2. Open each submission to check its details and file.
Step 3. Mark it verified or return it with remarks.
Step 4. Track the overall status in the list.

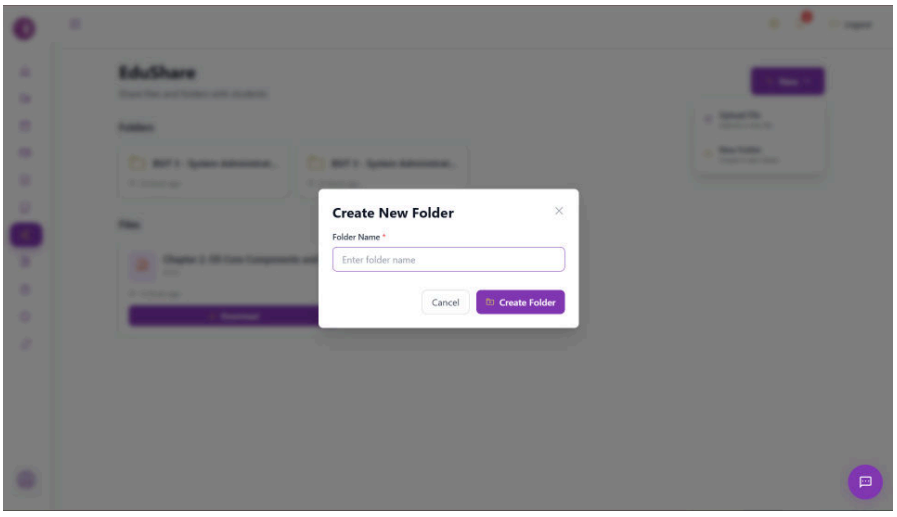


Figure 157. This figure shows the e-library resource upload modal of UniOrg.

Step 1. Go to the E-Library module.
Step 2. Click the Add or Upload Resource button.
Step 3. Enter the resource title, type, and description, then attach the file.
Step 4. Click the Save or Upload button.
Step 5. The resource is saved and appears in the e-library list for members to access.

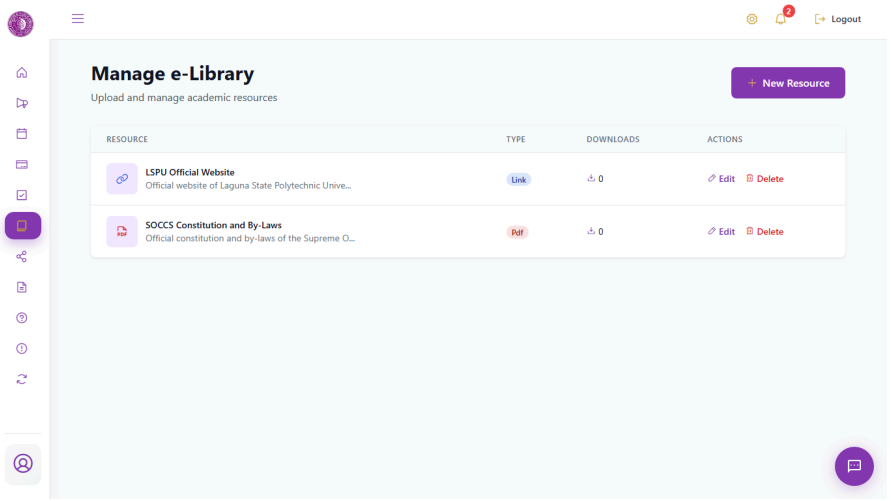


Figure 158. This figure shows the e-library management page of UniOrg.

Step 1. Go to E-Library to manage shared resources.
Step 2. Review the list of uploaded materials and their downloads.
Step 3. Add new resources or remove outdated ones.
Step 4. Check that members can access the published items.

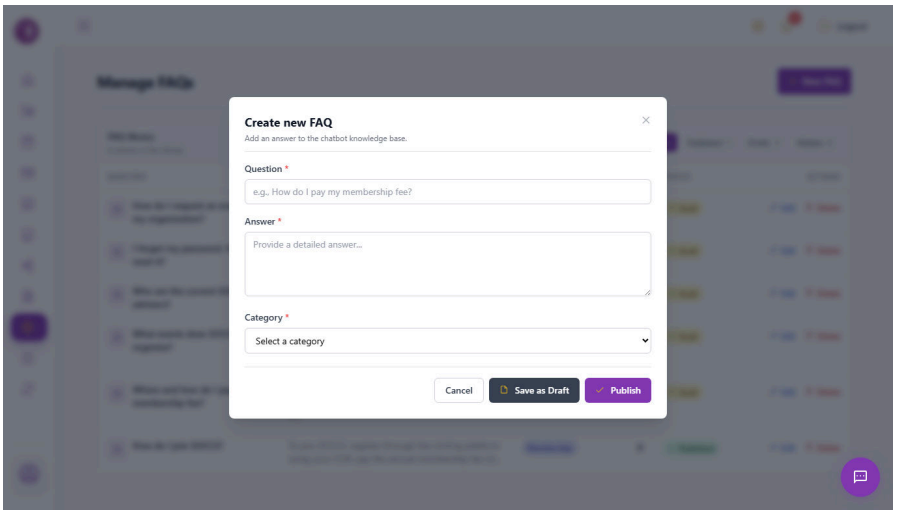


Figure 159. This figure shows the EduShare folder creation modal of UniOrg.

Step 1. Go to the EduShare module.
Step 2. Click the New Folder or Create Folder button.
Step 3. Enter the folder name and description.
Step 4. Click the Create button.
Step 5. The folder is created and you can now upload files into it for the members.

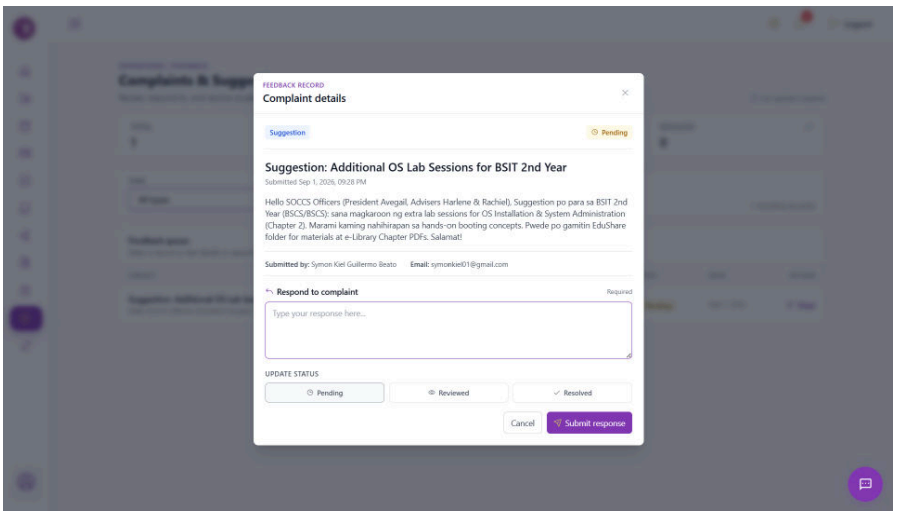


Figure 160. This figure shows the chatbot FAQ creation modal of UniOrg.

Step 1. Go to the Chatbot or FAQ management module.
Step 2. Click the New FAQ or Add FAQ button.
Step 3. Enter the frequently asked question and its answer.
Step 4. Click the Save button.
Step 5. The FAQ is saved and the chatbot uses it when members ask related questions.

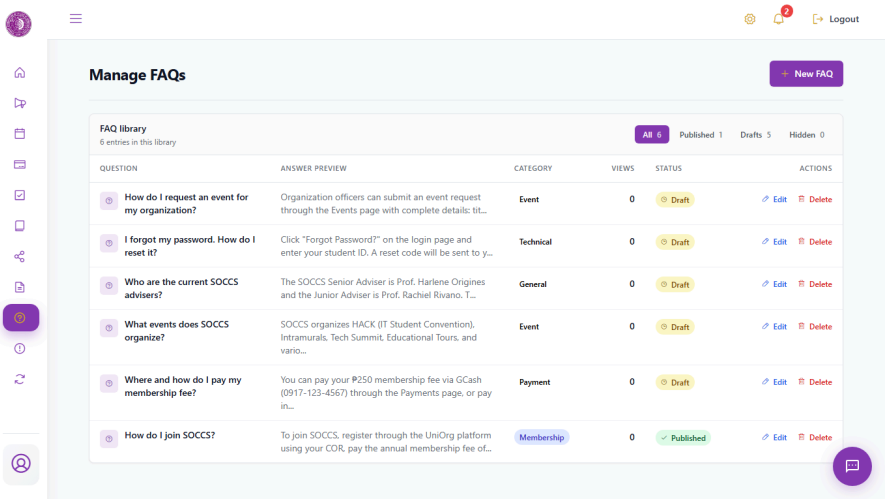


Figure 161. This figure shows the chatbot FAQs page of UniOrg.

Step 1. Go to the Chatbot or FAQ page to view all entries.
Step 2. Review the questions and answers stored.
Step 3. Add new FAQs or update existing answers.
Step 4. The chatbot uses these entries when members ask.

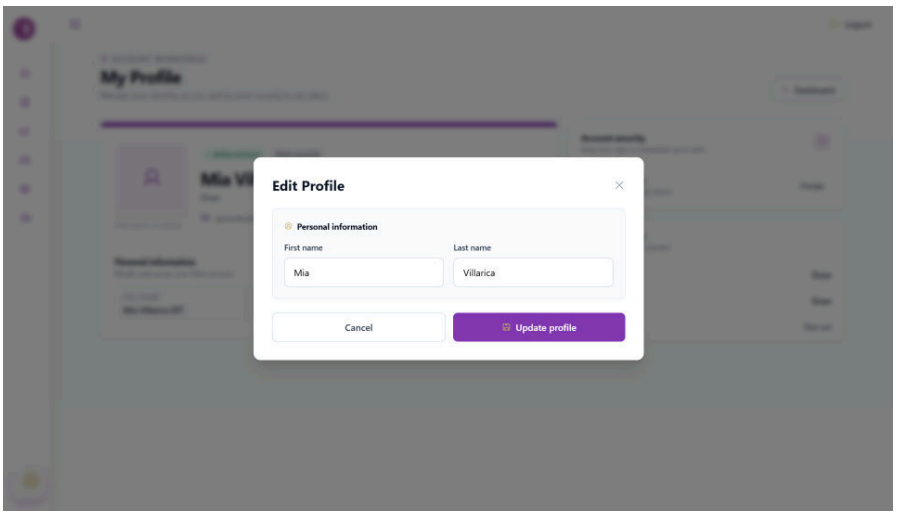


Figure 162. This figure shows the complaint response modal of UniOrg.

Step 1. Go to the Complaints module to view the list of filed complaints.
Step 2. Click a complaint to view its full details.
Step 3. Type your response or resolution in the respond modal.
Step 4. Click the Send or Resolve button.
Step 5. The complainant is notified of your response and the complaint status is updated.

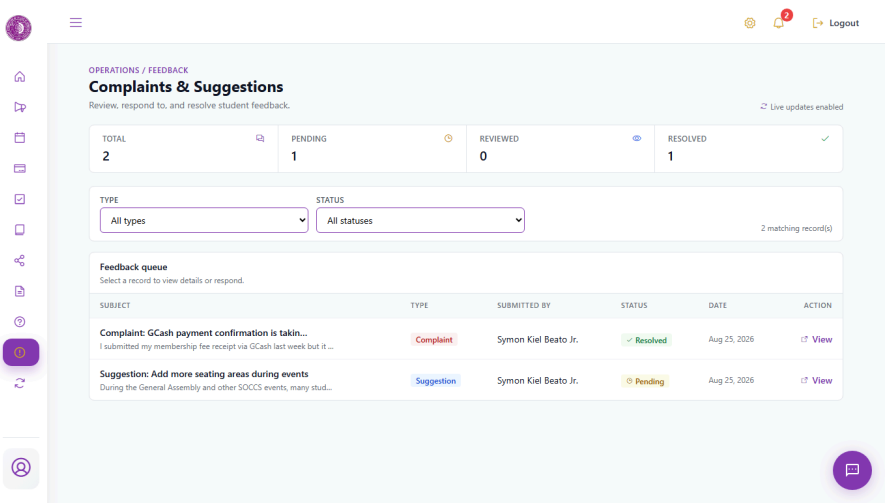


Figure 163. This figure shows the admin complaints list of UniOrg.

Step 1. Go to Complaints to view the filed complaints queue.
Step 2. Open a complaint to read its full details.
Step 3. Type your response or resolution in the respond modal.
Step 4. Send it so the complainant gets notified and the status updates.

Dean Users

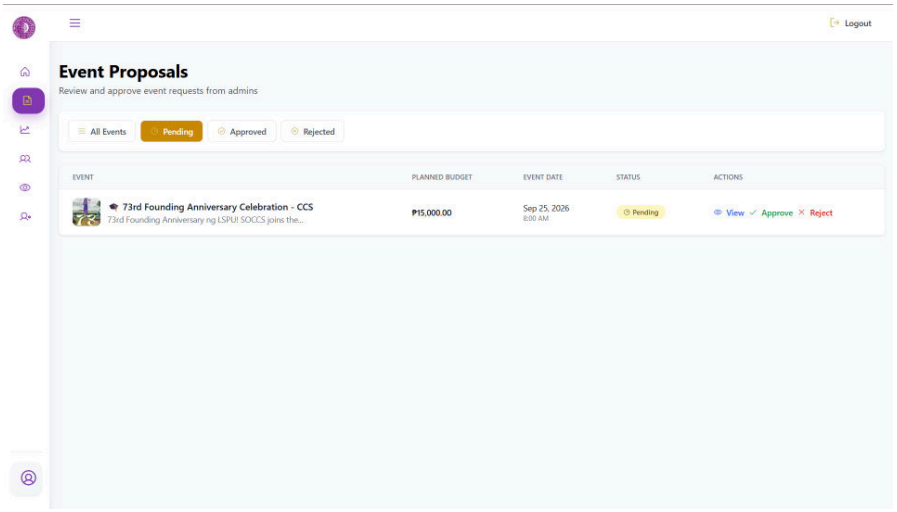


Figure 164. This figure shows the dean dashboard of UniOrg.

Step 1. Log in using your dean account to land on the dean dashboard.
Step 2. View the summary cards showing the number of organizations, members, pending proposals, and announcements in your college.
Step 3. Use the sidebar to navigate to User Management, Content Overview, Event Proposals, or Reports.
Step 4. Click any notification or pending item to go directly to the page that needs your action.

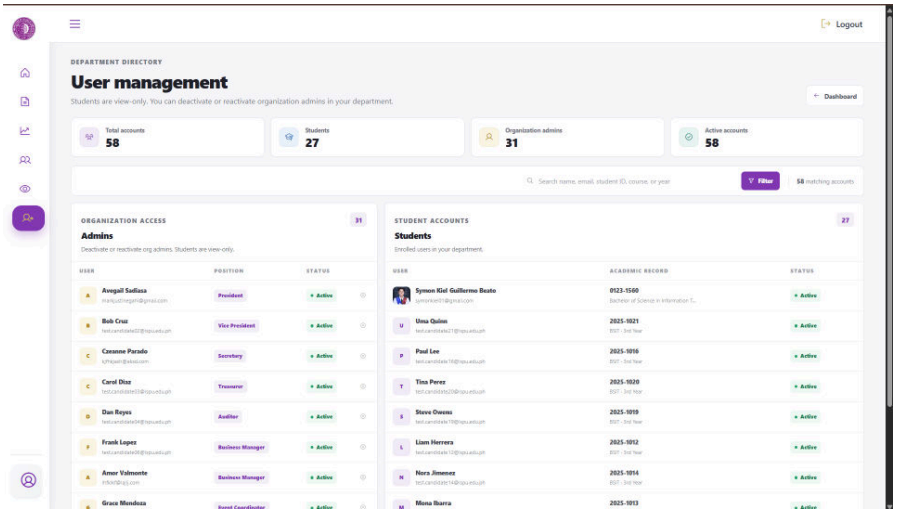


Figure 165. This figure shows the department user management list of UniOrg.

Step 1. Go to User Management to view department users.
Step 2. Search or filter to find a specific account.
Step 3. Open a user to review their profile and role.
Step 4. Update roles or deactivate accounts as needed.

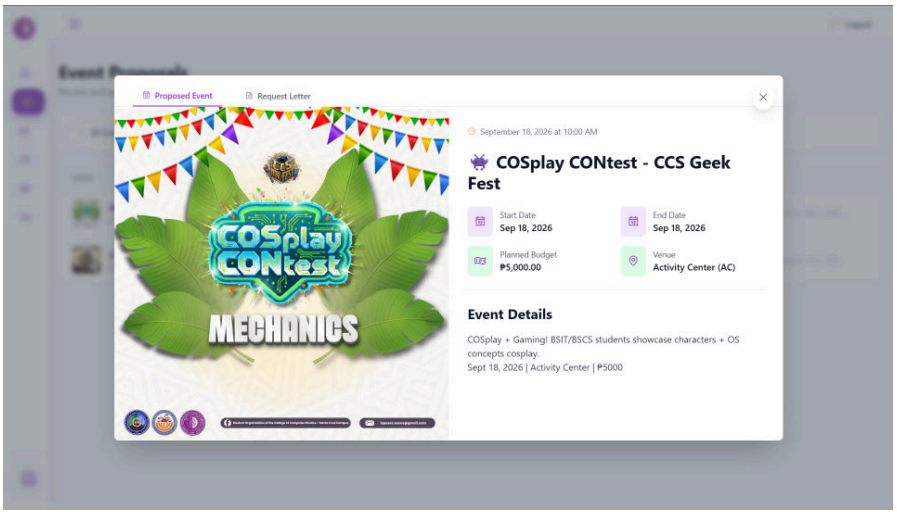


Figure 166. This figure shows the pending event proposals list of UniOrg.

Step 1. Go to the Event Proposals module and open the Pending tab.
Step 2. Review the list of event proposals submitted by the organizations for approval.
Step 3. Click a proposal to open its details and review the event information and attached documents.
Step 4. Click Approve to endorse the proposal or Reject with your remarks.
Step 5. The proposal status is updated and the organization is notified of your decision.

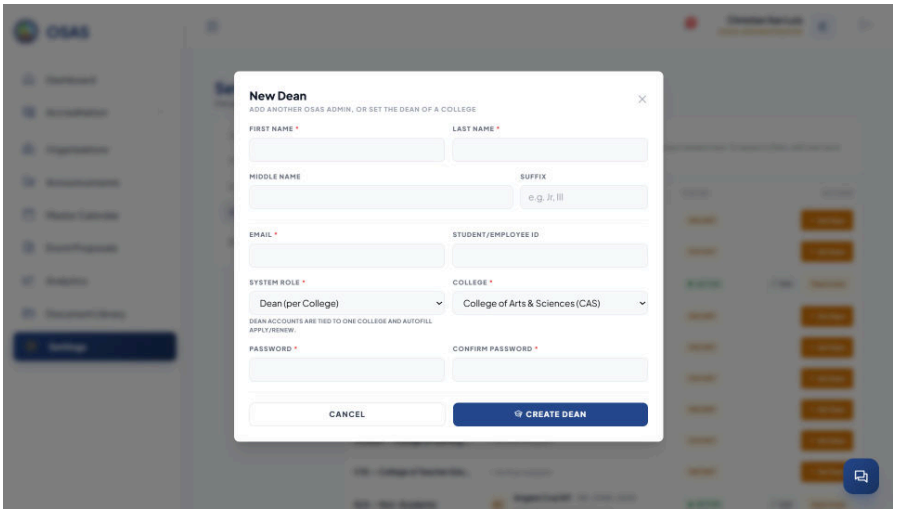


Figure 167. This figure shows the event proposal details modal of UniOrg.

Step 1. From the event proposals list, click the proposal you want to review.
Step 2. The details modal opens showing the event title, date, venue, description, and attached documents.

Step 3. Read through the proposal and verify that the details are complete and correct.
Step 4. Click the Approve button to endorse the event, or the Reject button with your remarks if revisions are needed.
Step 5. A confirmation message displays and the proposal moves to its updated status.

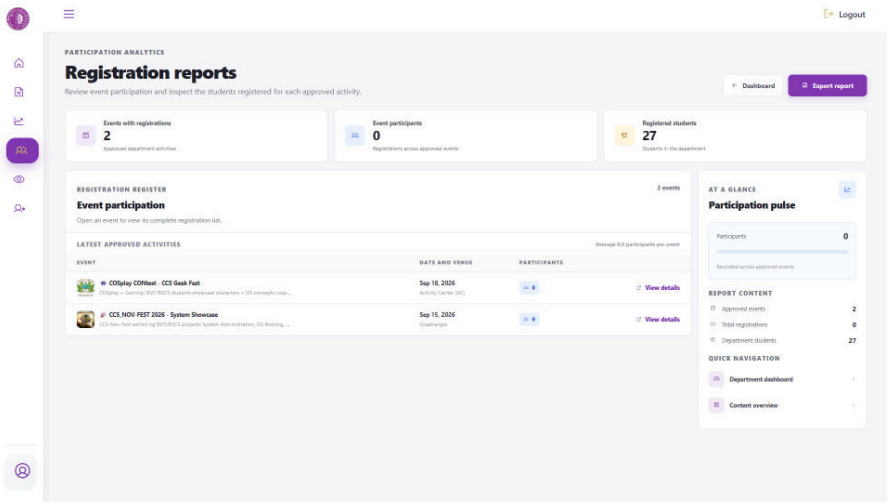


Figure 168. This figure shows the registration reports web list of UniOrg.

Step 1. Go to Reports and open the registration section.
Step 2. Select the event to view its registrants.
Step 3. Review the list shown on the page.
Step 4. Export or print the report if required.

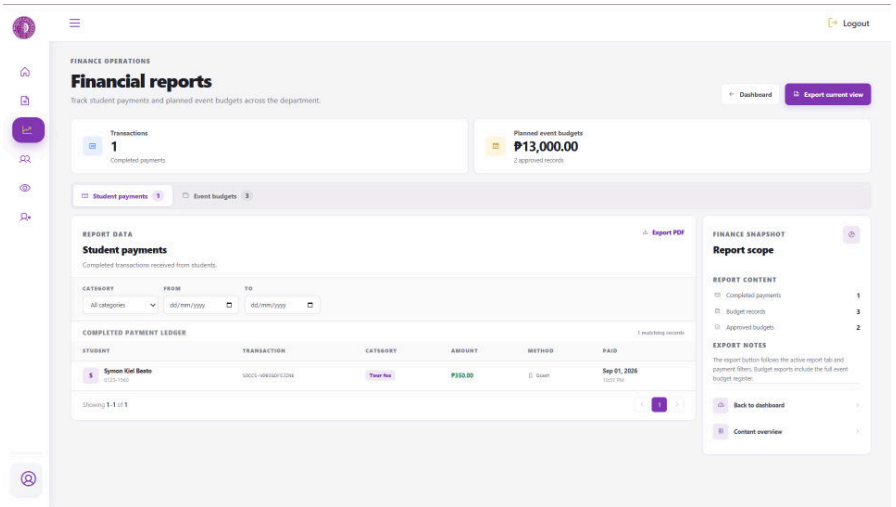


Figure 169. This figure shows the financial reports payments tab of UniOrg.

Step 1. Go to Reports and open the Finance section.
Step 2. Select the Payments tab to view collections and totals.
Step 3. Review the figures per organization or event.
Step 4. Export the summary for liquidation if needed.

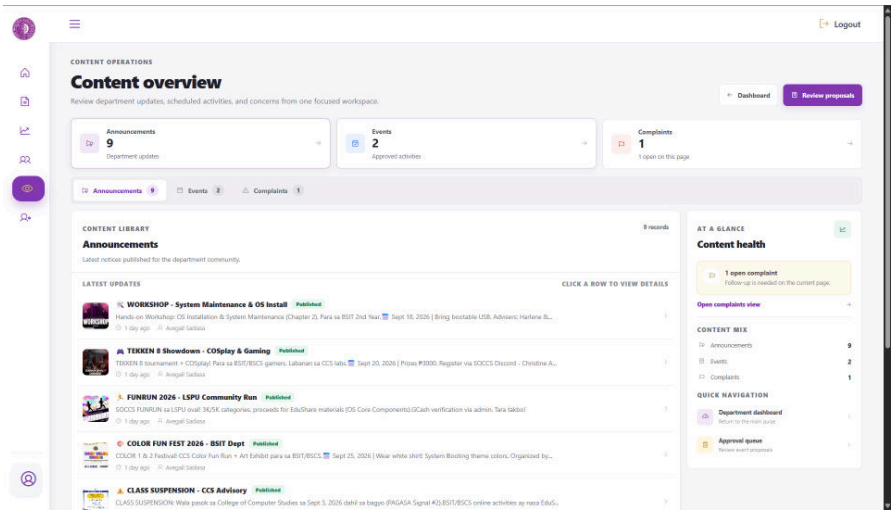


Figure 170. This figure shows the content overview announcements tab of UniOrg.

Step 1. Go to Content Overview as a dean.
Step 2. Open the Announcements tab to monitor posts.
Step 3. Review the content shared within your college.
Step 4. Flag any item that needs correction by the org.

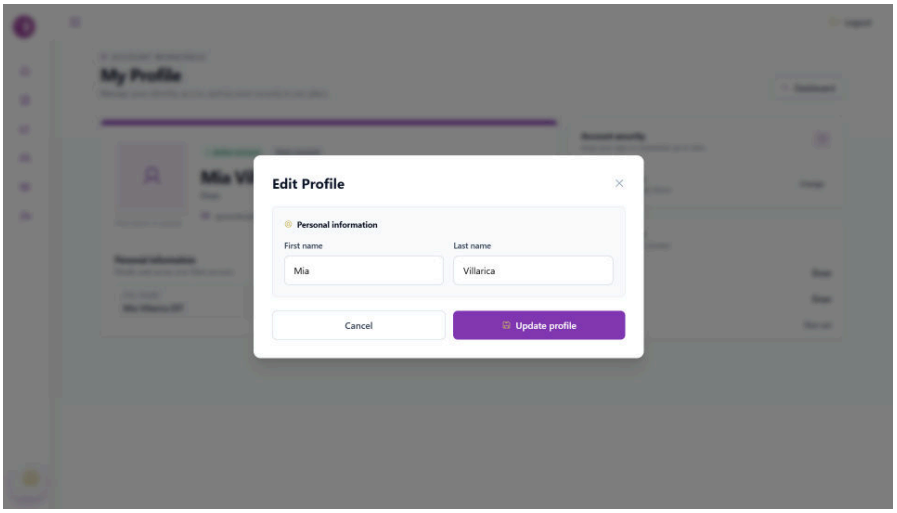


Figure 171. This figure shows the dean profile edit modal of UniOrg.

Step 1. Log in as a dean and go to your Profile page.
Step 2. Click the Edit Profile button to open the edit modal.
Step 3. Update your profile details as needed.
Step 4. Click the Save button.
Step 5. A confirmation message displays and your updated details appear on your profile.

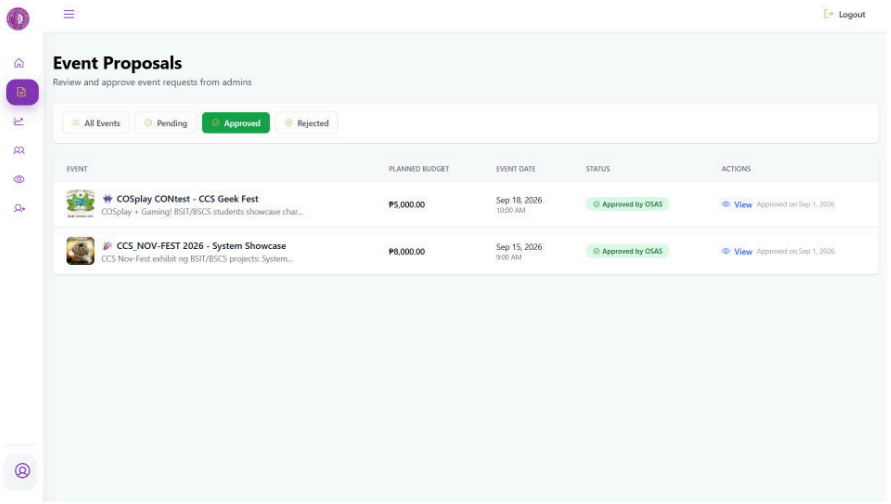


Figure 172. This figure shows the approved event proposals list of UniOrg.

Step 1. Go to Event Proposals and open the Approved tab.
Step 2. Review the proposals you already endorsed.
Step 3. Open any item to re-check its details.
Step 4. The organizations are notified of the approval.

OSAS Users

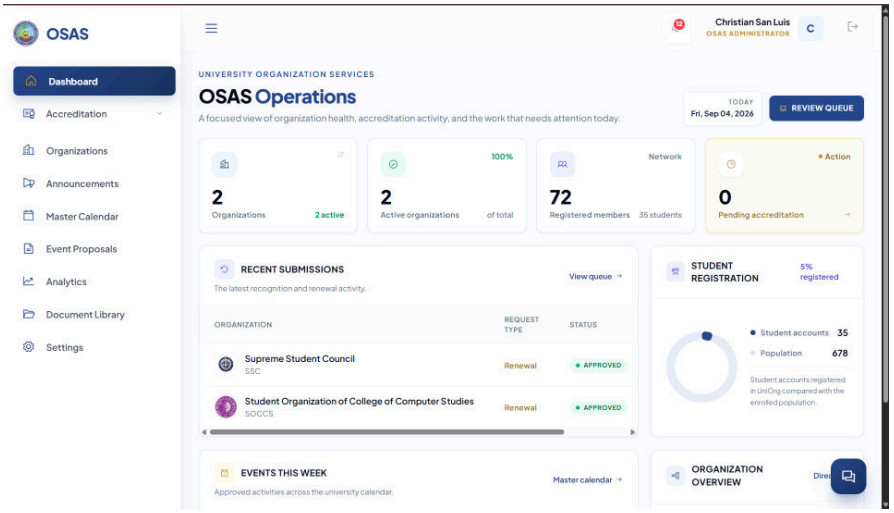


Figure 173. This figure shows the OSAS operations dashboard of UniOrg.

Step 1. Log in with your OSAS account to view the operations dashboard.
Step 2. Check the cards for accreditations, proposals, and organizations.
Step 3. Click any pending item to act on it directly.
Step 4. Use the sidebar to manage each area.

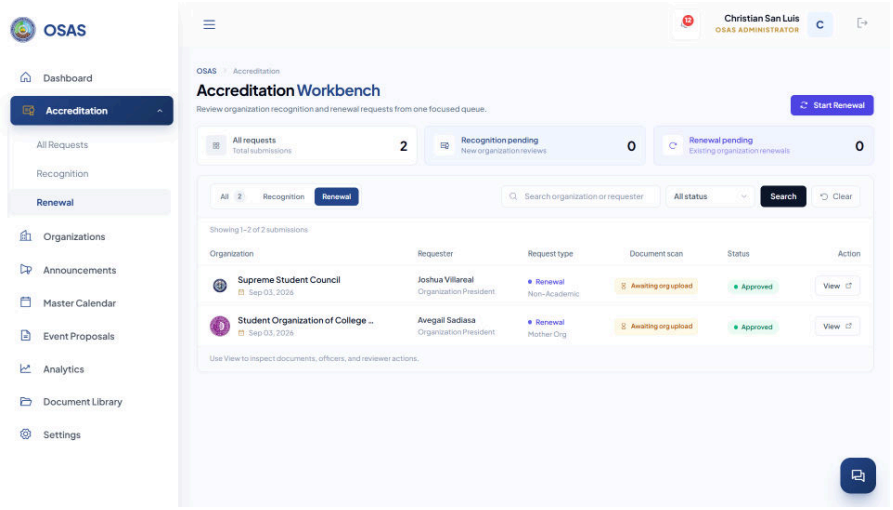


Figure 174. This figure shows the accreditation renewal list of UniOrg.

Step 1. Go to Accreditation to view renewal applications.
Step 2. Open an application to review its documents and details.
Step 3. Approve it or return it with remarks for compliance.
Step 4. The organization sees the updated status.

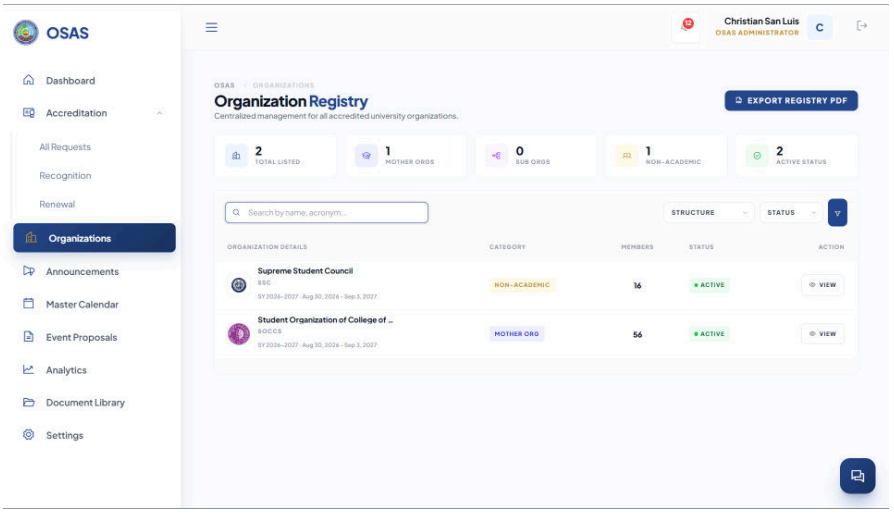


Figure 175. This figure shows the organization registry list of UniOrg.

Step 1. Go to Organizations to browse the registry.
Step 2. Search for an organization by name or college.
Step 3. Open its profile to view accreditation and details.
Step 4. Archive or update records as needed.

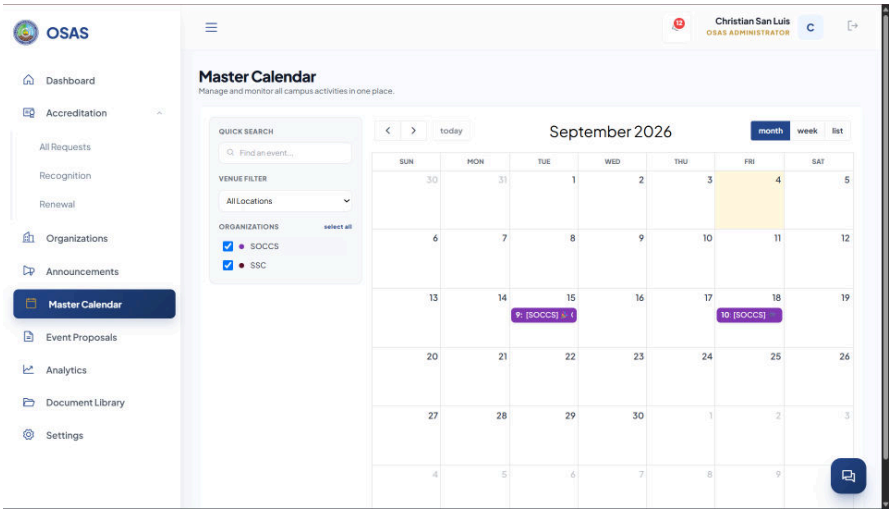


Figure 176. This figure shows the master calendar month view of UniOrg.

Step 1. Go to the Master Calendar to view all events.
Step 2. Browse the month view for university-wide activities.
Step 3. Filter by organization or date if needed.
Step 4. Click an event to view its full details.

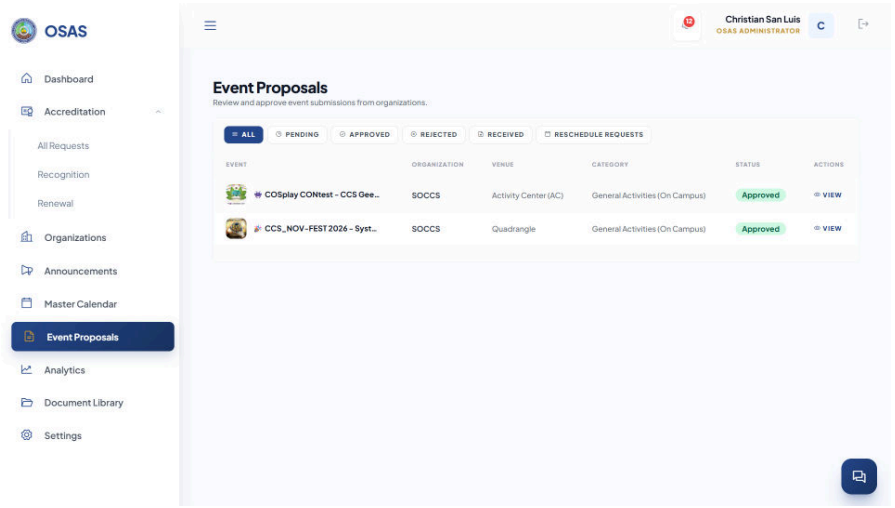


Figure 177. This figure shows the OSAS event proposals list of UniOrg.

Step 1. Go to Event Proposals to review submissions across organizations.
Step 2. Open a proposal to check its documents and details.
Step 3. Approve or return it with your remarks.
Step 4. The organization and dean are notified.

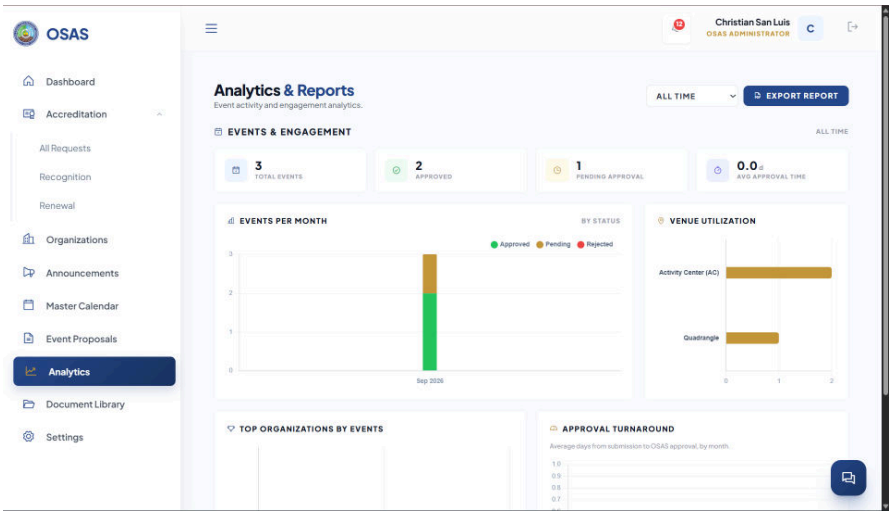


Figure 178. This figure shows the analytics and reports dashboard of UniOrg.

Step 1. Go to Analytics to view event and participation charts.
Step 2. Review the statistics shown per college or organization.
Step 3. Use the figures for planning and reports.
Step 4. Export the data if a hard copy is needed.

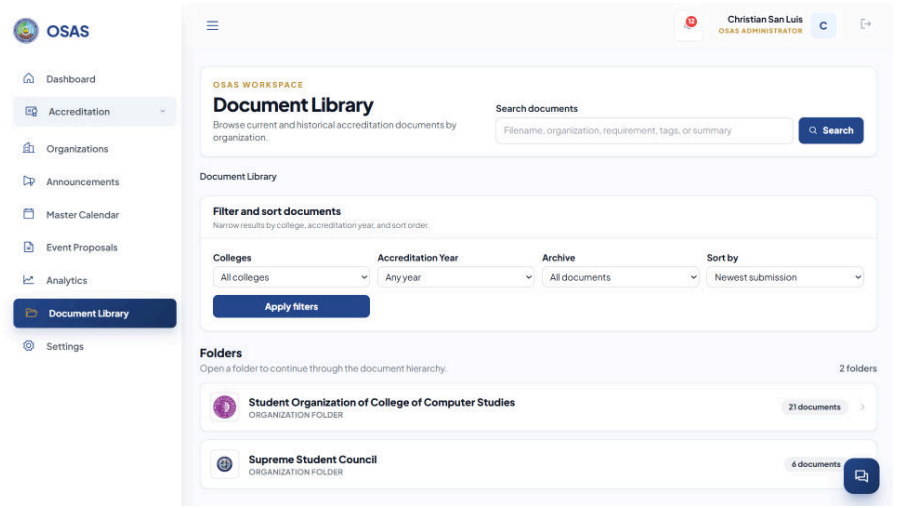


Figure 179. This figure shows the document library folders list of UniOrg.

Step 1. Go to the Document Library to browse folders per organization.
Step 2. Open a folder to view its archived documents.
Step 3. Use the AI summary and tags to find files fast.
Step 4. Download any file when needed.

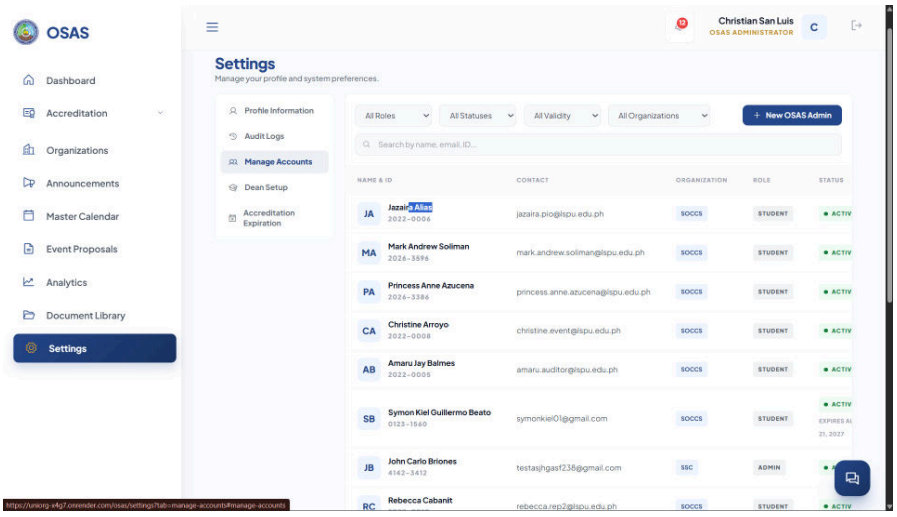


Figure 180. This figure shows the managed accounts list of UniOrg.

Step 1. Go to Settings and open Account Management.
Step 2. Browse the list of managed accounts.
Step 3. Search or filter by role or college.
Step 4. Create, update, or deactivate accounts as needed.

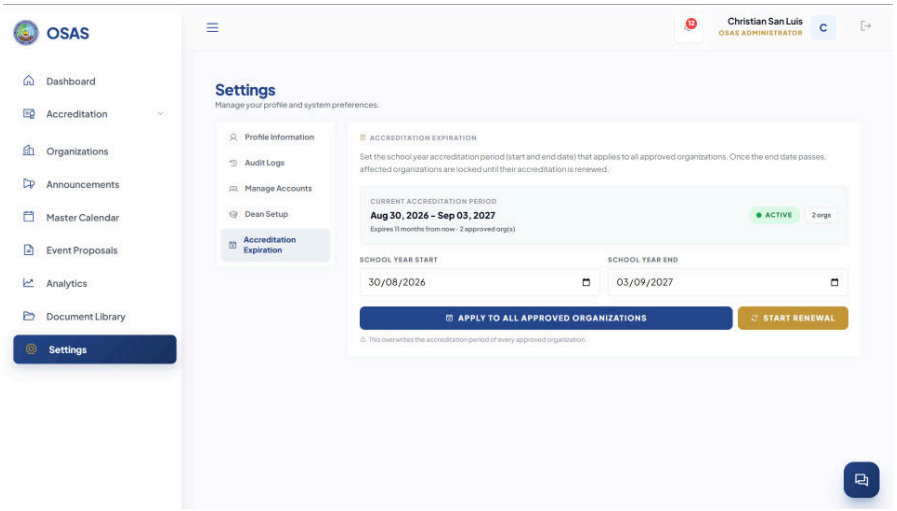


Figure 181. This figure shows the dean account creation modal of UniOrg.

Step 1. Log in as an OSAS user and go to Settings, then Account Management.
Step 2. Click the Create Dean or Add Dean button.
Step 3. Enter the dean details and assign the college.
Step 4. Click the Save button.
Step 5. The dean account is created and appears in the accounts list.

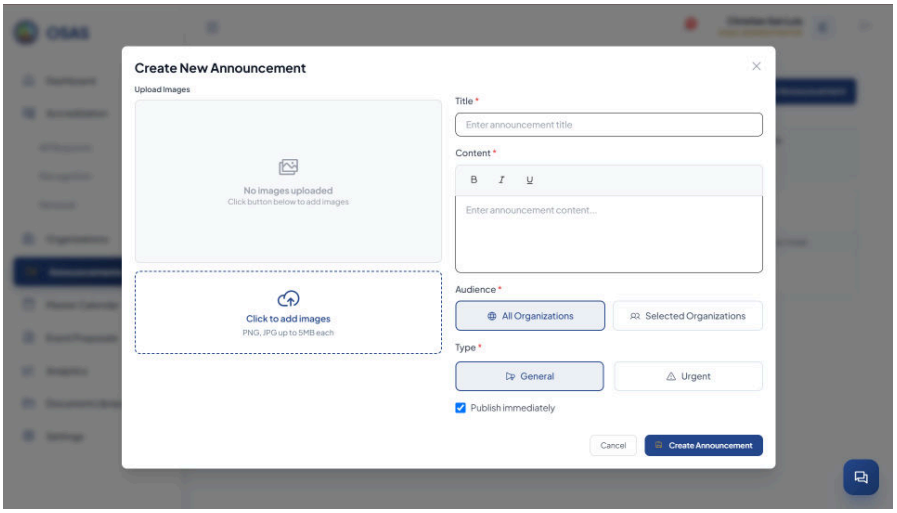


Figure 182. This figure shows the accreditation expiration period setting of UniOrg.

Step 1. Go to Settings in the OSAS panel.
Step 2. Locate the accreditation expiration period setting.
Step 3. Set the expiration period for the accreditation cycle.
Step 4. Click Apply to All and Save.
Step 5. The expiration setting is applied and is used to track the accreditation status of the organizations.

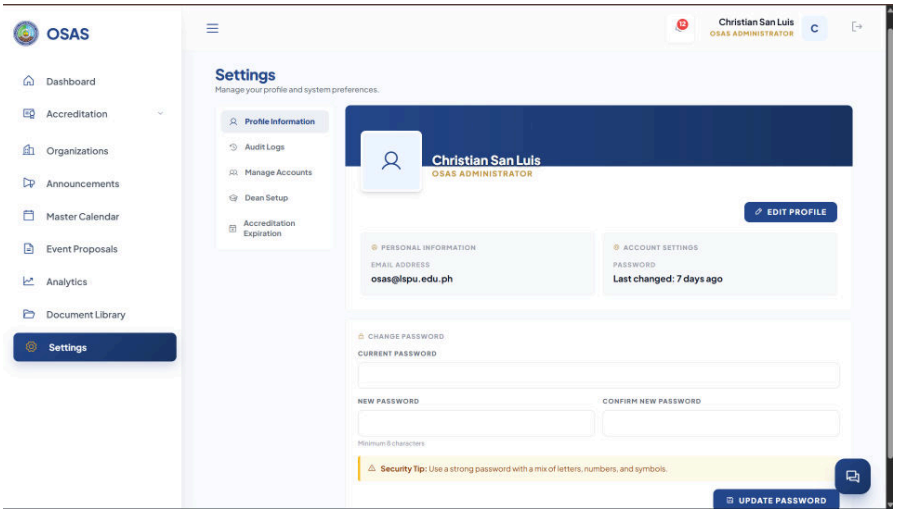


Figure 183. This figure shows the OSAS announcement creation modal of UniOrg.

Step 1. Go to the Announcements module.
Step 2. Click the Create or New Announcement button.
Step 3. Enter the announcement title, details, and target audience.
Step 4. Click the Post or Publish button.
Step 5. The announcement is posted university-wide and appears in the announcements list.

Account Settings (All Users)

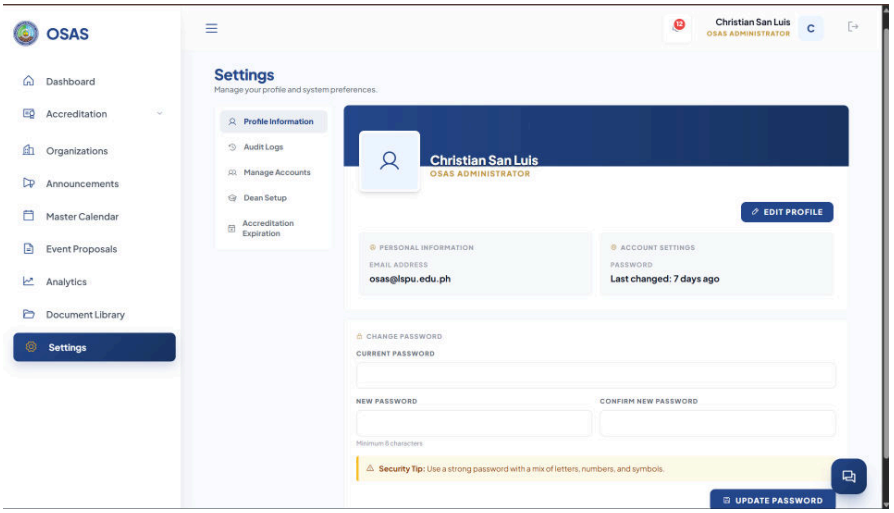


Figure 184. This figure shows the password change form of UniOrg.

Step 1. Go to your Profile or the Settings page.
Step 2. Enter your current password in the old password field.
Step 3. Enter your new password and confirm it.
Step 4. Click the Submit or Update button.
Step 5. Your password is updated and you can now use the new password when logging in.

Proof of Survey



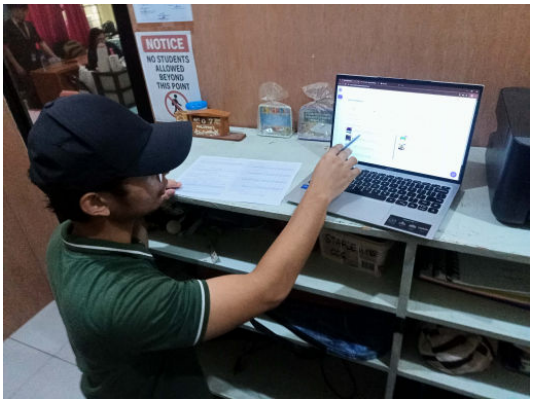
Figure 185. OSAS Survey





Figure 186. Deans Survey




NOTICE
NO STUDENTS
ALLOWED
BEYOND
THIS POINT








LAGUNA STATE POLYTECHNIC UNIVERSITY
STA. CRUZ CAMPUS


LAGUNA STATE POLYTECHNIC UNIVERSITY
STA. CRUZ CAMPUS



Figure 187. Organization advisers and Organization Officers Survey





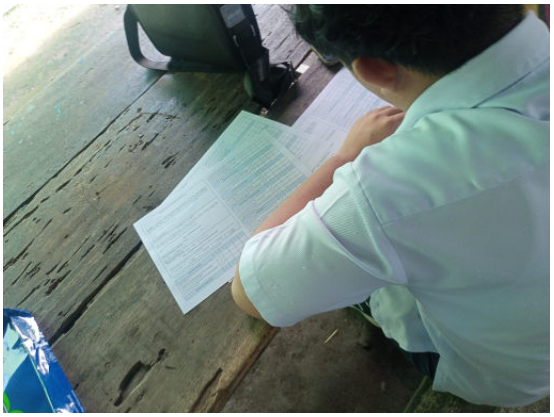






Figure 188. Students Survey

Blackbox Testing







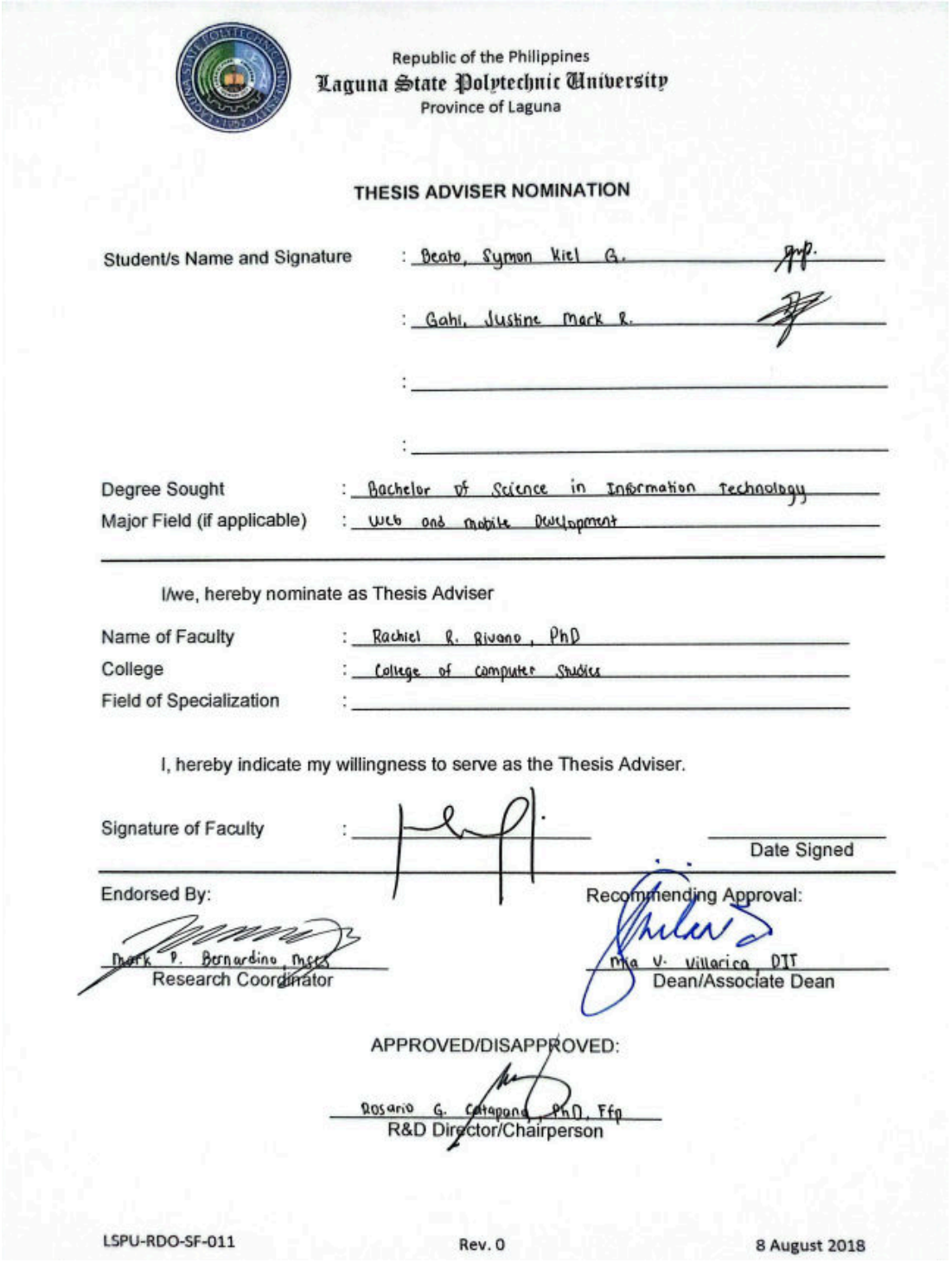

203
CEMA


203
CEMA

APPENDICES

(B. COMMUNICATION LETTERS AND FORMS)

ISO Forms



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : Beato, Symon Kiel G.

: Gahi, Justine Mark R.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : Web and Mobile Development

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Rachiel R. Rivano, PhD

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : Date Signed

Endorsed By:

Mark P. Bernardino, MSES
Research Coordinator

Recommending Approval:

Ma V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Beato, Symon Kiel G.

: Gahi, Justine Mark R.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : Web and Mobile Development

I/we, hereby nominate as Panel Member (Statistician)

Name of Faculty : Edward S. Flores, MSIT

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bermudino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villanica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FCP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Beato, Symon Kiel G.

: Gohi, Justine mark R.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : Web and mobile Development

I/we, hereby nominate as Panel Member (Technical Editor)

Name of Faculty : Reynalen C. Justo, LPT, DIT

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia U. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario C. Carapang, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Beato, Symon Kiel G.

: Gabi, Juskine Mark B.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : Web and Mobile Development

I/we, hereby nominate as Panel Member (Language Critic)

Name of Faculty : Ma. Cezanne O. Dimaculangan, MA

College : College of Arts and Sciences

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark B. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Calapoog, PhD, PEP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Beato, Symon Kiel G.

: Gahi, Justine Mark B.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : Web and Mobile Development

I/we, hereby nominate as Panel Member (System Expert)

Name of Faculty : Genes F. Masico, MSIT

College : College of Computer Studies

Field of Specialization : IT

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FEP
R&D Director/Chairperson

Request Letter

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: August 27, 2026

GREG R. REYES
OIC-Registrar
Office of the Registar
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request for the Total Student Population for the First Semester of A.Y. 2026-2027

Dear Mr. Reyes,

Warm greetings!

We are currently working on our thesis and preparing for our upcoming Final Oral Defense. As part of our data gathering and system requirements, we would like to formally request the **total student population for the First Semester of Academic Year 2026-2027** at the LSPU Santa Cruz Campus.

Thank you very much for your time, continuous support to students, and favorable response to this request.

Respectfully yours,

Beato, Symon Kiel G.
Bachelor of Science in Information Technology

Gahi, Justine Mark R.
Bachelor of Science in Information Technology

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser
College of Computer Studies

MIA V. VILLARICA, DIT
Associate Dean
College of Computer Studies

Approved by:

GREG R. REYES
OIC-Registrar

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: April 27, 2026

CHRISTIAN P. SAN LUIS, LPT, PhD
OSAS chairperson
Office of Student Affairs and Services
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request for Copies of Student Organization Bylaws *and Sample Documents*

Dear San Luis,

Warm greetings!

We are currently working on our thesis and preparing for our upcoming defense. As part of our data gathering and system requirements, we would like to formally request a copy of the **Constitutions and Bylaws of the recognized Mother Organizations** operating within the LSPU Santa Cruz Campus.

To ensure our project aligns with standard university formats, we would also like to kindly request sample copies or blank templates of the following organizational documents:

- Certification of Officers
- List of Members
- Plan of Activities
- Accreditation Form
- Renewal Form
- Adviser Commitment Form

Please be assured that these documents will be treated with utmost confidentiality and will be used strictly for academic and reference purposes only.

Thank you very much for your time, continuous support to students, and favorable response to this request.

Respectfully yours,

Beato, Symon Kiel G.
Bachelor of Science in Information Technology

Gahi, Justine Mark R.
Bachelor of Science in Information Technology

Noted by:

Rachiel R. Rivano, MSIT
Thesis Adviser
College of Computer Studies

Approved by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September __, 2026

FREDDIE S. PINUELA, EdD
Dean
College of Industrial Technology
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Dean Pinuela,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

We will personally administer the printed questionnaire within September 7–11, 2026, at your most convenient time and place. We will also briefly present and demonstrate the UNIORG system, particularly the Dean module, to provide context before answering. The entire session will take approximately 15–20 minutes. All responses shall be treated with utmost confidentiality under the Data Privacy Act of 2012 and used solely for academic purposes.

Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Freddie S. Pinuela, EdD
Dean, CIT

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September __, 2026

GLORIA N. RAMOS PhD
Dean
College of Nursing & Allied Health
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Dean Ramos,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

We will personally administer the printed questionnaire within September 7–11, 2026, at your most convenient time and place. We will also briefly present and demonstrate the UNIORG system, particularly the Dean module, to provide context before answering. The entire session will take approximately 15–20 minutes. All responses shall be treated with utmost confidentiality under the Data Privacy Act of 2012 and used solely for academic purposes.

Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours,

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Gloria N. Ramos, PhD
Dean, CONAH

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September __, 2026

DR. RANIEL LOUIE W. MENDIOLA
Dean
College of Engineering
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Dean Mendiola,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

We will personally administer the printed questionnaire within September 7–11, 2026, at your most convenient time and place. We will also briefly present and demonstrate the UNIORG system, particularly the Dean module, to provide context before answering. The entire session will take approximately 15–20 minutes. All responses shall be treated with utmost confidentiality under the Data Privacy Act of 2012 and used solely for academic purposes.

Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours,

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Dr. Raniel Louie W. Mendiola
Dean, COE

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September __, 2026

MARYJANE D. FUENTES, DPA
Dean
College of Business Administration & Accountancy
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Dean Fuentes,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

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Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours,

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Maryjane D. Fuentes, DPA
Dean, CBAA

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September 01, 2026

ENRICO R. RIVANO, PhD
Dean
College of Arts and Sciences
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Dean Rivano,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

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Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours,

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Enrico R. Rivano, PhD
Dean, CAS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: September __, 2026

RUSTOM JAY D. SANO
Associate Dean
College of Criminal Justice Education
Brgy. Bubukal Santa Cruz Laguna

SUBJECT: Request Permission to Conduct TAM Survey for Capstone Study (UNIORG)

Dear Associate Dean Sano,

Warm greetings!

We are 4th year BSIT students currently developing our capstone project entitled **"UniOrg: A Configurable University-Wide Student Organization Management Platform with OCR and NLP for Document Processing and Auto-Tagging"** in collaboration with the Office of Student Affairs and Services (OSAS) to benefit the OSAS, the college Dean offices, 39 student organizations, and the student body.

We respectfully request your permission to answer our Technology Acceptance Model (TAM) questionnaire. You are one of the nine (9) Deans selected as respondents given your oversight of student organizations under your college. While the Dean module in UNIORG is limited to approval and monitoring functions (accreditation review and event oversight), your perspective as academic head on its usefulness and acceptability is vital to our study.

We will personally administer the printed questionnaire within September 7–11, 2026, at your most convenient time and place. We will also briefly present and demonstrate the UNIORG system, particularly the Dean module, to provide context before answering. The entire session will take approximately 15–20 minutes. All responses shall be treated with utmost confidentiality under the Data Privacy Act of 2012 and used solely for academic purposes.

Thank you for your kind consideration. You may reach us at **justinemarkgahi@gmail.com** or **symonkiel01@gmail.com** for your preferred schedule.

Respectfully yours,

Beato, Symon Kiel G.
BSIT 4A

Gahi, Justine Mark R.
BSIT 4A

Noted by:

Rachiel R. Rivano, PhD
Thesis Adviser, CCS

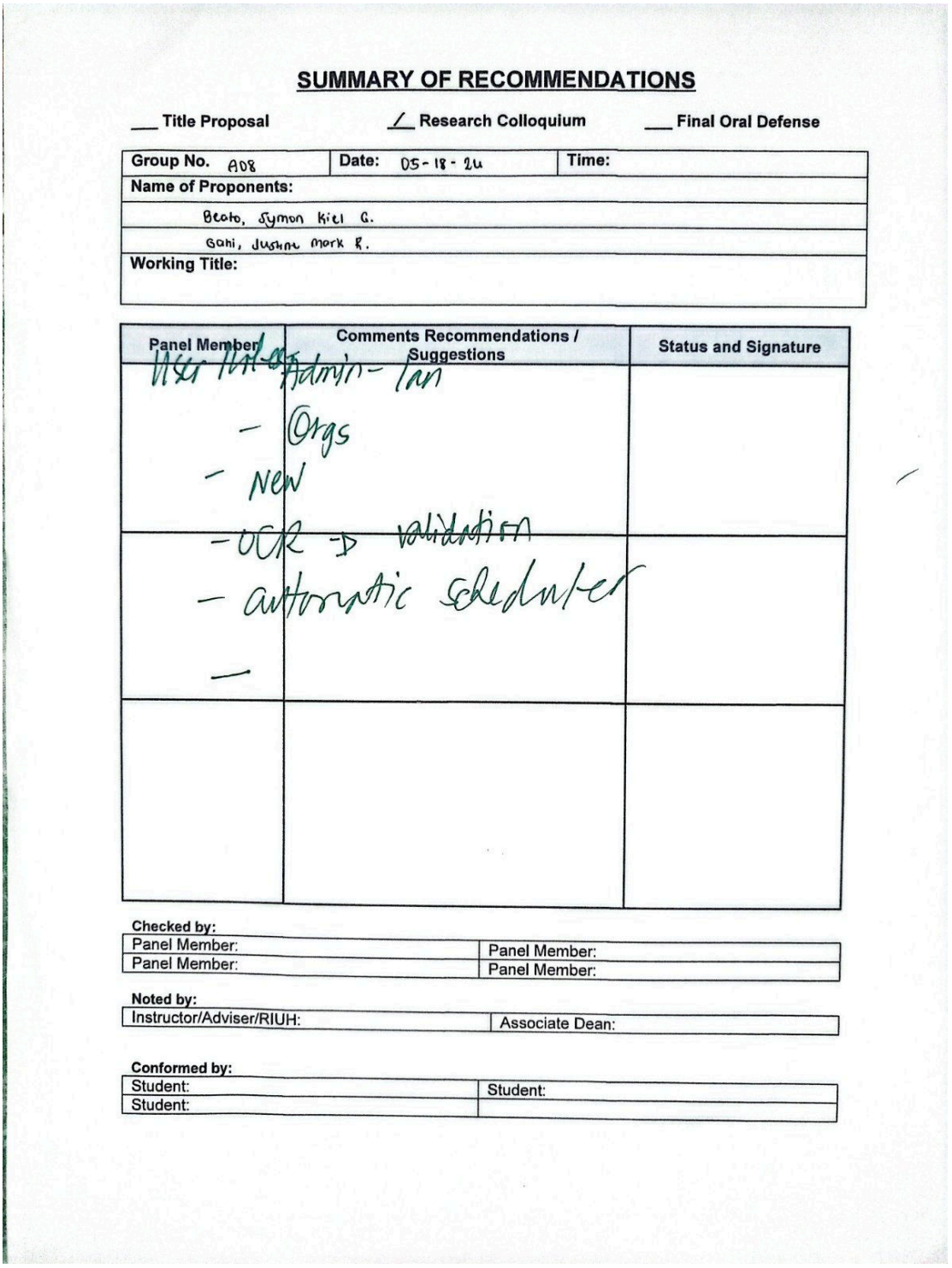
Endorsed by:

Christian P. San Luis, LPT, PhD
Chairperson, OSAS

Conforme:

Rustom Jay D. Sano
Associate Dean, CCJE

Summary of Recommendation



SUMMARY OF RECOMMENDATIONS

___ Title Proposal /_ Research Colloquium ___ Final Oral Defense

Group No. A08	Date: 05-18-24	Time:
Name of Proponents:		
Beato, Symon Kiel G.		
Gani, Joshua Mark R.		
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	Admin - Tan - Orgs - New - OCR → validation - automatic scheduler -	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

✓ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. A08	Date: March 15, 2024	Time:
Name of Proponents:	Beato, Symon kiel G.	
	Gahi, Justine Mark R.	
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	- Accreditation - Submission of FS Narratives - All activities of Org.	
	-	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

✓ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. A08	Date: March 18, 2026	Time:
Name of Proponents:	Beato, Symon Kiel G.	
	Gahi, Justine Mark R.	
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
R.J. Pielago	- Uni-ovy.	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

✓ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. A08	Date: March 18, 2026	Time:
Name of Proponents:	Beato, Symon Kiel G.	
	Gahi, Justine Mark R.	
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
G. Garcia	- follow the and consider all processes of all orgs. in ISPU-SCC - eliminate duplication of functions with the existing developed system	
	- Consult OSAS & SSC for the implementation of all orgs.	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

✓ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. A08	Date: March 18, 2026	Time:
Name of Proponents:	Beato, Symon Kiel G.	
	Gahi, Justine Mark B.	
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Reynaldo C. Ursua	1. UNI-ORG - follow the suggestion 2. uniqueness of your system	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

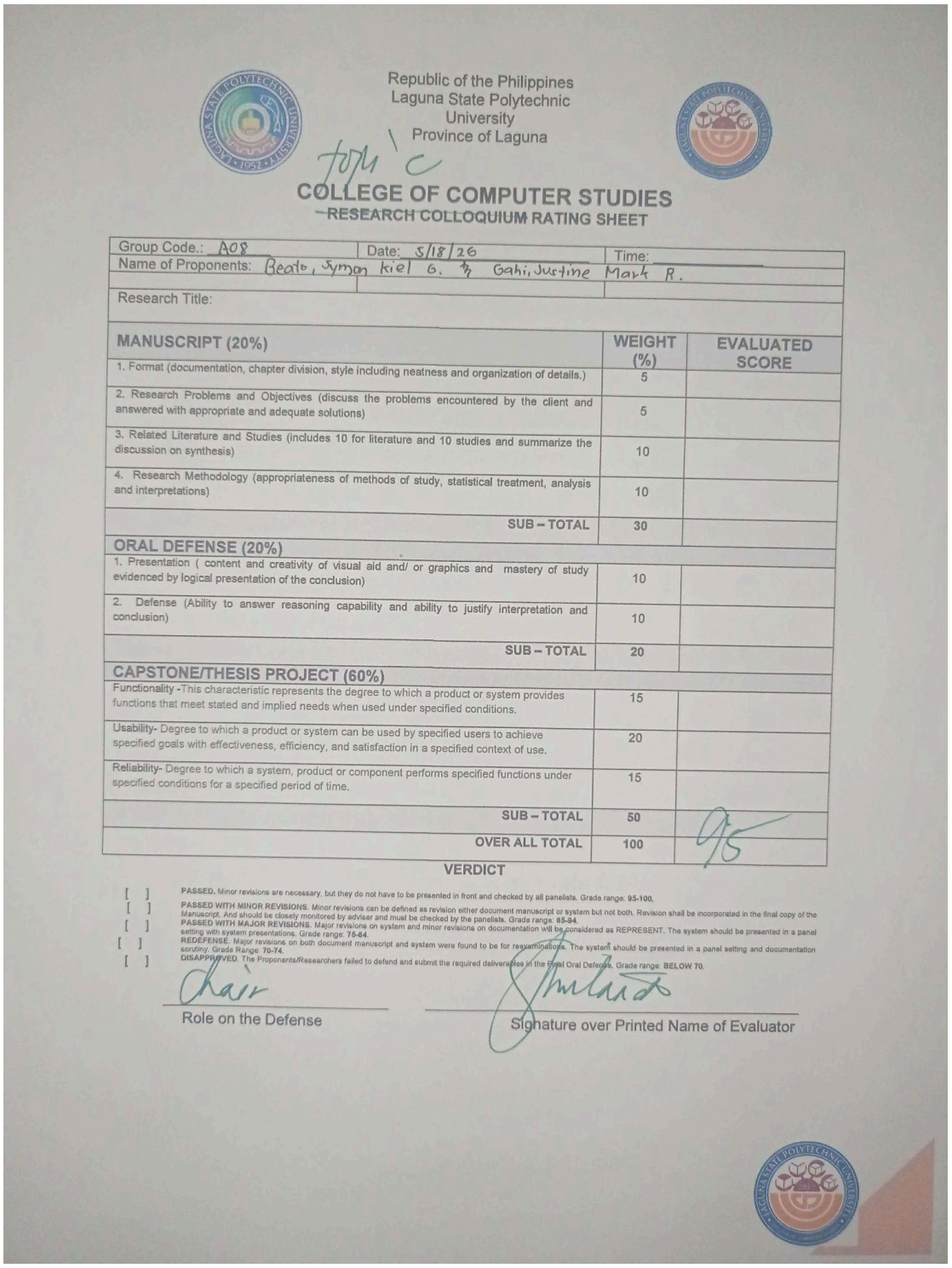
Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

Rating Sheets



Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
–RESEARCH COLLOQUIUM RATING SHEET

Group Code.: A08	Date: 5/18/26	Time:
Name of Proponents: Beato, Symon Kiel G. & Gahi, Justine Mark R.		
Research Title:		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	95

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**
[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**
[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**
[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**
[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Chair
Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: A08	Date: 05-18-2019	Time:
Name of Proponents: Beato, Symon Mel G. & Gabi, Dushne Mark R.		
Research Title:		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	83

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

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[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

SB

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: A08	Date: 5/18/26	Time:
Name of Proponents: Beato, Symon Kiel G. & Gahi, Justine Mark R.		
Research Title:		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	

VERDICT

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[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Chairman
Role on the Defense

Reynaldo G. Fuster
Signature over Printed Name of Evaluator

APPENDICES

(C. CURRICULUM VITAE OF SYSTEM TESTERS)



CHRISTIAN JAY B TANTAN

About Me

To work in a challenging surroundings demanding all my skills and efforts to explore and become accustomed myself in different fields, and realize my potential and add to the development of organization with inspiring performance.

09077554071/09684832840

tanchristianjay@gmail.com

Sitio 2 Brgy. Biñan Pagsanjan Laguna

LANGUAGE

- English
- Tagalog
- Ilocano

EXPERTISE

- Management Skills
- Graphic Design and Animation
- Website Development
- Android App Development
- Networking
- Knowledgeable in MS Office
- Computer Maintenance
- IOT Programming
- Computer Programming

EXPERIENCE

Registrar/ICT Teacher
2018-2021
Instructor -LSPU Sta Cruz Main Campus
2021-2022
IT Specialist- Phinma Education
May-August 2023
Instructor I- Laguna University
August 2022- December 2023
Part-Time Instructor -STI College Cruz
2022-2023
JNT Express Seasonal Rider
July 1-31 2023-up to present
Project Technical Assistant II(Quality Assurance)DOST-FPRDI
January 2024-up to present
Part-Time Instructor -Laguna University
Jan 2024- present

EDUCATION

Laguna State Polytechnic University Sta Cruz Laguna Main Campus
Master in Information Technology, Major in Knowledge Management Technology
ongoing 33 units earned

Laguna University
Bachelor of Science in Information Technology
2015-2018

Mangatarem National High School
2011-2012

Umangan Elementary School
2006-2007

SKILLS SUMMARY

Management Skills	100%
Graphic Design and Animation	100%
Website Development	90%
Android App Development	100%
Networking	100%
Knowledgeable in MS Office	100%
Computer Maintenance	100%
Programming-Php Laravel, Phyton.IOT.	100%



EDUCATIONAL BACKGROUND

Tertiary Education:
Bachelor of Science in Information Technology Major in Service Management Program,
Laguna State Polytechnic University
Sta.Cruz Main Campus, Sta. Cruz, Laguna
2019-2023

Secondary Education:
(Senior High School)
Laguna State Polytechnic University
Brgy. Bubukal Sta. Cruz, Laguna
2017-2019

(Junior High School)
Calumpang National High School
Brgy. Calumpang Nagcarlan, Laguna
2013-2017

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

Clydemarion P. Joya
Applicant

Clydemarion P. Joya
Brgy. Calumpang, Liliw, Laguna
Mobile Phone: 09456715001
E-Mail Address: clydemarionjoyase@gmail.com

CAREER OBJECTIVE

To obtain a challenging position where I can utilize my skills, knowledge, and experience while contributing to the growth and success of the organization and continuously developing my professional abilities.

PERSONAL INFORMATION

Age	:	25 years old
Birth date	:	September 4. 2000
Height	:	5'7
Weight	:	70 kgs.
Citizenship	:	Filipino
Civil Status	:	Single
Religion	:	Catholic
Language Spoken	:	English and Filipino
Father	:	Carmelo S. Joya
Mother	:	Marra P. Joya

TRAININGS AND SEMINAR

- **Y4IT-Youth Congress on Information Technology**
  November 31, 2019
- **Virtual Y4IT Youth Congress for Information Technology -20 Years of Inspiring future Technology Leaders**
  November 16.2022

WORK EXPERIENCE

On-The-Job Training(OJT)
IT Staff
Municipal of Liliw – Liliw Laguna
March 2023 - August 2023

Register Kit Operator
Philsys – San pablo, laguna
August 2024 – December 2024

IT Support
Synovatic Corp – Santa Cruz, Laguna
January 2024 – June 2024

Data Management staff
Dualtech Training Center Foundation Inc.
Carmelray Industrial Park 1, Canlubang, Calamba, Phillpines, 4028
March 2025 – January 2026

SKILLs

- **Computer and digitally literate**
- **Teamwork and Collaboration**
- **Multitasking**
- **Troubleshoot hardware**
- **Ability to Follow Instructions**
- **Reliability and Trustworthiness**
- **Driving**
- **Electrical**
- **Data Entry and Record Management**

JANSSEN REYES ESPINO



Committed graduate of Bachelor of Science in Information Technology in Laguna State Polytechnic University who has experience in the field of telecommunication activation and data encoding skills that I have learned in the program and internship to elevate self-growth and to contribute to future organizations. Passionate about precision and quality of work.

ACHIEVEMENTS AND PROFESSIONAL AFFILIATIONS

- Civil Service Exam Professional Passer, March 2025
- Cisco Networking Academy Operating Systems Basics - Completer, December 2024
- eLGU Admin Training and System Setup
- System Administrator in phpMyAdmin completer
- Graduated First Honorable Mention, 2015

EXPERIENCES

- **MUNICIPALITY OF SANTA CRUZ IT STAFF**
  JULY 2025-PRESENT
- **NESTLE BRAND AMBASSADOR**
  JULY 2022 - OCTOBER 2022
- **NATIONAL ELECTION SUPPORT STAFF**
  MAY 2022
- **BARANGAY ELECTION THIRD MEMBER**
  OCTOBER 2023
- **PLDT DATA ENCODER AND SERVICE ACTIVATION**
  JANUARY 2025-JUNE 2025
- **SENATORIAL ELECTION SUPPORT STAFF**
  MAY 2025

INTERNSHIP

CENTEGRA TECHNOLOGY INC. PLDT HOME DISTRIBUTOR
Banca-Banca, Victoria, Laguna

- Took internship in the company for 600 hours as data encoder and service activation.
- Validated customer's application before passing their service order to main office.
- Calling out customers to confirm the status of their internet and voice data.
- Data activation using third party apps.
- Manual service activation.

SKILLS

- **FOUNDATIONAL KNOWLEDGE IN MICROSOFT EXCEL**
- **DATA ENTRY AND ENCODING**
- **PROFICIENT IN COMPUTER APPLICATIONS**
- **ADOPTABLE TO NEW TECHNOLOGIES**
- **BASIC UNDERSTANDING OF COMPUTER MAINTENANCE AND REPAIR**
- **DETAIL ORIENTED**
- **COMMUNICATION SKILLS**
- **TIME MANAGEMENT**
- **ENTRY-LEVEL EXPERIENCE IN COMPUTER HARDWARE SERVICES**
- **COMPUTER TROUBLESHOOTING**

EDUCATION

Laguna State Polytechnic University, 2021-2025
Bachelor of Science in Information Technology

Laguna Senior High School, 2019-2021
Senior High School
Science, Technology, Engineering, and Mathematics

Pedro Guevara Memorial National High School 2015-2019
Junior High School

Bagumbayan Elementary School, 2009-2015
Grade 1-6

CONTACT INFORMATION

0975-353-7625
ssen.jre@gmail. com
Sitio 1 Bagumbayan Santa Cruz, Laguna

CHARACTER REFERENCE

Asher H. Pasco
0925-301-6826
Education Program Supervisor
San Pedro City Laguna

John Marvin A. Tabo

3192 Biya Street
Santo Angel Norte,
09754068759
itsotavs0099@gmail.com

Birthdate

May 29, 2001

EDUCATION

Santo Angel Norte Elementary School

June 2006 - April 2013

Academic Achiever (Grade 6)

Pedro Guevarra Memorial National High School (P.Gueverra Street, Sta Cruz Laguna. Junior High School)

June 2013- April 2017

Academic Achiever

Saint Therese Martin of Lisieux School and Business High School. - Sto Angel Sur, Santa Cruz, Laguna.(Information and Communication Technology)

June 2013- April 2017

Supreme Student Government President

With Honor

Laguna State Polytechnic University - Santa Cruz Campus (Bachelor of Science in Information Technology)

June 2019- July 2023

Dean`s Lister Second Semester AY 2022-2023

CSE PROFESSIONAL PASSER

SKILLS

Computer Literate

Knowledge in Microsoft Office

Hardware Configuration

Installation and Configuration

Computer Troubleshooting

AWARDS

Academic Achiever - Pedro Guevara Memorial National High School

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2017)

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2018)

Dean`s Lister - Laguna State Polytechnic University (2022)



JONAS T. SANTOS

Barangay. Jasaan, Santa Cruz, Laguna, Philippines, 4009
Jonassantos122113@gmail.com | 09675977572

- IT professional with three years of experience in technical support, system administration, and IT operations in the Municipal Government of Santa Cruz, Laguna. Skilled in troubleshooting hardware, software, printers, and network issues, providing IT support across multiple municipal offices, and maintaining systems to ensure efficient operations. Known for strong problem-solving abilities, adaptability, and delivering reliable technical solutions that improve productivity and service delivery.

Work Experience

- **MUNICIPAL GOVERNMENT OF SANTA CRUZ**

Management Information Systems office | MIS IT Staff **2025 – Present**

- Provide technical support for hardware, software, printers, and network issues across municipal offices.
- Install, configure, and maintain computer systems, peripherals, and user accounts.
- Perform system administration, preventive maintenance, and software updates.
- Troubleshoot network connectivity issues and support IT projects and system implementations.
- Maintain IT asset inventory, service records, and technical documentation.

Municipal Administrator's Office / Municipal ICT Office | Administrative Aide **2024 – 2025**

- Performed administrative and clerical functions to support daily office operations.
- Prepared and maintained records, reports, and official correspondence.
- Assisted with document processing, office coordination, and administrative support.

Municipal Government of Santa Cruz | DOLE - Government Intern **2023 – 2024**

- Municipal Administrator's Office / Municipal ICT Office

- **PROVINCIAL GOVERNMENT OF LAGUNA | ON-THE-JOB TRAINING** **2022**
  - Laguna Parole and Probation Office - R4A

Key responsibilities

- Provide technical and systems support for multiple LGU information systems, including the ***General Services Office Management System, Motorized Tricycle Operators Permit System***, and ***EBOSS Queueing System.***
- Troubleshoot and resolve system issues, perform database record updates and corrections, manage user accounts, and generate operational reports.
- Perform source code modifications, including layout adjustments (margins and page breaks), to improve system-generated reports and document formatting.
- Administer and maintain EBOSS Queueing System kiosks, ensuring reliable operation and resolving system-related concerns.
- Perform daily backups of LGU system databases to ensure data integrity, security, and disaster recovery readiness.
- Provide technical support for municipal offices by installing, maintaining, and troubleshooting computers, printers, network connectivity, and other IT equipment.
- Prepare and process Purchase Requests (PRs) for MIS office supplies, IT equipment, maintenance requirements, and other operational needs.
- Manage inventory and monitoring of office supplies, computer equipment, and IT tools to ensure availability and accountability.

Technical Skills & Abilities

- IT Technical Support
- Hardware Installation, Configuration & Troubleshooting
- Backup and recovery deleted files
- Software Installation & Maintenance
- Printer Installation & Troubleshooting
- Network Troubleshooting
- System Administration
- System User Account Creation and Management
- Operating System Installation & Configuration (Windows)
- Technical Documentation
- Computer System Maintenance
- Problem Solving & Troubleshooting
- Analytical Thinking
- Customer Service & End-User Support
- Communication Skills & Team Collaboration
- Time Management
- Attention to Detail
- Adaptability & Continuous Learning
- Organizational Skills
- Ability to Work Under Pressure

Education

- **TERTIARY LEVEL:** **2018 – 2022**
  Bachelor of Science in Information Technology
  Major in Service Management Program
  Laguna State Polytechnic University – Santa Cruz Campus
- **SENIOR HIGH SCHOOL:** **2016 – 2018**
  Science, Technology, Engineering and Mathematics – STEM
  AMA Computer Collage – Santa Cruz Campus

Civil Service Eligibility

- **Career Service Subprofessional Eligibility**
  Passed
  March 2026

Personal Information

- Age: 26 years old
- Birthdate: May 11, 2000
- Birthplace: Santa Cruz, Laguna
- Sex: Male
- Height: 4'11"
- Weight: 40kg
- Citizenship: Filipino
- Civil status: Single
- Religion: Roman Catholic
- Languages: English/Tagalog

Seminars & Training

• **Introduction to Networking**
Provincial Government of Laguna - Miso Training Room
April 30, 2026

• **E-LGU User's Training**
DICT- CALABARZON
April 23, 2026

• **Web Development (Laravel & MySQL) & Server Administration**
Provincial Government of Laguna - Miso Training Room
January 5 – 16, 2026

• **Data Analytics Essentials**
December 5 – 11, 2025
DICT-ITU DTC (Net. Acad. Program)

• **Gender Sensitivity and Safe Space Act of the Philippines**
LGU Santa Cruz
December 2, 2025

• **E-LGU Admin Training and System Set-Up**
DICT- CALABARZON
November 25 – 28, 2025

• **Google Workspace for Educators/LGU Employees**
DICT – CAR
October 27, 2025

• **Cybersecurity and PNPKI Awareness at LGU - Santa Cruz**
DICT - Calabarzon
August 11, 2025

• **Basic & Intermediate Geographic Information System (GIS) Training**
Uplink Integrated Solution Inc.
January 30 – February 1, 2025

• **City/Municipal-Level Training Coordination, and**
Field Visit of LGU on the Adoption of the Philippines Standard Industrial Classification
PHILLIPPINES STATISTICS AUTHORITY
July 6, 2024

• **Computer System Servicing Training**
Provincial Government of Laguna - Miso Training Room
October 16 – 20, 2023

• **Installing and Configuring Computer System & Introduction to Computer System Servicing**
TESDA Online Program
July 2022

Character Reference

Reference available upon request.

I hereby certify that the statement above are true and correct.



Jonas T. Santos



Kevin Adlaon

Analyst Programmer

Summary

Analyst Programmer with 10+ years of experience in analyzing and developing software solutions. Adept in working with other highly competent colleagues.

Experience

2022-11 - 2024-07

Data Scientist

Kasatria Technologies Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Built pipelines using Google Cloud Platform and Amazon Web Services
- Analyzed data and developed software solutions

2017-10 - 2022-11

Analyst Programmer

iTAc MSC Outsourcing Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Developed systems as innovative visualization tools for different data representation and analysis
- Created Google Ads using Google's platform

2016-07 - 2017-10

Systems Developer and Administrator

Management Information Systems Office, Provincial Government of Laguna, Philippines

2012-04 - 2016-07

Computer Programmer I

Management Information Systems Office, Provincial Government of Laguna, Philippines

2011-04 - 2011-05

On-the-job Training

Management Information Systems Office, Provincial Government of Laguna, Philippines

Personal Info

Address
3773 P. Zamora Street, Brgy. Santo Angel Sur, Santa Cruz, Laguna

Phone
+63 965 427 1050

E-mail
adlaon.kevin@gmail.com

Place of birth
Calamba City, Laguna, Philippines

Date of birth
1992-07-30

LinkedIn
https://linkedin.com/in/kevin-adlaon

Facebook
https://facebook.com/adlaon.kevin

Marital status
Widowed

Citizenship
Filipino

Skills

PHP
MySQL
CSS
jQuery
Ajax
JavaScript
Java
TypeScript
Python
Git
Google Analytics
Google Cloud Platform
Amazon Web Services
Salesforce Marketing Cloud

Education

2008 - 2012
ACTS Computer College
Sta. Cruz, Laguna, BS in Computer Science

2004 - 2008
Masico National High School
Pila, Laguna, Salutatorian

2004 - 1998
Mangumit Elementary School
Calamba City, Laguna

Certificates

- Google Cloud Certified Professional Data Engineer
- Salesforce Marketing Cloud Developer
- Salesforce Marketing Cloud Email Specialist

Languages

Filipino

English

References

Adoree Ramos
Professor
ACTS Computer College
Sta. Cruz, Laguna
(049) 501 1484

Shing Nien Fong
Director
Research & Development
Kasatria Technologies Sdn Bhd
+60 16 261 6060



MHARJUN V. TOBIAS

Graphic Artist / Technical Support

EDUCATION

2008 - 2013
BS INFORMATION TECHNOLOGY
Laguna State Polytechnic University

2004 - 2008
SECONDARY EDUCATION
Pedro Guevara Memorial National High School

SKILLS

- Networking Fundamentals
- Hardware Diagnostics
- Graphic Arts
- Microsoft Office Savvy
- GIS Mapping
- Technical Support

CONTACT

09171302395
tobiasmharjun666@gmail.com
Patimbao, Santa Cruz, Laguna
www.reallygreatsite.com

PROFILE INFO

Highly skilled professional with extensive experience in graphic design, technical support, and administrative operations. Proven ability to handle complex networking tasks and digital asset creation with precision and efficiency. Dedicated to delivering high-quality visual solutions and reliable technical troubleshooting.

EXPERIENCE

2025 - Present
CASUAL CLERK
MIS Office - MGO Santa Cruz, Laguna
Handled management information systems and administrative support for the local government unit.

2023 - 2025
CASUAL CLERK
Assessor's Office - MGO Santa Cruz, Laguna
Assisted in data management and office operations for tax assessment and land records.

2020 - 2022
GRAPHIC ARTIST / TECH SUPPORT
Mayor's Office - MGO Santa Cruz, Laguna
Provided digital creative solutions and technical maintenance for municipal operations.

2019 - 2019
ADMINISTRATIVE AIDE
PIO - MGO Santa Cruz, Laguna
Supported public information dissemination and media relations.

ACHIEVEMENTS

2022
CIVIL SERVICE PROFESSIONAL
Passed with a rating of 80.56.

2019
CIVIL SERVICE SUB-PROFESSIONAL
Passed with a rating of 82.26.



WILMAR T. MOROTA

09202603909

Lynville 8, Bagumbayan, Sta Cruz, Laguna

EDUCATION

College

AMA Computer College

Computer Programming – Under Graduate

2007–2012

Secondary

Pedro Guevarra Memorial High School

Sta Cruz, Laguna

2003–2007

Primary

Our Lady of Maulawin

1998–2003

REFERENCE

RYAN R. DUÑGO
09166738313
Bagumbayan, Sta Cruz, Laguna

JUNE DAE L. MORGA
09686058086
Bubukal, Sta Cruz, Laguna

PROFILE

Birthdate : December 8, 1990
Place of Birth : Sta Cruz, Laguna, Philippines
Civil Status : Married
Religion : Roman Catholic

WORK EXPERIENCE

Management Information Systems Office — **2025 - Present**

Job–Order

Economic Enterprice Management Office — **2022 - 2025**

Job–Order

- Daily submission of Collection Report of Market Collectors
- Daily Printout of Market Collector Reciepts

Freelance Encoder — **2017-2020**

OTHER ATTAINMENTS

CAREER SERVICE EXAMINATION PROFESSIONAL PASSER

MAY 2024



ROBIN PAOLO BONA

Bachelor of Science in Information Technology

To utilize experience and qualification in different and challenging position that professionally growth and enhance my future opportunities.

EDUCATION

2018

UNION COLLEGE OF LAGUNA

- Bachelor of Science in Information Technology

CERTIFICATE

2022

NATIONAL CERTIFICATION II

- Computer System Servicing NC II

SKILL

- Troubleshoot 100%
- Technical Support 90%
- Microsoft Office 100%
- Creative 90%
- Team Work 95%

EXPERIENCE

2019 - PRESENT

Municipal Government Santa Cruz, Laguna

Administrative Aide

- Information Technology

2018 - 2019

DEP-ED LAGUNA

Administrative Aide

- Information Technology

2018

SM Calamba

Stock Clerk

- Warehouse Stock Clerk

CONTACT

09650499411

Hushbona@gmail.com

229 Jamaica Subdivision, Santo Angel Central, Santa Cruz, Laguna

REFERENCE

BRYAN BOUVIER BONZA
Municipal Government Santa Cruz, Laguna
HRMO V
09177059405

MELVIN O. BONZA
Municipal Government Santa Cruz, Laguna
ADMINISTRATOR
09967079115



ASPIRING QA ENGINEER |
WEB DEVELOPER |
IT SUPPORT TECHNICIAN

Contact

+63 976 238 5693

jhayjustinlopez@gmail.com

Santa Cruz, Laguna, 4009, Phillipines

Skills

- Leadership & Team Collaboration
- Computer Literacy (MS Office, Email, Digital Filing)
- Visual Design & Editing
- Photo & Video Editing
- Basic UI/UX & Web Content Management

Language

- English
- Filipino

Justine Jhay Lopez

Education

(2022 - 2026)
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bachelor of Science in Information Technology
1st - 4th Year College

(2020 - 2022)
LAGUNA SENIOR HIGH SCHOOL
Science, Technology, Engineering, and Mathematics strand
Grade 11 - Grade 12

(2016 - 2019)
PEDRO GUEVARA MEMORIAL NATIONAL HIGH SCHOOL
Grade 7 - Grade 10

Experience

(2025 - Present)
MANAGEMENT INFORMATION SYSTEMS OFFICE
- MIS Programmer

(2024 - 2026)
MUNICIPAL SOCIAL WORKER AND DEVELOPMENT OFFICE
- Data Entry Clerk

Projects

- **IKNOWLGRC - KNOWLEDGE MANAGEMENT SYSTEM**
  -2026 (Full Stack Developer)
- **EXELENCIA ACADEMY PORTAL**
  -2024 (Full Stack Developer)
- **SHE WEAR (E-COMMERCE WEB)**
  -2024 (Programmer)
- **BARANAGAY RECORDING SYSTEM**
  -2024 (Full Stack Developer)
- **JAMAICA RECORDING SYSTEM**
  -2023 (Full Stack Developer)

APPENDICES

(D. CURRICULUM VITAE OF RESEARCHERS)



JUSTINE MARK GAHI

WEB DEVELOPER

AWARDS AND MEMBERSHIPS

- With honors
- Scholar
- November hunt second place
- Special Program for Employment of Students

PROFILE

Jack of all trades with a passion for web development and UI/UX design. A computer-savvy team player focused on mastering impactful solutions.

EDUCATION

2001 - 2004
Elementary
Cavinti Elementary School

1997 - 2000
Senior High-School
Cavinti Integrated National High School

SKILLS

- Software Troubleshooting
- Problem-Solving
- Time Management
- Programming
- Communication and teamwork
- Detail Orientation

LANGUAGES AND TECH

Python, Javascript, C, Java, CSS, HTML

CONTACT

+123-456-7890
justinemarkgahi@gmail.com
0123-1507
Jacinto St. Cavinti Laguna



SYMON KIEL BEATO

APP DEVELOPER

AWARDS AND MEMBERSHIPS

- With honors
- Scholar
- MPM Club

PROFILE

I am a problem-solver who thrives on teamwork and innovation, blending design, tools, and programming to craft user-centered, impactful solutions.

EDUCATION

2011 - 2017
Elementary
Bagumbayan Elementary School

2017 - 2023
Senior-High School
Liceo De Pila

SKILLS

- Mobile-Specific: Kotlin/Java (Android)
- Backend Development
- Problem-Solving
- Communication
- Critical Thinking
- Database Management

LANGUAGES AND TECH

Python, Java, C, Android Studio

CONTACT AND DETAILS

+123-456-7890
SYMONKIEL01@GMAIL.COM
0123-1560
Brgy. Labuin, Sta. Cruz, Laguna